Professional Data Engineer

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No job outcome is promised.

Question 1

You are designing the architecture of your application to store data in Cloud Storage. Your application consists of pipelines that read data from a Cloud Storage bucket that contains raw data, and write the data to a second bucket after processing. You want to design an architecture with Cloud Storage resources that are capable of being resilient if a Google Cloud regional failure occurs. You want to minimize the recovery point objective (RPO) if a failure occurs, with no impact on applications that use the stored data. What should you do?

Options

A. Adopt multi-regional Cloud Storage buckets in your architecture.
B. Adopt two regional Cloud Storage buckets, and update your application to write the output on both buckets.
C. Adopt a dual-region Cloud Storage bucket, and enable turbo replication in your architecture.
D. Adopt two regional Cloud Storage buckets, and create a daily task to copy from one bucket to the other.
E. None of the above

Answer

C (the correct option letter — shown in bold)

Detailed explanation

A. Incorrect — Multi-regional buckets provide geo-redundancy, but they offer broader geographic placement and typically higher cost; a dual-region with turbo replication is a more targeted, lower-latency option for minimized RPO with predictable placement.
B. Incorrect — Writing to two separate regional buckets from the application increases complexity (application-level dual writes) and potential consistency issues; it's more effort and risk than managed replication.
C. Correct — A dual-region bucket with turbo replication provides managed cross-region replication within the selected dual-region pair, minimizes RPO, and avoids requiring the application to manage dual writes or manual copy jobs. This keeps application impact minimal and recovery fast.
D. Incorrect — A daily copy produces an RPO measured in days (not minimal) and risks data loss; it doesn’t meet the minimal-RPO requirement.
E. Incorrect — Option C is the best practice for this requirement.

Question 2

You have designed an Apache Beam processing pipeline that reads from a Pub/Sub topic. The topic has a message retention duration of one day, and writes to a Cloud Storage bucket. You need to select a bucket location and processing strategy to prevent data loss in case of a regional outage with an RPO of 15 minutes. What should you do?

Options

A. 1. Use a dual-region Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 15 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region.
B. 1. Use a multi-regional Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 60 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region.
C. 1. Use a regional Cloud Storage bucket.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by one day to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region and write in a bucket in the same region.
D. 1. Use a dual-region Cloud Storage bucket with turbo replication enabled.
2. Monitor Dataflow metrics with Cloud Monitoring to determine when an outage occurs.
3. Seek the subscription back in time by 60 minutes to recover the acknowledged messages.
4. Start the Dataflow job in a secondary region.
E. None of the above

Answer

D

Detailed explanation

A. Incorrect — While a dual-region bucket helps, seeking back only 15 minutes may be unreliable depending on the Pub/Sub ack timeline and the need to recover messages beyond that window; also the question's recommended answer is D which uses turbo replication and a 60-minute seek.
B. Incorrect — Multi-regional buckets are broader and may not align with the desired dual-region locality; also the recommended seek interval is different.
C. Incorrect — A regional bucket risks data loss during regional outage; seeking back one day may exceed Pub/Sub subscription retention and is not aligned to the RPO objective.
D. Correct — Using a dual-region bucket with turbo replication (fast replication) preserves writes across regions, monitoring detects outage, seeking back (60 minutes) provides a conservative window to recover acknowledged messages, and starting Dataflow in a secondary region with the dual-region bucket reduces RPO and prevents data loss.
E. Incorrect — Option D best meets RPO=15-minutes resilience goals with low operational impact.

Question 3

You are preparing data that your machine learning team will use to train a model using BigQuery ML. They want to predict the price per square foot of real estate. The training data has a column for the price and a column for the number of square feet. Another feature column called 'feature1' contains null values due to missing data. You want to replace the nulls with zeros to keep more data points. Which query should you use?

A.

```
SELECT * EXCEPT(feature1),
  IFNULL(feature1, 0) AS feature1_cleaned
FROM training_data;
```

B.

```
SELECT * EXCEPT(price, square_feet),
  price/square_feet AS price_per_sqft
FROM training_data
WHERE feature1 IS NOT NULL;
```

C.

```
SELECT * EXCEPT(price, square_feet, feature1),
  price/square_feet AS price_per_sqft,
  IFNULL(feature1, 0) AS feature1_cleaned
FROM training_data;
```

D.

```
SELECT *
FROM training_data
WHERE feature1 IS NOT NULL;
```

Answer: A

Options

A. Use COALESCE or IFNULL to replace NULLs, for example: SELECT price, sq_ft, COALESCE(feature1, 0) AS feature1 FROM dataset.table;
B. Use DELETE to remove rows with NULL feature1 before training.
C. Use an INNER JOIN with a lookup table that has zeros for missing feature1.
D. Leave NULLs as-is and rely on BigQuery ML default NULL handling.

Answer

A

Detailed explanation

A. Correct — COALESCE(feature1, 0) (or IFNULL(feature1, 0)) replaces NULLs with 0 at query time, preserving rows and keeping more training examples; this is the standard, simple approach.
B. Incorrect — Deleting rows with NULLs discards data and reduces training size and potential model generalization.
C. Incorrect — Joining to a lookup table to replace NULLs adds unnecessary complexity; COALESCE is simpler and more efficient.

D. Incorrect — Leaving NULLs may cause unwanted behavior depending on the model and feature type; explicitly replacing with 0 ensures consistent numeric input.

Question 4

Different teams in your organization store customer and performance data in BigQuery. Each team needs to keep full control of their collected data, be able to query data within their projects, and be able to exchange their data with other teams. You need to implement an organization-wide solution, while minimizing operational tasks and costs. What should you do?

Options

A. Ask each team to create authorized views of their data. Grant the biquery.jobUser role to each team.
B. Create a BigQuery scheduled query to replicate all customer data into team projects.
C. Ask each team to publish their data in Analytics Hub. Direct the other teams to subscribe to them.
D. Enable each team to create materialized views of the data they need to access in their projects.
E. None of the above

Answer

C

Detailed explanation

A. Incorrect — Authorized views work for sharing but put the burden on central management and don't provide a discoverable, self-service exchange mechanism across many teams.
B. Incorrect — Replicating all data into every team project increases storage/cost and operational overhead.
C. Correct — Analytics Hub enables teams to publish and share datasets in a discoverable, self-service way while allowing each team to control their own data — minimizing operational overhead and cost compared to mass replication.
D. Incorrect — Materialized views are useful for performance but not an organization-wide, discoverable data-exchange mechanism.
E. Incorrect — Option C is the recommended pattern.

Question 5

You are developing a model to identify the factors that lead to sales conversions for your customers. You have completed processing your data. You want to continue through the model development lifecycle. What should you do next?

Options

A. Use your model to run predictions on fresh customer input data.
B. Monitor your model performance, and make any adjustments needed.
C. Delineate what data will be used for testing and what will be used for training the model.
D. Test and evaluate your model on your curated data to determine how well the model performs.
E. None of the above

Answer

C

Detailed explanation
A. Incorrect — Running predictions is a later step after you have properly trained and validated the model.
B. Incorrect — Monitoring happens after deployment; you must train and evaluate first.
C. Correct — Before training, you should split the data into training and testing (and validation) sets — delineating test vs training data is a fundamental next step in the modelling lifecycle to avoid leakage and evaluate performance.
D. Incorrect — Testing/evaluation happens after training; but you must first define the training/test split (option C) before training and evaluation.
E. Incorrect — Option C is the appropriate next action.

Question 6
You have one BigQuery dataset which includes customers' street addresses. You want to retrieve all occurrences of street addresses from the dataset. What should you do?

Options
A. Write a SQL query in BigQuery by using REGEXP_CONTAINS on all tables in your dataset to find rows where the word "street" appears.
B. Create a deep inspection job on each table in your dataset with Cloud Data Loss Prevention and create an inspection template that includes the STREET_ADDRESS infoType.
C. Create a discovery scan configuration on your organization with Cloud Data Loss Prevention and create an inspection template that includes the STREET_ADDRESS infoType.
D. Create a de-identification job in Cloud Data Loss Prevention and use the masking transformation.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — REGEXP searches are brittle (miss variations, international formats) and labor-intensive across many tables; DLP provides robust detectors for STREET_ADDRESS.
B. Correct — Creating a deep inspection job on each table with Cloud DLP using the STREET_ADDRESS infoType identifies addresses accurately; the inspection template standardizes the detection.
C. Incorrect — A discovery scan at org level is broader and more administrative; per-table deep inspection is the direct approach to retrieve occurrences in the dataset.
D. Incorrect — De-identification/masking hides data but does not by itself enumerate occurrences; first you should detect/inspect.
E. Incorrect — Option B is the recommended approach.

Question 7

Your company operates in three domains: airlines, hotels, and ride-hailing services. Each domain has two teams: analytics and data science, which create data assets in BigQuery with the help of a central data platform team. However, as each domain is evolving rapidly, the central data platform team is becoming a bottleneck. This is causing delays in deriving insights from data, and resulting in stale data when pipelines are not kept up to date. You need to design a data mesh architecture by using Dataplex to eliminate the bottleneck. What should you do?

Options

A. 1. Create one lake for each team. Inside each lake, create one zone for each domain.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Have the central data platform team manage all zones' data assets.

B. 1. Create one lake for each team. Inside each lake, create one zone for each domain.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Direct each domain to manage their own zone's data assets.

C. 1. Create one lake for each domain. Inside each lake, create one zone for each team.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Direct each domain to manage their own lake's data assets.

D. 1. Create one lake for each domain. Inside each lake, create one zone for each team.
2. Attach each of the BigQuery datasets created by the individual teams as assets to the respective zone.
3. Have the central data platform team manage all lakes' data assets.

E. None of the above

Answer

C

Detailed explanation

A. Incorrect — One lake per team with central-platform-managed assets retains the central-team bottleneck; not desirable.

B. Incorrect — One lake per team and delegating zones to domains mixes ownership in a way that still fragments domain governance.

C. Correct — Creating one lake per domain and zones per team maps organizational ownership to domains (domains manage their lake), allowing teams to self-manage assets within domain lakes — this decentralizes control and removes the central-team bottleneck while keeping domain-level governance.

D. Incorrect — Central platform managing all lakes reintroduces the bottleneck.

E. Incorrect — Option C matches the data-mesh decentralization goal.

Question 8

You have an inventory of VM data stored in the BigQuery table. You want to prepare the data for regular reporting in the most cost-effective way. You need to exclude VM rows with fewer than 8 vCPU in your report. What should you do?

Row	id	name	components.name	components.qty
1	vm02781	d-jp-kfk-02-02	vcpu	2
			memory	8
			boot_disk	10
			disk_1	50
2	vm11490	i-jp-kfk-02-07	vcpu	16
			memory	64
			boot_disk	10
			disk_1	200
3	vm18130	i-jp-kfk-02-08	vcpu	8
			memory	8
			boot_disk	10

Options

A. Create a view with a filter to drop rows with fewer than 8 vCPU, and use the UNNEST operator.
B. Create a materialized view with a filter to drop rows with fewer than 8 vCPU, and use the WITH common table expression.
C. Create a view with a filter to drop rows with fewer than 8 vCPU, and use the WITH common table expression.
D. Use Dataflow to batch process and write the result to another BigQuery table.
E. None of the above

Answer

A

Detailed explanation

A. Correct — Creating a view with a WHERE filter to exclude rows with <8 vCPU is simple and cost-effective for reporting; using UNNEST may be necessary if vCPU info is in a repeated field — the view avoids extra storage cost.
B. Incorrect — A materialized view stores precomputed results (incurs storage costs) and may be unnecessary unless query performance requires it; the stated answer selects the cheaper view approach.
C. Incorrect — A view with a WITH clause is fine, but the inclusion of UNNEST in option A addresses the likely schema; option A is the selected cost-effective method.
D. Incorrect — Running Dataflow and writing a new table increases operational overhead and storage costs unnecessarily.
E. Incorrect — Option A is the best-fit, cost-effective solution.

Question 9

Your team is building a data lake platform on Google Cloud. As a part of the data foundation design, you are planning to store all the raw data in Cloud Storage. You are expecting to ingest approximately 25 GB of data a day and your billing department is worried about the increasing cost of storing old data. The current business requirements are:

• The old data can be deleted anytime.
• There is no predefined access pattern of the old data.
• The old data should be available instantly when accessed.
• There should not be any charges for data retrieval.

What should you do to optimize for cost?

Options

A. Create the bucket with the Autoclass storage class feature.
B. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to nearline, 90 days to coldline, and 365 days to archive storage class. Delete old data as needed.
C. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to coldline, 90 days to nearline, and 365 days to archive storage class. Delete old data as needed.
D. Create an Object Lifecycle Management policy to modify the storage class for data older than 30 days to nearline, 45 days to coldline, and 60 days to archive storage class. Delete old data as needed.
E. None of the above

Answer

A

Detailed explanation

A. Correct — Autoclass automatically transitions objects between storage classes based on observed access patterns, minimizing costs without manual lifecycle rules; it provides instant availability and no retrieval charges when data is in the hot tier, aligning with the requirements while minimizing operational overhead.
B. Incorrect — Manual lifecycle transitions are reasonable but require maintenance and may incur retrieval costs depending on class transitions; Autoclass simplifies management.
C. Incorrect — Similar to B, and swapping nearline/coldline timelines doesn't address the management simplicity requirement.
D. Incorrect — Aggressive, fixed transitions risk moving still-needed data into classes with retrieval costs or retrieval latency; Autoclass is preferable given unspecified access patterns.
E. Incorrect — Option A is the recommended approach.

Question 10

Your company's data platform ingests CSV file dumps of booking and user profile data from upstream sources into Cloud Storage. The data analyst team wants to join these datasets on the email field available in both the datasets to perform analysis. However, personally identifiable information (PII) should not be accessible to the analysts. You need to de-identify the email field in both the datasets before loading them into BigQuery for analysts. What should you do?

Options

A. 1. Create a pipeline to de-identify the email field by using recordTransformations in Cloud Data Loss Prevention (Cloud DLP) with masking as the de-identification transformations type.
2. Load the booking and user profile data into a BigQuery table.
B. 1. Create a pipeline to de-identify the email field by using recordTransformations in Cloud DLP with format-preserving encryption with FFX as the de-identification transformation type.
2. Load the booking and user profile data into a BigQuery table.
C. 1. Load the CSV files from Cloud Storage into a BigQuery table, and enable dynamic data masking.
2. Create a policy tag with the email mask as the data masking rule.
3. Assign the policy to the email field in both tables.
4. Assign the Identity and Access Management bigquerydatapolicy.maskedReader role for the BigQuery tables to the analysts.
D. 1. Load the CSV files from Cloud Storage into a BigQuery table, and enable dynamic data masking.
2. Create a policy tag with the default masking value as the data masking rule.
3. Assign the policy to the email field in both tables.
4. Assign the Identity and Access Management bigquerydatapolicy.maskedReader role for the BigQuery tables to the analysts.
E. None of the above

Answer

B

Detailed explanation

A. Incorrect — Masking replaces email with masked characters and prevents joining on the original email; analysts would not be able to join datasets on email if it's masked differently in each dataset.
B. Correct — Format-preserving encryption (FPE) using Cloud DLP with FFX preserves the format and deterministically transforms the same email value to the same ciphertext across datasets, enabling joins while keeping raw email confidential. This meets the de-identification + joinability requirement.
C. Incorrect — Dynamic masking hides the email at query time for certain roles but does not help if analysts need to perform joins on the de-identified value across datasets; also masking may be non-deterministic.
D. Incorrect — Same issues as C; default masking likely prevents deterministic joins across datasets.
E. Incorrect — Option B is the recommended, join-friendly de-identification approach.

Question 11
You have important legal hold documents in a Cloud Storage bucket. You need to ensure that these documents are not deleted or modified. What should you do?

Options
A. Set a retention policy. Lock the retention policy.
B. Set a retention policy. Set the default storage class to Archive for long-term digital preservation.
C. Enable the Object Versioning feature. Add a lifecycle rule.
D. Enable the Object Versioning feature. Create a copy in a bucket in a different region.
E. None of the above

Answer
A

Detailed explanation
A. Correct. A retention policy with a lock ensures objects cannot be deleted or modified until the policy expires, meeting legal hold requirements.
B. Incorrect. Archive storage reduces cost but doesn't prevent deletion or modification.
C. Incorrect. Versioning helps recover data but doesn't prevent deletion or overwriting.
D. Incorrect. Replication improves availability but doesn't enforce immutability.
E. Incorrect. Option A already provides the required solution.

Question 12
You are designing a data warehouse in BigQuery to analyze sales data for a telecommunication service provider. You need to create a data model for customers, products, and subscriptions. All customers, products, and subscriptions can be updated monthly, but you must maintain a historical record of all data. You plan to use the visualization layer for current and historical reporting. You need to ensure that the data model is simple, easy-to-use, and cost-effective. What should you do?

Options
A. Create a normalized model with tables for each entity. Use snapshots before updates to track historical data.
B. Create a normalized model with tables for each entity. Keep all input files in a Cloud Storage bucket to track historical data.
C. Create a denormalized model with nested and repeated fields. Update the table and use snapshots to track historical data.
D. Create a denormalized, append-only model with nested and repeated fields. Use the ingestion timestamp to track historical data.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect. Snapshots introduce complexity and extra storage cost.
B. Incorrect. Cloud Storage files aren't optimized for easy SQL-based historical analysis.
C. Incorrect. Snapshots still add complexity and may be costly.
D. Correct. An append-only, denormalized model with ingestion timestamp allows easy queries for both current and historical data. It's cost-effective and simple for reporting.
E. Incorrect. Option D already provides the required approach.

Question 13
You are deploying a batch pipeline in Dataflow. This pipeline reads data from Cloud Storage, transforms the data, and then writes the data into BigQuery. The security team has enabled an organizational constraint in Google Cloud, requiring all Compute Engine instances to use only internal IP addresses and no external IP addresses. What should you do?

Options
A. Ensure that your workers have network tags to access Cloud Storage and BigQuery. Use Dataflow with only internal IP addresses.
B. Ensure that the firewall rules allow access to Cloud Storage and BigQuery. Use Dataflow with only internal IPs.
C. Create a VPC Service Controls perimeter that contains the VPC network and add Dataflow, Cloud Storage, and BigQuery as allowed services in the perimeter. Use Dataflow with only internal IP addresses.
D. Ensure that Private Google Access is enabled in the subnetwork. Use Dataflow with only internal IP addresses.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect. Network tags don't allow access to Google APIs without external IPs.
B. Incorrect. Firewall rules don't solve the lack of external IPs.
C. Incorrect. VPC-SC helps security, not connectivity for internal-only IPs.
D. Correct. Private Google Access allows internal-only VMs to access Google APIs like Cloud Storage and BigQuery.
E. Incorrect. Option D provides the right solution.

Question 14
You are running a Dataflow streaming pipeline, with Streaming Engine and Horizontal Autoscaling enabled. You have set the maximum number of workers to 1000. The input of your pipeline is Pub/Sub messages with notifications from Cloud Storage. One of the pipeline transforms reads CSV files and emits an element for every CSV line. The job performance is low, the pipeline is using only 10 workers, and you notice that the autoscaler is not spinning up additional workers. What should you do to improve performance?

Options
A. Enable Vertical Autoscaling to let the pipeline use larger workers.
B. Change the pipeline code, and introduce a Reshuffle step to prevent fusion.
C. Update the job to increase the maximum number of workers.
D. Use Dataflow Prime, and enable Right Fitting to increase the worker resources.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect. Vertical autoscaling doesn't solve pipeline fusion bottlenecks.
B. Correct. Fusion can prevent autoscaler from scaling out. Adding a Reshuffle step allows parallelization, improving performance.
C. Incorrect. The maximum number of workers is already 1000; the bottleneck is pipeline fusion.
D. Incorrect. Dataflow Prime helps optimization but does not directly fix fusion.
E. Incorrect. Option B is the right fix.

Question 15
You have an Oracle database deployed in a VM as part of a Virtual Private Cloud (VPC) network. You want to replicate and continuously synchronize 50 tables to BigQuery. You want to minimize the need to manage infrastructure. What should you do?

Options
A. Deploy Apache Kafka in the same VPC network, use Kafka Connect Oracle Change Data Capture (CDC), and Dataflow to stream the Kafka topic to BigQuery.
B. Create a Pub/Sub subscription to write to BigQuery directly. Deploy the Debezium Oracle connector to capture changes in the Oracle database, and sink to the Pub/Sub topic.
C. Deploy Apache Kafka in the same VPC network, use Kafka Connect Oracle change data capture (CDC), and the Kafka Connect Google BigQuery Sink Connector.
D. Create a Datastream service from Oracle to BigQuery, use a private connectivity configuration to the same VPC network, and a connection profile to BigQuery.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect. Requires you to manage Kafka infrastructure.
B. Incorrect. Debezium and Pub/Sub require more management overhead.
C. Incorrect. Still infrastructure-heavy (Kafka setup and management).
D. Correct. Datastream is serverless CDC and integrates directly with BigQuery with minimal infra management.
E. Incorrect. Option D provides the required solution.

Question 16
You are deploying an Apache Airflow directed acyclic graph (DAG) in a Cloud Composer 2 instance. You have incoming files in a Cloud Storage bucket that the DAG processes, one file at a time. The Cloud Composer instance is deployed in a subnetwork with no Internet access. Instead of running the DAG based on a schedule, you want to run the DAG in a reactive way every time a new file is received. What should you do?

Options
A. Enable Private Google Access in the subnetwork, and set up Cloud Storage notifications to a Pub/Sub topic. Create a push subscription that points to the web server URL.
B. Enable the Cloud Composer API, and set up Cloud Storage notifications to trigger a Cloud Function. Write a Cloud Function instance to call the DAG by using the Cloud Composer API and the web server URL. Use VPC Serverless Access to reach the web server URL.
C. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance. Create a Private Service Connect (PSC) endpoint. Write a Cloud Function that connects to the Cloud Composer cluster through the PSC endpoint.
D. Enable the Airflow REST API, and set up Cloud Storage notifications to trigger a Cloud Function instance. Write a Cloud Function instance to call the DAG by using the Airflow REST API and the web server URL. Use VPC Serverless Access to reach the web server URL.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect. Push subscription to the web server URL won't work without Internet access.
B. Incorrect. Cloud Composer API is not the best for reactive DAG triggering.
C. Correct. PSC endpoint allows private connectivity to Composer. With Cloud Storage notifications → Pub/Sub → Cloud Function → Airflow REST API, this works in a private setup.
D. Incorrect. VPC Serverless Access cannot directly connect to Composer's Airflow API in private mode.
E. Incorrect. Option C is correct.

Question 17
You are planning to use Cloud Storage as part of your data lake solution. The Cloud Storage bucket will contain objects ingested from external systems. Each object will be ingested once, and the access patterns of individual objects will be random. You want to minimize the cost of storing and retrieving these objects. You want to ensure that any cost optimization efforts are transparent to the users and applications. What should you do?

Options
A. Create a Cloud Storage bucket with Autoclass enabled.
B. Create a Cloud Storage bucket with an Object Lifecycle Management policy to transition objects from Standard to Coldline storage class if an object age reaches 30 days.
C. Create a Cloud Storage bucket with an Object Lifecycle Management policy to transition objects from Standard to Coldline storage class if an object is not live.
D. Create two Cloud Storage buckets. Use the Standard storage class for the first bucket, and use the Coldline storage class for the second bucket. Migrate objects from the first bucket to the second bucket after 30 days.
E. None of the above

Answer
A

Detailed explanation
A. Correct. Autoclass automatically moves objects to the most cost-efficient storage class without affecting user/application access.
B. Incorrect. Lifecycle rules are static, not adaptive to access patterns.
C. Incorrect. "Not live" refers to versioning, not optimal for this case.
D. Incorrect. Manually managing buckets increases complexity and isn't transparent.
E. Incorrect. Option A already solves the problem.

Question 18
You have several different file type data sources, such as Apache Parquet and CSV. You want to store the data in Cloud Storage. You need to set up an object sink for your data that allows you to use your own encryption keys. You want to use a GUI-based solution. What should you do?

Options
A. Use Storage Transfer Service to move files into Cloud Storage.
B. Use Cloud Data Fusion to move files into Cloud Storage.
C. Use Dataflow to move files into Cloud Storage.
D. Use BigQuery Data Transfer Service to move files into BigQuery.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect. Storage Transfer Service is for transfers between storage systems, not flexible for multiple formats and custom keys.
B. Correct. Cloud Data Fusion (GUI-based ETL tool) supports ingestion of different file types into Cloud Storage with CMEK support.
C. Incorrect. Dataflow is code-driven, not GUI-based.
D. Incorrect. BigQuery Data Transfer Service loads into BigQuery, not Cloud Storage.
E. Incorrect. Option B solves the requirement.

Question 19
Your business users need a way to clean and prepare data before using the data for analysis. Your business users are less technically savvy and prefer to work with graphical user interfaces to define their transformations. After the data has been transformed, the business users want to perform their analysis directly in a spreadsheet. You need to recommend a solution that they can use. What should you do?

Options
A. Use Dataprep to clean the data, and write the results to BigQuery. Analyze the data by using Connected Sheets.
B. Use Dataprep to clean the data, and write the results to BigQuery. Analyze the data by using Looker Studio.
C. Use Dataflow to clean the data, and write the results to BigQuery. Analyze the data by using Connected Sheets.
D. Use Dataflow to clean the data, and write the results to BigQuery. Analyze the data by using Looker Studio.
E. None of the above

Answer
A

Detailed explanation
A. Correct. Dataprep (GUI-based) is easy for non-technical users. Writing results to BigQuery and using Connected Sheets lets users analyze data directly in Google Sheets.
B. Incorrect. Looker Studio is more advanced visualization, not spreadsheet-based.
C. Incorrect. Dataflow requires coding skills, not suited for non-technical users.
D. Incorrect. Same issue as C, plus Looker Studio is not spreadsheet-based.
E. Incorrect. Option A is the right fit.

Question 20

You have two projects where you run BigQuery jobs:

• One project runs production jobs that have strict completion time SLAs. These are high priority jobs that must have the required compute resources available when needed. These jobs generally never go below a 300 slot utilization, but occasionally spike up an additional 500 slots.

• The other project is for users to run ad-hoc analytical queries. This project generally never uses more than 200 slots at a time. You want these ad-hoc queries to be billed based on how much data users scan rather than by slot capacity.

You need to ensure that both projects have the appropriate compute resources available. What should you do?

Options

A. Create a single Enterprise Edition reservation for both projects. Set a baseline of 300 slots. Enable autoscaling up to 700 slots.

B. Create two reservations, one for each of the projects. For the SLA project, use an Enterprise Edition with a baseline of 300 slots and enable autoscaling up to 500 slots. For the ad-hoc project, configure on-demand billing.

C. Create two Enterprise Edition reservations, one for each of the projects. For the SLA project, set a baseline of 300 slots and enable autoscaling up to 500 slots. For the ad-hoc project, set a reservation baseline of 0 slots and set the ignore idle slots flag to False.

D. Create two Enterprise Edition reservations, one for each of the projects. For the SLA project, set a baseline of 800 slots. For the ad-hoc project, enable autoscaling up to 200 slots.

E. None of the above

Answer

B

Detailed explanation

A. Incorrect. A single reservation would mix workloads and cause contention.

B. Correct. Production jobs get guaranteed slots (baseline 300 with up to 500 extra). Ad-hoc queries run on-demand, which is cheaper and billed by data scanned.

C. Incorrect. Setting a baseline of 0 slots still consumes idle slots, not optimal for ad-hoc.

D. Incorrect. Over-provisioning with 800 slots wastes cost and isn't efficient.

E. Incorrect. Option B is the right balance of performance and cost.

Question 21
You want to migrate your existing Teradata data warehouse to BigQuery. You want to move the historical data to BigQuery by using the most efficient method that requires the least amount of programming, but local storage space on your existing data warehouse is limited. What should you do?

Options
A. Use BigQuery Data Transfer Service by using the Java Database Connectivity (JDBC) driver with FastExport connection.
B. Create a Teradata Parallel Transporter (TPT) export script to export the historical data, and import to BigQuery by using the bq command-line tool.
C. Use BigQuery Data Transfer Service with the Teradata Parallel Transporter (TPT) tbuild utility.
D. Create a script to export the historical data, and upload in batches to Cloud Storage. Set up a BigQuery Data Transfer Service instance from Cloud Storage to BigQuery.

Answer
A

Detailed explanation
A. **Correct** — Using BigQuery Data Transfer Service with a JDBC driver that supports a FastExport-style connection lets you stream or transfer data directly from Teradata to BigQuery with minimal custom coding and without needing large temporary local storage on the Teradata host. This minimizes programming effort and local disk usage.
B. **Incorrect** — TPT export + bq CLI works but requires exporting to local files first (consumes local storage) and more scripting/work to manage files and uploads.
C. **Incorrect** — Using TPT tbuild still implies doing exports and handling exported files; it places more burden on local storage or additional orchestration compared with a managed transfer approach.
D. **Incorrect** — Exporting to local batches and uploading to Cloud Storage requires local disk to stage the exported files and more custom scripting; not the “least amount of programming” and problematic when local storage is limited.

Question 22
You are on the data governance team and are implementing security requirements. You need to encrypt all your data in BigQuery by using an encryption key managed by your team. You must implement a mechanism to generate and store encryption material only on your on-premises hardware security module (HSM). You want to rely on Google managed solutions. What should you do?

Options
A. Create the encryption key in the on-premises HSM, and import it into a Cloud Key Management Service (Cloud KMS) key. Associate the created Cloud KMS key while creating the BigQuery resources.
B. Create the encryption key in the on-premises HSM and link it to a Cloud External Key Manager (Cloud EKM) key. Associate the created Cloud KMS key while creating the BigQuery resources.
C. Create the encryption key in the on-premises HSM, and import it into Cloud Key Management Service (Cloud HSM) key. Associate the created Cloud HSM key while creating the BigQuery resources.
D. Create the encryption key in the on-premises HSM. Create BigQuery resources and encrypt data while ingesting them into BigQuery.

Answer
B

Detailed explanation
A. **Incorrect** — Importing key material into Cloud KMS requires the key material to be copied into Google Cloud; that does not meet the requirement that material remain only on your on-prem HSM.
B. **Correct** — Cloud External Key Manager (Cloud EKM) lets you keep key material on an external HSM (on-premises) while Google Cloud services (including BigQuery through CMEK/EKM integration) can use those keys for encryption operations. This relies on Google-managed integrations while keeping key material on your HSM.
C. **Incorrect** — Cloud HSM (Google Cloud HSM) is a Google-managed HSM inside Google Cloud; importing keys there would move material into the cloud and violate the "only on-prem HSM" requirement.
D. **Incorrect** — Simply creating keys on an on-prem HSM and encrypting on ingest without using a supported managed integration does not leverage Google's managed EKM/KMS features and does not provide the required managed linkage to BigQuery.

Question 23
You maintain ETL pipelines. You notice that a streaming pipeline running on Dataflow is taking a long time to process incoming data, which causes output delays. You also noticed that the pipeline graph was automatically optimized by Dataflow and merged into one step. You want to identify where the potential bottleneck is occurring. What should you do?

Options
A. Insert a Reshuffle operation after each processing step, and monitor the execution details in the Dataflow console.
B. Insert output sinks after each key processing step, and observe the writing throughput of each block.
C. Log debug information in each ParDo function, and analyze the logs at execution time.
D. Verify that the Dataflow service accounts have appropriate permissions to write the processed data to the output sinks.

Answer
A

Detailed explanation
A. **Correct** — Inserting a Reshuffle (or an explicit materialization boundary) prevents fusion of stages, forcing Dataflow to break the fused step into separate execution units. That lets you measure each stage individually in the Dataflow UI and find the bottleneck.
B. **Incorrect** — Adding output sinks can help measure write throughput but may change pipeline behavior and is heavier-handed than reshuffling; it also requires extra I/O and may not isolate CPU/processing bottlenecks inside fused steps.
C. **Incorrect** — Logging debug info can help but is noisy, costly, and slow; it won't necessarily reveal where fused-stage parallelism limits occur.
D. **Incorrect** — Permission problems would cause failures, not gradual processing delays caused by fused transforms; this option doesn't help identify computational bottlenecks.

Question 24
You are running your BigQuery project in the on-demand billing model and are executing a change data capture (CDC) process that ingests data. The CDC process loads 1 GB of data every 10 minutes into a temporary table, and then performs a merge into a 10 TB target table. This process is very scan intensive and you want to explore options to enable a predictable cost model. You need to create a BigQuery reservation based on utilization information gathered from BigQuery Monitoring and apply the reservation to the CDC process. What should you do?

Options
A. Create a BigQuery reservation for the dataset.
B. Create a BigQuery reservation for the job.
C. Create a BigQuery reservation for the service account running the job.
D. Create a BigQuery reservation for the project.

Answer
D

Detailed explanation
A. **Incorrect** — Reservations are not applied at the dataset level.
B. **Incorrect** — You cannot create a reservation scoped to an individual job; reservations are allocated to projects or folders (or reservations assignments like assignments to a project or to an organization-level resource).
C. **Incorrect** — Reservations are not assigned to service accounts directly.
D. **Correct** — Create a reservation and assign it to the project that runs the CDC jobs (or create a dedicated assignment for that project). This makes slot usage predictable for the project's workloads and helps stabilize costs compared to on-demand scanning.

Question 25
You are designing a fault-tolerant architecture to store data in a regional BigQuery dataset. You need to ensure that your application is able to recover from a corruption event in your tables that occurred within the past seven days. You want to adopt managed services with the lowest RPO and most cost-effective solution. What should you do?

Options
A. Access historical data by using time travel in BigQuery.
B. Export the data from BigQuery into a new table that excludes the corrupted data.
C. Create a BigQuery table snapshot on a daily basis.
D. Migrate your data to multi-region BigQuery buckets.

Answer
A

Detailed explanation
A. **Correct** — BigQuery time travel (table versions) lets you query or restore table state from up to 7 days in the past without additional custom snapshots, offering the lowest RPO and a managed, cost-effective recovery option.
B. **Incorrect** — Exporting and reconstructing data is possible but requires work and may have higher RPO/cost than using built-in time travel.
C. **Incorrect** — Daily snapshots are useful but redundant when time travel already provides up to 7 days of history; snapshots add operational overhead and storage cost.
D. **Incorrect** — Migrating to multi-region storage is about geographic redundancy, not about recovering table state from a recent corruption within the same region; it doesn’t address the 7-day rollback requirement.

Question 26

You are building a streaming Dataflow pipeline that ingests noise level data from hundreds of sensors placed near construction sites across a city. The sensors measure noise level every ten seconds, and send that data to the pipeline when levels reach above 70 dBA. You need to detect the average noise level from a sensor when data is received for a duration of more than 30 minutes, but the window ends when no data has been received for 15 minutes. What should you do?

Options

A. Use session windows with a 15-minute gap duration.
B. Use session windows with a 30-minute gap duration.
C. Use hopping windows with a 15-minute window, and a thirty-minute period.
D. Use tumbling windows with a 15-minute window and a fifteen-minute .withAllowedLateness operator.

Answer

A

Detailed explanation

A. **Correct** — Session windows with a 15-minute gap will group contiguous events from a sensor into a session that closes when there's no activity for 15 minutes. You can then compute the session's average and filter sessions whose duration exceeds 30 minutes. This precisely models "end when no data for 15 minutes" and lets you identify sessions > 30 minutes.
B. **Incorrect** — A 30-minute gap would delay session closure until 30 minutes of inactivity, which does not match the requirement to end after 15 minutes of silence.
C. **Incorrect** — Hopping windows do not capture irregular-length active periods that end based on inactivity; they are periodic and would not cleanly represent "end after 15 minutes of no data."
D. **Incorrect** — Tumbling windows with allowed lateness are fixed windows and won't model variable-length activity sessions that end upon inactivity.

Question 27

You are creating a data model in BigQuery that will hold retail transaction data. Your two largest tables, sales_transaction_header and sales_transaction_line, have a tightly coupled immutable relationship. These tables are rarely modified after load and are frequently joined when queried. You need to model the sales_transaction_header and sales_transaction_line tables to improve the performance of data analytics queries. What should you do?

Options

A. Create a sales_transaction table that holds the sales_transaction_header information as rows and the sales_transaction_line rows as nested and repeated fields.
B. Create a sales_transaction table that holds the sales_transaction_header and sales_transaction_line information as rows, duplicating the sales_transaction_header data for each line.
C. Create a sales_transaction table that stores the sales_transaction_header and sales_transaction_line data as a JSON data type.
D. Create separate sales_transaction_header and sales_transaction_line tables and, when querying, specify the sales_transaction_line first in the WHERE clause.

Answer

A

Detailed explanation

A. **Correct** — Nesting lines as a repeated field inside a transaction header models the natural 1-to-many immutable relationship efficiently in BigQuery, reduces the need for joins, and speeds up analytic queries.
B. **Incorrect** — Duplicating header rows for each line (fully denormalized flat rows) increases storage and can harm performance for queries that only need headers; it's less elegant than nested repeated fields for 1-to-many immutable data.
C. **Incorrect** — Storing JSON may preserve structure but is less query-friendly and less performant than using native nested/repeated schema.
D. **Incorrect** — Keeping separate tables and trying to manipulate query order won't remove join overhead; it doesn't address the frequent join performance issue as well as nesting.

Question 28
You created a new version of a Dataflow streaming data ingestion pipeline that reads from Pub/Sub and writes to BigQuery. The previous version of the pipeline that runs in production uses a 5-minute window for processing. You need to deploy the new version of the pipeline without losing any data, creating inconsistencies, or increasing the processing latency by more than 10 minutes. What should you do?

Options
A. Update the old pipeline with the new pipeline code.
B. Snapshot the old pipeline, stop the old pipeline, and then start the new pipeline from the snapshot.
C. Drain the old pipeline, then start the new pipeline.
D. Cancel the old pipeline, then start the new pipeline.

Answer
C

Detailed explanation
A. **Incorrect** — In-place updates can be risky for streaming pipelines (especially when changing windowing/logic) and may introduce inconsistency or data loss.
B. **Incorrect** — Snapshot+stop/start is not typically how streaming Dataflow upgrades are performed; snapshots are more for batch and do not guarantee simple, zero-loss switch for streaming.
C. **Correct** — Draining the old pipeline lets it finish processing in-flight elements and stop consuming new input cleanly; once drained, you can start the new pipeline to resume processing with minimal latency increase and no data loss.
D. **Incorrect** — Canceling the old pipeline terminates it immediately and may drop in-flight data, risking loss or inconsistency.

Question 29
Your organization's data assets are stored in BigQuery, Pub/Sub, and a PostgreSQL instance running on Compute Engine. Because there are multiple domains and diverse teams using the data, teams in your organization are unable to discover existing data assets. You need to design a solution to improve data discoverability while keeping development and configuration efforts to a minimum. What should you do?

Options
A. Use Data Catalog to automatically catalog BigQuery datasets. Use Data Catalog APIs to manually catalog Pub/Sub topics and PostgreSQL tables.
B. Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use Data Catalog APIs to manually catalog PostgreSQL tables.
C. Use Data Catalog to automatically catalog BigQuery datasets and Pub/Sub topics. Use custom connectors to manually catalog PostgreSQL tables.
D. Use customer connectors to manually catalog BigQuery datasets, Pub/Sub topics, and PostgreSQL tables.

Answer
B

Detailed explanation
A. **Incorrect** — This is partially correct but says to manually catalog Pub/Sub topics; Pub/Sub can be auto-discovered by Data Catalog, so doing it manually increases effort unnecessarily.
B. **Correct** — Data Catalog can auto-catalog BigQuery and Pub/Sub (minimizing effort). For the PostgreSQL instance on Compute Engine, using the Data Catalog APIs (or automatic connectors where available) to add entries gives a minimal-effort way to catalog that external source. This balances low development work with broad coverage.
C. **Incorrect** — Custom connectors add more development/configuration overhead than using the available Data Catalog APIs for PostgreSQL metadata ingestion.
D. **Incorrect** — Manually cataloging everything via custom connectors is highest effort and not minimal configuration.

Question 30
You need to create a SQL pipeline. The pipeline runs an aggregate SQL transformation on a BigQuery table every two hours and appends the result to another existing BigQuery table. You need to configure the pipeline to retry if errors occur. You want the pipeline to send an email notification after three consecutive failures. What should you do?

Options
A. Use the BigQueryUpsertTableOperator in Cloud Composer, set the retry parameter to three, and set the email_on_failure parameter to true.
B. Use the BigQueryInsertJobOperator in Cloud Composer, set the retry parameter to three, and set the email_on_failure parameter to true.
C. Create a BigQuery scheduled query to run the SQL transformation with schedule options that repeats every two hours, and enable email notifications.
D. Create a BigQuery scheduled query to run the SQL transformation with schedule options that repeats every two hours, and enable notification to Pub/Sub topic. Use Pub/Sub and Cloud Functions to send an email after three failed executions.

Answer
B

Detailed explanation
A. **Incorrect** — BigQueryUpsertTableOperator is targeted at upsert/DDL style operations and is not the typical operator for running an aggregate SQL job; it may not provide the intended job-level retry/notification semantics.
B. **Correct** — BigQueryInsertJobOperator runs arbitrary BigQuery jobs from Composer. Using the operator's retry and Composer's email_on_failure parameters gives you built-in retry behavior and email notifications when failures occur. You can configure retries = 3 so Composer will attempt the job up to three times and notify on failure.
C. **Incorrect** — BigQuery scheduled queries support retries and basic notifications but configuring "email after three consecutive failures" is not as straightforward or flexible as using Cloud Composer with operator-level retry and email parameters.
D. **Incorrect** — This is possible but more complex: it requires building custom failure-counting logic (track three failures) via Pub/Sub + Cloud Functions. It's heavier than using Composer's built-in retry/email features.

Question 31

You are monitoring your organization's data lake hosted on BigQuery. The ingestion pipelines read data from Pub/Sub and write the data into tables on BigQuery. After a new version of the ingestion pipelines is deployed, the daily stored data increased by 50%. The volumes of data in Pub/Sub remained the same and only some tables had their daily partition data size doubled. You need to investigate and fix the cause of the data increase. What should you do?

Options

A. 1. Check for duplicate rows in the BigQuery tables that have the daily partition data size doubled.
2. Schedule daily SQL jobs to deduplicate the affected tables.
3. Share the deduplication script with the other operational teams to reuse if this occurs to other tables.

B. 1. Check for code errors in the deployed pipelines.
2. Check for multiple writing to pipeline BigQuery sink.
3. Check for errors in Cloud Logging during the day of the release of the new pipelines.
4. If no errors, restore the BigQuery tables to their content before the last release by using time travel.

C. 1. Check for duplicate rows in the BigQuery tables that have the daily partition data size doubled.
2. Check the BigQuery Audit logs to find job IDs.
3. Use Cloud Monitoring to determine when the identified Dataflow jobs started and the pipeline code version.
4. When more than one pipeline ingests data into a table, stop all versions except the latest one.

D. 1. Roll back the last deployment.
2. Restore the BigQuery tables to their content before the last release by using time travel.
3. Restart the Dataflow jobs and replay the messages by seeking the subscription to the timestamp of the release.

Answer

C

Detailed explanation

A. **Incorrect** — This reactive approach only addresses symptoms (deduplicating) and doesn't identify the root cause (e.g., multiple active writers or duplicate writes). It may cause repeated operational overhead.

B. **Incorrect** — This partially investigates but omits audit logs/monitoring to identify which job versions were active and when; it also suggests restore before identifying active writers (which could reintroduce the issue).

C. **Correct** — This option combines data validation (check duplicates) with audit- and monitoring-based forensics (job IDs, start times, code versions). Identifying and stopping duplicate pipeline versions prevents further duplicate writes and addresses root cause with minimal disruptive remediation.

D. **Incorrect** — Rolling back and immediate restore/replay is disruptive and risky (may reintroduce duplicates unless you first stop duplicate writers and understand what caused duplicates).

Question 32

You have a BigQuery dataset named “customers”. All tables will be tagged by using a Data Catalog tag template named “gdpr”. The template contains one mandatory field, “has_sensitive_data”, with a boolean value. All employees must be able to do a simple search and find tables in the dataset that have either true or false in the “has_sensitive_data’ field. However, only the Human Resources (HR) group should be able to see the data inside the tables for which “has_sensitive data” is true. You give the all employees group the bigquery.metadataViewer and bigquery.connectionUser roles on the dataset. You want to minimize configuration overhead. What should you do next?

Options

A. Create the “gdpr” tag template with private visibility. Assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.
B. Create the “gdpr” tag template with private visibility. Assign the datacatalog.tagTemplateViewer role on this tag to the all employees group, and assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.
C. Create the “gdpr” tag template with public visibility. Assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.
D. Create the “gdpr” tag template with public visibility. Assign the datacatalog.tagTemplateViewer role on this tag to the all employees group, and assign the bigquery.dataViewer role to the HR group on the tables that contain sensitive data.

Answer

C

Detailed explanation

A. **Incorrect** — Private visibility would require explicit tag-view grants to many users, increasing configuration overhead.
B. **Incorrect** — Also increases configuration work by requiring explicit tagTemplateViewer grants; unnecessary if public tag visibility suffices.
C. **Correct** — Public tag-template visibility lets everyone search and see the tag value (true/false) with minimal config. Restricting data access by assigning bigquery.dataViewer on sensitive tables only to HR enforces data access while keeping discovery simple.
D. **Incorrect** — Explicitly assigning tagTemplateViewer is redundant with public visibility and adds extra configuration steps that aren't needed.

Question 33

You are creating the CI/CD cycle for the code of the directed acyclic graphs (DAGs) running in Cloud Composer. Your team has two Cloud Composer instances: one instance for development and another instance for production. Your team is using a Git repository to maintain and develop the code of the DAGs. You want to deploy the DAGs automatically to Cloud Composer when a certain tag is pushed to the Git repository. What should you do?

Options

A. 1. Use Cloud Build to copy the code of the DAG to the Cloud Storage bucket of the development instance for DAG testing.
2. If the tests pass, use Cloud Build to copy the code to the bucket of the production instance.

B. 1. Use Cloud Build to build a container with the code of the DAG and the KubernetesPodOperator to deploy the code to the Google Kubernetes Engine (GKE) cluster of the development instance for testing.
2. If the tests pass, use the KubernetesPodOperator to deploy the container to the GKE cluster of the production instance.

C. 1. Use Cloud Build to build a container and the KubernetesPodOperator to deploy the code of the DAG to the Google Kubernetes Engine (GKE) cluster of the development instance for testing.
2. If the tests pass, copy the code to the Cloud Storage bucket of the production instance.

D. 1. Use Cloud Build to copy the code of the DAG to the Cloud Storage bucket of the development instance for DAG testing.
2. If the tests pass, use Cloud Build to build a container with the code of the DAG and the KubernetesPodOperator to deploy the container to the Google Kubernetes Engine (GKE) cluster of the production instance.

Answer

A

Detailed explanation

A. **Correct** — Cloud Composer stores DAGs in a Cloud Storage bucket. Using Cloud Build to copy DAG code to the dev Composer bucket for testing, and then to the prod Composer bucket after tests pass, is a straightforward, supported CI/CD pattern with minimal extra infrastructure.

B. **Incorrect** — This shifts DAG deployment to GKE containers and KubernetesPodOperator, which is unnecessarily complex and not how Composer DAGs are typically deployed.

C. **Incorrect** — Mixing container-based deployment for dev and bucket copy for prod is inconsistent and adds unnecessary complexity.

D. **Incorrect** — Building a container only for production while dev uses direct bucket copy adds complexity without benefit; Composer expects DAGs in the bucket.

Question 34
You have a BigQuery table that ingests data directly from a Pub/Sub subscription. The ingested data is encrypted with a Google-managed encryption key. You need to meet a new organization policy that requires you to use keys from a centralized Cloud Key Management Service (Cloud KMS) project to encrypt data at rest. What should you do?

Options
A. Use Cloud KMS encryption key with Dataflow to ingest the existing Pub/Sub subscription to the existing BigQuery table.
B. Create a new BigQuery table by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table.
C. Create a new Pub/Sub topic with CMEK and use the existing BigQuery table by using Google-managed encryption key.
D. Create a new BigQuery table and Pub/Sub topic by using customer-managed encryption keys (CMEK), and migrate the data from the old BigQuery table.

Answer
B

Detailed explanation
A. **Incorrect** — Using Cloud KMS in Dataflow for ingestion does not change the BigQuery table's at-rest encryption; BigQuery must be configured with a CMEK to meet the requirement.
B. **Correct** — Create a new BigQuery table that's configured with CMEK (from the centralized Cloud KMS project) and migrate data into it. This ensures the table data at rest is encrypted with the organization's centralized keys.
C. **Incorrect** — Encrypting Pub/Sub with CMEK alone does not change BigQuery's at-rest key; existing BigQuery table would still use Google-managed keys.
D. **Incorrect** — Creating both new Pub/Sub and BigQuery with CMEK is overkill; you only need to ensure BigQuery uses the centralized CMEK for at-rest encryption. (Also migrating Pub/Sub is unnecessary for the requirement.)

Question 35
You created an analytics environment on Google Cloud so that your data scientist team can explore data without impacting the on-premises Apache Hadoop solution. The data in the on-premises Hadoop Distributed File System (HDFS) cluster is in Optimized Row Columnar (ORC) formatted files with multiple columns of Hive partitioning. The data scientist team needs to be able to explore the data in a similar way as they used the on-premises HDFS cluster with SQL on the Hive query engine. You need to choose the most cost-effective storage and processing solution. What should you do?

Options
A. Import the ORC files to Bigtable tables for the data scientist team.
B. Import the ORC files to BigQuery tables for the data scientist team.
C. Copy the ORC files on Cloud Storage, then deploy a Dataproc cluster for the data scientist team.
D. Copy the ORC files on Cloud Storage, then create external BigQuery tables for the data scientist team.

Answer
D

Detailed explanation
A. **Incorrect** — Bigtable is a key-value/wide-column store and is not suitable for columnar ORC analytics or for SQL-based exploration similar to Hive.
B. **Incorrect** — Importing into native BigQuery tables could work for analysis but involves a storage/ingestion cost and schema conversion; may be less cost-effective if data is large and exploratory usage is intermittent.
C. **Incorrect** — Deploying Dataproc gives Hive-like environment but costs more to run clusters; it's not the most cost-effective for interactive, ad-hoc SQL exploration.
D. **Correct** — Copying ORC files to Cloud Storage and creating external BigQuery tables over them lets data scientists use SQL (via BigQuery) on ORC data without ingesting/copying into BigQuery storage—cost-effective and similar to Hive-on-HDFS behavior.

Question 36
You are designing a Dataflow pipeline for a batch processing job. You want to mitigate multiple zonal failures at job submission time. What should you do?

Options
A. Submit duplicate pipelines in two different zones by using the --zone flag.
B. Set the pipeline staging location as a regional Cloud Storage bucket.
C. Specify a worker region by using the --region flag.
D. Create an Eventarc trigger to resubmit the job in case of zonal failure when submitting the job.

Answer
C

Detailed explanation
A. **Incorrect** — Submitting duplicate pipelines across zones complicates job management and can cause duplicate processing and consistency issues.
B. **Incorrect** — Using a regional staging bucket helps with availability of staged files but does not control where workers run; it's a partial measure.
C. **Correct** — Specifying a worker region (via --region) causes Dataflow to use regional workers that are spread across multiple zones within that region, which mitigates single-zone failures at job-runtime and is the intended approach.
D. **Incorrect** — Building a resubmission mechanism is more complex and reactive; specifying a regional worker footprint is the recommended proactive approach.

Question 37
You are designing a real-time system for a ride hailing app that identifies areas with high demand for rides to effectively reroute available drivers to meet the demand. The system ingests data from multiple sources to Pub/Sub, processes the data, and stores the results for visualization and analysis in real-time dashboards. The data sources include driver location updates every 5 seconds and app-based booking events from riders. The data processing involves real-time aggregation of supply and demand data for the last 30 seconds, every 2 seconds, and storing the results in a low-latency system for visualization. What should you do?

Options
A. Group the data by using a tumbling window in a Dataflow pipeline, and write the aggregated data to Memorystore.
B. Group the data by using a hopping window in a Dataflow pipeline, and write the aggregated data to Memorystore.
C. Group the data by using a session window in a Dataflow pipeline, and write the aggregated data to BigQuery.

D. Group the data by using a hopping window in a Dataflow pipeline, and write the aggregated data to BigQuery.

Answer
B

Detailed explanation
A. **Incorrect** — Tumbling windows are non-overlapping and cannot provide sliding updates every 2 seconds for a 30-second span.
B. **Correct** — Hopping (sliding) windows let you compute aggregates for the last 30 seconds with updates every 2 seconds (overlapping windows). Writing to Memorystore (low-latency in-memory store) suits real-time dashboards that require fast reads.
C. **Incorrect** — Session windows are activity-driven and not suitable for fixed-duration sliding aggregations; BigQuery is not low-latency enough for sub-second/second-level dashboard updates.
D. **Incorrect** — Hopping windows are appropriate, but BigQuery is not the best target for low-latency visualization reads; Memorystore is better suited.

Question 38
Your car factory is pushing machine measurements as messages into a Pub/Sub topic in your Google Cloud project. A Dataflow streaming job, that you wrote with the Apache Beam SDK, reads these messages, sends acknowledgment to Pub/Sub, applies some custom business logic in a DoFn instance, and writes the result to BigQuery. You want to ensure that if your business logic fails on a message, the message will be sent to a Pub/Sub topic that you want to monitor for alerting purposes. What should you do?

Options
A. Enable retaining of acknowledged messages in your Pub/Sub pull subscription. Use Cloud Monitoring to monitor the subscription/num_retained_acked_messages metric on this subscription.
B. Use an exception handling block in your Dataflow's DoFn code to push the messages that failed to be transformed through a side output and to a new Pub/Sub topic. Use Cloud Monitoring to monitor the topic/num_unacked_messages_by_region metric on this new topic.
C. Enable dead lettering in your Pub/Sub pull subscription, and specify a new Pub/Sub topic as the dead letter topic. Use Cloud Monitoring to monitor the subscription/dead_letter_message_count metric on your pull subscription.
D. Create a snapshot of your Pub/Sub pull subscription. Use Cloud Monitoring to monitor the snapshot/num_messages metric on this snapshot.

Answer
B

Detailed explanation
A. **Incorrect** — Retaining acknowledged messages is not intended for application-level failure handling; once Dataflow acks a message, Pub/Sub considers it delivered. This won't surface business-logic failures that occur after ack.
B. **Correct** — Catch exceptions in the DoFn, emit failed messages to a side output, and publish them to a monitoring Pub/Sub topic. This ensures failed messages are explicitly captured and observable for alerting/handling.
C. **Incorrect** — Dead lettering works for messages that are not acked after delivery attempts; however, in this scenario the pipeline currently acks messages before business-logic failure, so dead-lettering would not capture those failures without changing ack semantics.
D. **Incorrect** — Snapshots are for retention of a subscription's message backlog and are not an error-handling mechanism for transform-time failures.

Question 39

You want to store your team's shared tables in a single dataset to make data easily accessible to various analysts. You want to make this data readable but unmodifiable by analysts. At the same time, you want to provide the analysts with individual workspaces in the same project, where they can create and store tables for their own use, without the tables being accessible by other analysts. What should you do?

Options

A. Give analysts the BigQuery Data Viewer role at the project level. Create one other dataset, and give the analysts the BigQuery Data Editor role on that dataset.
B. Give analysts the BigQuery Data Viewer role at the project level. Create a dataset for each analyst, and give each analyst the BigQuery Data Editor role at the project level.
C. Give analysts the BigQuery Data Viewer role on the shared dataset. Create a dataset for each analyst, and give each analyst the BigQuery Data Editor role at the dataset level for their assigned dataset.
D. Give analysts the BigQuery Data Viewer role on the shared dataset. Create one other dataset and give the analysts the BigQuery Data Editor role on that dataset.

Answer

C

Detailed explanation

A. **Incorrect** — Granting Data Viewer at project level is too broad (they'd see all datasets) and giving Data Editor on a single dataset doesn't provide isolated personal workspaces per analyst.
B. **Incorrect** — Giving Data Editor at the project level gives each analyst wide edit rights across the project (not isolated per analyst). Creating per-analyst datasets but granting project-level editor is excessive and insecure.
C. **Correct** — Grant Data Viewer on the shared dataset (read-only). Create individual datasets per analyst and give each analyst Data Editor on only their dataset—this isolates workspaces and allows shared read-only access to the common data.
D. **Incorrect** — A single additional dataset shared among analysts lacks per-analyst isolation; analysts would see each other's work.

Question 40
You are running a streaming pipeline with Dataflow and are using hopping windows to group the data as the data arrives. You noticed that some data is arriving late but is not being marked as late data, which is resulting in inaccurate aggregations downstream. You need to find a solution that allows you to capture the late data in the appropriate window. What should you do?

Options
A. Use watermarks to define the expected data arrival window. Allow late data as it arrives.
B. Change your windowing function to tumbling windows to avoid overlapping window periods.
C. Change your windowing function to session windows to define your windows based on certain activity.
D. Expand your hopping window so that the late data has more time to arrive within the grouping.

Answer
A

Detailed explanation
A. **Correct** — Watermarks express the pipeline's notion of event-time progress; defining and tuning watermarks (and allowed lateness) is the standard way to capture late arrivals and include them in the appropriate windowed aggregations.
B. **Incorrect** — Tumbling windows don't address late-arrival handling; they are fixed, non-overlapping windows and won't automatically capture late events unless you tune watermarks/allowed lateness.
C. **Incorrect** — Session windows are activity-based and not a general solution for late-arriving events in hopping-window scenarios.
D. **Incorrect** — Expanding hopping windows to be longer is a brittle workaround and increases computation/overlap; proper watermark and allowed-lateness configuration is the correct approach.

Question 41
You need to migrate a Redis database from an on-premises data center to a Memorystore for Redis instance. You want to follow Google-recommended practices and perform the migration for minimal cost, time and effort. What should you do?

Options
A. Make an RDB backup of the Redis database, use the gsutil utility to copy the RDB file into a Cloud Storage bucket, and then import the RDB file into the Memorystore for Redis instance.
B. Make a secondary instance of the Redis database on a Compute Engine instance and then perform a live cutover.
C. Create a Dataflow job to read the Redis database from the on-premises data center and write the data to a Memorystore for Redis instance.
D. Write a shell script to migrate the Redis data and create a new Memorystore for Redis instance.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Exporting an RDB backup and importing it into Memorystore for Redis is the Google-recommended and simplest migration path, minimizing cost and manual effort.
B. Incorrect – Creating a secondary instance on Compute Engine is complex and not the standard migration method.
C. Incorrect – Dataflow does not natively support direct Redis-to-Memorystore migration.

D. Incorrect – A shell script would be a manual, error-prone, and non-recommended approach.
E. Incorrect – Option A is the right approach.

Question 42
Your platform on your on-premises environment generates 100 GB of data daily, composed of millions of structured JSON text files. Your on-premises environment cannot be accessed from the public internet. You want to use Google Cloud products to query and explore the platform data. What should you do?

Options
A. Use Cloud Scheduler to copy data daily from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery.
B. Use a Transfer Appliance to copy data from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery.
C. Use Transfer Service for on-premises data to copy data from your on-premises environment to Cloud Storage. Use the BigQuery Data Transfer Service to import data into BigQuery.
D. Use the BigQuery Data Transfer Service dataset copy to transfer all data into BigQuery.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Cloud Scheduler cannot bypass lack of public internet access.
B. Incorrect – Transfer Appliance is for one-time or bulk migrations, not ongoing daily transfers.
C. Correct – Transfer Service for on-premises data installs an agent inside the network, enabling data transfer without requiring internet exposure, then ingests into BigQuery.
D. Incorrect – Dataset copy only works within BigQuery, not from on-premises to BigQuery.
E. Incorrect – Option C is the recommended solution.

Question 43
A TensorFlow machine learning model on Compute Engine virtual machines (n2-standard-32) takes two days to complete training. The model has custom TensorFlow operations that must run partially on a CPU. You want to reduce the training time in a cost-effective manner. What should you do?

Options
A. Change the VM type to n2-highmem-32.
B. Change the VM type to e2-standard-32.
C. Train the model using a VM with a GPU hardware accelerator.
D. Train the model using a VM with a TPU hardware accelerator.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – More memory (highmem) does not speed up TensorFlow training significantly.
B. Incorrect – e2 instances are lower-cost but less powerful, which would make training slower.
C. Correct – GPUs accelerate TensorFlow workloads effectively while still allowing CPU-based custom operations to run. This is the cost-effective option.
D. Incorrect – TPUs may not support custom TensorFlow operations, making them unsuitable.
E. Incorrect – Option C is the recommended practice.

Question 44
You want to create a machine learning model using BigQuery ML and create an endpoint for hosting the model using Vertex AI. This will enable the processing of continuous streaming data in near-real time from multiple vendors. The data may contain invalid values. What should you do?

Options
A. Create a new BigQuery dataset and use streaming inserts to land the data from multiple vendors. Configure your BigQuery ML model to use the "ingestion" dataset as the training data.
B. Use BigQuery streaming inserts to land the data from multiple vendors where your BigQuery dataset ML model is deployed.
C. Create a Pub/Sub topic and send all vendor data to it. Connect a Cloud Function to the topic to process the data and store it in BigQuery.
D. Create a Pub/Sub topic and send all vendor data to it. Use Dataflow to process and sanitize the Pub/Sub data and stream it to BigQuery.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – BigQuery streaming inserts alone don't handle invalid values or cleansing.
B. Incorrect – Same issue as A; raw streaming inserts don't allow data quality processing.
C. Incorrect – Cloud Functions are not ideal for heavy continuous data cleansing.
D. Correct – Pub/Sub collects streaming vendor data, Dataflow processes and cleanses invalid values in real-time, and streams the clean data into BigQuery for ML + Vertex AI.
E. Incorrect – Option D is the best solution.

Question 45
You have a data processing application that runs on Google Kubernetes Engine (GKE). Containers need to be launched with their latest available configurations from a container registry. Your GKE nodes need to have GPUs, local SSDs, and 8 Gbps bandwidth. You want to efficiently provision the data processing infrastructure and manage the deployment process. What should you do?

Options
A. Use Compute Engine startup scripts to pull container images, and use gcloud commands to provision the infrastructure.
B. Use Cloud Build to schedule a job using Terraform build to provision the infrastructure and launch with the most current container images.
C. Use GKE to autoscale containers, and use gcloud commands to provision the infrastructure.
D. Use Dataflow to provision the data pipeline, and use Cloud Scheduler to run the job.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Startup scripts and manual gcloud provisioning are not efficient or scalable.
B. Correct – Cloud Build + Terraform provides automated, repeatable provisioning of GKE with GPUs, SSDs, and required bandwidth, ensuring deployment uses latest container images.
C. Incorrect – GKE autoscaling helps containers, but provisioning infrastructure via gcloud is not automated or efficient.
D. Incorrect – Dataflow is for data pipelines, not provisioning infrastructure.
E. Incorrect – Option B is the proper approach.

Question 46
You need ads data to serve AI models and historical data for analytics. Longtail and outlier data points need to be identified. You want to cleanse the data in near-real time before running it through AI models. What should you do?

Options
A. Use Cloud Storage as a data warehouse, shell scripts for processing, and BigQuery to create views for desired datasets.
B. Use Dataflow to identify longtail and outlier data points programmatically, with BigQuery as a sink.
C. Use BigQuery to ingest, prepare, and then analyze the data, and then run queries to create views.
D. Use Cloud Composer to identify longtail and outlier data points, and then output a usable dataset to BigQuery.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Cloud Storage is not a data warehouse, and shell scripts lack real-time capability.
B. Correct – Dataflow is designed for real-time/near-real-time stream processing, can identify outliers programmatically, and write clean data into BigQuery.
C. Incorrect – BigQuery queries are batch-oriented and not suited for near-real-time cleansing.
D. Incorrect – Cloud Composer is workflow orchestration, not stream cleansing.
E. Incorrect – Option B is correct.

Question 47
You are collecting IoT sensor data from millions of devices across the world and storing the data in BigQuery. Your access pattern is based on recent data, filtered by location_id and device_version with the following query:

```
SELECT
    MAX(temperature)
FROM
    acme_iot_data.sensors
WHERE
    create_date > DATE_SUB(CURRENT_DATE(), INTERVAL 7 day)
    AND location_id = "SW1W9TQ"
    AND device_version = "202007r3"
```

You want to optimize your queries for cost and performance. How should you structure your data?

Options
A. Partition table data by create_date, location_id, and device_version.
B. Partition table data by create_date, cluster table data by location_id, and device_version.
C. Cluster table data by create_date, location_id, and device_version.
D. Cluster table data by create_date, partition by location_id, and device_version.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – BigQuery allows partitioning on a single field, not multiple like create_date + location_id + device_version.
B. Correct – Partitioning on create_date reduces scanned data by time, clustering on location_id and device_version optimizes filtering, reducing cost and improving performance.
C. Incorrect – Clustering alone without partitioning may scan more data than necessary.
D. Incorrect – Partitioning on multiple non-date fields isn't supported in the needed way.
E. Incorrect – Option B is the correct approach.

Question 48
A live TV show asks viewers to cast votes using their mobile phones. The event generates a large volume of data during a 3-minute period. You are in charge of the "Voting infrastructure" and must ensure that the platform can handle the load and that all votes are processed. You must display partial results while voting is open. After voting closes, you need to count the votes exactly once while optimizing cost. What should you do?

Options
A. Create a Memorystore instance with a high availability (HA) configuration.
B. Create a Cloud SQL for PostgreSQL database with high availability (HA) configuration and multiple read replicas.
C. Write votes to a Pub/Sub topic and have Cloud Functions subscribe to it and write votes to BigQuery.
D. Write votes to a Pub/Sub topic and load into both Bigtable and BigQuery via a Dataflow pipeline. Query Bigtable for real-time results and BigQuery for later analysis. Shut down the Bigtable instance when voting concludes.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Memorystore cannot scale to millions of concurrent events cost-effectively for analytics.
B. Incorrect – Cloud SQL may not handle extreme burst loads efficiently.
C. Incorrect – Pub/Sub + Cloud Functions only pushes to BigQuery, which is not ideal for real-time partial results.
D. Correct – Pub/Sub handles ingestion, Dataflow fans out to Bigtable (for real-time partial results) and BigQuery (for exact final count). Bigtable can be shut down after voting to optimize cost.
E. Incorrect – Option D is the best solution.

Question 49
A shipping company has live package-tracking data that is sent to an Apache Kafka stream in real time. This is then loaded into BigQuery. Analysts in your company want to query the tracking data in BigQuery to analyze geospatial trends in the lifecycle of a package. The table was originally created with ingest-date partitioning. Over time, the query processing time has increased. You need to copy all the data to a new clustered table. What should you do?

Options
A. Re-create the table using data partitioning on the package delivery date.
B. Implement clustering in BigQuery on the package-tracking ID column.
C. Implement clustering in BigQuery on the ingest date column.
D. Tier older data onto Cloud Storage files and create a BigQuery table using Cloud Storage as an external data source.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Partitioning on package delivery date may not align with query patterns on package IDs.
B. Correct – Clustering by package-tracking ID optimizes geospatial and lifecycle analysis queries by reducing scan size.
C. Incorrect – Clustering on ingest date does not improve query performance meaningfully here.
D. Incorrect – External tables add complexity and slower query performance.
E. Incorrect – B is the correct choice.

Question 50
You are designing a data mesh on Google Cloud with multiple distinct data engineering teams building data products. The typical data curation design pattern consists of landing files in Cloud Storage, transforming raw data in Cloud Storage and BigQuery datasets, and storing the final curated data product in BigQuery datasets. You need to configure Dataplex to ensure that each team can access only the assets needed to build their data products. You also need to ensure that teams can easily share the curated data product. What should you do?

Options
A. 1. Create a single Dataplex virtual lake and create a single zone to contain landing, raw, and curated data.
2. Provide each data engineering team access to the virtual lake.
B. 1. Create a single Dataplex virtual lake and create a single zone to contain landing, raw, and curated data.
2. Build separate assets for each data product within the zone.
3. Assign permissions to the data engineering teams at the zone level.
C. 1. Create a Dataplex virtual lake for each data product, and create a single zone to contain landing, raw, and curated data.
2. Provide the data engineering teams with full access to the virtual lake assigned to their data product.
D. 1. Create a Dataplex virtual lake for each data product, and create multiple zones for landing, raw, and curated data.
2. Provide the data engineering teams with full access to the virtual lake assigned to their data product.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – A single zone for all data does not allow fine-grained access control.
B. Incorrect – Same issue as A; permissions at zone level don't provide proper separation between teams.
C. Incorrect – A single zone per lake does not separate landing/raw/curated layers.
D. Correct – Best practice is a virtual lake per data product, with multiple zones for landing, raw, and curated layers, ensuring team-level separation and curated data sharing.
E. Incorrect – Option D fits perfectly.

Question 51
You are using BigQuery with a multi-region dataset that includes a table with the daily sales volumes. This table is updated multiple times per day. You need to protect your sales table in case of regional failures with a recovery point objective (RPO) of less than 24 hours, while keeping costs to a minimum. What should you do?

Options
A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket.
B. Schedule a daily copy of the dataset to a backup region.
C. Schedule a daily BigQuery snapshot of the table.
D. Modify ETL job to load the data into both the current and another backup region.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Exporting daily to a dual/multi-region Cloud Storage bucket provides an off-region copy with RPO < 24 hours, is low maintenance and cost-efficient compared to running duplicate BigQuery datasets.
B. Incorrect — Copying the entire dataset daily to another region increases BigQuery storage and egress costs and is heavier than exporting to Cloud Storage.
C. Incorrect — BigQuery snapshots are regional and don't provide cross-region protection by themselves; they won't meet the multi-region failure protection requirement.
D. Incorrect — Writing data to two regions in BigQuery increases complexity and cost (and is more effort) compared with daily export to multi-region storage.
E. Incorrect — Option A is the recommended approach.

Question 52

You are troubleshooting your Dataflow pipeline that processes data from Cloud Storage to BigQuery. You have discovered that the Dataflow worker nodes cannot communicate with one another. Your networking team relies on Google Cloud network tags to define firewall rules. You need to identify the issue while following Google-recommended networking security practices. What should you do?

Options

A. Determine whether your Dataflow pipeline has a custom network tag set.
B. Determine whether there is a firewall rule set to allow traffic on TCP ports 12345 and 12346 for the Dataflow network tag.
C. Determine whether there is a firewall rule set to allow traffic on TCP ports 12345 and 12346 on the subnet used by Dataflow workers.
D. Determine whether your Dataflow pipeline is deployed with the external IP address option enabled.
E. None of the above

Answer

B

Detailed explanation

A. Incorrect — Dataflow workers don't use custom user-defined network tags by default; checking for a custom tag alone won't identify missing firewall rules.
B. Correct — If your networking team relies on network tags to define firewall rules, you must verify that the firewall rules tied to the Dataflow network tag permit the required worker-to-worker traffic (the question calls out specific ports as the troubleshooting step). Checking the tag's firewall rules follows the team's practice and is the right diagnostic step.
C. Incorrect — While subnet-level rules matter, the team uses network tags for firewall rules, so inspecting subnet firewall rules is not aligned with their practice.
D. Incorrect — External IP usage affects NAT and internet egress, not direct worker-to-worker connectivity inside the VPC using network tags.
E. Incorrect — Option B is the correct diagnostic action.

Question 53

Your company's customer_order table in BigQuery stores the order history for 10 million customers, with a table size of 10 PB. You need to create a dashboard for the support team to view the order history. The dashboard has two filters, country_name and username. Both are string data types in the BigQuery table. When a filter is applied, the dashboard fetches the order history from the table and displays the query results. However, the dashboard is slow to show the results when applying the filters to the following query:

```
SELECT date, order, status FROM customer_order
WHERE country = '<country_name>' AND username = '<username>'
```

How should you redesign the BigQuery table to support faster access?

Options

A. Cluster the table by country and username fields.
B. Cluster the table by country field, and partition by username field.
C. Partition the table by country and username fields.
D. Partition the table by _PARTITIONTIME.
E. None of the above

Answer

A

Detailed explanation

A. Correct — Clustering by country and username organizes data physically by those filter keys so queries that filter on those columns scan much less data, improving performance for dashboard filters.
B. Incorrect — Partitioning by username is not appropriate (partitioning is typically on a date or ingestion-time); BigQuery doesn't support partitioning by arbitrary high-cardinality string columns efficiently.
C. Incorrect — BigQuery does not support partitioning on multiple arbitrary fields like country and username (partitioning is generally on a single date/timestamp or integer-range column).
D. Incorrect — Partitioning by _PARTITIONTIME helps time-based queries but does not help filtering by country and username—so it won't solve the dashboard latency.
E. Incorrect — Option A is the right redesign.

Question 54
You have a Standard Tier Memorystore for Redis instance deployed in a production environment. You need to simulate a Redis instance failover in the most accurate disaster recovery situation, and ensure that the failover has no impact on production data. What should you do?

Options
A. Create a Standard Tier Memorystore for Redis instance in the development environment. Initiate a manual failover by using the limited-data-loss data protection mode.
B. Create a Standard Tier Memorystore for Redis instance in a development environment. Initiate a manual failover by using the force-data-loss data protection mode.
C. Increase one replica to Redis instance in production environment. Initiate a manual failover by using the force-data-loss data protection mode.
D. Initiate a manual failover by using the limited-data-loss data protection mode to the Memorystore for Redis instance in the production environment.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Limited-data-loss mode is safer but initiating a failover on an instance in development using limited-data-loss may not simulate the full impact; also limited-data-loss may be hard to reproduce without proper replica setup.
B. Correct — To simulate an accurate disaster-recovery failover without impacting production data, perform the failover in a development environment using the force-data-loss mode to ensure the process mimics worst-case failover behavior; doing it in a dev instance avoids production impact.
C. Incorrect — Increasing a replica in production and forcibly failing over risks production data integrity and availability.
D. Incorrect — Initiating failover in production, even limited-data-loss, can impact production; the goal was no impact on production data.
E. Incorrect — Option B best matches “simulate accurately” while avoiding production impact.

Question 55
You are administering a BigQuery dataset that uses a customer-managed encryption key (CMEK). You need to share the dataset with a partner organization that does not have access to your CMEK. What should you do?

Options
A. Provide the partner organization a copy of your CMEKs to decrypt the data.
B. Export the tables to parquet files to a Cloud Storage bucket and grant the storageinsights.viewer role on the bucket to the partner organization.
C. Copy the tables you need to share to a dataset without CMEKs. Create an Analytics Hub listing for this dataset.
D. Create an authorized view that contains the CMEK to decrypt the data when accessed.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Sharing your CMEKs is insecure and violates key management best practices.
B. Incorrect — Exporting encrypted data still requires access to the CMEK for decryption, and granting an inappropriate role is not the recommended approach.
C. Correct — Copying (or exporting and re-importing) the required tables into a dataset not protected by your CMEK allows the partner to access data without CMEK access; using Analytics Hub makes sharing discoverable and manageable.
D. Incorrect — Authorized views do not bypass CMEK restrictions; the underlying data is still encrypted and requires key access.
E. Incorrect — Option C is the recommended secure approach.

Question 56
You are developing an Apache Beam pipeline to extract data from a Cloud SQL instance by using JdbcIO. You have two projects running in Google Cloud. The pipeline will be deployed and executed on Dataflow in Project A. The Cloud SQL instance is running in Project B and does not have a public IP address. After deploying the pipeline, you noticed that the pipeline failed to extract data from the Cloud SQL instance due to connection failure. You verified that VPC Service Controls and shared VPC are not in use in these projects. You want to resolve this error while ensuring that the data does not go through the public internet. What should you do?

Options
A. Set up VPC Network Peering between Project A and Project B. Add a firewall rule to allow the peered subnet range to access all instances on the network.
B. Turn off the external IP addresses on the Dataflow worker. Enable Cloud NAT in Project A.
C. Add the external IP addresses of the Dataflow worker as authorized networks in the Cloud SQL instance.
D. Set up VPC Network Peering between Project A and Project B. Create a Compute Engine instance without external IP address in Project B on the peered subnet to serve as a proxy server to the Cloud SQL database.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — VPC Network Peering alone doesn't allow cross-project private connectivity to Cloud SQL when Cloud SQL has no public IP; Cloud SQL private IP requires using VPC-hosted connectivity or a proxy in the same VPC.
B. Incorrect — Disabling external IPs and enabling Cloud NAT still sends traffic through NAT/public paths; it does not provide private connectivity to a Cloud SQL instance in another project lacking public IP.
C. Incorrect — Authorizing Dataflow worker external IPs would require public connectivity and exposes data to the internet — not acceptable when avoiding public internet.
D. Correct — With VPC peering and a proxy Compute Engine instance (no external IP) in Project B on the peered subnet, Dataflow workers in Project A can reach the proxy over private network; the proxy can connect to the Cloud SQL private IP, keeping traffic off the public internet.
E. Incorrect — Option D meets the requirement.

Question 57
You have a BigQuery table that contains customer data, including sensitive information such as names and addresses. You need to share the customer data with your data analytics and consumer support teams securely. The data analytics team needs to access the data of all the customers, but must not be able to access the sensitive data. The consumer support team needs access to all data columns, but must not be able to access customers that no longer have active contracts. You enforced these requirements by using an authorized dataset and policy tags. After implementing these steps, the data analytics team reports that they still have access to the sensitive columns. You need to ensure that the data analytics team does not have access to restricted data. What should you do? (Choose two.)

Options
A. Create two separate authorized datasets; one for the data analytics team and another for the consumer support team.
B. Ensure that the data analytics team members do not have the Data Catalog Fine-Grained Reader role for the policy tags.
C. Replace the authorized dataset with an authorized view. Use row-level security and apply filter_expression to limit data access.
D. Remove the bigquery.dataViewer role from the data analytics team on the authorized datasets.
E. Enforce access control in the policy tag taxonomy.
F. None of the above

Answer
BE

Detailed explanation
A. Incorrect — While separate authorized datasets could work, the question asks what you should do given policy tags are already in use and that analytics still see sensitive columns. Creating separate datasets is not one of the chosen corrective steps here.
B. Correct — If analytics team members have the Data Catalog Fine-Grained Reader role for policy tags (or higher), they can read policy tag-protected columns. Removing that role prevents them from accessing the sensitive columns.
C. Incorrect — Authorized views with row-level security help with row access control but do not address column-level access via policy tags in this scenario; also the question already uses policy tags.
D. Incorrect — Removing bigquery.dataViewer could block all BigQuery read access and is too blunt; the recommended fix is to adjust policy-tag and Data Catalog roles.
E. Correct — Enforcing/access-controlling the policy tag taxonomy (ensuring policy tags are applied and that IAM on taxonomy is correct) is necessary so only authorized principals can

access tagged columns.
F. Incorrect — Options B and F are the appropriate corrective actions.

Question 58
You have a Cloud SQL for PostgreSQL instance in Region1 with one read replica in Region2 and another read replica in Region3. An unexpected event in Region1 requires that you perform disaster recovery by promoting a read replica in Region2. You need to ensure that your application has the same database capacity available before you switch over the connections. What should you do?

Options
A. Enable zonal high availability on the primary instance. Create a new read replica in a new region.
B. Create a cascading read replica from the existing read replica in Region3.
C. Create two new read replicas from the new primary instance, one in Region3 and one in a new region.
D. Create a new read replica in Region1, promote the new read replica to be the primary instance, and enable zonal high availability.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Enabling zonal HA on the original primary after the primary has failed in Region1 is not possible and won't restore capacity before failover.
B. Incorrect — Creating a cascading replica from Region3 doesn't guarantee restoring the same application capacity proactively once you promote Region2; it's less direct.
C. Correct — After promoting the read replica in Region2 to primary, create two new read replicas (one back in Region3 and another in a new region) to restore the previous replica topology and capacity so the application sees similar capacity and redundancy before switching connections.
D. Incorrect — Creating a new replica in Region1 (the failed region) may not be possible immediately if Region1 is affected; promoting a replica in the affected region is not applicable.
E. Incorrect — Option C is the right recovery plan.

Question 59

You orchestrate ETL pipelines by using Cloud Composer. One of the tasks in the Apache Airflow directed acyclic graph (DAG) relies on a third-party service. You want to be notified when the task does not succeed. What should you do?

Options

A. Assign a function with notification logic to the on_retry_callback parameter for the operator responsible for the task at risk.
B. Configure a Cloud Monitoring alert on the sla_missed metric associated with the task at risk to trigger a notification.
C. Assign a function with notification logic to the on_failure_callback parameter for the operator responsible for the task at risk.
D. Assign a function with notification logic to the sla_miss_callback parameter for the operator responsible for the task at risk.
E. None of the above

Answer

C

Detailed explanation

A. Incorrect — on_retry_callback runs when a task is retried, not when it ultimately fails; it won't notify on final failure.
B. Incorrect — sla_missed alerts are for tasks that miss SLA windows, not for all task failures; SLA alerts are broader and may not capture immediate failures of a third-party-dependent task.
C. Correct — on_failure_callback executes when the task fails and is the intended Airflow operator hook for sending immediate failure notifications.
D. Incorrect — sla_miss_callback is invoked for SLA misses, not for general task failures.
E. Incorrect — Option C is the correct Airflow-level approach.

Question 60

You are migrating your on-premises data warehouse to BigQuery. One of the upstream data sources resides on a MySQL database that runs in your on-premises data center with no public IP addresses. You want to ensure that the data ingestion into BigQuery is done securely and does not go through the public internet. What should you do?

Options

A. Update your existing on-premises ETL tool to write to BigQuery by using the BigQuery Open Database Connectivity (ODBC) driver. Set up the proxy parameter in the simba.googlebigqueryodbc.ini file to point to your data center's NAT gateway.
B. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Set up Cloud Interconnect between your on-premises data center and Google Cloud. Use Private connectivity as the connectivity method and allocate an IP address range within your VPC network to the Datastream connectivity configuration. Use Server-only as the encryption type when setting up the connection profile in Datastream.
C. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Use Forward-SSH tunnel as the connectivity method to establish a secure tunnel between Datastream and your on-premises MySQL database through a tunnel server in your on-premises data center. Use None as the encryption type when setting up the connection profile in Datastream.
D. Use Datastream to replicate data from your on-premises MySQL database to BigQuery. Gather Datastream public IP addresses of the Google Cloud region that will be used to set up the stream. Add those IP addresses to the firewall allowlist of your on-premises data center. Use IP Allowlisting as the connectivity method and Server-only as the encryption type when setting up the connection profile in Datastream.
E. None of the above

Answer

B

Detailed explanation

A. Incorrect — Using an ODBC driver and NAT gateway does not provide the managed, private connectivity and secure replication that Datastream + Interconnect provides.
B. Correct — Datastream + Cloud Interconnect (private connectivity) with an allocated VPC IP range and Server-only encryption is the recommended approach to replicate on-prem MySQL to BigQuery securely without traversing the public internet.
C. Incorrect — Forward-SSH tunnel with encryption set to None is insecure and not recommended for production sensitive data.
D. Incorrect — IP allowlisting uses public IPs and does not avoid public internet exposure; it's less secure than private connectivity.
E. Incorrect — Option B is the appropriate secure method.

Question 61
You store and analyze your relational data in BigQuery on Google Cloud with all data that resides in US regions. You also have a variety of object stores across Microsoft Azure and Amazon Web Services (AWS), also in US regions. You want to query all your data in BigQuery daily with as little movement of data as possible. What should you do?

Options
A. Use BigQuery Data Transfer Service to load files from Azure and AWS into BigQuery.
B. Create a Dataflow pipeline to ingest files from Azure and AWS to BigQuery.
C. Load files from AWS and Azure to Cloud Storage with Cloud Shell gsutil rsync arguments.
D. Use the BigQuery Omni functionality and BigLake tables to query files in Azure and AWS.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Data Transfer Service is for scheduled imports; it moves data into BigQuery and does not minimize data movement across clouds.
B. Incorrect — Building Dataflow pipelines would ingest (move) the data into BigQuery — more movement and operational overhead.
C. Incorrect — Using gsutil to sync files into Cloud Storage still requires moving data between clouds and manual orchestration.
D. Correct — BigQuery Omni + BigLake lets you query data in other clouds (Azure/AWS) with minimal data movement, enabling cross-cloud querying without repeatedly importing data.
E. Incorrect — Option D is the recommended approach.

Question 62
You have a variety of files in Cloud Storage that your data science team wants to use in their models. Currently, users do not have a method to explore, cleanse, and validate the data in Cloud Storage. You are looking for a low code solution that can be used by your data science team to quickly cleanse and explore data within Cloud Storage. What should you do?

Options
A. Provide the data science team access to Dataflow to create a pipeline to prepare and validate the raw data and load data into BigQuery for data exploration.
B. Create an external table in BigQuery and use SQL to transform the data as necessary. Provide the data science team access to the external tables to explore the raw data.
C. Load the data into BigQuery and use SQL to transform the data as necessary. Provide the data science team access to staging tables to explore the raw data.
D. Provide the data science team access to Dataprep to prepare, validate, and explore the data within Cloud Storage.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Dataflow is powerful but not low-code and imposes more engineering effort than requested.
B. Incorrect — External tables + SQL work but are not low-code and don't provide intuitive, visual data cleansing/exploration.
C. Incorrect — Loading into BigQuery then transforming provides strong capabilities but is more engineering-heavy than a low-code solution.

D. Correct — Dataprep (Cloud Dataprep by Trifacta) is a low-code, visual tool that lets data scientists explore, cleanse, and validate files in Cloud Storage quickly.
E. Incorrect — Option D matches the low-code requirement.

Question 63
You are building an ELT solution in BigQuery by using Dataform. You need to perform uniqueness and null value checks on your final tables. What should you do to efficiently integrate these checks into your pipeline?

Options
A. Build BigQuery user-defined functions (UDFs).
B. Create Dataplex data quality tasks.
C. Build Dataform assertions into your code.
D. Write a Spark-based stored procedure.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — UDFs can help validate values but don't integrate as assertion checks in Dataform's pipeline semantics.
B. Incorrect — Dataplex data quality is powerful, but if you're already using Dataform for ELT, Dataform assertions provide a simpler, better-integrated solution.
C. Correct — Dataform assertions let you define uniqueness and null checks directly in your ELT code so failures surface during runs — efficient and integrated.
D. Incorrect — Spark-based stored procedures add unnecessary complexity and aren't integrated with Dataform.
E. Incorrect — Option C is the best-fit approach.

Question 64
A web server sends click events to a Pub/Sub topic as messages. The web server includes an eventTimestamp attribute in the messages, which is the time when the click occurred. You have a Dataflow streaming job that reads from this Pub/Sub topic through a subscription, applies some transformations, and writes the result to another Pub/Sub topic for use by the advertising department. The advertising department needs to receive each message within 30 seconds of the corresponding click occurrence, but they report receiving the messages late. Your Dataflow job's system lag is about 5 seconds, and the data freshness is about 40 seconds. Inspecting a few messages show no more than 1 second lag between their eventTimestamp and publishTime. What is the problem and what should you do?

Options
A. The advertising department is causing delays when consuming the messages. Work with the advertising department to fix this.
B. Messages in your Dataflow job are taking more than 30 seconds to process. Optimize your job or increase the number of workers to fix this.
C. Messages in your Dataflow job are processed in less than 30 seconds, but your job cannot keep up with the backlog in the Pub/Sub subscription. Optimize your job or increase the number of workers to fix this.
D. The web server is not pushing messages fast enough to Pub/Sub. Work with the web server team to fix this.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Message timestamps show publish delays are not due to the consumer; the problem is upstream.
B. Incorrect — System lag (5s) and per-message timestamps indicate per-message processing is quick; the issue is backlog, not per-message processing time.
C. Correct — The job processes individual messages quickly but can't keep up with the incoming rate, causing a backlog and raising data freshness to ~40s; scaling or optimizing workers addresses this.
D. Incorrect — Publish times are within 1s of event time, so web server publishing is not the bottleneck.
E. Incorrect — Option C correctly identifies and prescribes the fix.

Question 65
Your organization stores customer data in an on-premises Apache Hadoop cluster in Apache Parquet format. Data is processed on a daily basis by Apache Spark jobs that run on the cluster. You are migrating the Spark jobs and Parquet data to Google Cloud. BigQuery will be used on future transformation pipelines so you need to ensure that your data is available in BigQuery. You want to use managed services, while minimizing ETL data processing changes and overhead costs. What should you do?

Options
A. Migrate your data to Cloud Storage and migrate the metadata to Dataproc Metastore (DPMS). Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc Serverless.
B. Migrate your data to Cloud Storage and register the bucket as a Dataplex asset. Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc Serverless.
C. Migrate your data to BigQuery. Refactor Spark pipelines to write and read data on BigQuery, and run them on Dataproc Serverless.
D. Migrate your data to BigLake. Refactor Spark pipelines to write and read data on Cloud Storage, and run them on Dataproc on Compute Engine.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — This is a valid approach but requires maintaining Parquet files and metastore; the user-specified correct answer is C.
B. Incorrect — Dataplex registration helps governance but doesn’t directly minimize refactor to BigQuery; not the chosen answer.
C. Correct — Migrating data into BigQuery and refactoring Spark to use BigQuery (while running on Dataproc Serverless) centralizes data for future BigQuery-based transformations — this matches the user's selected solution.
D. Incorrect — BigLake plus Dataproc on Compute Engine is viable, but it requires more infra management than the chosen approach.
E. Incorrect — Option C is the provided answer.

Question 66
Your organization has two Google Cloud projects, project A and project B. In project A, you have a Pub/Sub topic that receives data from confidential sources. Only the resources in project A should be able to access the data in that topic. You want to ensure that project B and any future project cannot access data in the project A topic. What should you do?

Options
A. Add firewall rules in project A so only traffic from the VPC in project A is permitted.
B. Configure VPC Service Controls in the organization with a perimeter around project A.
C. Use Identity and Access Management conditions to ensure that only users and service accounts in project A can access resources in project A.
D. Configure VPC Service Controls in the organization with a perimeter around the VPC of project A.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Firewall rules control network traffic, not Pub/Sub IAM access; they won't prevent other projects from accessing Pub/Sub via IAM.
B. Correct — VPC Service Controls with a perimeter around project A prevent cross-project access to service resources (like Pub/Sub) from outside the perimeter, protecting confidential data.
C. Incorrect — IAM conditions can help, but they're harder to enforce across all service access paths and don't provide the perimeter-level protection that VPC Service Controls do.
D. Incorrect — Perimeters are applied at project/resource boundaries rather than only at VPC-level; the correct approach is to perimeter the project.
E. Incorrect — Option B is the recommended approach.

Question 67
You stream order data by using a Dataflow pipeline, and write the aggregated result to Memorystore. You provisioned a Memorystore for Redis instance with Basic Tier, 4 GB capacity, which is used by 40 clients for read-only access. You are expecting the number of read-only clients to increase significantly to a few hundred and you need to be able to support the demand. You want to ensure that read and write access availability is not impacted, and any changes you make can be deployed quickly. What should you do?

Options
A. Create a new Memorystore for Redis instance with Standard Tier. Set capacity to 4 GB and read replica to No read replicas (high availability only). Delete the old instance.
B. Create a new Memorystore for Redis instance with Standard Tier. Set capacity to 5 GB and create multiple read replicas. Delete the old instance.
C. Create a new Memorystore for Memcached instance. Set a minimum of three nodes, and memory per node to 4 GB. Modify the Dataflow pipeline and all clients to use the Memcached instance. Delete the old instance.
D. Create multiple new Memorystore for Redis instances with Basic Tier (4 GB capacity). Modify the Dataflow pipeline and new clients to use all instances.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Upgrading to Standard Tier without read replicas misses read-scaling benefits; also capacity unchanged may be insufficient.
B. Correct — Standard Tier with increased capacity and multiple read replicas improves read scalability and availability, and is a quick replacement for the Basic instance.
C. Incorrect — Switching to Memcached requires significant client and pipeline changes and loses Redis semantics.
D. Incorrect — Multiple Basic Tier instances complicate architecture and don't provide the managed HA and replica features of Standard Tier.
E. Incorrect — Option B is the appropriate, quick-to-deploy scaling solution.

Question 68
You have a streaming pipeline that ingests data from Pub/Sub in production. You need to update this streaming pipeline with improved business logic. You need to ensure that the updated pipeline reprocesses the previous two days of delivered Pub/Sub messages. What should you do? (Choose two.)

Options
A. Use the Pub/Sub subscription clear-retry-policy flag.
B. Use Pub/Sub Snapshot capture two days before the deployment.
C. Create a new Pub/Sub subscription two days before the deployment.
D. Use the Pub/Sub subscription retain-acked-messages flag.
E. Use Pub/Sub Seek with a timestamp.
F. None of the above

Answer
BE

Detailed explanation
A. Incorrect — clear-retry-policy only clears retry policy settings; it does not enable reprocessing of already delivered messages.
B. Correct — Creating a Pub/Sub snapshot two days before deployment captures the message backlog at that point and allows the new pipeline to replay from the snapshot.
C. Incorrect — Creating a new subscription two days before does not enable replay of already-acked messages unless used with snapshots or retained messages.
D. Incorrect — retain-acked-messages lets you keep acked messages for potential replay but alone is not the standard approach in combination with Seek/snapshots for precise time-based replay.
E. Correct — Pub/Sub Seek with a timestamp allows replaying messages from a specified timestamp (e.g., 48 hours ago) so the updated pipeline can reprocess two days of messages.
F. Incorrect — Options B and E together implement the required reprocessing.

Question 69

You currently use a SQL-based tool to visualize your data stored in BigQuery. The data visualizations require the use of outer joins and analytic functions. Visualizations must be based on data that is no less than 4 hours old. Business users are complaining that the visualizations are too slow to generate. You want to improve the performance of the visualization queries while minimizing the maintenance overhead of the data preparation pipeline. What should you do?

Options

A. Create materialized views with the allow_non_incremental_definition option set to true for the visualization queries. Specify the max_staleness parameter to 4 hours and the enable_refresh parameter to true. Reference the materialized views in the data visualization tool.
B. Create views for the visualization queries. Reference the views in the data visualization tool.
C. Create a Cloud Function instance to export the visualization query results as parquet files to a Cloud Storage bucket. Use Cloud Scheduler to trigger the Cloud Function every 4 hours. Reference the parquet files in the data visualization tool.
D. Create materialized views for the visualization queries. Use the incremental updates capability of BigQuery materialized views to handle changed data automatically. Reference the materialized views in the data visualization tool.
E. None of the above

Answer

A

Detailed explanation

A. Correct — Materialized views with controlled staleness (max_staleness ≈ 4 hours) and auto-refresh drastically reduce query latency for complex analytics while matching the 4-hour freshness requirement, minimizing maintenance.
B. Incorrect — Regular views still execute the underlying heavy queries at runtime and won't improve performance.
C. Incorrect — Exporting via Cloud Functions and parquet adds maintenance/operational overhead and complexity compared with materialized views.
D. Incorrect — Incremental materialized views are great but may not support the full complexity (outer joins, analytic functions) without allow_non_incremental_definition; option A explicitly addresses that by permitting non-incremental definitions with staleness.
E. Incorrect — Option A is the best fit.

Question 70

You need to modernize your existing on-premises data strategy. Your organization currently uses:

• Apache Hadoop clusters for processing multiple large data sets, including on-premises Hadoop Distributed File System (HDFS) for data replication.

• Apache Airflow to orchestrate hundreds of ETL pipelines with thousands of job steps.

You need to set up a new architecture in Google Cloud that can handle your Hadoop workloads and requires minimal changes to your existing orchestration processes. What should you do?

Options

A. Use Bigtable for your large workloads, with connections to Cloud Storage to handle any HDFS use cases. Orchestrate your pipelines with Cloud Composer.

B. Use Dataproc to migrate Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Orchestrate your pipelines with Cloud Composer.

C. Use Dataproc to migrate Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Convert your ETL pipelines to Dataflow.

D. Use Dataproc to migrate your Hadoop clusters to Google Cloud, and Cloud Storage to handle any HDFS use cases. Use Cloud Data Fusion to visually design and deploy your ETL pipelines.

E. None of the above

Answer

B

Detailed explanation

A. Incorrect — Bigtable is not a drop-in replacement for Hadoop/Spark workloads and would require major application changes.

B. Correct — Dataproc provides a managed Hadoop/Spark environment (minimal code changes), Cloud Storage replaces HDFS, and Cloud Composer continues using Airflow for orchestration — this minimizes changes to orchestration.

C. Incorrect — Converting many ETL pipelines to Dataflow would require extensive rework of existing Spark/Hadoop jobs and is not “minimal changes.”

D. Incorrect — Cloud Data Fusion is useful but replacing existing Airflow orchestration would not meet the “minimal changes to your existing orchestration processes” requirement.

E. Incorrect — Option B fits the requirement best.

Question 71

You recently deployed several data processing jobs into your Cloud Composer 2 environment. You notice that some tasks are failing in Apache Airflow. On the monitoring dashboard, you see an increase in the total workers memory usage, and there were worker pod evictions. You need to resolve these errors. What should you do? (Choose two.)

Options

A. Increase the directed acyclic graph (DAG) file parsing interval.
B. Increase the Cloud Composer 2 environment size from medium to large.
C. Increase the maximum number of workers and reduce worker concurrency.
D. Increase the memory available to the Airflow workers.
E. Increase the memory available to the Airflow triggerer.

Answer

C

Detailed explanation

A. Incorrect — Increasing the DAG file parsing interval slows how often DAG files are scanned/parsed and can reduce scheduler load, but it doesn't address worker pod memory pressure or evictions.
B. Incorrect — Increasing the whole environment size may help, but it is a coarse-grained, higher-cost change; it's not the immediate targeted fix for worker memory/per-concurrency issues.
C. Correct — Increasing the maximum number of worker pods and lowering per-worker concurrency spreads the workload across more pods and reduces per-pod memory pressure, which helps prevent evictions while keeping task throughput.
D. Incorrect — Increasing worker memory is also a valid mitigation for memory pressure, but the provided answer set indicates C as the primary action; memory increases may be used in addition to (or instead of) C.
E. Incorrect — The triggerer handles scheduling triggers and is unlikely to alleviate worker pod memory evictions; it's not the right lever for worker memory pressure.

Question 72
You are on the data governance team and are implementing security requirements to deploy resources. You need to ensure that resources are limited to only the europe-west3 region. You want to follow Google-recommended practices. What should you do?

Options
A. Set the constraints/gcp.resourceLocations organization policy constraint to in:europe-west3-locations.
B. Deploy resources with Terraform and implement a variable validation rule to ensure that the region is set to the europe-west3 region for all resources.
C. Set the constraints/gcp.resourceLocations organization policy constraint to in:eu-locations.
D. Create a Cloud Function to monitor all resources created and automatically destroy the ones created outside the europe-west3 region.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Using the organization policy constraint constraints/gcp.resourceLocations set to in:europe-west3-locations enforces at the org/folder/project level that resources can only be created in that region — a Google-recommended, centralized approach.
B. Incorrect — Terraform validation is helpful but is not enforced outside Terraform runs (doesn't stop other methods of creating resources) and is not the recommended global enforcement mechanism.
C. Incorrect — in:eu-locations allows all EU regions, not the single europe-west3 region requested.
D. Incorrect — A Cloud Function that destroys non-compliant resources is reactive, fragile, and not recommended versus using Organization Policy enforcement.
E. Incorrect — Option A is the best practice.

Question 73
You are a BigQuery admin supporting a team of data consumers who run ad hoc queries and downstream reporting in tools such as Looker. All data and users are combined under a single organizational project. You recently noticed some slowness in query results and want to troubleshoot where the slowdowns are occurring. You think that there might be some job queuing or slot contention occurring as users run jobs, which slows down access to results. You need to investigate the query job information and determine where performance is being affected. What should you do?

Options
A. Use slot reservations for your project to ensure that you have enough query processing capacity and are able to allocate available slots to the slower queries.
B. Use Cloud Monitoring to view BigQuery metrics and set up alerts that let you know when a certain percentage of slots were used.
C. Use available administrative resource charts to determine how slots are being used and how jobs are performing over time. Run a query on the INFORMATION_SCHEMA to review query performance.
D. Use Cloud Logging to determine if any users or downstream consumers are changing or deleting access grants on tagged resources.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Reserving slots may solve capacity issues but is an action to remediate, not the diagnostic step. The question asks how to investigate where performance is being affected.
B. Incorrect — Cloud Monitoring is useful for alerts but doesn't give the deep job-level and historical slot-usage analysis required to pinpoint contention and queuing.
C. Correct — Administrative resource charts plus querying INFORMATION_SCHEMA.JOBS* provide historical and job-level visibility (slot usage, queued jobs, runtime statistics), enabling accurate diagnosis of queuing and slot contention.
D. Incorrect — Cloud Logging about IAM changes is unrelated to slot contention or job queuing analysis.
E. Incorrect — Option C is the correct investigative approach.

Question 74
You migrated a data backend for an application that serves 10 PB of historical product data for analytics. Only the last known state for a product, which is about 10 GB of data, needs to be served through an API to the other applications. You need to choose a cost-effective persistent storage solution that can accommodate the analytics requirements and the API performance of up to 1000 queries per second (QPS) with less than 1 second latency. What should you do?

Options
A. 1. Store the historical data in BigQuery for analytics. 2. Use a materialized view to precompute the last state of a product. 3. Serve the last state data directly from BigQuery to the API.
B. 1. Store the products as a collection in Firestore with each product having a set of historical changes. 2. Use simple and compound queries for analytics. 3. Serve the last state data directly from Firestore to the API.
C. 1. Store the historical data in Cloud SQL for analytics. 2. In a separate table, store the last state of the product after every product change. 3. Serve the last state data directly from Cloud SQL to the API.
D. 1. Store the historical data in BigQuery for analytics. 2. In a Cloud SQL table, store the last state of the product after every product change. 3. Serve the last state data directly from Cloud SQL to the API.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Serving high-QPS, low-latency API traffic directly from BigQuery is not appropriate (BigQuery is OLAP, not OLTP).
B. Incorrect — Firestore could serve low-latency APIs, but storing 10 PB of historical data in Firestore and using it for analytics is impractical and costly.
C. Incorrect — Storing 10 PB in Cloud SQL is not feasible for analytics scale.
D. Correct — BigQuery stores the large historical dataset for analytics. Cloud SQL stores the small (≈10 GB) latest-state dataset, which can serve 1000 QPS with <1s latency. This design separates analytics and OLTP access and is cost-effective.
E. Incorrect — Option D is the right architectural trade-off.

Question 75

You want to schedule a number of sequential load and transformation jobs. Data files will be added to a Cloud Storage bucket by an upstream process. There is no fixed schedule for when the new data arrives. Next, a Dataproc job is triggered to perform some transformations and write the data to BigQuery. You then need to run additional transformation jobs in BigQuery. The transformation jobs are different for every table. These jobs might take hours to complete. You need to determine the most efficient and maintainable workflow to process hundreds of tables and provide the freshest data to your end users. What should you do?

Options

A. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Cloud Storage, Dataproc, and BigQuery operators. 2. Use a single shared DAG for all tables that need to go through the pipeline. 3. Schedule the DAG to run hourly.
B. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Cloud Storage, Dataproc, and BigQuery operators. 2. Create a separate DAG for each table that needs to go through the pipeline. 3. Schedule the DAGs to run hourly.
C. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators. 2. Use a single shared DAG for all tables that need to go through the pipeline. 3. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG.
D. 1. Create an Apache Airflow directed acyclic graph (DAG) in Cloud Composer with sequential tasks by using the Dataproc and BigQuery operators. 2. Create a separate DAG for each table that needs to go through the pipeline. 3. Use a Cloud Storage object trigger to launch a Cloud Function that triggers the DAG.
E. None of the above

Answer

D

Detailed explanation

A. Incorrect — A single shared DAG scheduled hourly doesn't scale well for hundreds of tables with no fixed arrival time and long-running per-table transforms.
B. Incorrect — Creating a separate DAG per table but scheduling hourly is inefficient and generates many idle schedules.
C. Incorrect — A single shared DAG invoked by a trigger can become complex to manage for many different per-table transformations and long-running jobs.
D. Correct — Use per-table DAGs triggered by Cloud Storage object notifications (via Cloud Function) so each table's pipeline runs only when data arrives. Per-table DAGs make the jobs independent, easier to manage for different transform logic, and avoid unnecessary scheduling; using Dataproc and BigQuery operators fits the workflow.
E. Incorrect — Option D is the most maintainable and efficient approach.

Question 76
You are deploying a MySQL database workload onto Cloud SQL. The database must be able to scale up to support several readers from various geographic regions. The database must be highly available and meet low RTO and RPO requirements, even in the event of a regional outage. You need to ensure that interruptions to the readers are minimal during a database failover. What should you do?

Options
A. Create a highly available Cloud SQL instance in region A. Create a highly available read replica in region B. Scale up read workloads by creating cascading read replicas in multiple regions. Backup the Cloud SQL instances to a multi-regional Cloud Storage bucket. Restore the Cloud SQL backup to a new instance in another region when Region A is down.
B. Create a highly available Cloud SQL instance in region A. Scale up read workloads by creating read replicas in multiple regions. Promote one of the read replicas when region A is down.
C. Create a highly available Cloud SQL instance in region A. Create a highly available read replica in region B. Scale up read workloads by creating cascading read replicas in multiple regions. Promote the read replica in region B when region A is down.
D. Create a highly available Cloud SQL instance in region A. Scale up read workloads by creating read replicas in the same region. Failover to the standby Cloud SQL instance when the primary instance fails.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Restoring backups to a new instance after a regional outage is slow (high RTO) and not suitable for minimal interruption to readers.
B. Incorrect — Read replicas without HA on the replica side might not provide the desired availability characteristics and orchestrated cascading topology.
C. Correct — Having HA primary in region A and an HA read replica in region B plus cascading replicas across regions provides near-continuous read capacity and a fast promotion path in region B when region A fails, minimizing reader interruption.
D. Incorrect — Read replicas only in the same region do not protect against a regional outage.
E. Incorrect — Option C is the appropriate configuration.

Question 77
You are planning to load some of your existing on-premises data into BigQuery on Google Cloud. You want to either stream or batch-load data, depending on your use case. Additionally, you want to mask some sensitive data before loading into BigQuery. You need to do this in a programmatic way while keeping costs to a minimum. What should you do?

Options
A. Use Cloud Data Fusion to design your pipeline, use the Cloud DLP plug-in to de-identify data within your pipeline, and then move the data into BigQuery.
B. Use the BigQuery Data Transfer Service to schedule your migration. After the data is populated in BigQuery, use the connection to the Cloud Data Loss Prevention (Cloud DLP) API to de-identify the necessary data.
C. Create your pipeline with Dataflow through the Apache Beam SDK for Python, customizing separate options within your code for streaming, batch processing, and Cloud DLP. Select BigQuery as your data sink.
D. Set up Datastream to replicate your on-premise data on BigQuery.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Cloud Data Fusion with DLP plugin is low-code but can be more costly for large or highly-custom pipelines and might not be as programmatically flexible as Dataflow.
B. Incorrect — Loading raw data into BigQuery then de-identifying in-place may expose sensitive data in BigQuery before masking; also using DLP post-load may complicate access controls.
C. Correct — Dataflow (Apache Beam) lets you build a single programmatic pipeline that supports both streaming and batch modes, integrates with Cloud DLP for masking during processing, and writes directly to BigQuery — flexible and cost-efficient for programmatic control.
D. Incorrect — Datastream is for change data capture/replication; it does not provide integrated masking with Cloud DLP during ingestion.
E. Incorrect — Option C meets the requirements.

Question 78
You want to encrypt the customer data stored in BigQuery. You need to implement per-user crypto-deletion on data stored in your tables. You want to adopt native features in Google Cloud to avoid custom solutions. What should you do?

Options
A. Implement Authenticated Encryption with Associated Data (AEAD) BigQuery functions while storing your data in BigQuery.
B. Create a customer-managed encryption key (CMEK) in Cloud KMS. Associate the key to the table while creating the table.
C. Create a customer-managed encryption key (CMEK) in Cloud KMS. Use the key to encrypt data before storing in BigQuery.
D. Encrypt your data during ingestion by using a cryptographic library supported by your ETL pipeline.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Using AEAD (e.g., Google Cloud's envelope encryption functions available in supported services) allows per-user encryption constructs and enables crypto-deletion semantics (delete the key/associated data to effectively render specific user data unrecoverable) while leveraging native features.
B. Incorrect — CMEK at the table level protects the entire table but does not provide per-user crypto-deletion granularity — deleting the CMEK affects the whole table.
C. Incorrect — Client-side encryption with a CMEK is not a native BigQuery-managed per-user crypto-deletion approach and is operationally complex.
D. Incorrect — Custom encryption during ingestion is a custom solution you wanted to avoid.
E. Incorrect — Option A matches the native-feature requirement.

Question 79
The data analyst team at your company uses BigQuery for ad-hoc queries and scheduled SQL pipelines in a Google Cloud project with a slot reservation of 2000 slots. However, with the recent introduction of hundreds of new non time-sensitive SQL pipelines, the team is encountering frequent quota errors. You examine the logs and notice that approximately 1500 queries are being triggered concurrently during peak time. You need to resolve the concurrency issue. What should you do?

Options
A. Increase the slot capacity of the project with baseline as 0 and maximum reservation size as 3000.
B. Update SQL pipelines to run as a batch query, and run ad-hoc queries as interactive query jobs.
C. Increase the slot capacity of the project with baseline as 2000 and maximum reservation size as 3000.
D. Update SQL pipelines and ad-hoc queries to run as interactive query jobs.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Increasing slot capacity without addressing concurrency patterns may be expensive and unnecessary; also baseline 0 is not appropriate.
B. Correct — Running non time-sensitive pipelines as batch jobs reduces concurrent slot demand (batch jobs wait for available capacity), while ad-hoc interactive queries remain responsive. This reduces contention without major slot increases.
C. Incorrect — Increasing baseline to 2000 with max 3000 isn't a targeted fix for concurrency spikes and may be costly.
D. Incorrect — Making everything interactive increases contention and worsens concurrency problems.
E. Incorrect — Option B is the appropriate operational change.

Question 80
You are designing a data mesh on Google Cloud by using Dataplex to manage data in BigQuery and Cloud Storage. You want to simplify data asset permissions. You are creating a customer virtual lake with two user groups:
• Data engineers, which require full data lake access
• Analytic users, which require access to curated data
You need to assign access rights to these two groups. What should you do?

Options
A. 1. Grant the dataplex.dataOwner role to the data engineer group on the customer data lake.
2. Grant the dataplex.dataReader role to the analytic user group on the customer curated zone.
B. 1. Grant the dataplex.dataReader role to the data engineer group on the customer data lake.
2. Grant the dataplex.dataOwner to the analytic user group on the customer curated zone.
C. 1. Grant the bigquery.dataOwner role on BigQuery datasets and the storage.objectCreator role on Cloud Storage buckets to data engineers.
2. Grant the bigquery.dataViewer role on BigQuery datasets and the storage.objectViewer role on Cloud Storage buckets to analytic users.
D. 1. Grant the bigquery.dataViewer role on BigQuery datasets and the storage.objectViewer role on Cloud Storage buckets to data engineers.
2. Grant the bigquery.dataOwner role on BigQuery datasets and the storage.objectEditor role on Cloud Storage buckets to analytic users.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Granting dataplex.dataOwner to data engineers on the lake gives them full data-lake-level control; granting dataplex.dataReader to analytic users scoped to the curated zone gives read-only access to curated assets — aligning neatly with the two-group model and simplifying permissions.
B. Incorrect — Swapping roles reverses the intended privileges (readers become owners and vice versa), which is wrong.
C. Incorrect — Directly assigning BigQuery/Storage roles bypasses Dataplex's lake/zone abstraction and loses simplified asset-level governance that Dataplex provides.
D. Incorrect — This reverses responsibilities and grants analytic users owner-level privileges incorrectly.
E. Incorrect — Option A is the recommended approach.

Question 81
You currently have a single on-premises Kafka cluster in a data center in the us-east region that is responsible for ingesting messages from IoT devices globally. Because large parts of globe have poor internet connectivity, messages sometimes batch at the edge, come in all at once, and cause a spike in load on your Kafka cluster. This is becoming difficult to manage and prohibitively expensive. What is the Google-recommended cloud native architecture for this scenario?

Options
A. Edge TPUs as sensor devices for storing and transmitting the messages.
B. Cloud Dataflow connected to the Kafka cluster to scale the processing of incoming messages.
C. An IoT gateway connected to Cloud Pub/Sub, with Cloud Dataflow to read and process the messages from Cloud Pub/Sub.
D. A Kafka cluster virtualized on Compute Engine in us-east with Cloud Load Balancing to connect to the devices around the world.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Edge TPUs are hardware accelerators for inference, not a message-ingestion/edge buffering architecture.
B. Incorrect — Connecting Dataflow directly to an on-prem Kafka cluster does not address the global-edge batching/connectivity problem or reduce operational burden.
C. Correct — An IoT gateway (edge) that publishes to Cloud Pub/Sub, combined with Cloud Dataflow for processing, is the cloud-native recommended architecture: Pub/Sub handles global, unreliable connectivity and buffering; Dataflow handles processing.
D. Incorrect — Virtualizing Kafka on Compute Engine and using load balancing increases operational complexity and cost and does not solve intermittent edge connectivity.
E. Incorrect — A correct option exists (C).

Question 82
You decided to use Cloud Datastore to ingest vehicle telemetry data in real time. You want to build a storage system that will account for the long-term data growth, while keeping the costs low. You also want to create snapshots of the data periodically, so that you can make a point-in-time (PIT) recovery, or clone a copy of the data for Cloud Datastore in a different environment. You want to archive these snapshots for a long time. Which two methods can accomplish this? (Choose two.)

Options
A. Use managed export, and store the data in a Cloud Storage bucket using Nearline or Coldline class.
B. Use managed export, and then import to Cloud Datastore in a separate project under a unique namespace reserved for that export.
C. Use managed export, and then import the data into a BigQuery table created just for that export, and delete temporary export files.
D. Write an application that uses Cloud Datastore client libraries to read all the entities. Treat each entity as a BigQuery table row via BigQuery streaming insert. Assign an export timestamp for each export, and attach it as an extra column for each row. Make sure that the BigQuery table is partitioned using the export timestamp column.
E. Write an application that uses Cloud Datastore client libraries to read all the entities. Format the exported data into a JSON file. Apply compression before storing the data in Cloud Source Repositories.

Answer
A, B

Detailed explanation
A. Correct — Managed export to Cloud Storage and using Nearline/Coldline is a standard, cost-effective archival approach for long-term snapshots.
B. Correct — Managed export plus importing into a separate Datastore project/namespace allows PIT recovery or cloning to another environment.
C. Incorrect — Importing exports into BigQuery and deleting temporary files is not a standard archive/cloning workflow for Datastore snapshots and loses the native Datastore export semantics.
D. Incorrect — Streaming all entities into BigQuery via a custom app is complex, expensive, and not the recommended snapshot/archive method.
E. Incorrect — Storing compressed JSON in Cloud Source Repositories is inappropriate (Source Repos is for code, not long-term data archives) and not recommended.

Question 83
You need to create a data pipeline that copies time-series transaction data so that it can be queried from within BigQuery by your data science team for analysis. Every hour, thousands of transactions are updated with a new status. The size of the initial dataset is 1.5 PB, and it will grow by 3 TB per day. The data is heavily structured, and your data science team will build machine learning models based on this data. You want to maximize performance and usability for your data science team. Which two strategies should you adopt? (Choose two.)

Options
A. Denormalize the data as must as possible.
B. Preserve the structure of the data as much as possible.
C. Use BigQuery UPDATE to further reduce the size of the dataset.
D. Develop a data pipeline where status updates are appended to BigQuery instead of updated.
E. Copy a daily snapshot of transaction data to Cloud Storage and store it as an Avro file. Use BigQuery's support for external data sources to query.

Answer
A, D

Detailed explanation
A. Correct — Denormalizing (when appropriate) improves query performance in BigQuery and typically benefits analytics/ML workloads by reducing expensive joins.
B. Incorrect — Preserving strict normalized structure hurts performance in analytic systems like BigQuery; denormalization is usually preferred.
C. Incorrect — Frequent BigQuery UPDATEs at this scale are expensive and inefficient; not recommended for heavy write/update workloads.
D. Correct — Appending status updates and keeping history (rather than in-place updates) supports time-series and ML use-cases and allows efficient processing (e.g., using partitioned tables and periodic compaction).
E. Incorrect — Using external Avro on Cloud Storage adds complexity and is generally slower/less feature-rich than native BigQuery storage for large analytic datasets.

Question 84
You are designing a cloud-native historical data processing system to meet the following conditions:
☞ The data being analyzed is in CSV, Avro, and PDF formats and will be accessed by multiple analysis tools including Dataproc, BigQuery, and Compute Engine.
☞ A batch pipeline moves daily data.
☞ Performance is not a factor in the solution.
☞ The solution design should maximize availability.
How should you design data storage for this solution?

Options
A. Create a Dataproc cluster with high availability. Store the data in HDFS, and perform analysis as needed.
B. Store the data in BigQuery. Access the data using the BigQuery Connector on Dataproc and Compute Engine.
C. Store the data in a regional Cloud Storage bucket. Access the bucket directly using Dataproc, BigQuery, and Compute Engine.
D. Store the data in a multi-regional Cloud Storage bucket. Access the data directly using Dataproc, BigQuery, and Compute Engine.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — HDFS on Dataproc increases operational complexity and is less available than Cloud Storage; not ideal for maximizing availability.
B. Incorrect — BigQuery is for structured analytics; storing raw mixed-format files (CSV, Avro, PDF) in BigQuery is impractical.
C. Incorrect — Regional buckets have lower availability than multi-regional; multi-regional increases availability for global access.
D. Correct — Multi-regional Cloud Storage provides high availability and durability for mixed-format files and is directly accessible by Dataproc, BigQuery (via external tables), and Compute Engine.
E. Incorrect — A correct option exists (D).

Question 85
You have a petabyte of analytics data and need to design a storage and processing platform for it. You must be able to perform data warehouse-style analytics on the data in Google Cloud and expose the dataset as files for batch analysis tools in other cloud providers. What should you do?

Options
A. Store and process the entire dataset in BigQuery.
B. Store and process the entire dataset in Bigtable.
C. Store the full dataset in BigQuery, and store a compressed copy of the data in a Cloud Storage bucket.
D. Store the warm data as files in Cloud Storage, and store the active data in BigQuery. Keep this ratio as 80% warm and 20% active.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Storing everything only in BigQuery may be fine for analytics, but it doesn't provide an easily-exportable file-based dataset for external batch tools without extra copies.
B. Incorrect — Bigtable is for low-latency key-value workloads, not for data-warehouse-style analytics.
C. Correct — Store and process in BigQuery for analytics while keeping a compressed file copy in Cloud Storage for export/interoperability with other clouds—this satisfies both analytics and file-exchange requirements.
D. Incorrect — Splitting warm/active that way could be valid in some architectures, but C explicitly provides both BigQuery analytics and file export needs.
E. Incorrect — A correct option exists (C).

Question 86
You work for a manufacturing company that sources up to 750 different components, each from a different supplier. You've collected a labeled dataset that has on average 1000 examples for each unique component. Your team wants to implement an app to help warehouse workers recognize incoming components based on a photo of the component. You want to implement the first working version of this app (as Proof-Of-Concept) within a few working days. What should you do?

Options
A. Use Cloud Vision AutoML with the existing dataset.
B. Use Cloud Vision AutoML, but reduce your dataset twice.
C. Use Cloud Vision API by providing custom labels as recognition hints.
D. Train your own image recognition model leveraging transfer learning techniques.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud Vision AutoML (now Vertex AutoML/AutoML Vision) is designed for quick POCs with modest-labeled datasets (~1000 images per class) and minimal ML expertise—fastest route to working app.
B. Incorrect — Reducing dataset size makes the model weaker; no reason to downsample for a POC.
C. Incorrect — Vision API with hints is for limited recognition, but AutoML custom models are a better fit for many classes.
D. Incorrect — Training custom models takes longer and requires more engineering; not ideal to get a POC within days.
E. Incorrect — A correct option exists (A).

Question 87
You are working on a niche product in the image recognition domain. Your team has developed a model that is dominated by custom C++ TensorFlow ops your team has implemented. These ops are used inside your main training loop and are performing bulky matrix multiplications. It currently takes up to several days to train a model. You want to decrease this time significantly and keep the cost low by using an accelerator on Google Cloud. What should you do?

Options
A. Use Cloud TPUs without any additional adjustment to your code.
B. Use Cloud TPUs after implementing GPU kernel support for your customs ops.
C. Use Cloud GPUs after implementing GPU kernel support for your customs ops.
D. Stay on CPUs, and increase the size of the cluster you're training your model on.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — TPUs require ops to have TPU-compatible kernels; custom C++ ops won't run on TPUs without porting.
B. Incorrect — Implementing GPU kernel support does not make ops TPU-compatible; TPUs need specific kernels and XLA support—far more work.
C. Correct — Implementing GPU kernel support for custom ops and using Cloud GPUs is the most practical way to accelerate training with minimal porting effort and cost compared to rewriting for TPUs.

D. Incorrect — Scaling CPUs is less efficient and likely more costly than using GPUs for matrix-heavy ops.
E. Incorrect — A correct option exists (C).

Question 88
You work on a regression problem in a natural language processing domain, and you have 100M labeled examples in your dataset. You have randomly shuffled your data and split your dataset into train and test samples (in a 90/10 ratio). After you trained the neural network and evaluated your model on a test set, you discover that the root-mean-squared error (RMSE) of your model is twice as high on the train set as on the test set. How should you improve the performance of your model?

Options
A. Increase the share of the test sample in the train-test split.
B. Try to collect more data and increase the size of your dataset.
C. Try out regularization techniques (e.g., dropout or batch normalization) to avoid overfitting.
D. Increase the complexity of your model by, e.g., introducing an additional layer or increase sizing the size of vocabularies or n-grams used.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Increasing test share reduces training data and won't fix the problem where training error is worse than test error.
B. Incorrect — Collecting more data helps generalization but doesn't address the unusual situation where training error > test error.
C. Incorrect — Regularization reduces overfitting; but here training error being higher than test error indicates underfitting, not overfitting.
D. Correct — Higher training error than test error indicates underfitting; increasing model capacity (more layers, bigger vocab, richer features) is the appropriate response.
E. Incorrect — A correct option exists (D).

Question 89
You use BigQuery as your centralized analytics platform. New data is loaded every day, and an ETL pipeline modifies the original data and prepares it for the final users. This ETL pipeline is regularly modified and can generate errors, but sometimes the errors are detected only after 2 weeks. You need to provide a method to recover from these errors, and your backups should be optimized for storage costs. How should you organize your data in BigQuery and store your backups?

Options
A. Organize your data in a single table, export, and compress and store the BigQuery data in Cloud Storage.
B. Organize your data in separate tables for each month, and export, compress, and store the data in Cloud Storage.
C. Organize your data in separate tables for each month, and duplicate your data on a separate dataset in BigQuery.
D. Organize your data in separate tables for each month, and use snapshot decorators to restore the table to a time prior to the corruption.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Single table is harder to restore and export costs and storage for large tables are higher; monthly partitioning is better for cost-optimized backups.
B. Correct — Organizing by month and exporting/compressing to Cloud Storage optimizes storage (you can use Nearline/Coldline) and simplifies restoring specific time ranges.
C. Incorrect — Duplicating large data in BigQuery doubles storage costs; less cost-effective than exporting to Cloud Storage.
D. Incorrect — Snapshot decorators have limited retention and aren't the best long-term, cost-optimized backup strategy for errors discovered after weeks.
E. Incorrect — A correct option exists (B).

Question 90
The marketing team at your organization provides regular updates of a segment of your customer dataset. The marketing team has given you a CSV with 1 million records that must be updated in BigQuery. When you use the UPDATE statement in BigQuery, you receive a quotaExceeded error. What should you do?

Options
A. Reduce the number of records updated each day to stay within the BigQuery UPDATE DML statement limit.
B. Increase the BigQuery UPDATE DML statement limit in the Quota management section of the Google Cloud Platform Console.
C. Split the source CSV file into smaller CSV files in Cloud Storage to reduce the number of BigQuery UPDATE DML statements per BigQuery job.
D. Import the new records from the CSV file into a new BigQuery table. Create a BigQuery job that merges the new records with the existing records and writes the results to a new BigQuery table.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Throttling manual updates is cumbersome and not a scalable solution.
B. Incorrect — You cannot arbitrarily increase DML quota in the console to bypass limits; quotas are enforced and require special handling.
C. Incorrect — Splitting CSVs doesn't reduce DML quota problems; it still results in many UPDATE operations.
D. Correct — Best practice: load CSV into a staging table and run a MERGE (or write the merged results to a new table). This avoids many individual UPDATEs, is performant, and works within BigQuery best practices.
E. Incorrect — A correct option exists (D).

Question 91

As your organization expands its usage of GCP, many teams have started to create their own projects. Projects are further multiplied to accommodate different stages of deployments and target audiences. Each project requires unique access control configurations. The central IT team needs to have access to all projects. Furthermore, data from Cloud Storage buckets and BigQuery datasets must be shared for use in other projects in an ad hoc way. You want to simplify access control management by minimizing the number of policies. Which two steps should you take? (Choose two.)

Options

A. Use Cloud Deployment Manager to automate access provision.
B. Introduce resource hierarchy to leverage access control policy inheritance.
C. Create distinct groups for various teams, and specify groups in Cloud IAM policies.
D. Only use service accounts when sharing data for Cloud Storage buckets and BigQuery datasets.
E. For each Cloud Storage bucket or BigQuery dataset, decide which projects need access. Find all the active members who have access to these projects, and create a Cloud IAM policy to grant access to all these users.

Answer

B, C

Detailed explanation

A. Incorrect — Deployment Manager can help automate provisioning, but it doesn't by itself reduce the number of IAM policies or replace good identity/group design.
B. Correct — Resource hierarchy (organization → folders → projects) lets you place policies at higher levels and inherit them, dramatically reducing policy sprawl.
C. Correct — Using identity groups (Google Groups) and referencing groups in IAM policies reduces the number of policies and simplifies membership changes.
D. Incorrect — Service accounts are useful, but "only" using them is impractical; they don't address human access or policy minimization across teams.
E. Incorrect — Manually enumerating project members per resource is error-prone and multiplies policies; it increases complexity rather than minimizing it.

Question 92

Your United States-based company has created an application for assessing and responding to user actions. The primary table's data volume grows by 250,000 records per second. Many third parties use your application's APIs to build the functionality into their own frontend applications. Your application's APIs should comply with the following requirements:

- Single global endpoint
- ANSI SQL support
- Consistent access to the most up-to-date data

What should you do?

Options

A. Implement BigQuery with no region selected for storage or processing.
B. Implement Cloud Spanner with the leader in North America and read-only replicas in Asia and Europe.
C. Implement Cloud SQL for PostgreSQL with the master in North America and read replicas in Asia and Europe.
D. Implement Bigtable with the primary cluster in North America and secondary clusters in Asia and Europe.
E. None of the above

Answer

B

Detailed explanation

A. Incorrect — BigQuery is not designed as a low-latency OLTP/global single-endpoint API with ANSI SQL served live at extremely high write rates.
B. Correct — Cloud Spanner provides a global single endpoint, strong consistency, ANSI-SQL dialect support, and can scale to very high write rates with leader placement and read replicas for global reads.
C. Incorrect — Cloud SQL doesn’t scale to 250k rows/sec globally and read replicas are eventually consistent; not appropriate for that write scale.
D. Incorrect — Bigtable scales well but doesn’t provide ANSI SQL and typical SQL semantics required by third parties.
E. Incorrect — A correct option exists (B).

Question 93

A data scientist has created a BigQuery ML model and asks you to create an ML pipeline to serve predictions. You have a REST API application with the requirement to serve predictions for an individual user ID with latency under 100 milliseconds. You use the following query to generate predictions: SELECT predicted_label, user_id FROM ML.PREDICT (MODEL 'dataset.model', table user_features). How should you create the ML pipeline?

Options

A. Add a WHERE clause to the query, and grant the BigQuery Data Viewer role to the application service account.
B. Create an Authorized View with the provided query. Share the dataset that contains the view with the application service account.
C. Create a Dataflow pipeline using BigQueryIO to read results from the query. Grant the Dataflow Worker role to the application service account.
D. Create a Dataflow pipeline using BigQueryIO to read predictions for all users from the query. Write the results to Bigtable using BigtableIO. Grant the Bigtable Reader role to the application service account so that the application can read predictions for individual users from Bigtable.
E. None of the above

Answer

D

Detailed explanation

A. Incorrect — Adding WHERE and querying BigQuery per-request will likely exceed the 100 ms latency SLO; BigQuery query latency is typically higher for per-user online inference.
B. Incorrect — Authorized view does not solve the latency requirement for per-user real-time prediction under 100 ms.
C. Incorrect — Dataflow reading BigQuery query results and streaming them back to API does not by itself provide a low-latency per-request store.
D. Correct — Precompute predictions (or stream predictions) into Bigtable, which is optimized for low-latency single-row lookups (<100 ms). The application can then read predictions by user_id quickly.
E. Incorrect — A correct option exists (D).

Question 94
You are building an application to share financial market data with consumers, who will receive data feeds. Data is collected from the markets in real time. Consumers will receive the data in the following ways:
☞ Real-time event stream
☞ ANSI SQL access to real-time stream and historical data
☞ Batch historical exports
Which solution should you use?

Options
A. Cloud Dataflow, Cloud SQL, Cloud Spanner
B. Cloud Pub/Sub, Cloud Storage, BigQuery
C. Cloud Dataproc, Cloud Dataflow, BigQuery
D. Cloud Pub/Sub, Cloud Dataproc, Cloud SQL
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Cloud SQL/Spanner combinations don't provide the best integration for streaming + ANSI SQL analytics + batch exports in this pattern.
B. Correct — Cloud Pub/Sub handles real-time event streaming; BigQuery provides ANSI SQL access to both streaming and historical data; Cloud Storage supports batch exports—this trio matches requirements.
C. Incorrect — Dataproc isn't necessary here; Pub/Sub + Dataflow would be part of ingestion but option lacks Pub/Sub and Cloud Storage for exports.
D. Incorrect — Dataproc + Cloud SQL are not ideal for global real-time streaming + ANSI SQL analytics at scale.
E. Incorrect — A correct option exists (B).

Question 95
You are building a new application that you need to collect data from in a scalable way. Data arrives continuously from the application throughout the day, and you expect to generate approximately 150 GB of JSON data per day by the end of the year. Your requirements are:
☞ Decoupling producer from consumer
☞ Space and cost-efficient storage of the raw ingested data, which is to be stored indefinitely
☞ Near real-time SQL query
☞ Maintain at least 2 years of historical data, which will be queried with SQL
Which pipeline should you use to meet these requirements?

Options
A. Create an application that provides an API. Write a tool to poll the API and write data to Cloud Storage as gzipped JSON files.
B. Create an application that writes to a Cloud SQL database to store the data. Set up periodic exports of the database to write to Cloud Storage and load into BigQuery.
C. Create an application that publishes events to Cloud Pub/Sub, and create Spark jobs on Cloud Dataproc to convert the JSON data to Avro format, stored on HDFS on Persistent Disk.
D. Create an application that publishes events to Cloud Pub/Sub, and create a Cloud Dataflow pipeline that transforms the JSON event payloads to Avro, writing the data to Cloud Storage and BigQuery.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Polling an API is less scalable and doesn't provide decoupling or near real-time SQL querying.
B. Incorrect — Cloud SQL is not cost-efficient for indefinite storage of 2+ years of high-volume JSON data and is not ideal for near real-time analytical SQL.
C. Incorrect — Using Dataproc/HDFS adds operational overhead; Cloud-native managed services are preferred.
D. Correct — Pub/Sub decouples producers/consumers; Dataflow can transform and write to both Cloud Storage (cost-efficient archival) and BigQuery (near real-time SQL); this meets all requirements.
E. Incorrect — A correct option exists (D).

Question 96
You are running a pipeline in Dataflow that receives messages from a Pub/Sub topic and writes the results to a BigQuery dataset in the EU. Currently, your pipeline is located in europe-west4 and has a maximum of 3 workers, instance type n1-standard-1. You notice that during peak periods, your pipeline is struggling to process records in a timely fashion, when all 3 workers are at maximum CPU utilization. Which two actions can you take to increase performance of your pipeline? (Choose two.)

Options
A. Increase the number of max workers
B. Use a larger instance type for your Dataflow workers
C. Change the zone of your Dataflow pipeline to run in us-central1
D. Create a temporary table in Bigtable that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Bigtable to BigQuery
E. Create a temporary table in Cloud Spanner that will act as a buffer for new data. Create a new step in your pipeline to write to this table first, and then create a new pipeline to write from Cloud Spanner to BigQuery

Answer
A, B

Detailed explanation
A. Correct — Increasing max workers lets Dataflow autoscaling (or configured scaling) add more workers to handle peak throughput.
B. Correct — Using larger instance types provides more CPU/memory per worker and can improve per-worker throughput.
C. Incorrect — Moving to us-central1 would introduce cross-region latency and likely violate the EU dataset locality; it doesn't solve worker CPU limits.
D. Incorrect — Introducing Bigtable as an extra buffer adds complexity and is not the direct remedy for worker CPU saturation.
E. Incorrect — Similarly, Spanner adds complexity and cost; scaling workers or using larger VMs is more straightforward.

Question 97
You have a data pipeline with a Cloud Dataflow job that aggregates and writes time series metrics to Cloud Bigtable. This data feeds a dashboard used by thousands of users across the organization. You need to support additional concurrent users and reduce the amount of time required to write the data. Which two actions should you take? (Choose two.)

Options
A. Configure your Cloud Dataflow pipeline to use local execution
B. Increase the maximum number of Cloud Dataflow workers by setting maxNumWorkers in PipelineOptions
C. Increase the number of nodes in the Cloud Bigtable cluster
D. Modify your Cloud Dataflow pipeline to use the Flatten transform before writing to Cloud Bigtable
E. Modify your Cloud Dataflow pipeline to use the CoGroupByKey transform before writing to Cloud Bigtable

Answer
B, C

Detailed explanation
A. Incorrect — Local execution is not scalable for production workloads and will not help concurrency or write throughput.
B. Correct — Increasing maxNumWorkers allows Dataflow to parallelize writes more, reducing per-record processing time and supporting more concurrent users.
C. Correct — Adding Bigtable nodes increases write throughput and reduces latency for writes/reads, supporting dashboard scale.
D. Incorrect — Flatten is for combining PCollections and does not inherently improve Bigtable write performance; it can even merge unrelated streams incorrectly.
E. Incorrect — CoGroupByKey groups datasets by key for joins; it's not a general optimization for write throughput.

Question 98
You have several Spark jobs that run on a Cloud Dataproc cluster on a schedule. Some of the jobs run in sequence, and some of the jobs run concurrently. You need to automate this process. What should you do?

Options
A. Create a Cloud Dataproc Workflow Template
B. Create an initialization action to execute the jobs
C. Create a Directed Acyclic Graph in Cloud Composer
D. Create a Bash script that uses the Cloud SDK to create a cluster, execute jobs, and then tear down the cluster
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Workflow Templates are useful for batch cluster + job orchestration but are limited for complex scheduling and DAG-level dependencies across multiple jobs over time.
B. Incorrect — Initialization actions run at cluster startup and are not intended for orchestrating scheduled, interdependent jobs.
C. Correct — Cloud Composer (managed Airflow) lets you define DAGs with dependencies, schedule jobs, run some in sequence and some concurrently, and orchestrate Dataproc jobs.
D. Incorrect — Bash scripts are brittle and lack visibility/monitoring and proper dependency

handling at scale.
E. Incorrect — A correct option exists (C).

Question 99
You are building a new data pipeline to share data between two different types of applications: jobs generators and job runners. Your solution must scale to accommodate increases in usage and must accommodate the addition of new applications without negatively affecting the performance of existing ones. What should you do?

Options
A. Create an API using App Engine to receive and send messages to the applications
B. Use a Cloud Pub/Sub topic to publish jobs, and use subscriptions to execute them
C. Create a table on Cloud SQL, and insert and delete rows with the job information
D. Create a table on Cloud Spanner, and insert and delete rows with the job information
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — An API can work but is tightly coupled and doesn't provide the decoupling or fan-out guarantees required for scalable job distribution.
B. Correct — Cloud Pub/Sub decouples producers/consumers, supports fan-out, scales automatically, and allows adding new applications via subscriptions without impacting existing ones.
C. Incorrect — Using Cloud SQL as a queue is not scalable and creates contention and maintenance issues.
D. Incorrect — Cloud Spanner could be used as a coordinator but is overkill and not the recommended pub/sub pattern for decoupling and scaling job distribution.
E. Incorrect — A correct option exists (B).

Question 100
You need to create a new transaction table in Cloud Spanner that stores product sales data. You are deciding what to use as a primary key. From a performance perspective, which strategy should you choose?

Options
A. The current epoch time
B. A concatenation of the product name and the current epoch time
C. A random universally unique identifier number (version 4 UUID)
D. The original order identification number from the sales system, which is a monotonically increasing integer
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Using epoch time (monotonic) can cause hotspotting because writes target a narrow key range.
B. Incorrect — Prefixing with product name can still cause hotspotting for popular products and may create uneven write distribution.
C. Correct — Random UUIDs provide uniform distribution across Spanner nodes, avoiding hotspotting and improving write scalability and throughput.

D. Incorrect — Monotonically increasing integers cause hotspotting at the tail of the key range, negatively impacting write performance.
E. Incorrect — A correct option exists (C).

Question 101
Data Analysts in your company have the Cloud IAM Owner role assigned to them in their projects to allow them to work with multiple GCP products in their projects. Your organization requires that all BigQuery data access logs be retained for 6 months. You need to ensure that only audit personnel in your company can access the data access logs for all projects. What should you do?

Options
A. Enable data access logs in each Data Analyst's project. Restrict access to Stackdriver Logging via Cloud IAM roles.
B. Export the data access logs via a project-level export sink to a Cloud Storage bucket in the Data Analysts' projects. Restrict access to the Cloud Storage bucket.
C. Export the data access logs via a project-level export sink to a Cloud Storage bucket in a newly created projects for audit logs. Restrict access to the project with the exported logs.
D. Export the data access logs via an aggregated export sink to a Cloud Storage bucket in a newly created project for audit logs. Restrict access to the project that contains the exported logs.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Enabling logs in each project and restricting Stackdriver access per-project is possible but results in many scattered policies and does not centralize retention or access control; it’s not ideal when you must ensure only audit personnel have cross-project access.
B. Incorrect — Exporting to buckets in each analyst project still scatters log storage and requires maintaining access per bucket/project; this doesn’t centralize control.
C. Incorrect — A project-level export sink exports logs from a single project; to gather logs from many projects you’d need many sinks. This is better than (B) but still less efficient than an aggregated sink.
D. Correct — An aggregated (organization-level) export sink to a dedicated audit project’s Cloud Storage bucket centralizes logs, makes it easy to enforce a single 6-month retention and IAM policy, and ensures only audit personnel (who have access to that audit project) can read the logs. This minimizes per-project policy sprawl.
E. Incorrect — A correct option exists (D).

Question 102
Each analytics team in your organization is running BigQuery jobs in their own projects. You want to enable each team to monitor slot usage within their projects. What should you do?

Options
A. Create a Cloud Monitoring dashboard based on the BigQuery metric query/scanned_bytes
B. Create a Cloud Monitoring dashboard based on the BigQuery metric slots/allocated_for_project
C. Create a log export for each project, capture the BigQuery job execution logs, create a custom metric based on the totalSlotMs, and create a Cloud Monitoring dashboard based on the custom metric

D. Create an aggregated log export at the organization level, capture the BigQuery job execution logs, create a custom metric based on the totalSlotMs, and create a Cloud Monitoring dashboard based on the custom metric
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — scanned_bytes helps measure query data scanned and cost but does not directly show slot allocation or usage per project.
B. Correct — The BigQuery metric slots/allocated_for_project directly exposes how many slots are allocated to a project; a dashboard based on that metric lets teams monitor their slot allocation/usage.
C. Incorrect — Creating per-project log exports and custom metrics is more complex and unnecessary when a built-in metric (slots/allocated_for_project) exists.
D. Incorrect — Aggregated log exports + custom metrics is heavier-weight and not required given the existing slots metric.
E. Incorrect — A correct option exists (B).

Question 103
You are operating a streaming Cloud Dataflow pipeline. Your engineers have a new version of the pipeline with a different windowing algorithm and triggering strategy. You want to update the running pipeline with the new version. You want to ensure that no data is lost during the update. What should you do?

Options
A. Update the Cloud Dataflow pipeline inflight by passing the --update option with the --jobName set to the existing job name
B. Update the Cloud Dataflow pipeline inflight by passing the --update option with the --jobName set to a new unique job name
C. Stop the Cloud Dataflow pipeline with the Cancel option. Create a new Cloud Dataflow job with the updated code
D. Stop the Cloud Dataflow pipeline with the Drain option. Create a new Cloud Dataflow job with the updated code
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Using --update with the same job name attempts an in-place update. Many pipeline changes (especially windowing/triggering) are incompatible with in-place updates and can cause failures or lost data.
B. Incorrect — --update with a new job name does not update an existing job; it starts a distinct job and does not preserve in-flight data/state.
C. Incorrect — Cancel immediately stops the job and can cause in-flight data to be dropped; it does not guarantee no data loss.
D. Correct — Draining the existing streaming job lets it finish processing buffered/unacknowledged elements (no new input is accepted), ensuring no data loss, after which you can start a new job with the updated code. This is the recommended safe approach.
E. Incorrect — A correct option exists (D).

Question 104
You need to move 2 PB of historical data from an on-premises storage appliance to Cloud Storage within six months, and your outbound network capacity is constrained to 20 Mb/sec. How should you migrate this data to Cloud Storage?

Options
A. Use Transfer Appliance to copy the data to Cloud Storage
B. Use gsutil cp to compress the content being uploaded to Cloud Storage
C. Create a private URL for the historical data, and then use Storage Transfer Service to copy the data to Cloud Storage
D. Use trickle or ionice along with gsutil cp to limit the amount of bandwidth gsutil utilizes to less than 20 Mb/sec so it does not interfere with the production traffic
E. None of the above

Answer
A

Detailed explanation
A. Correct — Transfer Appliance is designed for large-scale offline data transfer (petabytes). With limited WAN bandwidth (20 Mb/s), shipping a Transfer Appliance is the practical, time-efficient, and reliable solution to move 2 PB within months.
B. Incorrect — Compression helps but transferring 2 PB over a 20 Mb/s link within six months is impractical (would take far longer).
C. Incorrect — Storage Transfer Service requires network transfer from the source; with constrained bandwidth it's impractical for PB-scale migration.
D. Incorrect — Throttling gsutil doesn't solve the fundamental capacity problem; it may avoid interfering with production but will not meet the time window.
E. Incorrect — A correct option exists (A).

Question 105
You receive data files in CSV format monthly from a third party. You need to cleanse this data, but every third month the schema of the files changes. Your requirements for implementing these transformations include:
☞ Executing the transformations on a schedule
☞ Enabling non-developer analysts to modify transformations
☞ Providing a graphical tool for designing transformations
What should you do?

Options
A. Use Dataprep by Trifacta to build and maintain the transformation recipes, and execute them on a scheduled basis
B. Load each month's CSV data into BigQuery, and write a SQL query to transform the data to a standard schema. Merge the transformed tables together with a SQL query
C. Help the analysts write a Dataflow pipeline in Python to perform the transformation. The Python code should be stored in a revision control system and modified as the incoming data's schema changes
D. Use Apache Spark on Dataproc to infer the schema of the CSV file before creating a Dataframe. Then implement the transformations in Spark SQL before writing the data out to Cloud Storage and loading into BigQuery
E. None of the above

Answer
A

Detailed explanation
A. Correct — Dataprep (Trifacta) provides a graphical UI, recipe-based transformations, and scheduling; it empowers non-developers to adjust recipes when schemas change and is designed for this use case.
B. Incorrect — SQL-based transformations are powerful but not a graphical tool and require developers/analysts comfortable with SQL; schema changes every three months make maintenance harder.
C. Incorrect — Writing and maintaining Dataflow code is developer-heavy and not friendly for non-developer analysts or quick schema changes.
D. Incorrect — Dataproc/Spark is heavierweight, developer-centric, and lacks the simple GUI recipe editing non-developers need.
E. Incorrect — A correct option exists (A).

Question 106
You want to migrate an on-premises Hadoop system to Cloud Dataproc. Hive is the primary tool in use, and the data format is Optimized Row Columnar (ORC). All ORC files have been successfully copied to a Cloud Storage bucket. You need to replicate some data to the cluster's local Hadoop Distributed File System (HDFS) to maximize performance. What are two ways to start using Hive in Cloud Dataproc? (Choose two.)

Options
A. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to HDFS. Mount the Hive tables locally.
B. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to any node of the Dataproc cluster. Mount the Hive tables locally.
C. Run the gsutil utility to transfer all ORC files from the Cloud Storage bucket to the master node of the Dataproc cluster. Then run the Hadoop utility to copy them to HDFS. Mount the Hive tables from HDFS.
D. Leverage Cloud Storage connector for Hadoop to mount the ORC files as external Hive tables. Replicate external Hive tables to the native ones.
E. Load the ORC files into BigQuery. Leverage BigQuery connector for Hadoop to mount the BigQuery tables as external Hive tables. Replicate external Hive tables to the native ones.

Answer
B, C

Detailed explanation
A. Incorrect — You can’t transfer "to HDFS" directly from gsutil; gsutil copies to local filesystem, then you must run Hadoop commands to put into HDFS. The option phrasing is imprecise.
B. Correct — Copying ORC files to any node (worker or master) and then mounting/using them locally (or copying into HDFS) is a valid approach; placing files on nodes and loading/attaching in Hive works.
C. Correct — Copying to the master node and then using hadoop fs -put (or similar) to move them into HDFS and mounting Hive tables from HDFS is a standard approach.
D. Incorrect — Using the Cloud Storage connector to create external Hive tables is valid too, but the option’s "Replicate external Hive tables to the native ones" adds extra steps; this option was not chosen as one of the two by the provided answer.
E. Incorrect — Loading ORC into BigQuery and then using BigQuery connector for Hive is an indirect and generally unnecessary path for Hive use; also BigQuery is columnar analytics and not a direct drop-in for Hive storage.

Question 107
You are implementing several batch jobs that must be executed on a schedule. These jobs have many interdependent steps that must be executed in a specific order. Portions of the jobs involve executing shell scripts, running Hadoop jobs, and running queries in BigQuery. The jobs are expected to run for many minutes up to several hours. If the steps fail, they must be retried a fixed number of times. Which service should you use to manage the execution of these jobs?

Options
A. Cloud Scheduler
B. Cloud Dataflow
C. Cloud Functions
D. Cloud Composer
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Cloud Scheduler can trigger jobs on a schedule but doesn't orchestrate multi-step DAGs with retries and rich dependency management.
B. Incorrect — Dataflow is for data processing pipelines, not general-purpose job orchestration across mixed step types.
C. Incorrect — Cloud Functions are event-driven short-lived functions, unsuitable for long-running, stateful multi-step pipelines with complex retries.
D. Correct — Cloud Composer (managed Airflow) is designed to orchestrate complex, long-running workflows, handle dependencies, retries, and integrate with Dataproc, BigQuery, GCS, and shell tasks.
E. Incorrect — A correct option exists (D).

Question 108
You work for a shipping company that has distribution centers where packages move on delivery lines to route them properly. The company wants to add cameras to the delivery lines to detect and track any visual damage to the packages in transit. You need to create a way to automate the detection of damaged packages and flag them for human review in real time while the packages are in transit. Which solution should you choose?

Options
A. Use BigQuery machine learning to be able to train the model at scale, so you can analyze the packages in batches.
B. Train an AutoML model on your corpus of images, and build an API around that model to integrate with the package tracking applications.
C. Use the Cloud Vision API to detect for damage, and raise an alert through Cloud Functions. Integrate the package tracking applications with this function.
D. Use TensorFlow to create a model that is trained on your corpus of images. Create a Python notebook in Cloud Datalab that uses this model so you can analyze for damaged packages.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — BigQuery ML is for tabular data and is not suited for image-based damage detection or real-time inference on video frames.
B. Correct — AutoML Vision (managed custom image model) enables you to train a model on your labeled images quickly and deploy it with a low-latency prediction endpoint (suitable for real-time inference and integration via an API). This is a good balance of speed-to-production and accuracy for a production POC.
C. Incorrect — Cloud Vision API provides general vision capabilities but is less suitable for a custom damage-detection classifier unless damage correlates with built-in detectors; AutoML lets you train a custom detector for your specific defects.
D. Incorrect — Building, training, and deploying a custom TensorFlow solution and integrating via notebooks is heavy-weight and slower to production compared with AutoML for a managed real-time endpoint.
E. Incorrect — A correct option exists (B).

Question 109
You are migrating your data warehouse to BigQuery. You have migrated all of your data into tables in a dataset. Multiple users from your organization will be using the data. They should only see certain tables based on their team membership. How should you set user permissions?

Options
A. Assign the users/groups data viewer access at the table level for each table
B. Create SQL views for each team in the same dataset in which the data resides, and assign the users/groups data viewer access to the SQL views
C. Create authorized views for each team in the same dataset in which the data resides, and assign the users/groups data viewer access to the authorized views
D. Create authorized views for each team in datasets created for each team. Assign the authorized views data viewer access to the dataset in which the data resides. Assign the users/groups data viewer access to the datasets in which the authorized views reside
E. None of the above

Answer
A

Detailed explanation
A. Correct — BigQuery supports table-level IAM; assigning Data Viewer at the table level restricts which tables users can read. This directly enforces the requirement that users only see certain tables.
B. Incorrect — Regular SQL views do not provide enforced row/table-level access isolation across datasets; they don't prevent access to underlying tables if users have direct table permissions.
C. Incorrect — Authorized views are a better pattern for controlled access with row/column filtering, but the provided answer indicates table-level grants (A). While authorized views are often recommended, the chosen answer here is A.
D. Incorrect — This option is a legitimate pattern (authorized views in separate datasets), but it's more complex; the provided answer selected A.
E. Incorrect — A correct option exists (A).

Question 110
You want to build a managed Hadoop system as your data lake. The data transformation process is composed of a series of Hadoop jobs executed in sequence. To accomplish the design of separating storage from compute, you decided to use the Cloud Storage connector to store all input data, output data, and intermediary data. However, you noticed that one Hadoop job runs very slowly with Cloud Dataproc, when compared with the on-premises bare-metal Hadoop environment (8-core nodes with 100-GB RAM). Analysis shows that this particular Hadoop job is disk I/O intensive. You want to resolve the issue. What should you do?

Options
A. Allocate sufficient memory to the Hadoop cluster, so that the intermediary data of that particular Hadoop job can be held in memory
B. Allocate sufficient persistent disk space to the Hadoop cluster, and store the intermediate data of that particular Hadoop job on native HDFS
C. Allocate more CPU cores of the virtual machine instances of the Hadoop cluster so that the networking bandwidth for each instance can scale up
D. Allocate additional network interface card (NIC), and configure link aggregation in the operating system to use the combined throughput when working with Cloud Storage
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — While more memory may help some jobs, if the job is specifically disk I/O bound and intermediates cannot fit in memory, increasing RAM won't solve the I/O bottleneck.
B. Correct — For disk-I/O–intensive jobs, using local persistent disks/HDFS (rather than Cloud Storage) reduces latency and increases I/O throughput. Copying the required intermediate files into HDFS on the cluster nodes improves performance for that job.
C. Incorrect — Increasing CPU won't fix disk I/O bottlenecks; CPU and network scaling don't address local disk throughput limitations.
D. Incorrect — Link aggregation affects network throughput to Cloud Storage but won't match HDFS local-disk performance required for disk-bound jobs; plus configuring NIC aggregation at VM level is complex and not the recommended fix.
E. Incorrect — A correct option exists (B).

Question 111
You work for an advertising company, and you've developed a Spark ML model to predict click-through rates at advertisement blocks. You've been developing everything at your on-premises data center, and now your company is migrating to Google Cloud. Your data center will be closing soon, so a rapid lift-and-shift migration is necessary. However, the data you've been using will be migrated to BigQuery. You periodically retrain your Spark ML models, so you need to migrate existing training pipelines to Google Cloud. What should you do?

Options
A. Use Vertex AI for training existing Spark ML models
B. Rewrite your models on TensorFlow, and start using Vertex AI
C. Use Dataproc for training existing Spark ML models, but start reading data directly from BigQuery
D. Spin up a Spark cluster on Compute Engine, and train Spark ML models on the data exported from BigQuery
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Vertex AI is optimized for TensorFlow / scikit-learn / XGBoost and managed training; it is not a drop-in for existing Spark ML training pipelines and would require refactoring.
B. Incorrect — Rewriting models in TensorFlow is a large rework and not appropriate for a rapid lift-and-shift when Spark ML pipelines already exist.
C. Correct — Dataproc provides managed Spark clusters, letting you lift-and-shift existing Spark ML jobs with minimal changes; Dataproc can read directly from BigQuery (via the BigQuery connector), which is ideal when data has been migrated to BigQuery.
D. Incorrect — Spinning up unmanaged Spark on Compute Engine increases operational burden; Dataproc is the managed, faster-to-migrate choice.
E. Incorrect — A correct option exists (C).

Question 112
You work for a global shipping company. You want to train a model on 40 TB of data to predict which ships in each geographic region are likely to cause delivery delays on any given day. The model will be based on multiple attributes collected from multiple sources. Telemetry data, including location in GeoJSON format, will be pulled from each ship and loaded every hour. You want to have a dashboard that shows how many and which ships are likely to cause delays within a region. You want to use a storage solution that has native functionality for prediction and geospatial processing. Which storage solution should you use?

Options
A. BigQuery
B. Cloud Bigtable
C. Cloud Datastore
D. Cloud SQL for PostgreSQL
E. None of the above

Answer
A

Detailed explanation
A. Correct — BigQuery supports large-scale analytics, ML (BigQuery ML), and built-in geospatial (GIS) functions. It is suited for 40 TB analytic workloads and integrates well with dashboards and predictive queries.
B. Incorrect — Bigtable is a wide-column low-latency store, not optimized for SQL analytics, geospatial processing, or built-in ML capabilities.
C. Incorrect — Datastore is for operational NoSQL workloads and is not designed for large-scale analytics, geospatial processing, or ML.
D. Incorrect — Cloud SQL (PostgreSQL) can do geospatial with PostGIS but does not scale as easily to 40 TB analytic workloads and lacks BigQuery's native ML/analytics scale.
E. Incorrect — A correct option exists (A).

Question 113

You operate an IoT pipeline built around Apache Kafka that normally receives around 5000 messages per second. You want to use Google Cloud Platform to create an alert as soon as the moving average over 1 hour drops below 4000 messages per second. What should you do?

Options

A. Consume the stream of data in Dataflow using Kafka IO. Set a sliding time window of 1 hour every 5 minutes. Compute the average when the window closes, and send an alert if the average is less than 4000 messages.
B. Consume the stream of data in Dataflow using Kafka IO. Set a fixed time window of 1 hour. Compute the average when the window closes, and send an alert if the average is less than 4000 messages.
C. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to Bigtable. Use Cloud Scheduler to run a script every hour that counts the number of rows created in Bigtable in the last hour. If that number falls below 4000, send an alert.
D. Use Kafka Connect to link your Kafka message queue to Pub/Sub. Use a Dataflow template to write your messages from Pub/Sub to BigQuery. Use Cloud Scheduler to run a script every five minutes that counts the number of rows created in BigQuery in the last hour. If that number falls below 4000, send an alert.
E. None of the above

Answer

A

Detailed explanation

A. Correct — Dataflow with KafkaIO and a sliding 1-hour window evaluated every 5 minutes produces a near-real-time moving average and can raise alerts promptly when the average drops below a threshold.
B. Incorrect — A fixed hourly window only yields one datapoint per hour, which is coarse and not suitable for timely alerting every 5 minutes.
C. Incorrect — Periodic batch scripts and Bigtable polling are higher-latency and more complex than an in-stream sliding window approach.
D. Incorrect — Periodically querying BigQuery is slower, more costly, and not as responsive for near-real-time alerting as Dataflow sliding windows.
E. Incorrect — A correct option exists (A).

Question 114
You plan to deploy Cloud SQL using MySQL. You need to ensure high availability in the event of a zone failure. What should you do?

Options
A. Create a Cloud SQL instance in one zone, and create a failover replica in another zone within the same region.
B. Create a Cloud SQL instance in one zone, and create a read replica in another zone within the same region.
C. Create a Cloud SQL instance in one zone, and configure an external read replica in a zone in a different region.
D. Create a Cloud SQL instance in a region, and configure automatic backup to a Cloud Storage bucket in the same region.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Creating a Cloud SQL instance with a failover replica in another zone within the same region enables automatic failover for zonal outages and provides HA.
B. Incorrect — Read replicas are for read scaling and are not automatic failover targets for primary instance availability.
C. Incorrect — External read replicas in other regions do not provide automatic zonal failover and are not the recommended HA configuration.
D. Incorrect — Backups help restore data but do not provide automatic high availability for live failover during zone failures.
E. Incorrect — A correct option exists (A).

Question 115
Your company is selecting a system to centralize data ingestion and delivery. You are considering messaging and data integration systems to address the requirements. The key requirements are:
- The ability to seek to a particular offset in a topic, possibly back to the start of all data ever captured
- Support for publish/subscribe semantics on hundreds of topics
- Retain per-key ordering

Which system should you choose?

Options
A. Apache Kafka
B. Cloud Storage
C. Dataflow
D. Firebase Cloud Messaging
E. None of the above

Answer
A

Detailed explanation
A. Correct — Apache Kafka natively supports seeking to offsets (including beginning), pub/sub semantics across many topics, and per-key partitioning that preserves ordering per key.
B. Incorrect — Cloud Storage is object storage, not a messaging system, and does not provide offset/seek or ordering semantics.
C. Incorrect — Dataflow is a processing engine, not a durable message bus that supports

offsets and seek semantics.
D. Incorrect — Firebase Cloud Messaging is for mobile push notifications; it does not meet the messaging semantics required here.
E. Incorrect — A correct option exists (A).

Question 116
You are planning to migrate your current on-premises Apache Hadoop deployment to the cloud. You need to ensure that the deployment is as fault-tolerant and cost-effective as possible for long-running batch jobs. You want to use a managed service. What should you do?

Options
A. Deploy a Dataproc cluster. Use a standard persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from hdfs:// to gs://
B. Deploy a Dataproc cluster. Use an SSD persistent disk and 50% preemptible workers. Store data in Cloud Storage, and change references in scripts from hdfs:// to gs://
C. Install Hadoop and Spark on a 10-node Compute Engine instance group with standard instances. Install the Cloud Storage connector, and store the data in Cloud Storage. Change references in scripts from hdfs:// to gs://
D. Install Hadoop and Spark on a 10-node Compute Engine instance group with preemptible instances. Store data in HDFS. Change references in scripts from hdfs:// to gs://
E. None of the above

Answer
A

Detailed explanation
A. Correct — Dataproc (managed) with standard persistent disks and a mix including preemptible workers (e.g., 50%) reduces cost while maintaining fault tolerance; storing data in Cloud Storage (gs://) decouples storage from compute and is resilient.
B. Incorrect — SSDs increase cost; for long-running batch jobs standard disks are typically cost-effective unless specific I/O needs justify SSDs.
C. Incorrect — Managing Hadoop/Spark on Compute Engine forfeits the benefits of managed Dataproc and increases operational overhead.
D. Incorrect — Using preemptible instances exclusively and storing data in HDFS is risky for long-running batch jobs (preemption can cause job failures) and contradicts the recommended gs:// approach.
E. Incorrect — A correct option exists (A).

Question 117
Your team is working on a binary classification problem. You have trained a support vector machine (SVM) classifier with default parameters, and received an area under the Curve (AUC) of 0.87 on the validation set. You want to increase the AUC of the model. What should you do?

Options
A. Perform hyperparameter tuning
B. Train a classifier with deep neural networks, because neural networks would always beat SVMs
C. Deploy the model and measure the real-world AUC; it's always higher because of generalization
D. Scale predictions you get out of the model (tune a scaling factor as a hyperparameter) in order to get the highest AUC
E. None of the above

Answer
A

Detailed explanation
A. Correct — Hyperparameter tuning (kernel choice, C, gamma, class weights, etc.) is a direct, standard approach to improve SVM performance and can raise AUC.
B. Incorrect — Neural networks do not always outperform SVMs; switching model families is not guaranteed and requires additional work and tuning.
C. Incorrect — Real-world AUC is not "always higher"; you must validate, and deployment without improvement is not a strategy to increase AUC.
D. Incorrect — Simply scaling predictions (calibration) affects decision thresholds but does not generally increase AUC substantially; tuning model hyperparameters is more impactful.
E. Incorrect — A correct option exists (A).

Question 118
You need to deploy additional dependencies to all nodes of a Cloud Dataproc cluster at startup using an existing initialization action. Company security policies require that Cloud Dataproc nodes do not have access to the Internet so public initialization actions cannot fetch resources. What should you do?

Options
A. Deploy the Cloud SQL Proxy on the Cloud Dataproc master
B. Use an SSH tunnel to give the Cloud Dataproc cluster access to the Internet
C. Copy all dependencies to a Cloud Storage bucket within your VPC security perimeter
D. Use Resource Manager to add the service account used by the Cloud Dataproc cluster to the Network User role
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Cloud SQL Proxy is unrelated to fetching initialization dependencies and does not provide a general package distribution mechanism.
B. Incorrect — SSH tunneling to provide Internet access violates the stated security policy (no Internet) and is not a recommended or manageable solution.
C. Correct — Staging dependencies in a Cloud Storage bucket that is accessible from the VPC (via VPC Service Controls / proper IAM and no public access) lets init actions pull required files without public Internet access.
D. Incorrect — Granting Network User to the service account does not provide a secure mechanism to fetch packages and is unrelated.
E. Incorrect — A correct option exists (C).

Question 119
You need to choose a database for a new project that has the following requirements:
☞ Fully managed
☞ Able to automatically scale up
☞ Transactionally consistent
☞ Able to scale up to 6 TB
☞ Able to be queried using SQL

Which database do you choose?

Options
A. Cloud SQL
B. Cloud Bigtable
C. Cloud Spanner
D. Cloud Datastore
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud SQL (managed relational, e.g., MySQL/Postgres) supports SQL, transactional consistency, and is a fully managed service. Many Cloud SQL configurations can scale to multi-TB sizes (note: consider performance and instance sizing). The provided answer selects Cloud SQL.
B. Incorrect — Bigtable is not a relational SQL database and does not provide transactional SQL semantics.
C. Incorrect — Cloud Spanner also fits many of these requirements (global scale, relational SQL + transactions) and is often chosen for large scale transactional SQL use cases; however the provided answer is Cloud SQL.
D. Incorrect — Datastore is NoSQL, not suited for SQL queries or full transactional SQL semantics required here.
E. Incorrect — A correct option exists (A).

Question 120
You work for a mid-sized enterprise that needs to move its operational system transaction data from an on-premises database to GCP. The database is about 20 TB in size. Which database should you choose?

Options
A. Cloud SQL
B. Cloud Bigtable
C. Cloud Spanner
D. Cloud Datastore
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud SQL is often used for migrating operational databases of this size when relational features and compatibility (MySQL/Postgres) are needed; the provided answer selects Cloud SQL.

B. Incorrect — Bigtable is a NoSQL wide-column store suited for high-throughput, low-latency workloads, not a direct replacement for general-purpose relational operational databases.
C. Incorrect — Spanner is a strong candidate for very large, globally distributed transactional systems; depending on requirements it might be preferable, but the provided answer was Cloud SQL.
D. Incorrect — Datastore is NoSQL and not a direct drop-in for relational operational transaction systems.
E. Incorrect — A correct option exists (A).

Question 122
Your company has recently grown rapidly and now ingesting data at a significantly higher rate than it was previously. You manage the daily batch MapReduce analytics jobs in Apache Hadoop. However, the recent increase in data has meant the batch jobs are falling behind. You were asked to recommend ways the development team could increase the responsiveness of the analytics without increasing costs. What should you recommend they do?

Options
A. Rewrite the job in Pig.
B. Rewrite the job in Apache Spark.
C. Increase the size of the Hadoop cluster.
D. Decrease the size of the Hadoop cluster but also rewrite the job in Hive.
E. Migrate the job to Cloud Dataproc and leave code unchanged.

Answer
B

Detailed explanation
A. Incorrect. Pig is a higher-level scripting abstraction over MapReduce and may simplify development but does not deliver the dramatic performance improvements of in-memory processing.
B. Correct. Apache Spark runs in-memory and is often much faster than MapReduce for iterative and interactive analytics; rewriting the job in Spark typically increases responsiveness without necessarily increasing cluster size (thus controlling cost).
C. Incorrect. Increasing cluster size increases cost; the question asks to improve responsiveness without increasing costs.
D. Incorrect. Decreasing cluster size would worsen performance; rewriting in Hive alone won't guarantee the needed speedup.
E. Incorrect. Migrating to Dataproc without changing the processing engine (still MapReduce) won't solve responsiveness—the key is using a faster processing engine like Spark.

Question 123

You work for a large fast food restaurant chain with over 400,000 employees. You store employee information in Google BigQuery in a Users table consisting of a FirstName field and a LastName field. A member of IT is building an application and asks you to modify the schema and data in BigQuery so the application can query a FullName field consisting of the value of the FirstName field concatenated with a space, followed by the value of the LastName field for each employee. How can you make that data available while minimizing cost?

Options

A. Create a view in BigQuery that concatenates the FirstName and LastName field values to produce the FullName.
B. Add a new column called FullName to the Users table. Run an UPDATE statement that updates the FullName column for each user with the concatenation of the FirstName and LastName values.
C. Create a Google Cloud Dataflow job that queries BigQuery for the entire Users table, concatenates the FirstName value and LastName value for each user, and loads the proper values for FirstName, LastName, and FullName into a new table in BigQuery.
D. Use BigQuery to export the data for the table to a CSV file. Create a Google Cloud Dataproc job to process the CSV file and output a new CSV file containing the proper values for FirstName, LastName and FullName. Run a BigQuery load job to load the new CSV file into BigQuery.
E. Use a scheduled query to populate a FullName column in a materialized view daily.

Answer

A

Detailed explanation

A. Correct. A view computes FullName on query time and avoids storing duplicate data—this minimizes storage and update cost while exposing FullName to the application.
B. Incorrect. Adding and populating a physical column duplicates data and increases storage and maintenance costs.
C. Incorrect. Using Dataflow to build a new table is overkill and more expensive than a simple view.
D. Incorrect. Exporting, processing externally, and reloading is complex and costlier than a view.
E. Incorrect. A scheduled query/materialized view that stores results would increase storage and maintenance; a plain view is cheaper and simpler.

Question 124

You are deploying a new storage system for your mobile application, which is a media streaming service. You decide the best fit is Google Cloud Datastore. You have entities with multiple properties, some of which can take on multiple values. For example, in the entity 'Movie' the property 'actors' and the property 'tags' have multiple values but the property 'date released' does not. A typical query would ask for all movies with actor=<actorname> ordered by date_released or all movies with tag=Comedy ordered by date_released. How should you avoid a combinatorial explosion in the number of indexes?

Options

A. Manually configure the index in your index config as follows: create composite indexes only for combinations you need (include actor/date_released and tag/date_released) and exclude other combinations from automatic indexing.
B. Manually configure the index in your index config as follows: include actor, tags, and date_released in a single multi-field index to cover all queries.
C. Set the following in your entity options: exclude_from_indexes = 'actors, tags'
D. Set the following in your entity options: exclude_from_indexes = 'date_published'
E. Use array-join denormalization to precompute actor-date combinations and index those instead.

Answer

A

Detailed explanation

A. Correct. Manually configuring only the composite indexes you need (e.g., actor + date_released and tag + date_released) prevents Datastore from building every possible combination and avoids exponential index growth.
B. Incorrect. One huge multi-field index including all array fields is not feasible and may not satisfy the variety of query patterns—also arrays create multiple index entries per element, causing explosion.
C. Incorrect. Excluding both actors and tags from indexing prevents the queries you need (you can’t query by actor/tag if they’re excluded).
D. Incorrect. Excluding date_published would prevent ordering by date, which is required by queries.
E. Incorrect. Denormalization can work but precomputing and indexing all combinations moves complexity to writes and can still grow large; targeted composite indexes are the recommended approach.

Question 125

You work for a manufacturing plant that batches application log files together into a single log file once a day at 2:00 AM. You have written a Google Cloud Dataflow job to process that log file. You need to make sure the log file is processed once per day as inexpensively as possible. What should you do?

Options

A. Change the processing job to use Google Cloud Dataproc instead.
B. Manually start the Cloud Dataflow job each morning when you get into the office.
C. Create a cron job with Google App Engine Cron Service to run the Cloud Dataflow job.
D. Configure the Cloud Dataflow job as a streaming job so that it processes the log data immediately.
E. Use Cloud Scheduler to trigger a Cloud Function that starts the Dataflow job at 2:05 AM daily.

Answer

C

Detailed explanation

A. Incorrect. Dataproc could work but introduces cluster management overhead and may be more costly for a simple once-per-day job.
B. Incorrect. Manual starts are error-prone and not cost-effective.
C. Correct. App Engine Cron (or equivalent scheduled trigger) running the Dataflow batch job daily automates the task at low cost.
D. Incorrect. Streaming is unnecessary for a once-per-day batch file and would be more expensive.
E. Incorrect. Although Cloud Scheduler + Cloud Function is a valid modern approach, the original correct answer provided was App Engine Cron; using App Engine Cron is simpler and directly supported in this scenario.

Question 126
You work for an economic consulting firm that helps companies identify economic trends as they happen. As part of your analysis, you use Google BigQuery to correlate customer data with the average prices of the 100 most common goods sold, including bread, gasoline, milk, and others. The average prices of these goods are updated every 30 minutes. You want to make sure this data stays up to date so you can combine it with other data in BigQuery as cheaply as possible. What should you do?

Options
A. Load the data every 30 minutes into a new partitioned table in BigQuery.
B. Store and update the data in a regional Google Cloud Storage bucket and create a federated data source in BigQuery.
C. Store the data in Google Cloud Datastore. Use Google Cloud Dataflow to query BigQuery and combine the data programmatically with the data stored in Cloud Datastore.
D. Store the data in a file in a regional Google Cloud Storage bucket. Use Cloud Dataflow to query BigQuery and combine the data programmatically with the data stored in Google Cloud Storage.
E. Keep the data in a small BigQuery table and use a scheduled MERGE every 30 minutes to update rows.

Answer
B

Detailed explanation
A. Incorrect. Frequent loads into BigQuery (every 30 minutes) can be more expensive than using a federated approach for small reference data.
B. Correct. Keeping the small frequently-updated dataset in Cloud Storage and using BigQuery federated access reads it on demand and is cost-effective for small lookup data that changes frequently.
C. Incorrect. Datastore and extra Dataflow processing add complexity and cost; unnecessary for small reference updates.
D. Incorrect. Using Dataflow to combine on the fly is more complex and costlier than federated access.
E. Incorrect. Regular MERGE jobs into BigQuery incur load and processing costs; federated storage is cheaper for small, frequently updated reference datasets.

Question 127
You are designing the database schema for a machine learning-based food ordering service that will predict what users want to eat. Here is some of the information you need to store:
☞ The user profile: What the user likes and doesn't like to eat
☞ The user account information: Name, address, preferred meal times
☞ The order information: When orders are made, from where, to whom
The database will be used to store all the transactional data of the product. You want to optimize the data schema. Which Google Cloud Platform product should you use?

Options
A. BigQuery
B. Cloud SQL
C. Cloud Bigtable
D. Cloud Datastore
E. Cloud Spanner

Answer
A

Detailed explanation
A. Correct. (Per the provided answer.) BigQuery is excellent for analytical workloads and large-scale model training queries; the supplied answer indicates BigQuery is the chosen product for storing and analyzing transactional data for ML.
B. Incorrect. Cloud SQL is transactional and fine for OLTP, but the given answer specified BigQuery.
C. Incorrect. Bigtable is good for time-series/wide-column workloads but was not the provided correct choice.
D. Incorrect. Datastore is schemaless and suited to some app data, but not the chosen answer.
E. Incorrect. Cloud Spanner is a globally consistent OLTP database—suitable for transactional workloads—but the stated answer was BigQuery.

Question 128
Your company is loading comma-separated values (CSV) files into Google BigQuery. The data is fully imported successfully; however, the imported data is not matching byte-to-byte to the source file. What is the most likely cause of this problem?

Options
A. The CSV data loaded in BigQuery is not flagged as CSV.
B. The CSV data has invalid rows that were skipped on import.
C. The CSV data loaded in BigQuery is not using BigQuery's default encoding.
D. The CSV data has not gone through an ETL phase before loading into BigQuery.
E. The CSV data contained different locale-specific decimal separators.

Answer
C

Detailed explanation
A. Incorrect. If the format were wrong BigQuery would likely error or mis-parse, but the symptom described is encoding mismatch.
B. Incorrect. Skipped rows would result in missing rows, not byte-to-byte differences in imported values.
C. Correct. Encoding mismatches (for example, source uses a different character encoding than BigQuery expects) commonly cause byte-level differences after import.
D. Incorrect. ETL omission may cause semantic issues but not byte-to-byte encoding differences.
E. Incorrect. Locale decimal separators affect numeric parsing but are not the primary cause of byte-to-byte mismatches at import.

Question 129

Your company produces 20,000 files every hour. Each data file is formatted as a comma separated values (CSV) file that is less than 4 KB. All files must be ingested on Google Cloud Platform before they can be processed. Your company site has a 200 ms latency to Google Cloud, and your Internet connection bandwidth is limited as 50 Mbps. You currently deploy a secure FTP (SFTP) server on a virtual machine in Google Compute Engine as the data ingestion point. A local SFTP client runs on a dedicated machine to transmit the CSV files as is. The goal is to make reports with data from the previous day available to the executives by 10:00 a.m. each day. This design is barely able to keep up with the current volume, even though the bandwidth utilization is rather low. You are told that due to seasonality, your company expects the number of files to double for the next three months. Which two actions should you take? (Choose two.)

Options

A. Introduce data compression for each file to increase the rate file of file transfer.
B. Contact your internet service provider (ISP) to increase your maximum bandwidth to at least 100 Mbps.
C. Redesign the data ingestion process to use gsutil tool to send the CSV files to a storage bucket in parallel.
D. Assemble 1,000 files into a tape archive (TAR) file. Transmit the TAR files instead, and disassemble the CSV files in the cloud upon receiving them.
E. Create an S3-compatible storage endpoint in your network, and use Google Cloud Storage Transfer Service to transfer on-premises data to the designated storage bucket.

Answer

C, D

Detailed explanation

A. Incorrect. Per-file compression may help but 20,000 tiny files have overheads—compression per small file yields limited benefit.
B. Incorrect. Increasing bandwidth is an option but may be slow/costly and not necessary if you reduce per-file overhead; question asks for options to handle doubling without necessarily increasing bandwidth.
C. Correct. Using gsutil (or parallel transfer tools) to push files directly to Cloud Storage in parallel reduces per-file transfer overhead and scales better than SFTP to a single VM.
D. Correct. Packing many small files into larger archives reduces protocol/latency overhead per file and makes more efficient use of available bandwidth; unpacking in the cloud is straightforward.
E. Incorrect. Setting up an S3-compatible endpoint and Storage Transfer Service is more complex and not the most direct fix compared to parallel uploads or batching files.

Question 130
You are choosing a NoSQL database to handle telemetry data submitted from millions of Internet-of-Things (IoT) devices. The volume of data is growing at 100 TB per year, and each data entry has about 100 attributes. The data processing pipeline does not require atomicity, consistency, isolation, and durability (ACID). However, high availability and low latency are required. You need to analyze the data by querying against individual fields. Which three databases meet your requirements? (Choose three.)

Options
A. Redis
B. HBase
C. MySQL
D. MongoDB
E. Cassandra
F. HDFS with Hive

Answer
B, D, E

Detailed explanation
A. Incorrect. Redis is an in-memory store optimized for low-latency access but not appropriate for 100 TB/year persistent telemetry storage at low cost.
B. Correct. HBase is a distributed wide-column store that can handle large volumes, supports field-level access patterns, and provides high availability when deployed properly.
C. Incorrect. MySQL is an RDBMS and does not scale cost-effectively to 100 TB/year telemetry workloads for field queries.
D. Correct. MongoDB supports flexible schemas, indexing on individual fields, and can scale horizontally for high availability and low latency.
E. Correct. Cassandra is designed for high write throughput, horizontal scale, high availability, and supports querying by indexed fields.
F. Incorrect. HDFS with Hive is batch-oriented (not low-latency) and not ideal for the real-time low-latency requirements described.

Question 131
You are training a spam classifier. You notice that you are overfitting the training data. Which three actions can you take to resolve this problem? (Choose three.)

Options
A. Get more training examples
B. Reduce the number of training examples
C. Use a smaller set of features
D. Use a larger set of features
E. Increase the regularization parameters
F. Decrease the regularization parameters

Answer
A, C, E

Detailed explanation
A. Correct. Getting more training examples usually reduces overfitting by giving the model more varied data to learn the true underlying patterns.
B. Incorrect. Reducing training examples typically makes overfitting worse because the model has less data to generalize from.
C. Correct. Using a smaller set of features (feature selection) reduces model complexity and lowers the chance of fitting noise.
D. Incorrect. Adding more features increases model complexity and can worsen overfitting unless they are highly informative and regularized.
E. Correct. Increasing regularization (stronger penalty) discourages complex weights and helps prevent overfitting.
F. Incorrect. Decreasing regularization relaxes constraints on the model and typically increases overfitting.

Question 132
You are implementing security best practices on your data pipeline. Currently, you are manually executing jobs as the Project Owner. You want to automate these jobs by taking nightly batch files containing non-public information from Google Cloud Storage, processing them with a Spark Scala job on a Google Cloud Dataproc cluster, and depositing the results into Google BigQuery. How should you securely run this workload?

Options
A. Restrict the Google Cloud Storage bucket so only you can see the files
B. Grant the Project Owner role to a service account, and run the job with it
C. Use a service account with the ability to read the batch files and to write to BigQuery
D. Use a user account with the Project Viewer role on the Cloud Dataproc cluster to read the batch files and write to BigQuery
E. Use your personal account but create a key and store it on the cluster

Answer
C

Detailed explanation
A. Incorrect. Restricting the bucket to a single user prevents automation and is not a secure automation practice.
B. Incorrect. Granting Project Owner to a service account is over-privileged and violates least-privilege principles.
C. Correct. Use a dedicated service account with just the necessary IAM permissions (read GCS, write BigQuery) — follows least privilege and is suitable for automated jobs.
D. Incorrect. A user account is not appropriate for automated service workloads and Project Viewer is insufficient for writes.
E. Incorrect. Using a personal account and embedding keys is insecure and not recommended.

Question 133
You are using Google BigQuery as your data warehouse. Your users report that the following simple query is running very slowly, no matter when they run the query:
SELECT country, state, city FROM [myproject:mydataset.mytable] GROUP BY country
You check the query plan and see data skew in the Read stage. What is the most likely cause of the delay for this query?

Options
A. Users are running too many concurrent queries in the system
B. The [myproject:mydataset.mytable] table has too many partitions
C. Either the state or the city columns in the [myproject:mydataset.mytable] table have too many NULL values
D. Most rows in the [myproject:mydataset.mytable] table have the same value in the country column, causing data skew
E. The table has too many columns

Answer
D

Detailed explanation
A. Incorrect. While concurrency can slow queries, the query plan specifically shows skew during the Read stage pointing to data distribution issues.
B. Incorrect. Too many partitions would affect planning or scanning but not typically cause skew where one worker receives most rows.
C. Incorrect. NULL values alone do not cause the data distribution skew observed in the Read stage.
D. Correct. If most rows share the same country value, the reduction/aggregation will be skewed toward a small number of workers handling the majority of data, causing slowdown.
E. Incorrect. Many columns increase I/O but don't explain the skew indicated in the plan.

Question 134
Your globally distributed auction application allows users to bid on items. Occasionally, users place identical bids at nearly identical times, and different application servers process those bids. Each bid event contains the item, amount, user, and timestamp. You want to collate those bid events into a single location in real time to determine which user bid first. What should you do?

Options
A. Create a file on a shared file and have the application servers write all bid events to that file. Process the file with Apache Hadoop to identify which user bid first.
B. Have each application server write the bid events to Cloud Pub/Sub as they occur. Push the events from Cloud Pub/Sub to a custom endpoint that writes the bid event information into Cloud SQL.
C. Set up a MySQL database for each application server to write bid events into. Periodically query each of those distributed MySQL databases and update a master MySQL database with bid event information.
D. Have each application server write the bid events to Google Cloud Pub/Sub as they occur. Use a pull subscription to pull the bid events using Google Cloud Dataflow. Give the bid for each item to the user in the bid event that is processed first.
E. Use a distributed filesystem to collect events and run batch reconciliation hourly.

Answer
B

Detailed explanation
A. Incorrect. Writing to a shared file and batch-processing with Hadoop is too slow for real-time determination.
B. Correct. Publishing events to Pub/Sub and pushing to an endpoint that writes to Cloud SQL provides low-latency persistence and a single authoritative store to determine ordering (assuming the endpoint handles idempotency and ordering appropriately).
C. Incorrect. Distributed per-server MySQL instances with periodic consolidation introduces complexity and latency.
D. Incorrect. While Pub/Sub + Dataflow is viable, using a pull/Dataflow pipeline and assigning the "first" bid based on processing order can be nondeterministic unless you have strict ordering guarantees; the user-provided answer indicates option B.
E. Incorrect. Hourly batch reconciles are too slow for real-time auctions.

Question 135
Your organization has been collecting and analyzing data in Google BigQuery for 6 months. The majority of the data analyzed is placed in a time-partitioned table named events_partitioned. To reduce the cost of queries, your organization created a view called events, which queries only the last 14 days of data. The view is described in legacy SQL. Next month, existing applications will be connecting to BigQuery to read the events data via an ODBC connection. You need to ensure the applications can connect. Which two actions should you take? (Choose two.)

Options
A. Create a new view over events using standard SQL
B. Create a new partitioned table using a standard SQL query
C. Create a new view over events_partitioned using standard SQL
D. Create a service account for the ODBC connection to use for authentication
E. Create a Google Cloud Identity and Access Management (Cloud IAM) role for the ODBC connection and shared events

Answer
C, D

Detailed explanation
A. Incorrect. Creating a view over events (which itself is a view) adds indirection and may not address legacy/ODBC compatibility; the required step is to use standard SQL directly on the partitioned table.
B. Incorrect. Creating a new partitioned table is unnecessary if events_partitioned already exists; the issue is SQL dialect and authentication for ODBC.
C. Correct. Creating a view over the partitioned table using **standard SQL** ensures compatibility with clients that expect standard SQL (ODBC drivers typically use standard SQL).
D. Correct. Creating a service account for ODBC authentication provides a stable, automatable credential for applications to connect securely.
E. Incorrect. Creating a custom IAM role is usually unnecessary; assign proper dataset/table permissions to the service account instead.

Question 136
You have enabled the free integration between Firebase Analytics and Google BigQuery. Firebase now automatically creates a new table daily in BigQuery in the format app_events_YYYYMMDD. You want to query all of the tables for the past 30 days in legacy SQL. What should you do?

Options
A. Use the TABLE_DATE_RANGE function
B. Use the WHERE_PARTITIONTIME pseudo column
C. Use WHERE date BETWEEN YYYY-MM-DD AND YYYY-MM-DD
D. Use SELECT IF.(date >= YYYY-MM-DD AND date <= YYYY-MM-DD)
E. Use table wildcard functions only in standard SQL

Answer
A

Detailed explanation
A. Correct. In legacy SQL, TABLE_DATE_RANGE is used to query multiple sharded daily tables by date range.
B. Incorrect. WHERE_PARTITIONTIME is for partitioned tables and standard SQL usage, not legacy sharded table wildcard queries.
C. Incorrect. A simple WHERE on a date column does not select across multiple daily tables in legacy SQL.
D. Incorrect. The IF expression is not how to target multiple daily tables in legacy SQL.
E. Incorrect. Table wildcard functions exist in legacy SQL (TABLE_DATE_RANGE); standard SQL has other mechanisms.

Question 137
Your company is currently setting up data pipelines for their campaign. For all the Google Cloud Pub/Sub streaming data, one of the important business requirements is to be able to periodically identify the inputs and their timings during their campaign. Engineers have decided to use windowing and transformation in Google Cloud Dataflow for this purpose. However, when testing this feature, they find that the Cloud Dataflow job fails for the all streaming insert. What is the most likely cause of this problem?

Options
A. They have not assigned the timestamp, which causes the job to fail
B. They have not set the triggers to accommodate the data coming in late, which causes the job to fail
C. They have not applied a global windowing function, which causes the job to fail when the pipeline is created
D. They have not applied a non-global windowing function, which causes the job to fail when the pipeline is created
E. They have not set the watermarking policy

Answer
D

Detailed explanation
A. Incorrect. Missing timestamps usually cause incorrect results or late-data handling issues, not immediate job failure.
B. Incorrect. Triggers affect when window results are emitted; lack of trigger configuration rarely causes job creation failure.
C. Incorrect. Global window is the default; not applying it is not a typical cause of pipeline failure.
D. Correct. If the pipeline expects non-global windowing (e.g., fixed/event-time windows) but none is applied, transforms that require windowing (like GroupByKey with windowing) can fail at pipeline creation.
E. Incorrect. Watermarking affects lateness handling, not typically immediate pipeline creation failure.

Question 138
You architect a system to analyze seismic data. Your extract, transform, and load (ETL) process runs as a series of MapReduce jobs on an Apache Hadoop cluster. The ETL process takes days to process a data set because some steps are computationally expensive. Then you discover that a sensor calibration step has been omitted. How should you change your ETL process to carry out sensor calibration systematically in the future?

Options
A. Modify the transform MapReduce jobs to apply sensor calibration before they do anything else.
B. Introduce a new MapReduce job to apply sensor calibration to raw data, and ensure all other MapReduce jobs are chained after this.
C. Add sensor calibration data to the output of the ETL process, and document that all users need to apply sensor calibration themselves.
D. Develop an algorithm through simulation to predict variance of data output from the last MapReduce job based on calibration factors, and apply the correction to all data.
E. Use ad-hoc scripts to apply calibration on demand.

Answer
B

Detailed explanation
A. Incorrect. Modifying each transform to include calibration duplicates logic and is error-prone.
B. Correct. Adding a dedicated calibration MapReduce job up-front that normalizes raw data, and then chaining downstream jobs ensures calibration is applied consistently and once.
C. Incorrect. Relying on downstream consumers to apply calibration is error-prone and will lead to inconsistent analyses.
D. Incorrect. Post-hoc predictive correction is complex and less reliable than calibrating raw input.
E. Incorrect. Ad-hoc scripts lack reproducibility and make process management difficult.

Question 139
An online retailer has built their current application on Google App Engine. A new initiative at the company mandates that they extend their application to allow their customers to transact directly via the application. They need to manage their shopping transactions and analyze combined data from multiple datasets using a business intelligence (BI) tool. They want to use only a single database for this purpose. Which Google Cloud database should they choose?

Options
A. BigQuery
B. Cloud SQL
C. Cloud Bigtable
D. Cloud Datastore
E. Cloud Spanner

Answer
B

Detailed explanation
A. Incorrect. BigQuery is analytical and not suitable as the single transactional database for customer transactions.
B. Correct. Cloud SQL provides transactional (OLTP) capabilities and can support application transactions while also allowing ETL into analytics systems as needed.
C. Incorrect. Bigtable is best for wide-column, high-throughput workloads but is not ideal for transactional shopping carts.
D. Incorrect. Datastore is schemaless and scalable but lacks full relational transactional semantics for complex shopping transactions.
E. Incorrect. Cloud Spanner could be used, but given the typical App Engine transactional use-case, Cloud SQL is the standard choice and matches the provided answer.

Question 140
You launched a new gaming app almost three years ago. You have been uploading log files from the previous day to a separate Google BigQuery table with the table name format LOGS_yyyymmdd. You have been using table wildcard functions to generate daily and monthly reports for all time ranges. Recently, you discovered that some queries that cover long date ranges are exceeding the limit of 1,000 tables and failing. How can you resolve this issue?

Options
A. Convert all daily log tables into date-partitioned tables
B. Convert the sharded tables into a single partitioned table
C. Enable query caching so you can cache data from previous months
D. Create separate views to cover each month, and query from these views
E. Move historic data to Cloud Storage and query via federated tables

Answer
B

Detailed explanation
A. Incorrect. Converting *all* daily tables into date-partitioned *tables* (plural) is ambiguous and not the succinct solution; the recommended action is to consolidate shards.
B. Correct. Converting the many sharded day-per-table pattern into a single partitioned table removes the 1,000-table wildcard limit and is the standard remedy.
C. Incorrect. Query caching does not address the table count / wildcard limits.
D. Incorrect. Creating many views would still reference many underlying tables and not necessarily avoid the 1,000-table limit.
E. Incorrect. Moving historical data to Cloud Storage adds complexity and does not directly solve the wildcard limit for BigQuery queries.

Question 141

Your analytics team wants to build a simple statistical model to determine which customers are most likely to work with your company again, based on a few different metrics. They want to run the model on Apache Spark, using data housed in Google Cloud Storage, and you have recommended using Google Cloud Dataproc to execute this job. Testing has shown that this workload can run in approximately 30 minutes on a 15-node cluster, outputting the results into Google BigQuery. The plan is to run this workload weekly. How should you optimize the cluster for cost?

Options

A. Migrate the workload to Google Cloud Dataflow
B. Use pre-emptible virtual machines (VMs) for the cluster
C. Use a higher-memory node so that the job runs faster
D. Use SSDs on the worker nodes so that the job can run faster
E. None of the above

Answer

B

Detailed explanation

A. Incorrect – Dataflow is more suited for streaming or parallelized ETL jobs, not Spark-based workloads. Migrating introduces unnecessary complexity.
B. Correct – Pre-emptible VMs are much cheaper and suitable for short-lived, fault-tolerant jobs like this weekly batch workload. Spark jobs can restart tasks if pre-emptibles are lost, making it the most cost-effective choice.
C. Incorrect – A higher-memory node reduces runtime slightly but increases cost significantly without improving overall efficiency for a weekly batch job.
D. Incorrect – SSDs improve I/O performance but cost more. This job completes fine in 30 minutes without SSDs, so it's not cost-efficient.
E. Incorrect – One of the listed answers is correct (B), so "None of the above" is not valid.

Question 142
Your company receives both batch- and stream-based event data. You want to process the data using Google Cloud Dataflow over a predictable time period. However, you realize that in some instances data can arrive late or out of order. How should you design your Cloud Dataflow pipeline to handle data that is late or out of order?

Options
A. Set a single global window to capture all the data
B. Set sliding windows to capture all the lagged data
C. Use watermarks and timestamps to capture the lagged data
D. Ensure every datasource type (stream or batch) has a timestamp, and use the timestamps to define the logic for lagged data
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – A single global window would collect everything but loses the ability to process late events properly in bounded time.
B. Incorrect – Sliding windows help in overlapping timeframes but do not inherently solve out-of-order or late-arriving data.
C. Correct – Watermarks with event-time timestamps are the standard way in Dataflow to handle late/out-of-order events predictably. They allow defining lateness tolerance.
D. Incorrect – While timestamps are needed, simply adding them doesn't manage lateness. You need watermarks in combination.
E. Incorrect – Since C is correct, "None of the above" is not valid.

Question 143
You have some data, which is shown in the graphic below. The two dimensions are X and Y, and the shade of each dot represents what class it is. You want to classify this data accurately using a linear algorithm. To do this you need to add a synthetic feature. What should the value of that feature be?

Options
A. X^2+Y^2
B. X^2
C. Y^2
D. cos(X)
E. None of the above

Answer
A

Detailed explanation
A. Correct – Adding X^2+Y^2 introduces a radial boundary (circle) feature, enabling a linear classifier to separate classes that are otherwise circularly distributed.
B. Incorrect – X^2 only captures quadratic variation in X, not the combined circular relationship.
C. Incorrect – Y^2 only captures quadratic variation in Y, insufficient for circular separation.
D. Incorrect – cos(X) adds periodicity but does not help capture the radial nature of the data.
E. Incorrect – A is correct, so "None of the above" is not valid.

Question 144

You are integrating one of your internal IT applications and Google BigQuery, so users can query BigQuery from the application's interface. You do not want individual users to authenticate to BigQuery and you do not want to give them access to the dataset. You need to securely access BigQuery from your IT application. What should you do?

Options

A. Create groups for your users and give those groups access to the dataset
B. Integrate with a single sign-on (SSO) platform, and pass each user's credentials along with the query request
C. Create a service account and grant dataset access to that account. Use the service account's private key to access the dataset
D. Create a dummy user and grant dataset access to that user. Store the username and password for that user in a file on the file system, and use those credentials to access the BigQuery dataset
E. None of the above

Answer

C

Detailed explanation

A. Incorrect – This would give dataset access directly to users, which violates the requirement.
B. Incorrect – Passing each user's credentials requires managing user-level access, not desired here.
C. Correct – The best practice is to use a service account with dataset access. The application authenticates with the service account, keeping users abstracted.
D. Incorrect – Creating a dummy user and storing credentials is insecure and against best practices.
E. Incorrect – Since C is correct, "None of the above" is not valid.

Question 145
You are building a data pipeline on Google Cloud. You need to prepare data using a casual method for a machine-learning process. You want to support a logistic regression model. You also need to monitor and adjust for null values, which must remain real-valued and cannot be removed. What should you do?

Options
A. Use Cloud Dataprep to find null values in sample source data. Convert all nulls to 'none' using a Cloud Dataproc job
B. Use Cloud Dataprep to find null values in sample source data. Convert all nulls to 0 using a Cloud Dataprep job
C. Use Cloud Dataflow to find null values in sample source data. Convert all nulls to 'none' using a Cloud Dataprep job
D. Use Cloud Dataflow to find null values in sample source data. Convert all nulls to 0 using a custom script
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Logistic regression requires numeric features, so converting nulls to 'none' (string) is invalid.
B. Correct – Replacing nulls with 0 preserves real values and ensures compatibility with logistic regression input requirements.
C. Incorrect – Same issue as A: 'none' is not numeric, and Dataflow is unnecessary for this task.
D. Incorrect – While replacing with 0 is correct, Dataprep is simpler for null handling than a custom Dataflow script.
E. Incorrect – Since B is correct, "None of the above" is not valid.

Question 146
You set up a streaming data insert into a Redis cluster via a Kafka cluster. Both clusters are running on Compute Engine instances. You need to encrypt data at rest with encryption keys that you can create, rotate, and destroy as needed. What should you do?

Options
A. Create a dedicated service account, and use encryption at rest to reference your data stored in your Compute Engine cluster instances as part of your API service calls
B. Create encryption keys in Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances
C. Create encryption keys locally. Upload your encryption keys to Cloud Key Management Service. Use those keys to encrypt your data in all of the Compute Engine cluster instances
D. Create encryption keys in Cloud Key Management Service. Reference those keys in your API service calls when accessing the data in your Compute Engine cluster instances
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Service accounts manage identity/permissions, not encryption key lifecycle.
B. Correct – Cloud KMS allows creating, rotating, and destroying encryption keys with full integration for Compute Engine disks. This directly addresses the requirement.
C. Incorrect – You can upload keys to KMS, but local key management adds unnecessary complexity.

D. Incorrect – Referencing keys in API calls is not how disk-level encryption works. KMS integrates automatically with storage services.
E. Incorrect – Since B is correct, “None of the above” is not valid.

Question 147
You are developing an application that uses a recommendation engine on Google Cloud. Your solution should display new videos to customers based on past views. Your solution needs to generate labels for the entities in videos that the customer has viewed. Your design must be able to provide very fast filtering suggestions based on data from other customer preferences on several TB of data. What should you do?

Options
A. Build and train a complex classification model with Spark MLlib to generate labels and filter the results. Deploy the models using Cloud Dataproc. Call the model from your application
B. Build and train a classification model with Spark MLlib to generate labels. Build and train a second classification model with Spark MLlib to filter results to match customer preferences. Deploy the models using Cloud Dataproc. Call the models from your application
C. Build an application that calls the Cloud Video Intelligence API to generate labels. Store data in Cloud Bigtable, and filter the predicted labels to match the user's viewing history to generate preferences
D. Build an application that calls the Cloud Video Intelligence API to generate labels. Store data in Cloud SQL, and join and filter the predicted labels to match the user's viewing history to generate preferences
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Spark MLlib is unnecessary; Google provides APIs for video labeling. Also, Dataproc introduces overhead.
B. Incorrect – Training two models is overcomplicated and still ignores fast querying requirements.
C. Correct – Cloud Video Intelligence API generates video labels efficiently. Storing results in Bigtable enables fast filtering on TB-scale data, meeting performance requirements.
D. Incorrect – Cloud SQL is not designed for TB-scale fast filtering. It would not scale well.
E. Incorrect – Since C is correct, “None of the above” is not valid.

Question 148
You are selecting services to write and transform JSON messages from Cloud Pub/Sub to BigQuery for a data pipeline on Google Cloud. You want to minimize service costs. You also want to monitor and accommodate input data volume that will vary in size with minimal manual intervention. What should you do?

Options
A. Use Cloud Dataproc to run your transformations. Monitor CPU utilization for the cluster. Resize the number of worker nodes in your cluster via the command line
B. Use Cloud Dataproc to run your transformations. Use the diagnose command to generate an operational output archive. Locate the bottleneck and adjust cluster resources
C. Use Cloud Dataflow to run your transformations. Monitor the job system lag with Stackdriver. Use the default autoscaling setting for worker instances
D. Use Cloud Dataflow to run your transformations. Monitor the total execution time for a sampling of jobs. Configure the job to use non-default Compute Engine machine types when needed
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Manual resizing adds operational overhead and does not minimize intervention.
B. Incorrect – Diagnosing bottlenecks and adjusting manually is not cost-efficient.
C. Correct – Cloud Dataflow with autoscaling and Stackdriver monitoring is cost-efficient and minimizes manual intervention for variable loads.
D. Incorrect – Using custom machine types manually adds overhead instead of automation.
E. Incorrect – Since C is correct, “None of the above” is not valid.

Question 149
Your infrastructure includes a set of YouTube channels. You have been tasked with creating a process for sending the YouTube channel data to Google Cloud for analysis. You want to design a solution that allows your world-wide marketing teams to perform ANSI SQL and other types of analysis on up-to-date YouTube channels log data. How should you set up the log data transfer into Google Cloud?

Options
A. Use Storage Transfer Service to transfer the offsite backup files to a Cloud Storage Multi-Regional storage bucket as a final destination
B. Use Storage Transfer Service to transfer the offsite backup files to a Cloud Storage Regional bucket as a final destination
C. Use BigQuery Data Transfer Service to transfer the offsite backup files to a Cloud Storage Multi-Regional storage bucket as a final destination
D. Use BigQuery Data Transfer Service to transfer the offsite backup files to a Cloud Storage Regional storage bucket as a final destination
E. None of the above

Answer
A

Detailed explanation
A. Correct – Storage Transfer Service is designed for ingesting external data into Cloud Storage. Multi-regional buckets provide global access, useful for worldwide teams.
B. Incorrect – A regional bucket restricts access locality, less optimal for global teams.
C. Incorrect – BigQuery Data Transfer Service is for specific SaaS connectors, not for generic offsite backup ingestion.
D. Incorrect – Same issue as C, plus regional restriction.
E. Incorrect – Since A is correct, “None of the above” is not valid.

Question 150
You are designing storage for very large text files for a data pipeline on Google Cloud. You want to support ANSI SQL queries. You also want to support compression and parallel load from the input locations using Google recommended practices. What should you do?

Options
A. Transform text files to compressed Avro using Cloud Dataflow. Use BigQuery for storage and query
B. Transform text files to compressed Avro using Cloud Dataflow. Use Cloud Storage and BigQuery permanent linked tables for query
C. Compress text files to gzip using the Grid Computing Tools. Use BigQuery for storage and query
D. Compress text files to gzip using the Grid Computing Tools. Use Cloud Storage, and then import into Cloud Bigtable for query
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – BigQuery does not store files directly; it uses its own storage format. Storing compressed Avro directly in BigQuery is not valid.
B. Correct – Avro is a columnar, splittable, compressed format ideal for BigQuery ingestion and ANSI SQL queries. Using linked tables allows efficient parallel loads from Cloud Storage.
C. Incorrect – Gzip is not splittable, so it limits parallel loading into BigQuery.
D. Incorrect – Cloud Bigtable is not for ANSI SQL queries.
E. Incorrect – Since B is correct, “None of the above” is not valid.

Question 151
You are developing an application on Google Cloud that will automatically generate subject labels for users' blog posts. You are under competitive pressure to add this feature quickly, and you have no additional developer resources. No one on your team has experience with machine learning. What should you do?

Options
A. Call the Cloud Natural Language API from your application. Process the generated Entity Analysis as labels
B. Call the Cloud Natural Language API from your application. Process the generated Sentiment Analysis as labels
C. Build and train a text classification model using TensorFlow. Deploy the model using Cloud Machine Learning Engine. Call the model from your application and process the results as labels
D. Build and train a text classification model using TensorFlow. Deploy the model using a Kubernetes Engine cluster. Call the model from your application and process the results as labels
E. None of the above

Answer
A

Detailed explanation
A. Correct – Cloud Natural Language API provides entity analysis out of the box, making it the fastest and easiest way without ML expertise. Entities can serve directly as labels.
B. Incorrect – Sentiment analysis measures positive/negative tone, not subject labeling.
C. Incorrect – Building and training a TensorFlow model requires expertise and time, which the team lacks.
D. Incorrect – Same as C, plus adds unnecessary infrastructure complexity with Kubernetes.
E. Incorrect – Since A is correct, "None of the above" is not valid.

Question 152
You are designing storage for 20 TB of text files as part of deploying a data pipeline on Google Cloud. Your input data is in CSV format. You want to minimize the cost of querying aggregate values for multiple users who will query the data in Cloud Storage with multiple engines. Which storage service and schema design should you use?

Options
A. Use Cloud Bigtable for storage. Install the HBase shell on a Compute Engine instance to query the Cloud Bigtable data
B. Use Cloud Bigtable for storage. Link as permanent tables in BigQuery for query
C. Use Cloud Storage for storage. Link as permanent tables in BigQuery for query
D. Use Cloud Storage for storage. Link as temporary tables in BigQuery for query
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Bigtable is not designed for SQL queries.
B. Incorrect – Linking Bigtable to BigQuery has limitations and higher costs for aggregation queries.
C. Correct – Storing in Cloud Storage and linking as permanent BigQuery tables allows cost-effective SQL queries across multiple engines.
D. Incorrect – Temporary tables require reloading and are inefficient for repeated queries.
E. Incorrect – Since C is correct, "None of the above" is not valid.

Question 153
You are designing storage for two relational tables that are part of a 10-TB database on Google Cloud. You want to support transactions that scale horizontally. You also want to optimize data for range queries on non-key columns. What should you do?

Options
A. Use Cloud SQL for storage. Add secondary indexes to support query patterns
B. Use Cloud SQL for storage. Use Cloud Dataflow to transform data to support query patterns
C. Use Cloud Spanner for storage. Add secondary indexes to support query patterns
D. Use Cloud Spanner for storage. Use Cloud Dataflow to transform data to support query patterns
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Cloud SQL does not scale horizontally; limited to single-region transactional workloads.
B. Incorrect – Transforming data does not solve lack of horizontal scaling in Cloud SQL.
C. Correct – Cloud Spanner supports horizontal scaling, strong consistency, and secondary indexes for efficient range queries.
D. Incorrect – Dataflow transformation is not needed for supporting range queries when Spanner indexes suffice.
E. Incorrect – Since C is correct, “None of the above” is not valid.

Question 154
Your financial services company is moving to cloud technology and wants to store 50 TB of financial time-series data in the cloud. This data is updated frequently and new data will be streaming in all the time. Your company also wants to move their existing Apache Hadoop jobs to the cloud to get insights into this data. Which product should they use to store the data?

Options
A. Cloud Bigtable
B. Google BigQuery
C. Google Cloud Storage
D. Google Cloud Datastore
E. None of the above

Answer
A

Detailed explanation
A. Correct – Bigtable is optimized for time-series data, scales well for continuous updates, and integrates with Hadoop jobs.
B. Incorrect – BigQuery is great for analytics but not optimized for frequent updates/streaming writes.
C. Incorrect – Cloud Storage is object storage, not suited for high-frequency updates or Hadoop integration.
D. Incorrect – Datastore is for transactional app data, not large time-series.
E. Incorrect – Since A is correct, “None of the above” is not valid.

Question 155
An organization maintains a Google BigQuery dataset that contains tables with user-level data. They want to expose aggregates of this data to other Google Cloud projects, while still controlling access to the user-level data. Additionally, they need to minimize their overall storage cost and ensure the analysis cost for other projects is assigned to those projects. What should they do?

Options
A. Create and share an authorized view that provides the aggregate results
B. Create and share a new dataset and view that provides the aggregate results
C. Create and share a new dataset and table that contains the aggregate results
D. Create dataViewer Identity and Access Management (IAM) roles on the dataset to enable sharing
E. None of the above

Answer
A

Detailed explanation
A. Correct – Authorized views allow exposing only aggregated data securely, minimize storage duplication, and ensure query costs are billed to the querying project.
B. Incorrect – A new dataset with views may work but requires duplication and more management overhead.
C. Incorrect – Creating new aggregate tables increases storage cost.
D. Incorrect – Giving dataViewer role exposes raw user-level data, violating requirements.
E. Incorrect – Since A is correct, “None of the above” is not valid.

Question 156
Government regulations in your industry mandate that you have to maintain an auditable record of access to certain types of data. Assuming that all expiring logs will be archived correctly, where should you store data that is subject to that mandate?

Options
A. Encrypted on Cloud Storage with user-supplied encryption keys. A separate decryption key will be given to each authorized user
B. In a BigQuery dataset that is viewable only by authorized personnel, with the Data Access log used to provide the auditability
C. In Cloud SQL, with separate database user names to each user. The Cloud SQL Admin activity logs will be used to provide the auditability
D. In a bucket on Cloud Storage that is accessible only by an AppEngine service that collects user information and logs the access before providing a link to the bucket
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Encrypting files with user-supplied keys does not provide detailed access audit logs.
B. Correct – BigQuery automatically generates Data Access logs for auditability, and access can be restricted to authorized users.
C. Incorrect – Cloud SQL audit logs are not as robust or scalable as BigQuery Data Access logs for regulatory needs.
D. Incorrect – Using App Engine adds unnecessary complexity and is not the standard audit approach.
E. Incorrect – Since B is correct, “None of the above” is not valid.

Question 157
Your neural network model is taking days to train. You want to increase the training speed. What can you do?

Options
A. Subsample your test dataset
B. Subsample your training dataset
C. Increase the number of input features to your model
D. Increase the number of layers in your neural network
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Reducing test data doesn't impact training time.
B. Correct – Subsampling the training dataset reduces data volume, directly lowering training time.
C. Incorrect – Adding more input features increases complexity and training time.
D. Incorrect – Adding layers makes the model deeper and slower to train.
E. Incorrect – Since B is correct, "None of the above" is not valid.

Question 158
You are responsible for writing your company's ETL pipelines to run on an Apache Hadoop cluster. The pipeline will require some checkpointing and splitting pipelines. Which method should you use to write the pipelines?

Options
A. PigLatin using Pig
B. HiveQL using Hive
C. Java using MapReduce
D. Python using MapReduce
E. None of the above

Answer
A

Detailed explanation
A. Correct – PigLatin (via Pig) is designed for ETL pipelines, supports checkpointing and splitting, and is simpler than raw MapReduce coding.
B. Incorrect – HiveQL is SQL-like, more for querying data, not building flexible ETL pipelines.
C. Incorrect – Java MapReduce works but is complex and time-consuming compared to Pig.
D. Incorrect – Python MapReduce has the same issues as Java MapReduce with additional overhead.
E. Incorrect – Since A is correct, "None of the above" is not valid.

Question 159
Your company maintains a hybrid deployment with GCP, where analytics are performed on your anonymized customer data. The data are imported to Cloud Storage from your data center through parallel uploads to a data transfer server running on GCP. Management informs you that the daily transfers take too long and have asked you to fix the problem. You want to maximize transfer speeds. Which action should you take?

Options
A. Increase the CPU size on your server
B. Increase the size of the Google Persistent Disk on your server
C. Increase your network bandwidth from your datacenter to GCP
D. Increase your network bandwidth from Compute Engine to Cloud Storage
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – CPU size does not resolve network transfer bottlenecks.
B. Incorrect – Persistent Disk size impacts storage capacity, not transfer speed.
C. Correct – The bottleneck is network throughput from the datacenter to GCP. Increasing bandwidth directly improves transfer speeds.
D. Incorrect – Compute Engine to Cloud Storage bandwidth is already optimized within GCP; the real issue is external network capacity.
E. Incorrect – Since C is correct, “None of the above” is not valid.

Question 161
You work for a large ecommerce company. You store your customer's order data in Bigtable. You have a garbage collection policy set to delete the data after 30 days and the number of versions is set to 1. When the data analysts run a query to report total customer spending, the analysts sometimes see customer data that is older than 30 days. You need to ensure that the analysts do not see customer data older than 30 days while minimizing cost and overhead. What should you do?

Options
A. Set the expiring values of the column families to 29 days and keep the number of versions to 1.
B. Use a timestamp range filter in the query to fetch the customer's data for a specific range.
C. Schedule a job daily to scan the data in the table and delete data older than 30 days.
D. Set the expiring values of the column families to 30 days and set the number of versions to 2.

Answer
B

Detailed explanation
A. **Incorrect** — Changing expiry to 29 days does not guarantee analysts won’t see data older than 30 days. Deletions are asynchronous.
B. **Correct** — Timestamp range filters ensure queries only return rows within the last 30 days regardless of GC lag, minimizing extra cost/overhead.
C. **Incorrect** — Running daily scan-and-delete jobs adds operational overhead and cost.
D. **Incorrect** — Increasing versions adds unnecessary storage and does not solve the issue of stale reads.

Question 162
You are using a Dataflow streaming job to read messages from a message bus that does not support exactly-once delivery. Your job then applies some transformations, and loads the result into BigQuery. You want to ensure that your data is being streamed into BigQuery with exactly-once delivery semantics. You expect your ingestion throughput into BigQuery to be about 1.5 GB per second. What should you do?

Options
A. Use the BigQuery Storage Write API and ensure that your target BigQuery table is regional.
B. Use the BigQuery Storage Write API and ensure that your target BigQuery table is multiregional.
C. Use the BigQuery Streaming API and ensure that your target BigQuery table is regional.
D. Use the BigQuery Streaming API and ensure that your target BigQuery table is multiregional.

Answer
A

Detailed explanation
A. **Correct** — The BigQuery Storage Write API supports exactly-once semantics, high throughput (GB/s scale), and requires regional tables. Best fit for 1.5 GB/s ingestion.
B. **Incorrect** — Multiregional tables are not supported with Storage Write API for exactly-once guarantees.
C. **Incorrect** — The Streaming API only supports at-least-once delivery; duplicates are possible.
D. **Incorrect** — Same issue as C, plus multiregion doesn't solve delivery semantics.

Question 163
You have created an external table for Apache Hive partitioned data that resides in a Cloud Storage bucket, which contains a large number of files. You notice that queries against this table are slow. You want to improve the performance of these queries. What should you do?

Options
A. Change the storage class of the Hive partitioned data objects from Coldline to Standard.
B. Create an individual external table for each Hive partition by using a common table name prefix. Use wildcard table queries to reference the partitioned data.
C. Upgrade the external table to a BigLake table. Enable metadata caching for the table.
D. Migrate the Hive partitioned data objects to a multi-region Cloud Storage bucket.

Answer
C

Detailed explanation
A. **Incorrect** — Storage class (Coldline vs Standard) affects cost and retrieval latency but not query execution performance at scale.
B. **Incorrect** — Creating many external tables increases management overhead and query complexity.
C. **Correct** — BigLake tables with metadata caching improve query performance over partitioned external data in Cloud Storage.
D. **Incorrect** — Multi-region buckets don't improve query scan performance; they improve availability.

Question 164
You have a network of 1000 sensors. The sensors generate time series data: one metric per sensor per second, along with a timestamp. You already have 1 TB of data, and expect the data to grow by 1 GB every day. You need to access this data in two ways. The first access pattern requires retrieving the metric from one specific sensor stored at a specific timestamp, with a median single-digit millisecond latency. The second access pattern requires running complex analytic queries on the data, including joins, once a day. How should you store this data?

Options
A. Store your data in BigQuery. Concatenate the sensor ID and timestamp, and use it as the primary key.
B. Store your data in Bigtable. Concatenate the sensor ID and timestamp and use it as the row key. Perform an export to BigQuery every day.
C. Store your data in Bigtable. Concatenate the sensor ID and metric, and use it as the row key. Perform an export to BigQuery every day.
D. Store your data in BigQuery. Use the metric as a primary key.

Answer
B

Detailed explanation
A. **Incorrect** — BigQuery cannot provide single-digit millisecond lookups; it's optimized for analytics not point reads.
B. **Correct** — Bigtable is ideal for low-latency point lookups with row keys structured as sensorID#timestamp. Daily export to BigQuery allows analytical queries.
C. **Incorrect** — Using metric in row key instead of timestamp breaks the natural access pattern and retrieval by time.
D. **Incorrect** — BigQuery does not support low-latency point reads for single sensor lookups.

Question 165
You have 100 GB of data stored in a BigQuery table. This data is outdated and will only be accessed one or two times a year for analytics with SQL. For backup purposes, you want to store this data to be immutable for 3 years. You want to minimize storage costs. What should you do?

Options
A. 1. Create a BigQuery table clone.
2. Query the clone when you need to perform analytics.
B. 1. Create a BigQuery table snapshot.
2. Restore the snapshot when you need to perform analytics.
C. 1. Perform a BigQuery export to a Cloud Storage bucket with archive storage class.
2. Enable versioning on the bucket.
3. Create a BigQuery external table on the exported files.
D. 1. Perform a BigQuery export to a Cloud Storage bucket with archive storage class.
2. Set a locked retention policy on the bucket.
3. Create a BigQuery external table on the exported files.

Answer
D

Detailed explanation
A. **Incorrect** — Clones still rely on BigQuery storage and incur ongoing costs.
B. **Incorrect** — Snapshots also remain in BigQuery storage, not the cheapest long-term backup.
C. **Incorrect** — Versioning increases storage cost unnecessarily for immutable data.
D. **Correct** — Exporting to Archive Cloud Storage with a locked retention policy ensures immutability, lowest cost, and can still be queried via external table.

Question 166
You have thousands of Apache Spark jobs running in your on-premises Apache Hadoop cluster. You want to migrate the jobs to Google Cloud. You want to use managed services to run your jobs instead of maintaining a long-lived Hadoop cluster yourself. You have a tight timeline and want to keep code changes to a minimum. What should you do?

Options
A. Move your data to BigQuery. Convert your Spark scripts to a SQL-based processing approach.
B. Rewrite your jobs in Apache Beam. Run your jobs in Dataflow.
C. Copy your data to Compute Engine disks. Manage and run your jobs directly on those instances.
D. Move your data to Cloud Storage. Run your jobs on Dataproc.

Answer
D

Detailed explanation
A. **Incorrect** — Rewriting Spark scripts to SQL requires significant code changes.
B. **Incorrect** — Rewriting into Beam pipelines is time-consuming and requires major refactoring.
C. **Incorrect** — Running Spark directly on Compute Engine loses managed benefits and increases ops burden.
D. **Correct** — Dataproc supports Spark with minimal code changes and uses Cloud Storage as storage layer, fitting the requirement of managed services and quick migration.

Question 167
You are administering shared BigQuery datasets that contain views used by multiple teams in your organization. The marketing team is concerned about the variability of their monthly BigQuery analytics spend using the on-demand billing model. You need to help the marketing team establish a consistent BigQuery analytics spend each month. What should you do?

Options
A. Create a BigQuery Enterprise reservation with a baseline of 250 slots and autoscaling set to 500 for the marketing team, and bill them back accordingly.
B. Establish a BigQuery quota for the marketing team, and limit the maximum number of bytes scanned each day.
C. Create a BigQuery reservation with a baseline of 500 slots with no autoscaling for the marketing team, and bill them back accordingly.
D. Create a BigQuery Standard pay-as-you go reservation with a baseline of 0 slots and autoscaling set to 500 for the marketing team, and bill them back accordingly.

Answer
C

Detailed explanation
A. **Incorrect** — Autoscaling reintroduces variability in cost.
B. **Incorrect** — Quotas limit usage but do not guarantee predictable spend.
C. **Correct** — A fixed reservation with baseline slots (e.g., 500) ensures consistent monthly billing regardless of query volume.
D. **Incorrect** — Pay-as-you-go with autoscaling also introduces cost variability.

Question 168
You are part of a healthcare organization where data is organized and managed by respective data owners in various storage services. As a result of this decentralized ecosystem, discovering and managing data has become difficult. You need to quickly identify and implement a cost-optimized solution to assist your organization with the following:
• Data management and discovery
• Data lineage tracking
• Data quality validation

How should you build the solution?

Options
A. Use BigLake to convert the current solution into a data lake architecture.
B. Build a new data discovery tool on Google Kubernetes Engine that helps with new source onboarding and data lineage tracking.
C. Use BigQuery to track data lineage, and use Dataprep to manage data and perform data quality validation.
D. Use Dataplex to manage data, track data lineage, and perform data quality validation.

Answer
D

Detailed explanation
A. **Incorrect** — BigLake helps unify lake/warehouse access but doesn't provide full governance, lineage, and quality features.
B. **Incorrect** — Building a custom tool on GKE increases cost/effort and duplicates built-in services.
C. **Incorrect** — BigQuery + Dataprep cover partial requirements but not full discovery and lineage across all storage services.
D. **Correct** — Dataplex is a managed service for data governance that supports management, discovery, lineage, and quality validation across decentralized data sources.

Question 169
You have data located in BigQuery that is used to generate reports for your company. You have noticed some weekly executive report fields do not correspond to format according to company standards. For example, report errors include different telephone formats and different country code identifiers. This is a frequent issue, so you need to create a recurring job to normalize the data. You want a quick solution that requires no coding. What should you do?

Options
A. Use Cloud Data Fusion and Wrangler to normalize the data, and set up a recurring job.
B. Use Dataflow SQL to create a job that normalizes the data, and that after the first run of the job, schedule the pipeline to execute recurrently.
C. Create a Spark job and submit it to Dataproc Serverless.
D. Use BigQuery and GoogleSQL to normalize the data, and schedule recurring queries in BigQuery.

Answer
A

Detailed explanation
A. **Correct** — Cloud Data Fusion + Wrangler allows no-code/low-code data transformation and can be scheduled for recurring normalization jobs.
B. **Incorrect** — Dataflow SQL requires more setup and is less “no coding” compared to Fusion + Wrangler.
C. **Incorrect** — Dataproc Spark jobs require coding and are overkill for this use case.
D. **Incorrect** — BigQuery SQL requires writing and maintaining SQL queries (not no-code).

Question 170
You are designing a messaging system by using Pub/Sub to process clickstream data with an event-driven consumer app that relies on a push subscription. You need to configure the messaging system that is reliable enough to handle temporary downtime of the consumer app. You also need the messaging system to store the input messages that cannot be consumed by the subscriber. The system needs to retry failed messages gradually, avoiding overloading the consumer app, and store the failed messages after a maximum of 10 retries in a topic. How should you configure the Pub/Sub subscription?

Options
A. Increase the acknowledgement deadline to 10 minutes.
B. Use immediate redelivery as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10.
C. Use exponential backoff as the subscription retry policy, and configure dead lettering to the same source topic with maximum delivery attempts set to 10.
D. Use exponential backoff as the subscription retry policy, and configure dead lettering to a different topic with maximum delivery attempts set to 10.

Answer
D

Detailed explanation
A. **Incorrect** — Extending ack deadline doesn’t address retries, backoff, or dead lettering.
B. **Incorrect** — Immediate redelivery may overload the consumer during downtime; not recommended.
C. **Incorrect** — Dead lettering to the same source topic risks infinite retry loops.
D. **Correct** — Exponential backoff avoids overloading the consumer, and dead lettering to a separate topic after 10 retries preserves failed messages for later inspection.

Question 171
You designed a data warehouse in BigQuery to analyze sales data. You want a self-serving, low-maintenance, and cost- effective solution to share the sales dataset to other business units in your organization. What should you do?

Options
A. Create an Analytics Hub private exchange, and publish the sales dataset.
B. Enable the other business units’ projects to access the authorized views of the sales dataset.
C. Create and share views with the users in the other business units.
D. Use the BigQuery Data Transfer Service to create a schedule that copies the sales dataset to the other business units’ projects.

Answer
A

Detailed explanation
A. **Correct** — Analytics Hub (private exchange) is designed for self-service, discoverable, managed dataset sharing across projects/organizations with minimal maintenance and centralized control; it's cost-effective and supports subscriptions.
B. **Incorrect** — Authorized views work but require per-project configuration and maintenance; less self-serving and scalable than Analytics Hub for many teams.
C. **Incorrect** — Sharing views directly to users requires more manual ACL management and lacks the discoverability/ subscription model Analytics Hub provides.
D. **Incorrect** — Copying datasets via Data Transfer Service creates duplicate data (storage cost) and operational overhead; not low maintenance or cost-effective.

Question 172
You have terabytes of customer behavioral data streaming from Google Analytics into BigQuery daily. Your customers' information, such as their preferences, is hosted on a Cloud SQL for MySQL database. Your CRM database is hosted on a Cloud SQL for PostgreSQL instance. The marketing team wants to use your customers' information from the two databases and the customer behavioral data to create marketing campaigns for yearly active customers. You need to ensure that the marketing team can run the campaigns over 100 times a day on typical days and up to 300 during sales. At the same time, you want to keep the load on the Cloud SQL databases to a minimum. What should you do?

Options
A. Create BigQuery connections to both Cloud SQL databases. Use BigQuery federated queries on the two databases and the Google Analytics data on BigQuery to run these queries.
B. Create a job on Apache Spark with Dataproc Serverless to query both Cloud SQL databases and the Google Analytics data on BigQuery for these queries.
C. Create streams in Datastream to replicate the required tables from both Cloud SQL databases to BigQuery for these queries.
D. Create a Dataproc cluster with Trino to establish connections to both Cloud SQL databases and BigQuery, to execute the queries.

Answer
C

Detailed explanation
A. **Incorrect** — Federated queries hit Cloud SQL at query time, imposing significant load and latency; not suitable for hundreds of runs per day with low DB load.
B. **Incorrect** — Dataproc Serverless still queries Cloud SQL at runtime unless you replicate—this would keep load on Cloud SQL.
C. **Correct** — Datastream replicates required Cloud SQL tables into BigQuery (near real-time), enabling marketing to run many queries against BigQuery without loading Cloud SQL. This minimizes DB load and supports high query volume.
D. **Incorrect** — Trino on Dataproc would query Cloud SQL directly and would increase load on the source databases.

Question 173
Your organization is modernizing their IT services and migrating to Google Cloud. You need to organize the data that will be stored in Cloud Storage and BigQuery. You need to enable a data mesh approach to share the data between sales, product design, and marketing departments. What should you do?

Options
A. 1. Create a project for storage of the data for each of your departments.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. Create user groups for authorized readers for each bucket and dataset.
4. Enable the IT team to administer the user groups to add or remove users as the departments' request.
B. 1. Create multiple projects for storage of the data for each of your departments' applications.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. Publish the data that each department shared in Analytics Hub.
4. Enable all departments to discover and subscribe to the data they need in Analytics Hub.
C. 1. Create a project for storage of the data for your organization.
2. Create a central Cloud Storage bucket with three folders to store the files for each department.
3. Create a central BigQuery dataset with tables prefixed with the department name.
4. Give viewer rights for the storage project for the users of your departments.
D. 1. Create multiple projects for storage of the data for each of your departments' applications.
2. Enable each department to create Cloud Storage buckets and BigQuery datasets.
3. In Dataplex, map each department to a data lake and the Cloud Storage buckets, and map the BigQuery datasets to zones.
4. Enable each department to own and share the data of their data lakes.

Answer
D

Detailed explanation
A. **Incorrect** — While per-department projects are reasonable, delegating IAM to IT for group management increases central bottlenecks; lacks governance/discovery features of Dataplex.
B. **Incorrect** — Analytics Hub helps sharing and discovery but doesn't provide per-department data-mesh governance, lineage, and unified policy enforcement.
C. **Incorrect** — Centralized single-project approach becomes a bottleneck and violates data mesh ownership principles; poor for autonomy and scalability.
D. **Correct** — Multiple projects per department/app plus Dataplex mapping (zones/lakes) lets each domain own/share data, provides unified governance, discovery, and policy enforcement—aligns with data mesh principles.

Question 174
You work for a large ecommerce company. You are using Pub/Sub to ingest the clickstream data to Google Cloud for analytics. You observe that when a new subscriber connects to an existing topic to analyze data, they are unable to subscribe to older data. For an upcoming yearly sale event in two months, you need a solution that, once implemented, will enable any new subscriber to read the last 30 days of data. What should you do?

Options
A. Create a new topic, and publish the last 30 days of data each time a new subscriber connects to an existing topic.
B. Set the topic retention policy to 30 days.
C. Set the subscriber retention policy to 30 days.
D. Ask the source system to re-push the data to Pub/Sub, and subscribe to it.

Answer
B

Detailed explanation
A. **Incorrect** — Creating a new topic and republishing on each new subscriber is impractical and costly.
B. **Correct** — Setting topic message retention to 30 days ensures new subscriptions can pull messages published in the last 30 days (subject to subscription settings), enabling new subscribers to read historical messages.
C. **Incorrect** — Subscriber (subscription) retention affects how long acknowledged messages are retained for that subscription, not enabling new subscriptions to access past topic messages.
D. **Incorrect** — Asking source to re-push is operationally heavy and unreliable compared to configuring topic retention.

Question 175
You are designing the architecture to process your data from Cloud Storage to BigQuery by using Dataflow. The network team provided you with the Shared VPC network and subnetwork to be used by your pipelines. You need to enable the deployment of the pipeline on the Shared VPC network. What should you do?

Options
A. Assign the compute.networkUser role to the Dataflow service agent.
B. Assign the compute.networkUser role to the service account that executes the Dataflow pipeline.
C. Assign the dataflow.admin role to the Dataflow service agent.
D. Assign the dataflow.admin role to the service account that executes the Dataflow pipeline.

Answer
A

Detailed explanation
A. **Correct** — Dataflow uses its service agent to create and manage network resources in a Shared VPC; granting the Dataflow service agent compute.networkUser on the Shared VPC project/subnet allows pipeline deployment there.
B. **Incorrect** — The worker service account needing networkUser is sometimes required for worker VM creation, but the primary necessary grant to use Shared VPC is for the Dataflow service agent.
C. **Incorrect** — dataflow.admin alone does not grant the network permissions required to use a Shared VPC.

D. **Incorrect** — Granting dataflow.admin to the pipeline service account is unnecessary for Shared VPC network attachment and over-privileges; compute.networkUser for the service agent is the standard requirement.

Question 176

Your infrastructure team has set up an interconnect link between Google Cloud and the on-premises network. You are designing a high-throughput streaming pipeline to ingest data in streaming from an Apache Kafka cluster hosted on-premises. You want to store the data in BigQuery, with as minimal latency as possible. What should you do?

Options

A. Setup a Kafka Connect bridge between Kafka and Pub/Sub. Use a Google-provided Dataflow template to read the data from Pub/Sub, and write the data to BigQuery.
B. Use a proxy host in the VPC in Google Cloud connecting to Kafka. Write a Dataflow pipeline, read data from the proxy host, and write the data to BigQuery.
C. Use Dataflow, write a pipeline that reads the data from Kafka, and writes the data to BigQuery.
D. Setup a Kafka Connect bridge between Kafka and Pub/Sub. Write a Dataflow pipeline, read the data from Pub/Sub, and write the data to BigQuery.

Answer

C

Detailed explanation

A. **Incorrect** — Introducing Pub/Sub adds an extra hop and potential latency; while workable, it's not minimal latency.
B. **Incorrect** — Proxy hosts complicate architecture and can be a bottleneck; not the recommended managed approach.
C. **Correct** — Dataflow can directly read from Kafka (on-prem via interconnect) and write to BigQuery, offering a managed, low-latency, high-throughput pipeline with minimal hops.
D. **Incorrect** — Using Pub/Sub introduces additional latency compared to direct Kafka input into Dataflow.

Question 177
You migrated your on-premises Apache Hadoop Distributed File System (HDFS) data lake to Cloud Storage. The data scientist team needs to process the data by using Apache Spark and SQL. Security policies need to be enforced at the column level. You need a cost-effective solution that can scale into a data mesh. What should you do?

Options
A. 1. Deploy a long-living Dataproc cluster with Apache Hive and Ranger enabled.
2. Configure Ranger for column level security.
3. Process with Dataproc Spark or Hive SQL.
B. 1. Define a BigLake table.
2. Create a taxonomy of policy tags in Data Catalog.
3. Add policy tags to columns.
4. Process with the Spark-BigQuery connector or BigQuery SQL.
C. 1. Load the data to BigQuery tables.
2. Create a taxonomy of policy tags in Data Catalog.
3. Add policy tags to columns.
4. Process with the Spark-BigQuery connector or BigQuery SQL.
D. 1. Apply an Identity and Access Management (IAM) policy at the file level in Cloud Storage.
2. Define a BigQuery external table for SQL processing.
3. Use Dataproc Spark to process the Cloud Storage files.

Answer
B

Detailed explanation
A. **Incorrect** — Long-living clusters and Ranger add operational overhead and cost; not the most cost-effective or scalable data-mesh friendly approach.
B. **Correct** — BigLake + Data Catalog policy tags provide column-level access control across data lake and lakehouse patterns. Using Spark-BigQuery connector or BigQuery SQL lets data scientists run Spark/SQL while enforcing policy tags—cost-effective and scalable.
C. **Incorrect** — Loading all data into BigQuery may be more expensive; BigLake allows using Cloud Storage with BigQuery governance features without full ingestion.
D. **Incorrect** — File-level IAM is coarse-grained and cannot implement column-level policies.

Question 178
One of your encryption keys stored in Cloud Key Management Service (Cloud KMS) was exposed. You need to reencrypt all of your CMEK-protected Cloud Storage data that used that key, and then delete the compromised key. You also want to reduce the risk of objects getting written without customer-managed encryption key (CMEK) protection in the future. What should you do?

Options
A. Rotate the Cloud KMS key version. Continue to use the same Cloud Storage bucket.
B. Create a new Cloud KMS key. Set the default CMEK key on the existing Cloud Storage bucket to the new one.
C. Create a new Cloud KMS key. Create a new Cloud Storage bucket. Copy all objects from the old bucket to the new one bucket while specifying the new Cloud KMS key in the copy command.
D. Create a new Cloud KMS key. Create a new Cloud Storage bucket configured to use the new key as the default CMEK key. Copy all objects from the old bucket to the new bucket without specifying a key.

Answer
D

Detailed explanation
A. **Incorrect** — Rotating the compromised key version does not fix exposure; the compromised key material remains accessible.
B. **Incorrect** — Setting a default CMEK on the existing bucket does not reencrypt existing objects already encrypted with the compromised key.
C. **Incorrect** — Copying to a new bucket while specifying the new key works, but if the new bucket is not configured with a default CMEK it's possible future writes might omit CMEK.
D. **Correct** — Creating a new bucket with the new key set as the default CMEK and copying objects into it ensures objects are reencrypted with the new key and future objects inherit CMEK automatically, reducing risk of non-CMEK writes.

Question 179
You have an upstream process that writes data to Cloud Storage. This data is then read by an Apache Spark job that runs on Dataproc. These jobs are run in the us-central1 region, but the data could be stored anywhere in the United States. You need to have a recovery process in place in case of a catastrophic single region failure. You need an approach with a maximum of 15 minutes of data loss (RPO=15 mins). You want to ensure that there is minimal latency when reading the data. What should you do?

Options
A. 1. Create two regional Cloud Storage buckets, one in the us-central1 region and one in the us-south1 region.
2. Have the upstream process write data to the us-central1 bucket. Use the Storage Transfer Service to copy data hourly from the us-central1 bucket to the us-south1 bucket.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in that region.
4. In case of regional failure, redeploy your Dataproc clusters to the us-south1 region and read from the bucket in that region instead.
B. 1. Create a Cloud Storage bucket in the US multi-region.
2. Run the Dataproc cluster in a zone in the us-central1 region, reading data from the US multi-region bucket.
3. In case of a regional failure, redeploy the Dataproc cluster to the us-central2 region and continue reading from the same bucket.
C. 1. Create a dual-region Cloud Storage bucket in the us-central1 and us-south1 regions.
2. Enable turbo replication.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in the us-south1 region.
4. In case of a regional failure, redeploy your Dataproc cluster to the us-south1 region and continue reading from the same bucket.
D. 1. Create a dual-region Cloud Storage bucket in the us-central1 and us-south1 regions.
2. Enable turbo replication.
3. Run the Dataproc cluster in a zone in the us-central1 region, reading from the bucket in the same region.
4. In case of a regional failure, redeploy the Dataproc clusters to the us-south1 region and read from the same bucket.

Answer
D

Detailed explanation
A. **Incorrect** — Hourly Storage Transfer would exceed the 15-minute RPO requirement.
B. **Incorrect** — US multi-region gives redundancy but is less explicit about consistent dual-region locality and performance guarantees; not the optimal pattern for low RPO with regional failover.
C. **Incorrect** — Reading from the remote replica (us-south1) while running Dataproc in us-central1 would add unnecessary latency.
D. **Correct** — Dual-region bucket (us-central1 + us-south1) with turbo replication provides fast, synchronous replication and low-latency reads in the primary region; in a regional failure, Dataproc can redeploy to the other region with minimal data loss (<15 mins when combined with fast replication and operational readiness).

Question 180
You currently have transactional data stored on-premises in a PostgreSQL database. To modernize your data environment, you want to run transactional workloads and support analytics needs with a single database. You need to move to Google Cloud without changing database management systems, and minimize cost and complexity. What should you do?

Options
A. Migrate and modernize your database with Cloud Spanner.
B. Migrate your workloads to AlloyDB for PostgreSQL.
C. Migrate to BigQuery to optimize analytics.
D. Migrate your PostgreSQL database to Cloud SQL for PostgreSQL.

Answer
D

Detailed explanation
A. **Incorrect** — Cloud Spanner is a different DBMS (not PostgreSQL-compatible) and requires schema/application changes.
B. **Incorrect** — AlloyDB is PostgreSQL-compatible and powerful, but compared to Cloud SQL it can be more costly/complex for straightforward lift-and-shift transactional workloads.
C. **Incorrect** — BigQuery is for analytics (OLAP), not transactional workloads; migrating transactional DB to BigQuery requires redesign.
D. **Correct** — Cloud SQL for PostgreSQL preserves the DBMS, minimizes application changes, and is the lowest-effort, lower-complexity path to run transactional workloads in Google Cloud while still supporting analytics (with replication/ETL to analytical systems as needed).

Question 181
You are architecting a data transformation solution for BigQuery. Your developers are proficient with SQL and want to use the ELT development technique. In addition, your developers need an intuitive coding environment and the ability to manage SQL as code. You need to identify a solution for your developers to build these pipelines. What should you do?

Options
A. Use Dataform to build, manage, and schedule SQL pipelines.
B. Use Dataflow jobs to read data from Pub/Sub, transform the data, and load the data to BigQuery.
C. Use Data Fusion to build and execute ETL pipelines.
D. Use Cloud Composer to load data and run SQL pipelines by using the BigQuery job operators.

Answer
A

Detailed explanation
A. **Correct** — Dataform (now part of BigQuery ecosystem) is designed for SQL-first ELT: it provides an intuitive SQL-based authoring environment, dependency management, scheduling, and treats SQL as code (supports CI/CD). It aligns exactly with developers proficient in SQL and ELT workflows.
B. **Incorrect** — Dataflow is for stream/batch code-based pipelines (Apache Beam) and is not a SQL-first ELT development environment; it requires more programming.
C. **Incorrect** — Data Fusion is a no-code/low-code ETL visual tool (more ETL than ELT) and while powerful, it's not focused on SQL-as-code and the developer experience described.
D. **Incorrect** — Cloud Composer (Airflow) orchestrates jobs and can run SQL operators, but it doesn't provide the SQL-as-code authoring and dependency management that Dataform provides natively.

Question 182
You work for a farming company. You have one BigQuery table named sensors, which is about 500 MB and contains the list of your 5000 sensors, with columns for id, name, and location. This table is updated every hour. Each sensor generates one metric every 30 seconds along with a timestamp, which you want to store in BigQuery. You want to run an analytical query on the data once a week for monitoring purposes. You also want to minimize costs. What data model should you use?

Options
A. 1. Create a metrics column in the sensors table. 2. Set RECORD type and REPEATED mode for the metrics column. 3. Use an UPDATE statement every 30 seconds to add new metrics.
B. 1. Create a metrics column in the sensors table. 2. Set RECORD type and REPEATED mode for the metrics column. 3. Use an INSERT statement every 30 seconds to add new metrics.
C. 1. Create a metrics table partitioned by timestamp. 2. Create a sensorId column in the metrics table, that points to the id column in the sensors table. 3. Use an INSERT statement every 30 seconds to append new metrics to the metrics table. 4. Join the two tables, if needed, when running the analytical query.
D. 1. Create a metrics table partitioned by timestamp. 2. Create a sensorId column in the metrics table, which points to the id column in the sensors table. 3. Use an UPDATE statement every 30 seconds to append new metrics to the metrics table. 4. Join the two tables, if needed, when running the analytical query.

Answer
C

Detailed explanation
A. **Incorrect** — Updating a single sensors table every 30 seconds with repeated RECORDs is inefficient (high write amplification) and UPDATEs are expensive; not cost-minimizing.
B. **Incorrect** — INSERTing into a nested REPEATED field inside the sensors table is not how BigQuery typically handles high-frequency append workloads; per-row nested growth is unwieldy.
C. **Correct** — Use a separate partitioned metrics table (by timestamp) and append metrics via INSERTs. This supports high write throughput, keeps sensor metadata small and cheap to update hourly, enables partition pruning for weekly analytics, and minimizes cost.
D. **Incorrect** — Using UPDATEs to append time-series data is inefficient and not recommended for frequent appends; INSERT append into a partitioned table is the right approach.

Question 183
You are managing a Dataplex environment with raw and curated zones. A data engineering team is uploading JSON and CSV files to a bucket asset in the curated zone but the files are not being automatically discovered by Dataplex. What should you do to ensure that the files are discovered by Dataplex?

Options
A. Move the JSON and CSV files to the raw zone.
B. Enable auto-discovery of files for the curated zone.
C. Use the bg command-line tool to load the JSON and CSV files into BigQuery tables.
D. Grant object level access to the CSV and JSON files in Cloud Storage.

Answer
A

Detailed explanation
A. **Correct** — Dataplex patterns expect raw data to land in raw zones for discovery and ingestion flows; curated zones are typically for cleaned/managed assets. Moving new raw files to the raw zone enables Dataplex auto-discovery and downstream processing into curated zones.
B. **Incorrect** — Curated zones normally contain managed assets; enabling auto-discovery on curated zones is not the recommended pattern and may conflict with governance workflows.
C. **Incorrect** — Manually loading via bg (or other tools) bypasses Dataplex discovery; it doesn't fix the automatic discovery behavior.
D. **Incorrect** — Object-level access controls affect permissions, not discovery; if permissions were the issue you'd get errors, but the recommended practice is to place raw files in raw zone.

Question 184
You have a table that contains millions of rows of sales data, partitioned by date. Various applications and users query this data many times a minute. The query requires aggregating values by using AVG, MAX, and SUM, and does not require joining to other tables. The required aggregations are only computed over the past year of data, though you need to retain full historical data in the base tables. You want to ensure that the query results always include the latest data from the tables, while also reducing computation cost, maintenance overhead, and duration. What should you do?

Options
A. Create a materialized view to aggregate the base table data. Include a filter clause to specify the last one year of partitions.
B. Create a materialized view to aggregate the base table data. Configure a partition expiration on the base table to retain only the last one year of partitions.
C. Create a view to aggregate the base table data. Include a filter clause to specify the last year of partitions.
D. Create a new table that aggregates the base table data. Include a filter clause to specify the last year of partitions. Set up a scheduled query to recreate the new table every hour.

Answer
A

Detailed explanation
A. **Correct** — Materialized views precompute aggregation results and automatically refresh to include recent data (near real-time). Adding a filter for the last year limits the materialized view to the relevant partitions, which reduces compute and speeds queries while retaining historical base data.
B. **Incorrect** — Partition-expiring the base table would delete historical data, which you must retain; also unnecessary if only the view needs last-year data.
C. **Incorrect** — A standard view runs the aggregation at query time each time, causing high compute cost and latency for frequent queries.
D. **Incorrect** — Scheduled queries + tables require maintenance and may introduce staleness; materialized views provide lower maintenance and fresher results automatically.

Question 185
Your organization uses a multi-cloud data storage strategy, storing data in Cloud Storage, and data in Amazon Web Services' (AWS) S3 storage buckets. All data resides in US regions. You want to query up-to-date data by using BigQuery, regardless of which cloud the data is stored in. You need to allow users to query the tables from BigQuery without giving direct access to the data in the storage buckets. What should you do?

Options
A. Setup a BigQuery Omni connection to the AWS S3 bucket data. Create BigLake tables over the Cloud Storage and S3 data and query the data using BigQuery directly.
B. Set up a BigQuery Omni connection to the AWS S3 bucket data. Create external tables over the Cloud Storage and S3 data and query the data using BigQuery directly.
C. Use the Storage Transfer Service to copy data from the AWS S3 buckets to Cloud Storage buckets. Create BigLake tables over the Cloud Storage data and query the data using BigQuery directly.
D. Use the Storage Transfer Service to copy data from the AWS S3 buckets to Cloud Storage buckets. Create external tables over the Cloud Storage data and query the data using BigQuery directly.

Answer
A

Detailed explanation
A. **Correct** — BigQuery Omni + BigLake lets you query S3 data in-place (via Omni) and Cloud Storage in-place through BigLake, providing up-to-date queries without copying data or granting direct bucket access; BigLake provides unified access and governance.
B. **Incorrect** — While Omni supports external tables, BigLake over Omni gives better unified governance and performance characteristics; external tables alone are less feature-rich for a cross-cloud mesh.
C. **Incorrect** — Copying data adds operational overhead, storage cost, and delays; you wanted up-to-date queries without granting direct access.
D. **Incorrect** — Same drawbacks as C: copying data is less desirable than querying in-place across clouds with Omni + BigLake.

Question 186
You are preparing an organization-wide dataset. You need to preprocess customer data stored in a restricted bucket in Cloud Storage. The data will be used to create consumer analyses. You need to comply with data privacy requirements. What should you do?

Options
A. Use Dataflow and the Cloud Data Loss Prevention API to mask sensitive data. Write the processed data in BigQuery.
B. Use customer-managed encryption keys (CMEK) to directly encrypt the data in Cloud Storage. Use federated queries from BigQuery. Share the encryption key by following the principle of least privilege.
C. Use the Cloud Data Loss Prevention API and Dataflow to detect and remove sensitive fields from the data in Cloud Storage. Write the filtered data in BigQuery.
D. Use Dataflow and Cloud KMS to encrypt sensitive fields and write the encrypted data in BigQuery. Share the encryption key by following the principle of least privilege.

Answer
A

Detailed explanation
A. **Correct** — Dataflow + Cloud DLP is a standard, automated pattern to detect/mask (tokenize/redact) sensitive fields at scale and write privacy-preserving data into BigQuery; it directly meets privacy requirements and produces usable analytics data.
B. **Incorrect** — CMEK protects at-rest encryption but does not remove or mask sensitive fields for analytics usage; sharing keys adds complexity and doesn't meet the requirement to preprocess/mask data for privacy.
C. **Incorrect** — Removing sensitive fields (deletion) may be overly destructive; the question asks preprocessing for analyses — masking (A) is more appropriate than outright removal (though removal can be valid in some cases).
D. **Incorrect** — Encrypting fields with KMS preserves data confidentiality but requires key management and doesn't produce directly usable analytic data unless decrypted, which reintroduces privacy risk.

Question 187
You need to connect multiple applications with dynamic public IP addresses to a Cloud SQL instance. You configured users with strong passwords and enforced the SSL connection to your Cloud SQL instance. You want to use Cloud SQL public IP and ensure that you have secured connections. What should you do?

Options
A. Add CIDR 0.0.0.0/0 network to Authorized Network. Use Identity and Access Management (IAM) to add users.
B. Add all application networks to Authorized Network and regularly update them.
C. Leave the Authorized Network empty. Use Cloud SQL Auth proxy on all applications.
D. Add CIDR 0.0.0.0/0 network to Authorized Network. Use Cloud SQL Auth proxy on all applications.

Answer
C

Detailed explanation
A. **Incorrect** — Opening 0.0.0.0/0 is insecure; IAM does not replace network-level restrictions for Cloud SQL public IP access.
B. **Incorrect** — Maintaining dynamic public IPs in authorized networks is operationally heavy and error-prone.
C. **Correct** — Leaving Authorized Network empty (no IP whitelisting) and using Cloud SQL Auth proxy ensures secure, authenticated connections regardless of client IPs; proxy handles IAM-based auth and secures the connection.
D. **Incorrect** — Adding 0.0.0.0/0 is insecure even if combined with the proxy; unnecessary exposure.

Question 188
You are migrating a large number of files from a public HTTPS endpoint to Cloud Storage. The files are protected from unauthorized access using signed URLs. You created a TSV file that contains the list of object URLs and started a transfer job by using Storage Transfer Service. You notice that the job has run for a long time and eventually failed. Checking the logs of the transfer job reveals that the job was running fine until one point, and then it failed due to HTTP 403 errors on the remaining files. You verified that there were no changes to the source system. You need to fix the problem to resume the migration process. What should you do?

Options
A. Set up Cloud Storage FUSE, and mount the Cloud Storage bucket on a Compute Engine instance. Remove the completed files from the TSV file. Use a shell script to iterate through the TSV file and download the remaining URLs to the FUSE mount point.
B. Renew the TLS certificate of the HTTPS endpoint. Remove the completed files from the TSV file and rerun the Storage Transfer Service job.
C. Create a new TSV file for the remaining files by generating signed URLs with a longer validity period. Split the TSV file into multiple smaller files and submit them as separate Storage Transfer Service jobs in parallel.
D. Update the file checksums in the TSV file from using MD5 to SHA256. Remove the completed files from the TSV file and rerun the Storage Transfer Service job.

Answer
C

Detailed explanation
A. **Incorrect** — Mounting buckets with GCS FUSE and custom scripts is operationally complex and not necessary when Storage Transfer Service can handle signed URLs properly.
B. **Incorrect** — TLS certificate issues would cause different errors; you verified the source system unchanged and errors are 403 (access denied), which likely indicates signed URL expiry.
C. **Correct** — 403 errors for signed URLs commonly indicate URL expiry. Regenerate signed URLs with longer validity for remaining files, split into smaller TSV batches and run parallel transfer jobs to avoid long-running single jobs and repeated expiry.
D. **Incorrect** — Checksums don't cause 403 errors; changing checksum algorithm won't fix signed URL expiry issues.

Question 189
You work for an airline and you need to store weather data in a BigQuery table. Weather data will be used as input to a machine learning model. The model only uses the last 30 days of weather data. You want to avoid storing unnecessary data and minimize costs. What should you do?

Options
A. Create a BigQuery table where each record has an ingestion timestamp. Run a scheduled query to delete all the rows with an ingestion timestamp older than 30 days.
B. Create a BigQuery table partitioned by datetime value of the weather date. Set up partition expiration to 30 days.
C. Create a BigQuery table partitioned by ingestion time. Set up partition expiration to 30 days.
D. Create a BigQuery table with a datetime column for the day the weather data refers to. Run a scheduled query to delete rows with a datetime value older than 30 days.

Answer
B

Detailed explanation
A. **Incorrect** — Periodic deletes are more expensive and slower than using built-in partition expiration.
B. **Correct** — Partitioning by the event weather date (the datetime value that the weather refers to) and applying 30-day partition expiration ensures data older than 30 days is automatically deleted with minimal cost and operational overhead.
C. **Incorrect** — Partitioning by ingestion time can work, but if ingestion is delayed or backfilled, partitioning by event date is more accurate for models that rely on the weather date.
D. **Incorrect** — Scheduled deletes are operationally heavier than partition expiration and subject to failure/latency.

Question 190
You need to look at BigQuery data from a specific table multiple times a day. The underlying table you are querying is several petabytes in size, but you want to filter your data and provide simple aggregations to downstream users. You want to run queries faster and get up-to-date insights quicker. What should you do?

Options
A. Run a scheduled query to pull the necessary data at specific intervals daily.
B. Use a cached query to accelerate time to results.
C. Limit the query columns being pulled in the final result.
D. Create a materialized view based off of the query being run.

Answer
D

Detailed explanation
A. **Incorrect** — Scheduled queries can help, but they may add latency and maintenance; materialized views provide automatic incremental maintenance and faster reads.
B. **Incorrect** — Cached queries speed repeated identical queries but may be invalidated frequently if underlying data changes; they aren't a long-term optimization for frequent up-to-date aggregations.
C. **Incorrect** — Limiting columns reduces bytes scanned but doesn't address repeated aggregation cost or provide the fastest up-to-date results.
D. **Correct** — Materialized views precompute and incrementally maintain aggregation results, dramatically improving latency and cost for frequent queries while keeping data up to date.

Question 191
Your chemical company needs to manually check documentation for customer order. You use a pull subscription in Pub/Sub so that sales agents get details from the order. You must ensure that you do not process orders twice with different sales agents and that you do not add more complexity to this workflow. What should you do?

Options
A. Use a Deduplicate PTransform in Dataflow before sending the messages to the sales agents.
B. Create a transactional database that monitors the pending messages.
C. Use Pub/Sub exactly-once delivery in your pull subscription.
D. Create a new Pub/Sub push subscription to monitor the orders processed in the agent's system.

Answer
C

Detailed explanation
A. **Incorrect** — A Deduplicate PTransform adds moving parts and complexity to the workflow; it requires processing in Dataflow and doesn't remove the need for deduplication guarantees in Pub/Sub delivery.
B. **Incorrect** — Adding a transactional database is heavy-weight and increases complexity and operational overhead.
C. **Correct** — Pub/Sub exactly-once delivery for pull subscriptions prevents duplicates without adding extra workflow complexity; it ensures each message is delivered and processed only once.
D. **Incorrect** — Creating an extra push subscription increases complexity and does not inherently prevent duplicate processing between agents.

Question 192
You are migrating your on-premises data warehouse to BigQuery. As part of the migration, you want to facilitate cross-team collaboration to get the most value out of the organization's data. You need to design an architecture that would allow teams within the organization to securely publish, discover, and subscribe to read-only data in a self-service manner. You need to minimize costs while also maximizing data freshness. What should you do?

Options
A. Use Analytics Hub to facilitate data sharing.
B. Create authorized datasets to publish shared data in the subscribing team's project.
C. Create a new dataset for sharing in each individual team's project. Grant the subscribing team the bigquery.dataViewer role on the dataset.
D. Use BigQuery Data Transfer Service to copy datasets to a centralized BigQuery project for sharing.

Answer
A

Detailed explanation
A. **Correct** — Analytics Hub (private exchange) enables secure, discoverable, self-service dataset publishing and subscription with minimal duplication and supports fresh data sharing; it's cost-effective compared with copying data.
B. **Incorrect** — Authorized datasets require per-project setup and are less self-service and scalable than Analytics Hub.
C. **Incorrect** — Creating separate datasets per team and granting viewers is manual, less discoverable, and more administrative overhead.
D. **Incorrect** — Copying datasets increases storage costs and adds latency; it's not optimal for freshness or cost.

Question 193
You want to migrate an Apache Spark 3 batch job from on-premises to Google Cloud. You need to minimally change the job so that the job reads from Cloud Storage and writes the result to BigQuery. Your job is optimized for Spark, where each executor has 8 vCPU and 16 GB memory, and you want to be able to choose similar settings. You want to minimize installation and management effort to run your job. What should you do?

Options
A. Execute the job as part of a deployment in a new Google Kubernetes Engine cluster.
B. Execute the job from a new Compute Engine VM.
C. Execute the job in a new Dataproc cluster.
D. Execute as a Dataproc Serverless job.

Answer
D

Detailed explanation
A. **Incorrect** — Running on GKE requires setting up Spark containers and managing the cluster — more operational overhead.
B. **Incorrect** — A VM-based approach requires manual provisioning and management; not managed and more ops effort.
C. **Incorrect** — A full Dataproc cluster works and supports fine-grained instance sizing but requires cluster management; more maintenance than serverless.
D. **Correct** — Dataproc Serverless runs Spark jobs with minimal management, supports configurable resources to match executor sizing needs, and requires minimal installation/ops effort — best for minimal changes and reduced management.

Question 194
You are configuring networking for a Dataflow job. The data pipeline uses custom container images with the libraries that are required for the transformation logic preinstalled. The data pipeline reads the data from Cloud Storage and writes the data to BigQuery. You need to ensure cost-effective and secure communication between the pipeline and Google APIs and services. What should you do?

Options
A. Disable external IP addresses from worker VMs and enable Private Google Access.
B. Leave external IP addresses assigned to worker VMs while enforcing firewall rules.
C. Disable external IP addresses and establish a Private Service Connect endpoint IP address.
D. Enable Cloud NAT to provide outbound internet connectivity while enforcing firewall rules.

Answer
A

Detailed explanation
A. **Correct** — Disabling external IPs and enabling Private Google Access lets workers securely reach Google APIs (Cloud Storage, BigQuery) without public IPs, reducing exposure and often lowering NAT/cloud egress costs — a recommended secure and cost-effective pattern.
B. **Incorrect** — Leaving external IPs increases attack surface and is less secure; firewall rules don't eliminate public IP exposure.
C. **Incorrect** — Private Service Connect is used for private connectivity to specific services but is more complex; Private Google Access is the simpler, standard approach for Dataflow workers to access Google APIs privately.
D. **Incorrect** — Cloud NAT allows outbound internet for non-public IP workers but still relies on NAT for connectivity; Private Google Access avoids NAT for Google APIs and is more cost-effective for this use case.

Question 195
You are using Workflows to call an API that returns a 1KB JSON response, apply some complex business logic on this response, wait for the logic to complete, and then perform a load from a Cloud Storage file to BigQuery. The Workflows standard library does not have sufficient capabilities to perform your complex logic, and you want to use Python's standard library instead. You want to optimize your workflow for simplicity and speed of execution. What should you do?

Options
A. Create a Cloud Composer environment and run the logic in Cloud Composer.
B. Create a Dataproc cluster, and use PySpark to apply the logic on your JSON file.
C. Invoke a Cloud Function instance that uses Python to apply the logic on your JSON file.
D. Invoke a subworkflow in Workflows to apply the logic on your JSON file.

Answer
C

Detailed explanation
A. **Incorrect** — Cloud Composer (Airflow) is heavyweight for a small piece of Python logic and adds orchestration complexity.
B. **Incorrect** — Dataproc + PySpark is heavy and overkill for small JSON-processing tasks.
C. **Correct** — Invoking a Cloud Function (Python) is lightweight, fast to execute, easy to integrate with Workflows, and supports Python standard library — ideal for simple, quick logic execution.
D. **Incorrect** — A subworkflow in Workflows still uses Workflows' capabilities and doesn't add native Python standard library support; it doesn't solve the lack of Python runtime.

Question 196
You are administering a BigQuery on-demand environment. Your business intelligence tool is submitting hundreds of queries each day that aggregate a large (50 TB) sales history fact table at the day and month levels. These queries have a slow response time and are exceeding cost expectations. You need to decrease response time, lower query costs, and minimize maintenance. What should you do?

Options
A. Build authorized views on top of the sales table to aggregate data at the day and month level.
B. Enable BI Engine and add your sales table as a preferred table.
C. Build materialized views on top of the sales table to aggregate data at the day and month level.
D. Create a scheduled query to build sales day and sales month aggregate tables on an hourly basis.

Answer
C

Detailed explanation
A. **Incorrect** — Authorized views control access but do not improve query performance or reduce scan costs for heavy aggregations.
B. **Incorrect** — BI Engine accelerates dashboard queries but has capacity limits and requires configuration; materialized views directly reduce scan cost and speed for repeated aggregations.
C. **Correct** — Materialized views precompute and incrementally maintain aggregates, reducing scan volume and latency for repeated day/month aggregations, with low maintenance overhead.
D. **Incorrect** — Scheduled queries to build aggregate tables work but add maintenance and can have staleness windows; materialized views are simpler and more automatic.

Question 197
You have several different unstructured data sources, within your on-premises data center as well as in the cloud. The data is in various formats, such as Apache Parquet and CSV. You want to centralize this data in Cloud Storage. You need to set up an object sink for your data that allows you to use your own encryption keys. You want to use a GUI-based solution. What should you do?

Options
A. Use BigQuery Data Transfer Service to move files into BigQuery.
B. Use Storage Transfer Service to move files into Cloud Storage
C. Use Dataflow to move files into Cloud Storage
D. Use Cloud Data Fusion to move files into Cloud Storage.

Answer
B

Detailed explanation
A. **Incorrect** — BigQuery Data Transfer Service moves data into BigQuery, not into Cloud Storage, and is not the GUI-focused object sink tool for arbitrary files.
B. **Correct** — Storage Transfer Service supports GUI-based configuration to transfer objects from on-premises or other cloud sources into Cloud Storage, and supports specifying CMEK for the destination buckets. It's the appropriate GUI-friendly object sink.
C. **Incorrect** — Dataflow can move and transform data but is developer-centric (code) rather than GUI-based.
D. **Incorrect** — Cloud Data Fusion provides a GUI ETL experience but is more for data transformation pipelines; Storage Transfer Service is the native GUI tool for bulk file transfer to Cloud Storage and supports CMEK.

Question 198
You are using BigQuery with a regional dataset that includes a table with the daily sales volumes. This table is updated multiple times per day. You need to protect your sales table in case of regional failures with a recovery point objective (RPO) of less than 24 hours, while keeping costs to a minimum. What should you do?

Options
A. Schedule a daily export of the table to a Cloud Storage dual or multi-region bucket.
B. Schedule a daily copy of the dataset to a backup region.
C. Schedule a daily BigQuery snapshot of the table.
D. Modify ETL job to load the data into both the current and another backup region.

Answer
A

Detailed explanation
A. **Correct** — Daily export to a dual- or multi-region Cloud Storage bucket provides durable off-region backups with low storage cost and meets RPO < 24 hours; restoring from export is feasible and cost-effective.
B. **Incorrect** — Copying datasets daily into another BigQuery region increases BigQuery storage cost and is heavier than exporting to multi-region Cloud Storage.
C. **Incorrect** — BigQuery snapshots are regional and may not protect against regional failure; snapshots alone do not replicate off-region.
D. **Incorrect** — Modifying ETL to write to two regions doubles ingestion complexity and cost; heavier than needed for <24h RPO.

Question 199
You are preparing an organization-wide dataset. You need to preprocess customer data stored in a restricted bucket in Cloud Storage. The data will be used to create consumer analyses. You need to follow data privacy requirements, including protecting certain sensitive data elements, while also retaining all of the data for potential future use cases. What should you do?

Options
A. Use the Cloud Data Loss Prevention API and Dataflow to detect and remove sensitive fields from the data in Cloud Storage. Write the filtered data in BigQuery.
B. Use customer-managed encryption keys (CMEK) to directly encrypt the data in Cloud Storage. Use federated queries from BigQuery. Share the encryption key by following the principle of least privilege.
C. Use Dataflow and the Cloud Data Loss Prevention API to mask sensitive data. Write the processed data in BigQuery.
D. Use Dataflow and Cloud KMS to encrypt sensitive fields and write the encrypted data in BigQuery. Share the encryption key by following the principle of least privilege.

Answer
C

Detailed explanation
A. **Incorrect** — Removing sensitive fields loses potentially valuable data for future use cases; the requirement is to protect while retaining data.
B. **Incorrect** — CMEK protects at-rest but does not provide usable masked data for analysts; it also complicates sharing and access patterns.
C. **Correct** — Masking (tokenization/redaction) via Dataflow + Cloud DLP preserves privacy while retaining the data structure for future use; analysts get usable, privacy-preserving data.
D. **Incorrect** — Encrypting fields leaves them unusable without decryption and increases key management complexity; masking is preferable for analysis while retaining usable data.

Question 200
Suppose you have a table that includes a nested column called "city" inside a column called "person", but when you try to submit the following query in BigQuery, it gives you an error.
SELECT person FROM project1.example.table1 WHERE city = "London"
How would you correct the error?

Options
A. Add ", UNNEST(person)" before the WHERE clause.
B. Change "person" to "person.city".
C. Change "person" to "city.person".
D. Add ", UNNEST(city)" before the WHERE clause.

Answer
B

Detailed explanation
A. **Incorrect** — UNNEST(person) would attempt to flatten the entire person record; the problem is the WHERE clause referencing city directly—UNNESTing person may be unnecessary and syntactically wrong here.
B. **Correct** — Using the nested field path person.city in the WHERE clause correctly references the nested column. For example: SELECT person FROM \project1.example.table1` WHERE person.city = "London". C. **Incorrect** — city.personreverses the nesting order and is invalid because city is nested inside person, not the other way around. D. **Incorrect** — UNNEST(city)is invalid becausecityis not an array field at the top level; the correct approach is to reference the nested field viaperson.city`.

Question 201
Your company built a TensorFlow neutral-network model with a large number of neurons and layers. The model fits well for the training data. However, when tested against new data, it performs poorly. What method can you employ to address this?

Options
A. Threading
B. Serialization
C. Dropout Methods
D. Dimensionality Reduction

Answer
C

Detailed explanation
A. Incorrect - Threading is a compute/parallelism technique, not a regularization method for overfitting.
B. Incorrect - Serialization deals with saving/loading models, not generalization.
C. Correct - Dropout is a regularization technique that randomly disables neurons during training to reduce overfitting and improve generalization.
D. Incorrect - Dimensionality reduction can help in some cases, but dropout is the direct method to combat overfitting in large neural nets.

Question 202
You are building a model to make clothing recommendations. You know a user's fashion preference is likely to change over time, so you build a data pipeline to stream new data back to the model as it becomes available. How should you use this data to train the model?

Options
A. Continuously retrain the model on just the new data.
B. Continuously retrain the model on a combination of existing data and the new data.
C. Train on the existing data while using the new data as your test set.
D. Train on the new data while using the existing data as your test set.

Answer
B

Detailed explanation
A. Incorrect - Training only on new data risks catastrophic forgetting and bias toward recent items.
B. Correct - Combine historical and recent data (with appropriate weighting/windowing) so the model retains long-term patterns while adapting to new trends.
C. Incorrect - Using new data only as test prevents the model from learning recent behavior.
D. Incorrect - Using existing data only as test prevents learning necessary long-term patterns.

Question 203
You designed a database for patient records as a pilot project to cover a few hundred patients in three clinics. Your design used a single database table to represent all patients and their visits, and you used self-joins to generate reports. The server resource utilization was at 50%. Since then, the scope of the project has expanded. The database must now store 100 times more patient records. You can no longer run the reports, because they either take too long or they encounter errors with insufficient compute resources. How should you adjust the database design?

Options
A. Add capacity (memory and disk space) to the database server by the order of 200.
B. Shard the tables into smaller ones based on date ranges, and only generate reports with prespecified date ranges.
C. Normalize the master patient-record table into the patient table and the visits table, and create other necessary tables to avoid self-join.
D. Partition the table into smaller tables, with one for each clinic. Run queries against the smaller table pairs, and use unions for consolidated reports.

Answer
C

Detailed explanation
A. Incorrect - Simply adding huge capacity is expensive and postpones addressing inefficient schema/queries.
B. Incorrect - Sharding by date is a possible optimization but doesn't fix the fundamental schema inefficiency and complicates queries.
C. Correct - Normalizing into separate patient and visit tables (proper relational schema) removes expensive self-joins, enables indexes and optimized queries, and scales far better.
D. Incorrect - Per-clinic tables partition data but still preserves poor schema; normalization is the cleaner, principle-based fix.

Question 204
You create an important report for your large team in Google Data Studio 360. The report uses Google BigQuery as its data source. You notice that visualizations are not showing data that is less than 1 hour old. What should you do?

Options
A. Disable caching by editing the report settings.
B. Disable caching in BigQuery by editing table details.
C. Refresh your browser tab showing the visualizations.
D. Clear your browser history for the past hour then reload the tab showing the virtualizations.

Answer
A

Detailed explanation
A. Correct - Data Studio caches query results; disabling or reducing caching interval in the report settings ensures fresher data appears.
B. Incorrect - BigQuery table settings don't control Data Studio's client-side/report-level cache.
C. Incorrect - Browser refresh won't bypass Data Studio caching of query results.
D. Incorrect - Clearing browser history is unrelated to Data Studio's data caching.

Question 205
An external customer provides you with a daily dump of data from their database. The data flows into Google Cloud Storage GCS as comma-separated values (CSV) files. You want to analyze this data in Google BigQuery, but the data could have rows that are formatted incorrectly or corrupted. How should you build this pipeline?

Options
A. Use federated data sources, and check data in the SQL query.
B. Enable BigQuery monitoring in Google Stackdriver and create an alert.
C. Import the data into BigQuery using the gcloud CLI and set max_bad_records to 0.
D. Run a Google Cloud Dataflow batch pipeline to import the data into BigQuery, and push errors to another dead-letter table for analysis.

Answer
D

Detailed explanation
A. Incorrect - Federated queries against raw CSVs complicate validation and can be slow; better to preprocess.
B. Incorrect - Monitoring alerts don't address validation and handling of corrupted rows.
C. Incorrect - Setting max_bad_records=0 will fail the load on any bad row — not robust for daily external dumps.
D. Correct - Dataflow enables robust parsing/validation, can route bad rows to a dead-letter table for inspection, and only load clean data into BigQuery.

Question 206
Your weather app queries a database every 15 minutes to get the current temperature. The frontend is powered by Google App Engine and server millions of users. How should you design the frontend to respond to a database failure?

Options
A. Issue a command to restart the database servers.
B. Retry the query with exponential backoff, up to a cap of 15 minutes.
C. Retry the query every second until it comes back online to minimize staleness of data.
D. Reduce the query frequency to once every hour until the database comes back online.

Answer
B

Detailed explanation
A. Incorrect - App frontends shouldn't control or restart DB servers; that's ops responsibility.
B. Correct - Exponential backoff avoids hammering the DB while eventually retrying within the acceptable freshness window (cap to 15 minutes matches expected interval).
C. Incorrect - Retrying every second causes thundering-herd and worsens outage impact.
D. Incorrect - Reducing queries to hourly increases staleness unacceptably.

Question 207
You are creating a model to predict housing prices. Due to budget constraints, you must run it on a single resource-constrained virtual machine. Which learning algorithm should you use?

Options
A. Linear regression
B. Logistic classification
C. Recurrent neural network
D. Feedforward neural network

Answer
A

Detailed explanation
A. Correct - Linear regression is simple, efficient, and well-suited for numeric prediction on constrained hardware.
B. Incorrect - Logistic classification is for binary classification, not predicting continuous prices.
C. Incorrect - RNNs are heavy and designed for sequence data; overkill and resource-intensive.
D. Incorrect - Feedforward neural networks may be feasible but are more computationally expensive than linear regression.

Question 208
You are building new real-time data warehouse for your company and will use Google BigQuery streaming inserts. There is no guarantee that data will only be sent in once but you do have a unique ID for each row of data and an event timestamp. You want to ensure that duplicates are not included while interactively querying data. Which query type should you use?

Options
A. Include ORDER BY DESK on timestamp column and LIMIT to 1.
B. Use GROUP BY on the unique ID column and timestamp column and SUM on the values.
C. Use the LAG window function with PARTITION by unique ID along with WHERE LAG IS NOT NULL.
D. Use the ROW_NUMBER window function with PARTITION by unique ID along with WHERE row equals 1.

Answer
D

Detailed explanation
A. Incorrect - ORDER/LIMIT alone won't deduplicate correctly across streaming data.
B. Incorrect - GROUP BY with timestamp may aggregate multiple events incorrectly.
C. Incorrect - LAG can detect previous rows but isn't the cleanest deduplication method for selecting a single canonical row.
D. Correct - ROW_NUMBER() OVER (PARTITION BY unique_id ORDER BY event_timestamp DESC) then filter WHERE row_number = 1 selects the latest unique record and removes duplicates interactively.

Question 209
Your company is using WILDCARD tables to query data across multiple tables with similar names. The SQL statement is currently failing with the following error:

```
# Syntax error : Expected end of statement but got "-" at [4:11]
SELECT age
FROM
    bigquery-public-data.noaa_gsod.gsod
WHERE
    age != 99
    AND_TABLE_SUFFIX = '1929'
ORDER BY
    age DESC
```

Which table name will make the SQL statement work correctly?

Options
A. 'bigquery-public-data.noaa_gsod.gsod'
B. bigquery-public-data.noaa_gsod.gsod*
C. 'bigquery-public-data.noaa_gsod.gsod'*
D. 'bigquery-public-data.noaa_gsod.gsod*`

Answer
D

Detailed explanation
A. Incorrect - Quoted full table name is a single table, not a wildcard pattern.
B. Incorrect - Unquoted with * lacks correct quoting/backtick for fully qualified wildcard.
C. Incorrect - Misplaced quoting/asterisk makes it invalid syntax.
D. Correct - (Per your provided answer) the variant with the wildcard/backtick is the option indicated to be syntactically acceptable in this question context.

Question 210
Your company is in a highly regulated industry. One of your requirements is to ensure individual users have access only to the minimum amount of information required to do their jobs. You want to enforce this requirement with Google BigQuery. Which three approaches can you take? (Choose three.)

Options
A. Disable writes to certain tables.
B. Restrict access to tables by role.
C. Ensure that the data is encrypted at all times.
D. Restrict BigQuery API access to approved users.
E. Segregate data across multiple tables or databases.
F. Use Google Stackdriver Audit Logging to determine policy violations.

Answer
B D E

Detailed explanation
A. Incorrect - Disabling writes controls data mutability but doesn't enforce least-privilege access for reads.
B. Correct - Table-level IAM (roles/permissions) enforces who can read specific datasets/tables.
C. Incorrect - Encryption is important for protection-at-rest/in-transit but does not by itself restrict user access to minimal data.
D. Correct - Restricting API access reduces who can call BigQuery and complements IAM-based access control.

E. Correct - Segregating sensitive data into separate tables/projects makes it easier to grant least privilege and control access.
F. Incorrect - Audit logs help detect violations but do not enforce least privilege by themselves.

Question 211
You are designing a basket abandonment system for an ecommerce company. The system will send a message to a user based on these rules:
☞ No interaction by the user on the site for 1 hour
☞ Has added more than $30 worth of products to the basket
☞ Has not completed a transaction
You use Google Cloud Dataflow to process the data and decide if a message should be sent. How should you design the pipeline?

Options
A. Use a fixed-time window with a duration of 60 minutes.
B. Use a sliding time window with a duration of 60 minutes.
C. Use a session window with a gap time duration of 60 minutes.
D. Use a global window with a time based trigger with a delay of 60 minutes.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect. Fixed windows group data strictly by time intervals, not inactivity, so they don't capture "no interaction" rules.
B. Incorrect. Sliding windows overlap in time but still don't model user inactivity properly.
C. Correct. Session windows are designed for periods of user activity separated by inactivity. A 60-minute gap session window exactly matches the "no interaction for 1 hour" requirement.
D. Incorrect. Global windows with triggers can delay processing, but they are not efficient for user-specific inactivity detection.
E. Incorrect. The right answer exists (C).

Question 212
Your company handles data processing for a number of different clients. Each client prefers to use their own suite of analytics tools, with some allowing direct query access via Google BigQuery. You need to secure the data so that clients cannot see each other's data. You want to ensure appropriate access to the data. Which three steps should you take? (Choose three.)

Options
A. Load data into different partitions.
B. Load data into a different dataset for each client.
C. Put each client's BigQuery dataset into a different table.
D. Restrict a client's dataset to approved users.
E. Only allow a service account to access the datasets.
F. Use the appropriate identity and access management (IAM) roles for each client's users.

Answer
BDF

Detailed explanation
A. Incorrect. Partitions are for data management, not security boundaries. Clients could still see each other's data.
B. Correct. Different datasets provide strong isolation and allow dataset-level IAM control per client.
C. Incorrect. Multiple tables in the same dataset still share the same access control, so clients could see all tables.
D. Correct. Restricting datasets to approved users ensures that only the right client has dataset access.
E. Incorrect. Service accounts manage app access, not per-client user access control.
F. Correct. IAM roles enforce least privilege access and let you control who can query or modify each client's dataset.

Question 213
You want to process payment transactions in a point-of-sale application that will run on Google Cloud Platform. Your user base could grow exponentially, but you do not want to manage infrastructure scaling. Which Google database service should you use?

Options
A. Cloud SQL
B. BigQuery
C. Cloud Bigtable
D. Cloud Datastore
E. Memorystore

Answer
D

Detailed explanation
A. Incorrect. Cloud SQL requires manual scaling and is not ideal for exponential growth.
B. Incorrect. BigQuery is an analytical database, not suited for transactional workloads.
C. Incorrect. Bigtable is good for scale but doesn't support ACID transactions needed for payments.
D. Correct. Datastore (Firestore in Datastore mode) is fully managed, scales automatically, and supports transactional consistency — ideal for payments.
E. Incorrect. Memorystore is for caching, not as a primary database.

Question 214
You want to use a database of information about tissue samples to classify future tissue samples as either normal or mutated. You are evaluating an unsupervised anomaly detection method for classifying the tissue samples. Which two characteristics support this method? (Choose two.)

Options
A. There are very few occurrences of mutations relative to normal samples.
B. There are roughly equal occurrences of both normal and mutated samples in the database.
C. You expect future mutations to have different features from the mutated samples in the database.
D. You expect future mutations to have similar features to the mutated samples in the database.
E. You already have labels for which samples are mutated and which are normal in the database.

Answer
A, C

Detailed explanation
A. Correct. Anomaly detection is best when anomalies (mutations) are rare compared to normal samples.
B. Incorrect. Equal occurrences suggest supervised learning instead of anomaly detection.
C. Correct. Unsupervised methods work when future anomalies differ from known data patterns.
D. Incorrect. If new mutations are similar to old ones, supervised learning would be more effective.
E. Incorrect. Having labels supports supervised learning, not anomaly detection.

Question 215
You need to store and analyze social media postings in Google BigQuery at a rate of 10,000 messages per minute in near real-time. Initially, design the application to use streaming inserts for individual postings. Your application also performs data aggregations right after the streaming inserts. You discover that the queries after streaming inserts do not exhibit strong consistency, and reports from the queries might miss in-flight data. How can you adjust your application design?

Options
A. Re-write the application to load accumulated data every 2 minutes.
B. Convert the streaming insert code to batch load for individual messages.
C. Load the original message to Google Cloud SQL, and export the table every hour to BigQuery via streaming inserts.
D. Estimate the average latency for data availability after streaming inserts, and always run queries after waiting twice as long.
E. Use BigQuery BI Engine to guarantee consistency.

Answer
D

Detailed explanation
A. Incorrect. Loading every 2 minutes reduces freshness and is unnecessary.
B. Incorrect. Batch load for individual messages defeats the purpose of real-time ingestion.
C. Incorrect. Adding Cloud SQL increases latency and complexity.
D. Correct. Streaming inserts take some time before data is consistent. Estimating latency and waiting ensures accurate queries while maintaining near real-time.
E. Incorrect. BI Engine improves performance but does not solve consistency delays.

Question 216
Your startup has never implemented a formal security policy. Currently, everyone in the company has access to the datasets stored in Google BigQuery. Teams have freedom to use the service as they see fit, and they have not documented their use cases. You have been asked to secure the data warehouse. You need to discover what everyone is doing. What should you do first?

Options
A. Use Google Stackdriver Audit Logs to review data access.
B. Get the identity and access management (IAM) policy of each table.
C. Use Stackdriver Monitoring to see the usage of BigQuery query slots.
D. Use the Google Cloud Billing API to see what account the warehouse is being billed to.
E. Create dataset-level monitoring policies in BigQuery.

Answer
A

Detailed explanation
A. Correct. Audit logs show exactly who accessed what data, giving the required visibility.
B. Incorrect. IAM policies show permissions, not actual usage.
C. Incorrect. Query slot monitoring shows performance, not data usage.
D. Incorrect. Billing data shows cost, not data access or behavior.
E. Incorrect. Dataset-level monitoring doesn't give historical usage detail.

Question 217
Your company is migrating their 30-node Apache Hadoop cluster to the cloud. They want to re-use Hadoop jobs they have already created and minimize the management of the cluster as much as possible. They also want to be able to persist data beyond the life of the cluster. What should you do?

Options
A. Create a Google Cloud Dataflow job to process the data.
B. Create a Google Cloud Dataproc cluster that uses persistent disks for HDFS.
C. Create a Hadoop cluster on Google Compute Engine that uses persistent disks.
D. Create a Cloud Dataproc cluster that uses the Google Cloud Storage connector.
E. Create a Hadoop cluster on Google Compute Engine that uses Local SSD disks.

Answer
D

Detailed explanation
A. Incorrect. Dataflow requires rewriting jobs, which they don't want.
B. Incorrect. Persistent disks tie storage to cluster nodes, limiting persistence.
C. Incorrect. Running Hadoop directly on Compute Engine requires more management.
D. Correct. Using Dataproc with Cloud Storage connector allows jobs to run without rewriting and data persists in GCS independent of cluster lifecycle.
E. Incorrect. Local SSDs are ephemeral and unsuitable for long-term storage.

Question 218
Business owners at your company have given you a database of bank transactions. Each row contains the user ID, transaction type, transaction location, and transaction amount. They ask you to investigate what type of machine learning can be applied to the data. Which three machine learning applications can you use? (Choose three.)

Options
A. Supervised learning to determine which transactions are most likely to be fraudulent.
B. Unsupervised learning to determine which transactions are most likely to be fraudulent.
C. Clustering to divide the transactions into N categories based on feature similarity.
D. Supervised learning to predict the location of a transaction.
E. Reinforcement learning to predict the location of a transaction.
F. Unsupervised learning to predict the location of a transaction.

Answer
B, C, D

Detailed explanation
A. Incorrect. Fraud detection with supervised learning requires labeled fraud data, which isn't mentioned.
B. Correct. Unsupervised anomaly detection can identify fraud when no labels exist.
C. Correct. Clustering groups transactions into meaningful categories for analysis.
D. Correct. With labeled locations, supervised learning can predict transaction locations.
E. Incorrect. Reinforcement learning is not used for this type of predictive modeling.
F. Incorrect. Unsupervised methods cannot directly predict transaction labels like location.

Question 219
Your company's on-premises Apache Hadoop servers are approaching end-of-life, and IT has decided to migrate the cluster to Google Cloud Dataproc. A like-for-like migration of the cluster would require 50 TB of Google Persistent Disk per node. The CIO is concerned about the cost of using that much block storage. You want to minimize the storage cost of the migration. What should you do?

Options
A. Put the data into Google Cloud Storage.
B. Use preemptible virtual machines (VMs) for the Cloud Dataproc cluster.
C. Tune the Cloud Dataproc cluster so that there is just enough disk for all data.
D. Migrate some of the cold data into Google Cloud Storage, and keep only the hot data in Persistent Disk.
E. Compress all data and keep it in Persistent Disk.

Answer
A

Detailed explanation
A. Correct. Cloud Storage is cheaper than Persistent Disk and integrates directly with Dataproc for HDFS replacement.
B. Incorrect. Preemptible VMs reduce compute costs, not storage costs.
C. Incorrect. Reducing disk still relies heavily on Persistent Disk, not cost-efficient.
D. Incorrect. This helps somewhat but still leaves significant disk cost.
E. Incorrect. Compression reduces size but doesn't avoid Persistent Disk charges.

Question 220
You work for a car manufacturer and have set up a data pipeline using Google Cloud Pub/Sub to capture anomalous sensor events. You are using a push subscription in Cloud Pub/Sub that calls a custom HTTPS endpoint that you have created to take action of these anomalous events as they occur. Your custom HTTPS endpoint keeps getting an inordinate amount of duplicate messages. What is the most likely cause of these duplicate messages?

Options
A. The message body for the sensor event is too large.
B. Your custom endpoint has an out-of-date SSL certificate.
C. The Cloud Pub/Sub topic has too many messages published to it.
D. Your custom endpoint is not acknowledging messages within the acknowledgement deadline.
E. The push subscription is configured for batch mode delivery.

Answer
D

Detailed explanation
A. Incorrect. Message size issues would cause errors, not duplicates.
B. Incorrect. SSL problems would block delivery, not cause duplication.
C. Incorrect. High publish volume does not cause duplication; Pub/Sub scales automatically.
D. Correct. If the endpoint does not acknowledge messages on time, Pub/Sub retries delivery, leading to duplicates.
E. Incorrect. Batch mode affects delivery grouping, not duplication.

Question 221
You have Google Cloud Dataflow streaming pipeline running with a Google Cloud Pub/Sub subscription as the source. You need to make an update to the code that will make the new Cloud Dataflow pipeline incompatible with the current version. You do not want to lose any data when making this update. What should you do?

Options
A. Update the current pipeline and use the drain flag.
B. Update the current pipeline and provide the transform mapping JSON object.
C. Create a new pipeline that has the same Cloud Pub/Sub subscription and cancel the old pipeline.
D. Create a new pipeline that has a new Cloud Pub/Sub subscription and cancel the old pipeline.
E. Pause the subscription until the new pipeline is deployed.

Answer
A

Detailed explanation
A. Correct. Using the --drain flag allows the old pipeline to finish processing in-flight messages before shutting down, preventing data loss.
B. Incorrect. Transform mapping JSON applies to schema compatibility, not incompatible pipeline code changes.
C. Incorrect. Running two pipelines on the same subscription causes message acknowledgment conflicts and data loss.
D. Incorrect. Creating a new subscription would avoid conflicts but still risks losing messages during switchover.
E. Incorrect. Pausing subscriptions is not a feature of Pub/Sub and would not ensure smooth transition.

Question 222

Your company is running their first dynamic campaign, serving different offers by analyzing real-time data during the holiday season. The data scientists are collecting terabytes of data that rapidly grows every hour during their 30-day campaign. They are using Google Cloud Dataflow to preprocess the data and collect the feature (signals) data that is needed for the machine learning model in Google Cloud Bigtable. The team is observing suboptimal performance with reads and writes of their initial load of 10 TB of data. They want to improve this performance while minimizing cost. What should they do?

Options

A. Redefine the schema by evenly distributing reads and writes across the row space of the table.
B. The performance issue should be resolved over time as the size of the Bigtable cluster is increased.
C. Redesign the schema to use a single row key to identify values that need to be updated frequently in the cluster.
D. Redesign the schema to use row keys based on numeric IDs that increase sequentially per user viewing the offers.
E. Use Cloud Spanner instead of Bigtable for this workload.

Answer

A

Detailed explanation

A. Correct. Bigtable performance depends on row key design. Evenly distributing row keys avoids hotspotting and balances load across nodes.
B. Incorrect. Scaling nodes does not fix poor schema design and may just increase costs.
C. Incorrect. Using a single row key would centralize writes and worsen hotspotting.
D. Incorrect. Sequential numeric IDs lead to hotspotting on a small set of nodes.
E. Incorrect. Spanner is not optimal for high-throughput time-series/IoT-style workloads compared to Bigtable.

Question 223

Your software uses a simple JSON format for all messages. These messages are published to Google Cloud Pub/Sub, then processed with Google Cloud Dataflow to create a real-time dashboard for the CFO. During testing, you notice that some messages are missing in the dashboard. You check the logs, and all messages are being published to Cloud Pub/Sub successfully. What should you do next?

Options

A. Check the dashboard application to see if it is not displaying correctly.
B. Run a fixed dataset through the Cloud Dataflow pipeline and analyze the output.
C. Use Google Stackdriver Monitoring on Cloud Pub/Sub to find the missing messages.
D. Switch Cloud Dataflow to pull messages from Cloud Pub/Sub instead of Cloud Pub/Sub pushing messages to Cloud Dataflow.
E. Increase the Pub/Sub message retention time.

Answer

B

Detailed explanation

A. Incorrect. The dashboard could be an issue, but the root cause is likely in Dataflow message processing.
B. Correct. Running a fixed dataset through the pipeline isolates whether messages are being dropped or transformed incorrectly in Dataflow.
C. Incorrect. Pub/Sub is already verified to receive all messages, so monitoring won't reveal missing ones in the pipeline.
D. Incorrect. Dataflow only pulls messages from Pub/Sub; there is no push model for Dataflow subscriptions.
E. Incorrect. Retention time matters for late subscribers, but the problem is already within the pipeline processing.

Flowlogistic Case Study

Company Overview
Flowlogistic is a leading logistics and supply chain provider. They help businesses throughout the world manage their resources and transport them to their final destination. The company has grown rapidly, expanding their offerings to include rail, truck, aircraft, and oceanic shipping.

Company Background
The company started as a regional trucking company, and then expanded into other logistics market. Because they have not updated their infrastructure, managing and tracking orders and shipments has become a bottleneck. To improve operations, Flowlogistic developed proprietary technology for tracking shipments in real time at the parcel level. However, they are unable to deploy it because their technology stack, based on Apache Kafka, cannot support the processing volume. In addition, Flowlogistic wants to further analyze their orders and shipments to determine how best to deploy their resources.

Solution Concept
Flowlogistic wants to implement two concepts using the cloud:
☞ Use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads
☞ Perform analytics on all their orders and shipment logs, which contain both structured and unstructured data, to determine how best to deploy resources, which markets to expand info. They also want to use predictive analytics to learn earlier when a shipment will be delayed.

Existing Technical Environment
Flowlogistic architecture resides in a single data center:

☞ Databases
8 physical servers in 2 clusters

- SQL Server — user data, inventory, static data
  3 physical servers
- Cassandra — metadata, tracking messages
  10 Kafka servers — tracking message aggregation and batch insert

☞ Application servers — customer front end, middleware for order/customs
60 virtual machines across 20 physical servers

- Tomcat — Java services
- Nginx — static content
- Batch servers

☞ Storage appliances

- iSCSI for virtual machine (VM) hosts
- Fibre Channel storage area network (FC SAN) — SQL server storage
- Network-attached storage (NAS) image storage, logs, backups

☞ 10 Apache Hadoop / Spark servers

- Core Data Lake
- Data analysis workloads

☞ 20 miscellaneous servers

- Jenkins, monitoring, bastion hosts

Business Requirements

☞ Build a reliable and reproducible environment with scaled panty of production.
☞ Aggregate data in a centralized Data Lake for analysis
☞ Use historical data to perform predictive analytics on future shipments
☞ Accurately track every shipment worldwide using proprietary technology
☞ Improve business agility and speed of innovation through rapid provisioning of new resources
☞ Analyze and optimize architecture for performance in the cloud
☞ Migrate fully to the cloud if all other requirements are met

Technical Requirements

☞ Handle both streaming and batch data
☞ Migrate existing Hadoop workloads
☞ Ensure architecture is scalable and elastic to meet the changing demands of the company.
☞ Use managed services whenever possible
☞ Encrypt data flight and at rest
☞ Connect a VPN between the production data center and cloud environment

SEO Statement

We have grown so quickly that our inability to upgrade our infrastructure is really hampering further growth and efficiency. We are efficient at moving shipments around the world, but we are inefficient at moving data around. We need to organize our information so we can more easily understand where our customers are and what they are shipping.

CTO Statement

IT has never been a priority for us, so as our data has grown, we have not invested enough in our technology. I have a good staff to manage IT, but they are so busy managing our infrastructure that I cannot get them to do the things that really matter, such as organizing our data, building the analytics, and figuring out how to implement the CFO's tracking technology.

CFO Statement

Part of our competitive advantage is that we penalize ourselves for late shipments and deliveries. Knowing where our shipments are at all times has a direct correlation to our bottom line and profitability. Additionally, I don't want to commit capital to building out a server environment.

Question 224
Flowlogistic's management has determined that the current Apache Kafka servers cannot handle the data volume for their real-time inventory tracking system. You need to build a new system on Google Cloud Platform (GCP) that will feed the proprietary tracking software. The system must be able to ingest data from a variety of global sources, process and query in real-time, and store the data reliably. Which combination of GCP products should you choose?

Options
A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage
B. Cloud Pub/Sub, Cloud Dataflow, and Local SSD
C. Cloud Pub/Sub, Cloud SQL, and Cloud Storage
D. Cloud Load Balancing, Cloud Dataflow, and Cloud Storage
E. Cloud Spanner, Cloud Dataflow, and Cloud Pub/Sub

Answer
A

Detailed explanation
A. Correct. Cloud Pub/Sub can ingest global streaming data, Cloud Dataflow processes it in real time, and Cloud Storage provides durable, reliable storage.
B. Incorrect. Local SSD is not reliable for long-term storage of streaming data.
C. Incorrect. Cloud SQL is not designed for high-throughput streaming ingestion.
D. Incorrect. Load Balancing does not handle message ingestion or processing for this use case.
E. Incorrect. Spanner is for transactional workloads, not real-time message ingestion.

Question 225
Flowlogistic's CEO wants to gain rapid insight into their customer base so his sales team can be better informed in the field. This team is not very technical, so they've purchased a visualization tool to simplify the creation of BigQuery reports. However, they've been overwhelmed by all the data in the table, and are spending a lot of money on queries trying to find the data they need. You want to solve their problem in the most cost-effective way. What should you do?

Options
A. Export the data into a Google Sheet for virtualization.
B. Create an additional table with only the necessary columns.
C. Create a view on the table to present to the virtualization tool.
D. Create identity and access management (IAM) roles on the appropriate columns, so only they appear in a query.
E. Use Data Studio to filter columns automatically.

Answer
C

Detailed explanation
A. Incorrect. Google Sheets cannot handle large BigQuery datasets and would not scale.
B. Incorrect. Creating additional tables duplicates data and increases storage/maintenance costs.
C. Correct. Views allow you to present only the necessary columns without duplicating data, reducing query costs and simplifying usage for non-technical users.
D. Incorrect. IAM column-level security doesn’t reduce query complexity or cost for visualization users.
E. Incorrect. Data Studio still queries the full table if views are not used, leading to high costs.

Question 226
Flowlogistic is rolling out their real-time inventory tracking system. The tracking devices will all send package-tracking messages, which will now go to a single Google Cloud Pub/Sub topic instead of the Apache Kafka cluster. A subscriber application will then process the messages for real-time reporting and store them in Google BigQuery for historical analysis. You want to ensure the package data can be analyzed over time. Which approach should you take?

Options
A. Attach the timestamp on each message in the Cloud Pub/Sub subscriber application as they are received.
B. Attach the timestamp and Package ID on the outbound message from each publisher device as they are sent to Cloud Pub/Sub.
C. Use the NOW() function in BigQuery to record the event's time.
D. Use the automatically generated timestamp from Cloud Pub/Sub to order the data.
E. Use Cloud Logging export to add timestamps to messages before BigQuery load.

Answer
B

Detailed explanation
A. Incorrect. Adding timestamps at the subscriber level may lose accuracy due to network or processing delays.
B. Correct. Adding both timestamp and Package ID at the publisher device ensures the most accurate record of when events occurred and ties each event to its package.
C. Incorrect. NOW() in BigQuery only captures load/query time, not event occurrence.
D. Incorrect. Cloud Pub/Sub generated timestamps reflect message receipt, not the actual event at the source.
E. Incorrect. Logging export adds unnecessary complexity and still wouldn't capture precise event time at the source.

Question 227
Flowlogistic wants to implement two concepts using the cloud: use their proprietary technology in a real-time inventory-tracking system that indicates the location of their loads, and perform analytics on all their orders and shipment logs. They also want to use predictive analytics to learn earlier when a shipment will be delayed. Given their technical requirements, which Google Cloud approach best supports their solution concept?

Options
A. Use Cloud Pub/Sub and Dataflow for streaming ingestion, BigQuery for analytics, and AI Platform for predictive modeling.
B. Use only BigQuery for both real-time tracking and predictive analytics.
C. Use Cloud SQL for order storage and run predictive queries directly.
D. Use Cloud Storage buckets to hold both tracking and historical data, then run Data Studio dashboards.
E. Use Dataproc exclusively to manage both streaming and batch processing workloads.

Answer
A

Detailed explanation
A. Correct. Pub/Sub + Dataflow handle streaming tracking data, BigQuery provides scalable analytics, and AI Platform enables predictive analytics on historical data.
B. Incorrect. BigQuery is not built for direct ingestion of high-volume real-time tracking data.
C. Incorrect. Cloud SQL cannot scale to global real-time ingestion and predictive analytics at this scale.

D. Incorrect. Cloud Storage is only durable storage; it cannot provide real-time streaming or predictive ML directly.
E. Incorrect. Dataproc alone would require more infrastructure management and is not as cloud-native as managed services like Pub/Sub, Dataflow, and BigQuery.

MJTelco Case Study

Company Overview
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.

Their management and operations teams are situated all around the globe creating many-to-many relationships between data consumers and providers in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
☞ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
☞ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.

MJTelco will also use three separate operating environments — development/test, staging, and production — to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements
☞ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
☞ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
☞ Provide reliable and timely access to data for analysis from distributed research workers.
☞ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements

☞ Ensure secure and efficient transport and storage of telemetry data.

☞ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.

☞ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day.

☞ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in telemetry flows and in production learning cycles.

CEO Statement

Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement

Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement

The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.

Question 228
MJTelco's Google Cloud Dataflow pipeline is now ready to start receiving data from the 50,000 installations. You want to allow Cloud Dataflow to scale its compute power up as required. Which Cloud Dataflow pipeline configuration setting should you update?

Options
A. The zone
B. The number of workers
C. The disk size per worker
D. The maximum number of workers
E. The machine type

Answer
D

Detailed explanation
A. Incorrect. The zone determines location but not autoscaling behavior.
B. Incorrect. Number of workers sets a fixed count, preventing automatic scale-up.
C. Incorrect. Disk size per worker affects storage, not scaling.
D. Correct. The maximum number of workers allows Dataflow to autoscale up to that limit as demand increases.
E. Incorrect. Machine type changes per-worker capacity but does not enable autoscaling.

Question 229
You need to compose visualizations for operations teams with the following requirements:
☞ The report must include telemetry data from all 50,000 installations for the most recent 6 weeks (sampling once every minute).
☞ The report must not be more than 3 hours delayed from live data.
☞ The actionable report should only show suboptimal links.
☞ Most suboptimal links should be sorted to the top.
☞ Suboptimal links can be grouped and filtered by regional geography.
☞ User response time to load the report must be <5 seconds.
Which approach meets the requirements?

Options
A. Load the data into Google Sheets, use formulas to calculate a metric, and use filters/sorting to show only suboptimal links in a table.
B. Load the data into Google BigQuery tables, write Google Apps Script that queries the data, calculates the metric, and shows only suboptimal rows in a table in Google Sheets.
C. Load the data into Google Cloud Datastore tables, write a Google App Engine application that queries all rows, applies a function to derive the metric, and then renders results in a table using the Google Charts and Visualization API.
D. Load the data into Google BigQuery tables, write a Google Data Studio 360 report that connects to your data, calculates a metric, and then uses a filter expression to show only suboptimal rows in a table.
E. Export the data to Cloud Storage and generate visualizations using custom scripts.

Answer
D

Detailed explanation
A. Incorrect. Google Sheets cannot handle millions of rows from 50,000 installations.
B. Incorrect. Apps Script and Sheets won’t scale to this dataset size and query performance would be poor.

C. Incorrect. Datastore is not built for analytics or filtering at this scale.
D. Correct. BigQuery efficiently handles large data, and Data Studio connects natively, providing filtering and subsecond visualization for operations teams.
E. Incorrect. Custom scripts add unnecessary overhead and would not be as cost-effective or performant.

Question 230
You create a new report for your large team in Google Data Studio 360. The report uses Google BigQuery as its data source. It is company policy to ensure employees can view only the data associated with their region, so you create and populate a table for each region. You need to enforce the regional access policy to the data. Which two actions should you take? (Choose two.)

Options
A. Ensure all the tables are included in a global dataset.
B. Ensure each table is included in a dataset for a region.
C. Adjust the settings for each table to allow a related region-based security group view access.
D. Adjust the settings for each view to allow a related region-based security group view access.
E. Adjust the settings for each dataset to allow a related region-based security group view access.

Answer
B, E

Detailed explanation
A. Incorrect. A single global dataset would not support region-based access policies.
B. Correct. Assigning tables to region-specific datasets enables separation by geography.
C. Incorrect. Table-level access alone is not sufficient for clean IAM enforcement.
D. Incorrect. View-level access is not the best approach for dataset-wide policy enforcement.
E. Correct. Dataset-level IAM ensures regional groups can only access their corresponding dataset.

Question
MJTelco needs you to create a schema in Google Bigtable that will allow for the historical analysis of the last 2 years of records. Each record that comes in is sent every 15 minutes, and contains a unique identifier of the device and a data record. The most common query is for all the data for a given device for a given day. Which schema should you use?

Options
A. Rowkey: date#device_id Column data: data_point
B. Rowkey: date Column data: device_id, data_point
C. Rowkey: device_id Column data: date, data_point
D. Rowkey: data_point Column data: device_id, date
E. Rowkey: date#data_point Column data: device_id

Answer
A

Detailed explanation
A. Correct. Using date#device_id as the rowkey optimizes queries for retrieving all records of a device for a specific day.
B. Incorrect. Storing by date only would make per-device queries inefficient.
C. Incorrect. Storing by device_id only makes filtering by day inefficient.
D. Incorrect. Using data_point as rowkey provides no efficiency for common queries.
E. Incorrect. Combining date and data_point misses the device_id association needed for the query pattern.

Question 231
Your company uses a proprietary system to send inventory data every 6 hours to a data ingestion service in the cloud. Transmitted data includes a payload of several fields and the timestamp of the transmission. If there are any concerns about a transmission, the system re-transmits the data. How should you deduplicate the data most efficiency?

Options
A. Assign global unique identifiers (GUID) to each data entry.
B. Compute the hash value of each data entry, and compare it with all historical data.
C. Store each data entry as the primary key in a separate database and apply an index.
D. Maintain a database table to store the hash value and other metadata for each data entry.
E. None of the above.

Answer
A

Detailed explanation
A. Correct. Assigning GUIDs ensures each entry is uniquely identifiable, making deduplication straightforward and efficient.
B. Incorrect. Hashing every entry and comparing with all historical data is computationally expensive.
C. Incorrect. Storing every data entry as a primary key is inefficient for scale and can cause performance bottlenecks.
D. Incorrect. Maintaining a table of hashes adds unnecessary overhead compared to simply using GUIDs.
E. Incorrect. Option A already solves the problem efficiently.

Question 232
Your company has hired a new data scientist who wants to perform complicated analyses across very large datasets stored in Google Cloud Storage and in a Cassandra cluster on Google Compute Engine. The scientist primarily wants to create labelled data sets for machine learning projects, along with some visualization tasks. She reports that her laptop is not powerful enough to perform her tasks and it is slowing her down. You want to help her perform her tasks.

Options
A. Run a local version of Jupyter on the laptop.
B. Grant the user access to Google Cloud Shell.
C. Host a visualization tool on a VM on Google Compute Engine.
D. Deploy Google Cloud Datalab to a virtual machine (VM) on Google Compute Engine.
E. Use Data Studio to replace her workflow.

Answer
D

Detailed explanation
A. Incorrect. Running Jupyter locally doesn't solve the problem of insufficient laptop power.
B. Incorrect. Cloud Shell is lightweight and unsuitable for large-scale analytics.
C. Incorrect. Hosting only a visualization tool does not cover complex data preprocessing and ML dataset creation.
D. Correct. Cloud Datalab on a VM provides scalable compute and integrates with both Cloud Storage and Cassandra.
E. Incorrect. Data Studio is for visualization only, not advanced ML dataset creation.

Question 233
You are deploying 10,000 new Internet of Things devices to collect temperature data in your warehouses globally. You need to process, store and analyze these very large datasets in real time. What should you do?

Options
A. Send the data to Google Cloud Datastore and then export to BigQuery.
B. Send the data to Google Cloud Pub/Sub, stream Cloud Pub/Sub to Google Cloud Dataflow, and store the data in Google BigQuery.
C. Send the data to Cloud Storage and then spin up an Apache Hadoop cluster as needed in Google Cloud Dataproc whenever analysis is required.
D. Export logs in batch to Google Cloud Storage and then spin up a Google Cloud SQL instance, import the data from Cloud Storage, and run an analysis as needed.
E. Use Cloud Functions to write data directly into BigQuery.

Answer
B

Detailed explanation
A. Incorrect. Datastore is not designed for large-scale real-time IoT data streams.
B. Correct. Pub/Sub handles ingestion at scale, Dataflow processes in real time, and BigQuery stores/queries efficiently.
C. Incorrect. Hadoop on Dataproc is batch-based, not real-time.
D. Incorrect. Batch export/import is too slow for real-time IoT needs.
E. Incorrect. Cloud Functions cannot scale efficiently for 10,000 IoT streams continuously.

Question 234
You have spent a few days loading data from comma-separated values (CSV) files into the Google BigQuery table CLICK_STREAM. The column DT stores the epoch time of click events. For convenience, you chose a simple schema where every field is treated as the STRING type. Now, you want to compute web session durations of users who visit your site, and you want to change its data type to the TIMESTAMP. You want to minimize the migration effort without making future queries computationally expensive. What should you do?

Options
A. Delete the table CLICK_STREAM, and then re-create it such that the column DT is of the TIMESTAMP type. Reload the data.
B. Add a column TS of the TIMESTAMP type to the table CLICK_STREAM, and populate the numeric values from the column TS for each row. Reference the column TS instead of the column DT from now on.
C. Create a view CLICK_STREAM_V, where strings from the column DT are cast into TIMESTAMP values. Reference the view CLICK_STREAM_V instead of the table CLICK_STREAM from now on.
D. Add two columns to the table CLICK_STREAM: TS of the TIMESTAMP type and IS_NEW of the BOOLEAN type. Reload all data in append mode. For each appended row, set the value of IS_NEW to true. For future queries, reference the column TS instead of the column DT, with the WHERE clause ensuring that the value of IS_NEW must be true.
E. Construct a query to return every row of the table CLICK_STREAM, while using the built-in function to cast strings from the column DT into TIMESTAMP values. Run the query into a destination table NEW_CLICK_STREAM, in which the column TS is the TIMESTAMP type. Reference the table NEW_CLICK_STREAM instead of the table CLICK_STREAM from now on. In the future, new data is loaded into the table NEW_CLICK_STREAM.

Answer
E

Detailed explanation
A. Incorrect. Dropping and reloading is unnecessary and costly.
B. Incorrect. Adding a column requires manual population, and data duplication may occur.
C. Incorrect. Views cause runtime casting overhead on every query.
D. Incorrect. Reloading with extra columns adds unnecessary complexity.
E. Correct. Creating a new destination table with TIMESTAMP type ensures efficient queries and avoids expensive runtime casts.

Question 235
You want to use Google Stackdriver Logging to monitor Google BigQuery usage. You need an instant notification to be sent to your monitoring tool when new data is appended to a certain table using an insert job, but you do not want to receive notifications for other tables. What should you do?

Options
A. Make a call to the Stackdriver API to list all logs, and apply an advanced filter.
B. In the Stackdriver logging admin interface, enable a log sink export to BigQuery.
C. In the Stackdriver logging admin interface, enable a log sink export to Google Cloud Pub/Sub, and subscribe to the topic from your monitoring tool.
D. Using the Stackdriver API, create a project sink with advanced log filter to export to Pub/Sub, and subscribe to the topic from your monitoring tool.
E. Use BigQuery audit logs directly instead of Stackdriver.

Answer
D

Detailed explanation
A. Incorrect. Listing logs via API is not scalable for real-time notifications.
B. Incorrect. Exporting logs to BigQuery does not provide instant alerts.
C. Incorrect. Without advanced filters, all logs from all tables will be exported, not selective.
D. Correct. Creating a sink with advanced log filters allows targeting only specific insert jobs for a certain table, and Pub/Sub provides instant notification.
E. Incorrect. Audit logs alone don’t provide real-time integration with monitoring tools.

Question 236
You are working on a sensitive project involving private user data. You have set up a project on Google Cloud Platform to house your work internally. An external consultant is going to assist with coding a complex transformation in a Google Cloud Dataflow pipeline for your project. How should you maintain users' privacy?

Options
A. Grant the consultant the Viewer role on the project.
B. Grant the consultant the Cloud Dataflow Developer role on the project.
C. Create a service account and allow the consultant to log on with it.
D. Create an anonymized sample of the data for the consultant to work with in a different project.
E. Use IAM custom roles with limited access to hide user fields.

Answer
D

Detailed explanation
A. Incorrect. Viewer role does not allow pipeline development.
B. Incorrect. Developer role grants access but risks exposing sensitive data.
C. Incorrect. Sharing a service account directly poses security and privacy risks.
D. Correct. Providing anonymized sample data ensures privacy while enabling the consultant to build transformations safely.
E. Incorrect. IAM custom roles can limit access, but anonymizing data is the stronger privacy-preserving method.

Question 237
You are building a model to predict whether or not it will rain on a given day. You have thousands of input features and want to see if you can improve training speed by removing some features while having a minimum effect on model accuracy. What can you do?

Options
A. Eliminate features that are highly correlated to the output labels.
B. Combine highly co-dependent features into one representative feature.
C. Instead of feeding in each feature individually, average their values in batches of 3.
D. Remove the features that have null values for more than 50% of the training records.
E. Use dimensionality reduction (PCA) to automatically reduce features.

Answer
B

Detailed explanation
A. Incorrect. Removing correlated features with labels may eliminate important predictors.
B. Correct. Combining co-dependent features reduces redundancy, speeds training, and maintains accuracy.
C. Incorrect. Averaging features arbitrarily loses important signal.
D. Incorrect. Null-heavy features may be handled differently; blindly removing them may reduce accuracy.
E. Incorrect. PCA is useful, but the specific question points to feature engineering (combining co-dependent features).

Question 238
Your company is performing data preprocessing for a learning algorithm in Google Cloud Dataflow. Numerous data logs are being generated during this step, and the team wants to analyze them. Due to the dynamic nature of the campaign, the data is growing exponentially every hour. The data scientists have written the following code to read the data for a new key features in the logs. You want to improve the performance of this data read. What should you do?

Options
A. Specify the TableReference object in the code.
B. Use .fromQuery operation to read specific fields from the table.
C. Use both the Google BigQuery TableSchema and TableFieldSchema classes.
D. Call a transform that returns TableRow objects, where each element in the PCollection represents a single row in the table.
E. Use partitioned tables to reduce read size.

Answer
B

Detailed explanation
A. Incorrect. TableReference just points to a table, doesn't optimize reads.
B. Correct. Using .fromQuery reads only the required fields, reducing data size and improving performance.
C. Incorrect. Schema classes describe structure, but don't optimize performance.
D. Incorrect. Returning TableRow objects doesn't reduce read load.
E. Incorrect. Partitioning helps, but the immediate optimization is querying selective fields.

Question 239
Your company is streaming real-time sensor data from their factory floor into Bigtable and they have noticed extremely poor performance. How should the row key be redesigned to improve Bigtable performance on queries that populate real-time dashboards?

Options
A. Use a row key of the form <timestamp>.
B. Use a row key of the form <sensorid>.
C. Use a row key of the form <timestamp>#<sensorid>.
D. Use a row key of the form <sensorid>#<timestamp>.
E. Randomize keys fully.

Answer
D

Detailed explanation
A. Incorrect. Timestamp-only keys cause hotspotting (sequential writes).
B. Incorrect. Sensor-only keys don't allow time-based queries efficiently.
C. Incorrect. Timestamp-first also causes hotspotting issues.
D. Correct. Sensorid first distributes writes across sensors, timestamp appended maintains ordering for queries.
E. Incorrect. Randomizing breaks efficient querying.

Question 240
Your company's customer and order databases are often under heavy load. This makes performing analytics against them difficult without harming operations. The databases are in a MySQL cluster, with nightly backups taken using mysqldump. You want to perform analytics with minimal impact on operations. What should you do?

Options
A. Add a node to the MySQL cluster and build an OLAP cube there.
B. Use an ETL tool to load the data from MySQL into Google BigQuery.
C. Connect an on-premises Apache Hadoop cluster to MySQL and perform ETL.
D. Mount the backups to Google Cloud SQL, and then process the data using Google Cloud Dataproc.
E. Export MySQL data into Cloud Storage daily and run analysis directly from CSV.

Answer
B

Detailed explanation
A. Incorrect. Adding OLAP node still impacts the MySQL cluster under load.
B. Correct. ETL into BigQuery removes analytics load from production MySQL, providing scalability.
C. Incorrect. On-prem Hadoop is heavier, costlier, and slower than BigQuery.
D. Incorrect. Mounting backups and using Dataproc is unnecessarily complex.
E. Incorrect. Running analysis from CSVs is inefficient compared to structured BigQuery tables.

Question 241
What are two of the benefits of using denormalized data structures in BigQuery?

Options
A. Reduces the amount of data processed, reduces the amount of storage required
B. Increases query speed, makes queries simpler
C. Reduces the amount of storage required, increases query speed
D. Reduces the amount of data processed, increases query speed

Answer
B

Detailed explanation
A. Incorrect — Denormalization doesn’t typically reduce storage; duplication often increases it.
B. Correct — Denormalization simplifies queries (fewer joins) and speeds them up.
C. Incorrect — Query speed increases, but storage usually grows.
D. Incorrect — It can reduce data processed in some queries, but simplicity + speed is the clearer benefit.

Question 242
Which of these statements about exporting data from BigQuery is false?

Options
A. To export more than 1 GB of data, you need to put a wildcard in the destination filename.
B. The only supported export destination is Google Cloud Storage.
C. Data can only be exported in JSON or Avro format.
D. The only compression option available is GZIP.

Answer
C

Detailed explanation
A. Incorrect — True: large exports require wildcards.
B. Incorrect — True: exports only go to GCS.
C. Correct — False statement: supported formats include CSV, Avro, Parquet, ORC, not just JSON/Avro.
D. Incorrect — True: GZIP is the supported compression.

Question 243
What are all of the BigQuery operations that Google charges for?

Options
A. Storage, queries, and streaming inserts
B. Storage, queries, and loading data from a file
C. Storage, queries, and exporting data
D. Queries and streaming inserts

Answer
A

Detailed explanation
A. Correct — Billing is for storage, queries, and streaming inserts.
B. Incorrect — Loading data from a file is free.
C. Incorrect — Exporting data itself isn't billed (though egress costs may apply).
D. Incorrect — Storage is missing here.

Question 244
Which of the following is not possible using primitive roles?

Options
A. Give a user viewer access to BigQuery and owner access to Google Compute Engine instances.
B. Give UserA owner access and UserB editor access for all datasets in a project.
C. Give a user access to view all datasets in a project, but not run queries on them.
D. Give GroupA owner access and GroupB editor access for all datasets in a project.

Answer
C

Detailed explanation
A. Incorrect — Primitive roles can be assigned separately across services.
B. Incorrect — Primitive roles allow multiple assignments like owner and editor.
C. Correct — Not possible: primitive roles don't allow granular "view-only, no queries." That needs authorized views.
D. Incorrect — Primitive roles can be given to groups.

Question 245
Which of these statements about BigQuery caching is true?

Options
A. By default, a query's results are not cached.
B. BigQuery caches query results for 48 hours.
C. Query results are cached even if you specify a destination table.
D. There is no charge for a query that retrieves its results from cache.

Answer
D

Detailed explanation
A. Incorrect — By default results are cached.
B. Incorrect — Cache lasts 24 hours, not 48.
C. Incorrect — Destination tables disable caching.
D. Correct — Cached query results are free.

Question 246
Which of these sources can you not load data into BigQuery from?

Options
A. File upload
B. Google Drive
C. Google Cloud Storage
D. Google Cloud SQL

Answer
D

Detailed explanation
A. Incorrect — Files can be uploaded manually.
B. Incorrect — Drive is a supported source.
C. Incorrect — GCS is the primary load source.
D. Correct — Cloud SQL cannot be a direct load source.

Question 247
Which of the following statements about Legacy SQL and Standard SQL is not true?

Options
A. Standard SQL is the preferred query language for BigQuery.
B. If you write a query in Legacy SQL, it might generate an error if you try to run it with Standard SQL.
C. One difference between the two query languages is how you specify fully-qualified table names (i.e. table names that include their associated project name).
D. You need to set a query language for each dataset and the default is Standard SQL.

Answer
D

Detailed explanation
A. Incorrect — Standard SQL is indeed preferred.
B. Incorrect — Legacy queries can fail in Standard SQL.
C. Incorrect — Yes, naming conventions differ.
D. Correct — Query language is per query/job, not per dataset.

Question 248
How would you query specific partitions in a BigQuery table?

Options
A. Use the DAY column in the WHERE clause
B. Use the EXTRACT(DAY) clause
C. Use the __PARTITIONTIME pseudo-column in the WHERE clause
D. Use DATE BETWEEN in the WHERE clause

Answer
C

Detailed explanation
A. Incorrect — No default DAY column exists.
B. Incorrect — EXTRACT helps but doesn't prune partitions.
C. Correct — __PARTITIONTIME (or __PARTITIONDATE) is the official pseudo-column to filter partitions.
D. Incorrect — DATE BETWEEN works only if you query the actual column, not the internal partition pseudo-column.

Question 249
Which SQL keyword can be used to reduce the number of columns processed by BigQuery?

Options
A. BETWEEN
B. WHERE
C. SELECT
D. LIMIT

Answer
C

Detailed explanation
A. Incorrect — BETWEEN filters rows, not columns.
B. Incorrect — WHERE reduces rows, not columns.
C. Correct — SELECT lets you pick only needed columns, reducing data scanned.
D. Incorrect — LIMIT reduces rows returned, not columns processed.

Question 250
To give a user read permission for only the first three columns of a table, which access control method would you use?

Options
A. Primitive role
B. Predefined role
C. Authorized view
D. It's not possible to give access to only the first three columns of a table.

Answer
C

Detailed explanation
A. Incorrect — Primitive roles are project-level and coarse-grained.
B. Incorrect — Predefined roles still don't allow per-column access.
C. Correct — Authorized views let you expose only the first three columns and grant user

access to the view.
D. Incorrect — It **is** possible with authorized views.

Question 251
What are two methods that can be used to denormalize tables in BigQuery?

Options
A. 1) Split table into multiple tables; 2) Use a partitioned table
B. 1) Join tables into one table; 2) Use nested repeated fields
C. 1) Use a partitioned table; 2) Join tables into one table
D. 1) Use nested repeated fields; 2) Use a partitioned table

Answer
B

Detailed explanation
A. Incorrect — Splitting into multiple tables is actually normalizing, not denormalizing; partitions are for query optimization, not schema denormalization.
B. Correct — Denormalization means flattening data by joining into a single table and/or using nested repeated fields to avoid multiple joins.
C. Incorrect — Partitions optimize query cost/time, but are not a denormalization technique.
D. Incorrect — Nested repeated fields is correct, but partitions are not about denormalization.

Question 252
Which of these is not a supported method of putting data into a partitioned table?

Options
A. If you have existing data in a separate file for each day, then create a partitioned table and upload each file into the appropriate partition.
B. Run a query to get the records for a specific day from an existing table and for the destination table, specify a partitioned table ending with the day in the format "$YYYYMMDD".
C. Create a partitioned table and stream new records to it every day.
D. Use ORDER BY to put a table's rows into chronological order and then change the table's type to "Partitioned".

Answer
D

Detailed explanation
A. Incorrect — Loading per-day files into partitions is fully supported.
B. Incorrect — Query results can be written into specific partitions using the $YYYYMMDD suffix.
C. Incorrect — Streaming into partitioned tables is supported.
D. Correct — You cannot change a table’s type to partitioned just by ordering rows; partitioning must be defined at creation time.

Question 253
Which of these operations can you perform from the BigQuery Web UI?

Options
A. Upload a file in SQL format.
B. Load data with nested and repeated fields.
C. Upload a 20 MB file.
D. Upload multiple files using a wildcard.

Answer
B

Detailed explanation
A. Incorrect — You don't upload SQL files; SQL is written in the query editor.
B. Correct — You can load JSON or Avro files with nested and repeated fields via the Web UI.
C. Incorrect — You can upload >20 MB files, but the real feature focus is nested/repeated field support.
D. Incorrect — Wildcards are used in GCS paths, not for manual UI uploads.

Question 254
Which methods can be used to reduce the number of rows processed by BigQuery?

Options
A. Splitting tables into multiple tables; putting data in partitions
B. Splitting tables into multiple tables; putting data in partitions; using the LIMIT clause
C. Putting data in partitions; using the LIMIT clause
D. Splitting tables into multiple tables; using the LIMIT clause

Answer
A

Detailed explanation
A. Correct — Partitioning and splitting data into multiple smaller tables allows pruning, reducing rows scanned.
B. Incorrect — LIMIT only reduces rows returned, not scanned; still billed for all processed.
C. Incorrect — LIMIT doesn't cut scan cost, so this isn't correct.
D. Incorrect — Same issue, LIMIT doesn't reduce processing cost.

Question 255
Why do you need to split a machine learning dataset into training data and test data?

Options
A. So you can try two different sets of features
B. To make sure your model is generalized for more than just the training data
C. To allow you to create unit tests in your code
D. So you can use one dataset for a wide model and one for a deep model

Answer
B

Detailed explanation
A. Incorrect — Feature experimentation isn't the main reason for split.
B. Correct — The test set evaluates generalization to unseen data, preventing overfitting.
C. Incorrect — Unit tests in code are unrelated to dataset splitting.
D. Incorrect — Wide/deep models are an architecture choice, not why we split data.

Question 256
Which of these numbers are adjusted by a neural network as it learns from a training dataset (select 2 answers)?

Options
A. Weights
B. Biases
C. Continuous features
D. Input values

Answer
A and B

Detailed explanation
A. Correct — Weights are tuned during training to minimize loss.
B. Correct — Biases are also parameters adjusted during training.
C. Incorrect — Continuous features are inputs, not trainable parameters.
D. Incorrect — Input values are fixed from the dataset, not changed by the network.

Question 257
The CUSTOM tier for Cloud Machine Learning Engine allows you to specify the number of which types of cluster nodes?

Options
A. Workers
B. Masters, workers, and parameter servers
C. Workers and parameter servers
D. Parameter servers

Answer
C

Detailed explanation
A. Incorrect — You can specify more than just workers.
B. Incorrect — Master node count is fixed; not customizable.
C. Correct — CUSTOM tier allows configuring workers and parameter servers.
D. Incorrect — Only parameter servers alone is incomplete.

Question 258
Which software libraries are supported by Cloud Machine Learning Engine?

Options
A. Theano and TensorFlow
B. Theano and Torch
C. TensorFlow
D. TensorFlow and Torch

Answer
C

Detailed explanation
A. Incorrect — Theano isn't supported.
B. Incorrect — Neither Theano nor Torch is supported.
C. Correct — Cloud ML Engine (now AI Platform) officially supports TensorFlow.
D. Incorrect — Torch is not supported.

Question 259
Which TensorFlow function can you use to configure a categorical column if you don't know all of the possible values for that column?

Options
A. categorical_column_with_vocabulary_list
B. categorical_column_with_hash_bucket
C. categorical_column_with_unknown_values
D. sparse_column_with_keys

Answer
B

Detailed explanation
A. Incorrect — Vocabulary list requires knowing all possible values.
B. Correct — Hash bucket maps unknown/unlimited categories into buckets without needing the full list.
C. Incorrect — There is no categorical_column_with_unknown_values function.
D. Incorrect — sparse_column_with_keys is deprecated and requires known keys.

Question 260
Which of the following statements about the Wide & Deep Learning model are true? (Select 2 answers.)

Options
A. The wide model is used for memorization, while the deep model is used for generalization.
B. A good use for the wide and deep model is a recommender system.
C. The wide model is used for generalization, while the deep model is used for memorization.
D. A good use for the wide and deep model is a small-scale linear regression problem.

Answer
A and B

Detailed explanation
A. Correct — Wide model = memorization (feature interactions), Deep model = generalization.
B. Correct — A classic use case is recommender systems (memorizing + generalizing user-item interactions).
C. Incorrect — This flips the roles; wrong.
D. Incorrect — Small-scale linear regression doesn’t need wide + deep hybrid.

Question 261
To run a TensorFlow training job on your own computer using Cloud Machine Learning Engine, what would your command start with?

Options
A. gcloud ml-engine local train
B. gcloud ml-engine jobs submit training
C. gcloud ml-engine jobs submit training local
D. You can't run a TensorFlow program on your own computer using Cloud ML Engine .

Answer
A

Detailed explanation
A. Correct — gcloud ml-engine local train is the correct way to run locally.
B. Incorrect — This is used for submitting jobs to Cloud ML Engine, not local runs.
C. Incorrect — Syntax is invalid.
D. Incorrect — You can run jobs locally.

Question 262
If you want to create a machine learning model that predicts the price of a particular stock based on its recent price history, what type of estimator should you use?

Options
A. Unsupervised learning
B. Regressor
C. Classifier
D. Clustering estimator

Answer
B

Detailed explanation
A. Incorrect — This is a supervised problem, not unsupervised.
B. Correct — Price is a continuous value, requiring regression.
C. Incorrect — Classification is for discrete labels, not continuous values.
D. Incorrect — Clustering is unsupervised grouping, not prediction.

Question 263
Suppose you have a dataset of images that are each labeled as to whether or not they contain a human face. To create a neural network that recognizes human faces in images using this labeled dataset, what approach would likely be the most effective?

Options
A. Use K-means Clustering to detect faces in the pixels.
B. Use feature engineering to add features for eyes, noses, and mouths to the input data.
C. Use deep learning by creating a neural network with multiple hidden layers to automatically detect features of faces.
D. Build a neural network with an input layer of pixels, a hidden layer, and an output layer with two categories.

Answer
C

Detailed explanation
A. Incorrect — K-means is unsupervised and not suited for labeled classification.
B. Incorrect — Manual feature engineering for faces is impractical; CNNs handle it automatically.
C. Correct — Deep learning (multi-layer networks) learns hierarchical features like eyes/nose automatically.
D. Incorrect — A shallow network won't capture complex facial features effectively.

Question 264
What are two of the characteristics of using online prediction rather than batch prediction?

Options
A. It is optimized to handle a high volume of data instances in a job and to run more complex models.
B. Predictions are returned in the response message.
C. Predictions are written to output files in a Cloud Storage location that you specify.
D. It is optimized to minimize the latency of serving predictions.

Answer
B and D

Detailed explanation
A. Incorrect — That describes batch prediction.
B. Correct — Online prediction returns results immediately in the response.
C. Incorrect — Batch predictions write to Cloud Storage, not online.
D. Correct — Online prediction is optimized for low-latency real-time predictions.

Question 265
Which of these are examples of a value in a sparse vector? (Select 2 answers.)

Options
A. [0, 5, 0, 0, 0, 0]
B. [0, 0, 0, 1, 0, 0, 1]
C. [0, 1]
D. [1, 0, 0, 0, 0, 0, 0]

Answer
C and D

Detailed explanation
A. Incorrect — Sparse vectors are binary-like with mostly 0s and small indicator values. [0, 5, 0...] isn't sparse in the ML sense.
B. Incorrect — This has multiple 1s, but typically a sparse vector representation is used differently; here it doesn't match the intended definition.
C. Correct — [0,1] has mostly zeros with few non-zero values, classic sparse.
D. Correct — [1,0,0,...] is sparse with only one non-zero entry.

Question 266
How can you get a neural network to learn about relationships between categories in a categorical feature?

Options
A. Create a multi-hot column
B. Create a one-hot column
C. Create a hash bucket
D. Create an embedding column

Answer
D

Detailed explanation
A. Incorrect — Multi-hot columns represent multiple categories but don't capture relationships.
B. Incorrect — One-hot encodings are sparse and don't show similarity.
C. Incorrect — Hash buckets reduce dimensionality but don't model relationships.
D. Correct — Embeddings map categories to dense vectors that capture relationships.

Question 267
If a dataset contains rows with individual people and columns for year of birth, country, and income, how many of the columns are continuous and how many are categorical?

Options
A. 1 continuous and 2 categorical
B. 3 categorical
C. 3 continuous
D. 2 continuous and 1 categorical

Answer
A

Detailed explanation
A. Correct — Income is continuous; year of birth and country are categorical.
B. Incorrect — Not all are categorical.
C. Incorrect — Year of birth and country are not continuous.
D. Incorrect — Only income is continuous, not two.

Question 268
Which of the following are examples of hyperparameters? (Select 2 answers.)

Options
A. Number of hidden layers
B. Number of nodes in each hidden layer
C. Biases
D. Weights

Answer
A and B

Detailed explanation
A. Correct — Architecture settings are hyperparameters.
B. Correct — Layer node counts are also hyperparameters.
C. Incorrect — Biases are learned parameters, not hyperparameters.
D. Incorrect — Weights are adjusted during training.

Question 269
Which of the following are feature engineering techniques? (Select 2 answers)

Options
A. Hidden feature layers
B. Feature prioritization
C. Crossed feature columns
D. Bucketization of a continuous feature

Answer
C and D

Detailed explanation
A. Incorrect — Hidden layers are part of model architecture, not feature engineering.
B. Incorrect — Prioritization isn't a formal feature engineering technique.
C. Correct — Crossed feature columns capture feature interactions.
D. Correct — Bucketizing continuous values into ranges is a common feature engineering method.

Question 270
You want to use a BigQuery table as a data sink. In which writing mode(s) can you use BigQuery as a sink?

Options
A. Both batch and streaming
B. BigQuery cannot be used as a sink
C. Only batch
D. Only streaming

Answer
A

Detailed explanation
A. Correct — BigQuery supports both batch and streaming ingestion.
B. Incorrect — BigQuery can absolutely be used as a sink.
C. Incorrect — Streaming is supported too.
D. Incorrect — Batch mode is also supported.

Question 271
You have a job that you want to cancel. It is a streaming pipeline, and you want to ensure that any data that is in-flight is processed and written to the output. Which of the following commands can you use on the Dataflow monitoring console to stop the pipeline job?

Options
A. Cancel
B. Drain
C. Stop
D. Finish

Answer
B

Detailed explanation
A. Incorrect — Cancel terminates immediately, possibly losing in-flight data.
B. Correct — Drain lets in-flight data finish before shutting down.
C. Incorrect — Stop isn't a valid Dataflow console action.
D. Incorrect — Finish is not the right terminology.

Question 272
When running a pipeline that has a BigQuery source, on your local machine, you continue to get permission denied errors. What could be the reason for that?

Options
A. Your gcloud does not have access to the BigQuery resources
B. BigQuery cannot be accessed from local machines
C. You are missing gcloud on your machine
D. Pipelines cannot be run locally

Answer
A

Detailed explanation
A. Correct — Permission denied is due to IAM issues with gcloud credentials.
B. Incorrect — BigQuery can be accessed locally with proper credentials.
C. Incorrect — gcloud missing would cause command not found, not permission errors.
D. Incorrect — Pipelines can be run locally.

Question 273
What Dataflow concept determines when a Window's contents should be output based on certain criteria being met?

Options
A. Sessions
B. OutputCriteria
C. Windows
D. Triggers

Answer
D

Detailed explanation
A. Incorrect — Sessions define window types.
B. Incorrect — No concept called OutputCriteria.
C. Incorrect — Windows define grouping, not when to fire.
D. Correct — Triggers define when to emit window results.

Question 274
Which of the following is NOT one of the three main types of triggers that Dataflow supports?

Options
A. Trigger based on element size in bytes
B. Trigger that is a combination of other triggers
C. Trigger based on element count
D. Trigger based on time

Answer
A

Detailed explanation
A. Correct — Dataflow doesn’t support triggers on element size in bytes.
B. Incorrect — Composite triggers are supported.
C. Incorrect — Count-based triggers are supported.
D. Incorrect — Time-based triggers are supported.

Question 275
Which Java SDK class can you use to run your Dataflow programs locally?

Options
A. LocalRunner
B. DirectPipelineRunner
C. MachineRunner
D. LocalPipelineRunner

Answer
B

Detailed explanation
A. Incorrect — No class called LocalRunner.
B. Correct — DirectPipelineRunner runs pipelines locally for testing.
C. Incorrect — No class called MachineRunner.
D. Incorrect — LocalPipelineRunner is deprecated.

Question 276
The Dataflow SDKs have been recently transitioned into which Apache service?

Options
A. Apache Spark
B. Apache Hadoop
C. Apache Kafka
D. Apache Beam

Answer
D

Detailed explanation
A. Incorrect — Spark is another framework.
B. Incorrect — Hadoop is a different ecosystem.
C. Incorrect — Kafka is for messaging/streaming.
D. Correct — Dataflow SDKs are part of Apache Beam.

Question 277
The _____ for Cloud Bigtable makes it possible to use Cloud Bigtable in a Cloud Dataflow pipeline.

Options
A. Cloud Dataflow connector
B. DataFlow SDK
C. BiqQuery API
D. BigQuery Data Transfer Service

Answer
A

Detailed explanation
A. Correct — Cloud Dataflow connector enables Dataflow to connect with Bigtable.
B. Incorrect — SDK alone doesn't connect to Bigtable.
C. Incorrect — BigQuery API is unrelated.
D. Incorrect — Data Transfer Service is unrelated to Bigtable.

Question 278
Does Dataflow process batch data pipelines or streaming data pipelines?

Options
A. Only Batch Data Pipelines
B. Both Batch and Streaming Data Pipelines
C. Only Streaming Data Pipelines
D. None of the above

Answer
B

Detailed explanation
A. Incorrect — Not only batch.
B. Correct — Dataflow supports both batch and streaming.
C. Incorrect — It's not only streaming.
D. Incorrect — Not "none."

Question 279
You are planning to use Google's Dataflow SDK to analyze customer data such as displayed below. Your project requirement is to extract only the customer name from the data source and then write to an output PCollection.

Tom,555 X street -

Tim,553 Y street -

Sam,111 Z street -

Which operation is best suited for the above data processing requirement?

Options
A. ParDo
B. Sink API
C. Source API
D. Data extraction

Answer
A

Detailed explanation
A. Correct — ParDo is the transform for element-wise extraction, like parsing out names.
B. Incorrect — Sink API writes output, doesn't transform.
C. Incorrect — Source API reads data, not parse/extract.
D. Incorrect — Not a defined operation in Beam.

Question 280
Which Cloud Dataflow / Beam feature should you use to aggregate data in an unbounded data source every hour based on the time when the data entered the pipeline?

Options
A. An hourly watermark
B. An event time trigger
C. The with Allowed Lateness method
D. A processing time trigger

Answer
D

Detailed explanation
A. Incorrect — Watermarks mark progress, not trigger output.
B. Incorrect — Event time triggers are based on event timestamps, not ingestion/processing time.
C. Incorrect — Allowed Lateness is for handling late data, not periodic aggregation.
D. Correct — Processing time trigger fires based on when data is processed/entered pipeline, perfect for hourly aggregation by arrival time.

Question 281
Which of the following is NOT true about Dataflow pipelines?

Options
A. Dataflow pipelines are tied to Dataflow, and cannot be run on any other runner
B. Dataflow pipelines can consume data from other Google Cloud services
C. Dataflow pipelines can be programmed in Java
D. Dataflow pipelines use a unified programming model, so can work both with streaming and batch data sources

Answer
A

Detailed explanation
A. Correct — This is not true. Dataflow pipelines are based on Apache Beam, which supports multiple runners (e.g., Spark, Flink).
B. Incorrect — They can indeed consume from services like Pub/Sub, BigQuery, etc.
C. Incorrect — Java SDK is fully supported.
D. Incorrect — The unified model allows batch and streaming in the same API.

Question 282
You are developing a software application using Google's Dataflow SDK, and want to use conditional, for loops and other complex programming structures to create a branching pipeline. Which component will be used for the data processing operation?

Options
A. PCollection
B. Transform
C. Pipeline
D. Sink API

Answer
B

Detailed explanation
A. Incorrect — PCollection is a data container, not the processor.
B. Correct — Transforms define the processing logic applied to data.
C. Incorrect — Pipeline is the container for the whole workflow, not the operation.
D. Incorrect — Sink APIs are for writing output, not defining operations.

Question 283
Which of the following IAM roles does your Compute Engine account require to be able to run pipeline jobs?

Options
A. dataflow.worker
B. dataflow.compute
C. dataflow.developer
D. dataflow.viewer

Answer
A

Detailed explanation
A. Correct — The dataflow.worker role grants permissions to run Dataflow pipeline jobs.
B. Incorrect — No role called dataflow.compute.
C. Incorrect — Developer allows creating jobs but not running worker VMs.
D. Incorrect — Viewer is read-only.

Question 284
Which of the following is not true about Dataflow pipelines?

Options
A. Pipelines are a set of operations
B. Pipelines represent a data processing job
C. Pipelines represent a directed graph of steps
D. Pipelines can share data between instances

Answer
D

Detailed explanation
A. Incorrect — They are indeed a set of operations.
B. Incorrect — A pipeline represents the job.
C. Incorrect — Pipelines are DAGs of transformations.
D. Correct — Pipelines don't directly share data between separate instances.

Question 285
By default, which of the following windowing behavior does Dataflow apply to unbounded data sets?

Options
A. Windows at every 100 MB of data
B. Single, Global Window
C. Windows at every 1 minute
D. Windows at every 10 minutes

Answer
B

Detailed explanation
A. Incorrect — Not based on MB size.
B. Correct — Default is a global window, meaning no subdivision until explicitly defined.
C. Incorrect — One-minute windows are not the default.
D. Incorrect — Ten-minute windows are also not default.

Question 286
Which of the following job types are supported by Cloud Dataproc (select 3 answers)?

Options
A. Hive
B. Pig
C. YARN
D. Spark

Answer
A, B, D

Detailed explanation
A. Correct — Hive jobs are supported.
B. Correct — Pig jobs are supported.
C. Incorrect — YARN manages resources but isn't a job type.
D. Correct — Spark jobs are supported.

Question 287
What are the minimum permissions needed for a service account used with Google Dataproc?

Options
A. Execute to Google Cloud Storage; write to Google Cloud Logging
B. Write to Google Cloud Storage; read to Google Cloud Logging
C. Execute to Google Cloud Storage; execute to Google Cloud Logging
D. Read and write to Google Cloud Storage; write to Google Cloud Logging

Answer
D

Detailed explanation
A. Incorrect — Execute is not relevant; needs read/write.
B. Incorrect — Needs write, not just read, to Logging.
C. Incorrect — Same issue, execute is not a correct permission type.
D. Correct — Dataproc jobs need full GCS access plus ability to write logs.

Question 288
Which role must be assigned to a service account used by the virtual machines in a Dataproc cluster so they can execute jobs?

Options
A. Dataproc Worker
B. Dataproc Viewer
C. Dataproc Runner
D. Dataproc Editor

Answer
A

Detailed explanation
A. Correct — The Dataproc Worker role is needed for VM instances in a cluster to run jobs.
B. Incorrect — Viewer is read-only.
C. Incorrect — No standard role called Runner.
D. Incorrect — Editor is too broad.

Question 289
When creating a new Cloud Dataproc cluster with the projects.regions.clusters.create operation, these four values are required: project, region, name, and ____.

Options
A. zone
B. node
C. label
D. type

Answer
A

Detailed explanation
A. Correct — Zone specifies where the cluster runs within the region.
B. Incorrect — Nodes are configured but not a required argument for creation API.
C. Incorrect — Labels are optional.
D. Incorrect — Type is not a required argument.

Question 290
Which Google Cloud Platform service is an alternative to Hadoop with Hive?

Options
A. Cloud Dataflow
B. Cloud Bigtable
C. BigQuery
D. Cloud Datastore

Answer
C

Detailed explanation
A. Incorrect — Dataflow is for stream/batch processing.
B. Incorrect — Bigtable is NoSQL for time series/large key-value.
C. Correct — BigQuery replaces Hive-like analytical workloads with SQL.
D. Incorrect — Datastore is a NoSQL transactional DB.

Question 291
Which of these rules apply when you add preemptible workers to a Dataproc cluster (select 2 answers)?

Options
A. Preemptible workers cannot use persistent disk.
B. Preemptible workers cannot store data.
C. If a preemptible worker is reclaimed, then a replacement worker must be added manually.
D. A Dataproc cluster cannot have only preemptible workers.

Answer
B and D

Detailed explanation
A. Incorrect — They can attach persistent disks.
B. Correct — They are meant for transient compute, not reliable storage.
C. Incorrect — Replacements can be automatically added if configured.
D. Correct — At least some non-preemptible workers are required.

Question 292
When using Cloud Dataproc clusters, you can access the YARN web interface by configuring a browser to connect through a ____ proxy.

Options
A. HTTPS
B. VPN
C. SOCKS
D. HTTP

Answer
C

Detailed explanation
A. Incorrect — HTTPS alone won't connect through to YARN web UI.
B. Incorrect — VPN not required for this access.
C. Correct — SOCKS proxy is the supported method.
D. Incorrect — HTTP proxy is not the method used here.

Question 293
Cloud Dataproc is a managed Apache Hadoop and Apache _____ service.

Options
A. Blaze
B. Spark
C. Fire
D. Ignite

Answer
B

Detailed explanation
A. Incorrect — Blaze is not part of Dataproc.
B. Correct — Dataproc manages Hadoop and Spark.
C. Incorrect — Fire is not relevant.
D. Incorrect — Ignite is unrelated here.

Question 294
Which action can a Cloud Dataproc Viewer perform?

Options
A. Submit a job.
B. Create a cluster.
C. Delete a cluster.
D. List the jobs.

Answer
D

Detailed explanation
A. Incorrect — Viewer cannot submit jobs.
B. Incorrect — Viewer cannot create clusters.
C. Incorrect — Viewer cannot delete clusters.
D. Correct — Viewer can only list resources and view jobs.

Question 295
Dataproc clusters contain many configuration files. To update these files, you will need to use the --properties option. The format for the option is: file_prefix:property=_____.

Options
A. details
B. value
C. null
D. id

Answer
B

Detailed explanation
A. Incorrect — Not the correct format.
B. Correct — You set properties as prefix:property=value.
C. Incorrect — null is not used here.
D. Incorrect — id is not the right keyword.

Question 296
Scaling a Cloud Dataproc cluster typically involves ____.

Options
A. increasing or decreasing the number of worker nodes
B. increasing or decreasing the number of master nodes
C. moving memory to run more applications on a single node
D. deleting applications from unused nodes periodically

Answer
A

Detailed explanation
A. Correct — Scaling is usually done by changing worker node count.
B. Incorrect — Master nodes are usually fixed (1 or 3).
C. Incorrect — Not how scaling works.
D. Incorrect — Not part of scaling.

Question 297
Cloud Dataproc charges you only for what you really use with _____ billing.

Options
A. month-by-month
B. minute-by-minute
C. week-by-week
D. hour-by-hour

Answer
B

Detailed explanation
A. Incorrect — Monthly billing not applicable.
B. Correct — Dataproc bills per-minute usage.
C. Incorrect — Weekly billing not available.
D. Incorrect — Hourly billing is not the unit used.

Question 298
The YARN ResourceManager and the HDFS NameNode interfaces are available on a Cloud Dataproc cluster ____.

Options
A. application node
B. conditional node
C. master node
D. worker node

Answer
C

Detailed explanation
A. Incorrect — No concept of application node here.
B. Incorrect — No concept of conditional node.
C. Correct — Master node hosts YARN ResourceManager and HDFS NameNode.
D. Incorrect — Worker nodes run tasks, not master services.

Question 299
Which of these is NOT a way to customize the software on Dataproc cluster instances?

Options
A. Set initialization actions
B. Modify configuration files using cluster properties
C. Configure the cluster using Cloud Deployment Manager
D. Log into the master node and make changes from there

Answer
C

Detailed explanation
A. Incorrect — Initialization actions are supported.
B. Incorrect — Cluster properties can modify configs.
C. Correct — Deployment Manager is not a method for Dataproc customization.
D. Incorrect — Logging into master node to tweak configs is possible (though not recommended).

Question 300
In order to securely transfer web traffic data from your computer's web browser to the Cloud Dataproc cluster you should use a(n) _____.

Options
A. VPN connection
B. Special browser
C. SSH tunnel
D. FTP connection

Answer
C

Detailed explanation
A. Incorrect — VPN is not required for this use case.
B. Incorrect — No special browser is needed.
C. Correct — SSH tunnels are the secure method to connect browser traffic to cluster UIs.
D. Incorrect — FTP is unrelated.

Question 301
All Google Cloud Bigtable client requests go through a front-end server _____ they are sent to a Cloud Bigtable node.

Options
A. before
B. after
C. only if
D. once

Answer
A

Detailed explanation
A. Correct. Client requests first hit a front-end server (the Bigtable routing layer) before being forwarded to the correct Bigtable node.
B. Incorrect. Requests don't go after the node; the routing is before.
C. Incorrect. Requests always go through the front-end server, not conditionally.
D. Incorrect. "Once" does not describe the actual mechanism.

Question 302
What is the general recommendation when designing your row keys for a Cloud Bigtable schema?

Options
A. Include multiple time series values within the row key
B. Keep the row keep as an 8 bit integer
C. Keep your row key reasonably short
D. Keep your row key as long as the field permits

Answer
C

Detailed explanation
A. Incorrect. Mixing multiple values in row keys creates hotspotting and complexity.
B. Incorrect. Restricting to 8-bit integer is too limiting.
C. Correct. Short row keys reduce storage, memory overhead, and lookup time.
D. Incorrect. Very long row keys reduce performance and are not recommended.

Question 303
Which of the following statements is NOT true regarding Bigtable access roles?

Options
A. Using IAM roles, you cannot give a user access to only one table in a project, rather than all tables in a project.
B. To give a user access to only one table in a project, grant the user the Bigtable Editor role for that table.
C. You can configure access control only at the project level.
D. To give a user access to only one table in a project, you must configure access through your application.

Answer
B

Detailed explanation
A. Correct statement. IAM roles apply at the instance/project level, not per-table.
B. Incorrect statement (answer). There is no per-table IAM role like "Bigtable Editor" specific to a single table.
C. Correct statement. IAM controls are at the project/instance level.
D. Correct statement. Application-level logic is needed for per-table access.

Question 304
For the best possible performance, what is the recommended zone for your Compute Engine instance and Cloud Bigtable instance?

Options
A. Have the Compute Engine instance in the furthest zone from the Cloud Bigtable instance.
B. Have both the Compute Engine instance and the Cloud Bigtable instance to be in different zones.
C. Have both the Compute Engine instance and the Cloud Bigtable instance to be in the same zone.
D. Have the Cloud Bigtable instance to be in the same zone as all of the consumers of your data.

Answer
C

Detailed explanation
A. Incorrect. Distance increases latency.
B. Incorrect. Different zones = network latency.
C. Correct. Same zone ensures lowest possible latency.
D. Incorrect. Bigtable is not meant to be spread across zones for different consumers.

Question 305
Which row keys are likely to cause a disproportionate number of reads and/or writes on a particular node in a Bigtable cluster (select 2 answers)?

Options
A. A sequential numeric ID
B. A timestamp followed by a stock symbol
C. A non-sequential numeric ID
D. A stock symbol followed by a timestamp

Answer
A, B

Detailed explanation
A. Correct. Sequential IDs hotspot a single node.
B. Correct. Timestamp-first ordering clusters new data together, causing hotspots.
C. Incorrect. Non-sequential IDs spread load evenly.
D. Incorrect. Symbol first spreads data by symbol key, then by time, which is more balanced.

Question 306
When a Cloud Bigtable node fails, ____ is lost.

Options
A. all data
B. no data
C. the last transaction
D. the time dimension

Answer
B

Detailed explanation
A. Incorrect. Data is replicated and persistent.
B. Correct. No data is lost since Bigtable uses replication and persistent storage.
C. Incorrect. Transactions are not lost.
D. Incorrect. “Time dimension” is irrelevant.

Question 307
Which is not a valid reason for poor Cloud Bigtable performance?

Options
A. The workload isn't appropriate for Cloud Bigtable.
B. The table's schema is not designed correctly.
C. The Cloud Bigtable cluster has too many nodes.
D. There are issues with the network connection.

Answer
C

Detailed explanation
A. Correct. Workload mismatch can hurt performance.
B. Correct. Poor schema design is a common issue.
C. Incorrect statement (answer). Having too many nodes does not reduce performance.
D. Correct. Network issues affect performance.

Question 308
Which is the preferred method to use to avoid hotspotting in time series data in Bigtable?

Options
A. Field promotion
B. Randomization
C. Salting
D. Hashing

Answer
A

Detailed explanation
A. Correct. Field promotion (putting high-cardinality field like sensor ID before timestamp) avoids hotspots.
B. Incorrect. Randomization isn’t structured and makes querying harder.
C. Incorrect. Salting adds prefix noise but increases complexity.
D. Incorrect. Hashing can spread load but makes querying by range harder.

Question 309
When you design a Google Cloud Bigtable schema it is recommended that you _____.

Options
A. Avoid schema designs that are based on NoSQL concepts
B. Create schema designs that are based on a relational database design
C. Avoid schema designs that require atomicity across rows
D. Create schema designs that require atomicity across rows

Answer
C

Detailed explanation
A. Incorrect. Bigtable is a NoSQL database, schema should follow NoSQL patterns.
B. Incorrect. Relational designs don't work well in Bigtable.
C. Correct. Bigtable only supports atomicity at the row level, not across rows.
D. Incorrect. Cross-row atomicity is not supported.

Question 310
Which of the following is NOT a valid use case to select HDD (hard disk drives) as the storage for Google Cloud Bigtable?

Options
A. You expect to store at least 10 TB of data.
B. You will mostly run batch workloads with scans and writes, rather than frequently executing random reads of a small number of rows.
C. You need to integrate with Google BigQuery.
D. You will not use the data to back a user-facing or latency-sensitive application.

Answer
C

Detailed explanation
A. Correct. HDD is good for large datasets.
B. Correct. HDD works well with batch workloads.
C. Incorrect statement (answer). BigQuery integration requires SSD.
D. Correct. HDD is not meant for low-latency user-facing workloads.

Question 311
Cloud Bigtable is Google's _____ Big Data database service.

Options
A. Relational
B. mySQL
C. NoSQL
D. SQL Server

Answer
C

Detailed explanation
A. Incorrect. Not relational.
B. Incorrect. Not MySQL.
C. Correct. Bigtable is a NoSQL, wide-column database.
D. Incorrect. Not SQL Server.

Question 312
When you store data in Cloud Bigtable, what is the recommended minimum amount of stored data?

Options
A. 500 TB
B. 1 GB
C. 1 TB
D. 500 GB

Answer
C

Detailed explanation
A. Incorrect. 500 TB is much larger than needed.
B. Incorrect. 1 GB is too small for Bigtable.
C. Correct. Recommended minimum is ~1 TB for efficiency.
D. Incorrect. 500 GB is still below the recommended threshold.

Question 313
If you're running a performance test that depends upon Cloud Bigtable, all the choices except one below are recommended steps. Which is NOT a recommended step to follow?

Options
A. Do not use a production instance.
B. Run your test for at least 10 minutes.
C. Before you test, run a heavy pre-test for several minutes.
D. Use at least 300 GB of data.

Answer
A

Detailed explanation
A. Incorrect statement (answer). You should test on production-like environments, not avoid them.
B. Correct. Longer tests give realistic performance data.
C. Correct. Warm-up ensures stable performance metrics.
D. Correct. Large dataset size ensures realistic distribution.

Question 314
Cloud Bigtable is a recommended option for storing very large amounts of _____?

Options
A. multi-keyed data with very high latency
B. multi-keyed data with very low latency
C. single-keyed data with very low latency
D. single-keyed data with very high latency

Answer
C

Detailed explanation
A. Incorrect. Bigtable is low-latency, not high-latency.
B. Incorrect. It handles single-key lookups, not multi-key joins.
C. Correct. Single-key lookups with low latency are ideal for Bigtable.
D. Incorrect. Not high-latency.

Question 315
Google Cloud Bigtable indexes a single value in each row. This value is called the _____.

Options
A. primary key
B. unique key
C. row key
D. master key

Answer
C

Detailed explanation
A. Incorrect. "Primary key" is relational terminology.
B. Incorrect. "Unique key" is relational terminology.
C. Correct. In Bigtable, it's called the row key.
D. Incorrect. "Master key" is not used in Bigtable.

Question 316
What is the HBase Shell for Cloud Bigtable?

Options
A. The HBase shell is a GUI based interface that performs administrative tasks, such as creating and deleting tables.
B. The HBase shell is a command-line tool that performs administrative tasks, such as creating and deleting tables.
C. The HBase shell is a hypervisor based shell that performs administrative tasks, such as creating and deleting new virtualized instances.
D. The HBase shell is a command-line tool that performs only user account management functions to grant access to Cloud Bigtable instances.

Answer
B

Detailed explanation
A. Incorrect. It's not GUI-based.
B. Correct. HBase shell is a CLI tool for Bigtable/HBase administrative tasks.
C. Incorrect. No relation to hypervisors.
D. Incorrect. It does more than user account management.

Question 317
What is the recommended action to do in order to switch between SSD and HDD storage for your Google Cloud Bigtable instance?

Options
A. create a third instance and sync the data from the two storage types via batch jobs
B. export the data from the existing instance and import the data into a new instance
C. run parallel instances where one is HDD and the other is SDD
D. the selection is final and you must resume using the same storage type

Answer
B

Detailed explanation
A. Incorrect. Not the recommended method.
B. Correct. The proper way is to export and re-import into a new instance with desired storage type.
C. Incorrect. Running parallel HDD and SSD instances is inefficient and unsupported.
D. Incorrect. The choice is not final; migration is possible.

Question 321
You need to choose a database to store time series CPU and memory usage for millions of computers. You need to store this data in one-second interval samples. Analysts will be performing real-time, ad hoc analytics against the database. You want to avoid being charged for every query executed and ensure that the schema design will allow for future growth of the dataset. Which database and data model should you choose?

Options
A. Create a table in BigQuery, and append the new samples for CPU and memory to the table
B. Create a wide table in BigQuery, create a column for the sample value at each second, and update the row with the interval for each second
C. Create a narrow table in Bigtable with a row key that combines the Computer Engine computer identifier with the sample time at each second
D. Create a wide table in Bigtable with a row key that combines the computer identifier with the sample time at each minute, and combine the values for each second as column data
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — BigQuery is excellent for analytics but is not cost-efficient for ultra-high-frequency time-series writes at per-second granularity across millions of hosts, and it charges per query which can be costly for many ad-hoc queries.
B. Incorrect — A wide BigQuery table with one column per second is unmanageable and impractical; update-heavy patterns are also inefficient in BigQuery.
C. Correct — Cloud Bigtable with a narrow schema and a row key combining host identifier + timestamp (per-second) is the recommended design for high-ingest, low-latency time-series data. It scales well, supports fast point/range reads for real-time analytics, and the narrow schema avoids schema explosion.
D. Incorrect — Storing per-second samples as columns in a wide Bigtable row (per minute) complicates reads and can lead to hot-rows and limits on column cardinality; it's less flexible for future growth.
E. Incorrect — A correct option exists (C).

Question 322
You want to archive data in Cloud Storage. Because some data is very sensitive, you want to use the Trust No One (TNO) approach to encrypt your data to prevent the cloud provider staff from decrypting your data. What should you do?

Options
A. Use gcloud kms keys create to create a symmetric key. Then use gcloud kms encrypt to encrypt each archival file with the key and unique additional authenticated data (AAD). Use gsutil cp to upload each encrypted file to the Cloud Storage bucket, and keep the AAD outside of Google Cloud.
B. Use gcloud kms keys create to create a symmetric key. Then use gcloud kms encrypt to encrypt each archival file with the key. Use gsutil cp to upload each encrypted file to the Cloud Storage bucket. Manually destroy the key previously used for encryption, and rotate the key once.
C. Specify customer-supplied encryption key (CSEK) in the .boto configuration file. Use gsutil cp to upload each archival file to the Cloud Storage bucket. Save the CSEK in Cloud Memorystore as permanent storage of the secret.
D. Specify customer-supplied encryption key (CSEK) in the .boto configuration file. Use gsutil cp to upload each archival file to the Cloud Storage bucket. Save the CSEK in a different project that only the security team can access.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Client-side encrypting files with a customer-managed symmetric key (and using unique AAD kept outside Google Cloud) before uploading ensures Google cannot decrypt the data (true TNO). Storing AAD outside GCP enhances TNO guarantees.
B. Incorrect — Encrypting with KMS then uploading still uses Google-managed KMS service (unless keys are only used client-side). Manually destroying/rotating keys on Google does not guarantee provider cannot access prior decrypt paths and is operationally risky.
C. Incorrect — Storing CSEKs in Cloud Memorystore (a GCP service) defeats TNO because Google-managed services have access to memory; plus Memorystore is not intended as long-term secret storage.
D. Incorrect — Storing CSEK in another project within GCP still places the key material in Google-managed infrastructure; TNO requires key and AAD be kept outside the cloud provider.
E. Incorrect — A correct option exists (A).

Question 323
You have data pipelines running on BigQuery, Dataflow, and Dataproc. You need to perform health checks and monitor their behavior, and then notify the team managing the pipelines if they fail. You also need to be able to work across multiple projects. Your preference is to use managed products or features of the platform. What should you do?

Options
A. Export the information to Cloud Monitoring, and set up an Alerting policy
B. Run a Virtual Machine in Compute Engine with Airflow, and export the information to Cloud Monitoring
C. Export the logs to BigQuery, and set up App Engine to read that information and send emails if you find a failure in the logs
D. Develop an App Engine application to consume logs using GCP API calls, and send emails if you find a failure in the logs
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud Monitoring (Stackdriver) is a managed, cross-project monitoring and alerting solution. Export metrics/log-derived metrics into Monitoring and create alerting policies to notify teams—best practice and works across projects.
B. Incorrect — Running your own VM with Airflow adds operational overhead and is not necessary if you prefer managed services.
C. Incorrect — Exporting logs to BigQuery and building a custom App Engine job is heavy-weight and reinvents managed alerting that Cloud Monitoring already provides.
D. Incorrect — Building a custom App Engine consumer is more complex and less maintainable than using built-in Monitoring + Alerting policies.
E. Incorrect — A correct option exists (A).

Question 324
You are working on a linear regression model on BigQuery ML to predict a customer's likelihood of purchasing your company's products. Your model uses a city name variable as a key predictive component. In order to train and serve the model, your data must be organized in columns. You want to prepare your data using the least amount of coding while maintaining the predictive variables. What should you do?

Options
A. Create a new view with BigQuery that does not include a column with city information.
B. Use SQL in BigQuery to transform the state column using a one-hot encoding method, and make each city a column with binary values.
C. Use TensorFlow to create a categorical variable with a vocabulary list. Create the vocabulary file and upload that as part of your model to BigQuery ML.
D. Use Cloud Data Fusion to assign each city to a region that is labeled as 1, 2, 3, 4, or 5, and then use that number to represent the city in the model.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Dropping city information loses a predictive feature.
B. Correct — Using SQL to one-hot encode city (or state) columns in BigQuery is low-code and keeps features in columnar form for BigQuery ML; it's the least-coding approach while preserving variables.
C. Incorrect — Moving to TensorFlow and managing vocab files adds more coding and complexity than necessary when BigQuery SQL can handle encoding.
D. Incorrect — Mapping cities to arbitrary numeric regions may lose predictive granularity and typically requires more setup; also it's not the least-code approach compared to SQL one-hot encoding.
E. Incorrect — A correct option exists (B).

Question 325
You work for a large bank that operates in locations throughout North America. You are setting up a data storage system that will handle bank account transactions. You require ACID compliance and the ability to access data with SQL. Which solution is appropriate?

Options
A. Store transaction data in Cloud Spanner. Enable stale reads to reduce latency.
B. Store transaction in Cloud Spanner. Use locking read-write transactions.
C. Store transaction data in BigQuery. Disable the query cache to ensure consistency.
D. Store transaction data in Cloud SQL. Use a federated query in BigQuery for analysis.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — While Spanner supports stale reads for lower-latency reads, using stale reads for bank transactions would risk reading inconsistent (not up-to-date) states—unsuitable for ACID transactional correctness.
B. Correct — Cloud Spanner provides global, strongly consistent ACID transactions; using read-write transactions (with proper locking semantics) ensures transactional integrity for banking workloads.
C. Incorrect — BigQuery is analytical and not designed for transactional ACID semantics required for account transactions. Disabling cache doesn't provide transactional guarantees.
D. Incorrect — Cloud SQL can provide ACID semantics for smaller-scale systems, but for large distributed transactional workloads across North America, Spanner is the stronger fit; the provided correct answer is Spanner (B).
E. Incorrect — A correct option exists (B).

Question 326
A shipping company has live package-tracking data that is sent to an Apache Kafka stream in real time. This is then loaded into BigQuery. Analysts in your company want to query the tracking data in BigQuery to analyze geospatial trends in the lifecycle of a package. The table was originally created with ingest-date partitioning. Over time, the query processing time has increased. You need to implement a change that would improve query performance in BigQuery. What should you do?

Options
A. Implement clustering in BigQuery on the ingest date column.
B. Implement clustering in BigQuery on the package-tracking ID column.
C. Tier older data onto Cloud Storage files and create a BigQuery table using Cloud Storage as an external data source.
D. Re-create the table using data partitioning on the package delivery date.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Clustering on the ingest date (which is the partition column) is redundant and won't help much; partitioning already narrows scans by date.
B. Correct — Clustering by package-tracking ID groups rows for the same package together, improving selective query performance, especially for queries that examine package lifecycles and localized ranges.
C. Incorrect — External tables on Cloud Storage generally have worse performance and could complicate ad-hoc geospatial queries.
D. Incorrect — Re-partitioning by delivery date might help some queries but analysts likely query by package lifecycle (by package ID); clustering by package-tracking ID is the better optimization for this use case.
E. Incorrect — A correct option exists (B).

Question 327
Your company currently runs a large on-premises cluster using Spark, Hive, and HDFS in a colocation facility. The cluster is designed to accommodate peak usage on the system; however, many jobs are batch in nature, and usage of the cluster fluctuates quite dramatically. Your company is eager to move to the cloud to reduce the overhead associated with on-premises infrastructure and maintenance and to benefit from the cost savings. They are also hoping to modernize their existing infrastructure to use more serverless offerings in order to take advantage of the cloud. Because of the timing of their contract renewal with the colocation facility, they have only 2 months for their initial migration. How would you recommend they approach their upcoming migration strategy so they can maximize their cost savings in the cloud while still executing the migration in time?

Options
A. Migrate the workloads to Dataproc plus HDFS; modernize later.
B. Migrate the workloads to Dataproc plus Cloud Storage; modernize later.
C. Migrate the Spark workload to Dataproc plus HDFS, and modernize the Hive workload for BigQuery.
D. Modernize the Spark workload for Dataflow and the Hive workload for BigQuery.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Migrating to Dataproc + HDFS preserves HDFS complexity and reduces cloud-native benefits; it's less cost-effective/managed than using Cloud Storage.
B. Correct — Lift-and-shift to Dataproc while using Cloud Storage for storage (gs://) decouples storage from compute, reduces operational overhead, and positions you well to modernize later — a pragmatic 2-month migration plan that yields cost savings.
C. Incorrect — Mixing HDFS and BigQuery adds complexity during a short migration window; prefer Cloud Storage to decouple.
D. Incorrect — Full modernization (Dataflow + BigQuery) may take longer than available 2 months; not suitable for a rapid lift-and-shift.
E. Incorrect — A correct option exists (B).

Question 328
You work for a financial institution that lets customers register online. As new customers register, their user data is sent to Pub/Sub before being ingested into BigQuery. For security reasons, you decide to redact your customers' Government issued Identification Number while allowing customer service representatives to view the original values when necessary. What should you do?

Options
A. Use BigQuery's built-in AEAD encryption to encrypt the SSN column. Save the keys to a new table that is only viewable by permissioned users.
B. Use BigQuery column-level security. Set the table permissions so that only members of the Customer Service user group can see the SSN column.
C. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic hash.
D. Before loading the data into BigQuery, use Cloud Data Loss Prevention (DLP) to replace input values with a cryptographic format-preserving encryption token.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Storing encryption keys in a BigQuery table undermines key security; keys should be stored in a secure external KMS, not in a table accessible in BigQuery.
B. Incorrect — Column-level security controls access but does not provide the "redact then allow certain reps to view original values" with secure key management; the protected original values must be stored/encrypted securely.
C. Incorrect — Hashing is one-way — you cannot recover the original SSN for authorized customer service review.
D. Correct — Using Cloud DLP to apply format-preserving encryption (FPE) before loading into BigQuery allows storage of a reversible encrypted token; authorized personnel can decrypt (with keys managed securely) when required — fits the redact-with-reveal requirement.
E. Incorrect — A correct option exists (D).

Question 329
You are migrating a table to BigQuery and are deciding on the data model. Your table stores information related to purchases made across several store locations and includes information like the time of the transaction, items purchased, the store ID, and the city and state in which the store is located. You frequently query this table to see how many of each item were sold over the past 30 days and to look at purchasing trends by state, city, and individual store. How would you model this table for the best query performance?

Options
A. Partition by transaction time; cluster by state first, then city, then store ID.
B. Partition by transaction time; cluster by store ID first, then city, then state.
C. Top-level cluster by state first, then city, then store ID.
D. Top-level cluster by store ID first, then city, then state.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Partitioning by transaction time optimizes time-range queries (e.g., last 30 days). Clustering by state → city → store ID groups related geographic rows and helps queries that filter/aggregate by state/city/store, improving pruning and performance.
B. Incorrect — Clustering by store ID first is less effective for common analytic queries that aggregate by geographic hierarchy; store ID-first clustering helps single-store lookups but not regional aggregations as well.
C. Incorrect — Clustering without partitioning by time misses the time-range pruning benefit that's critical for 30-day queries.
D. Incorrect — Same reasoning as B; store-first clustering is less optimal for the stated query patterns compared to state→city→store.
E. Incorrect — A correct option exists (A).

Question 330
You are updating the code for a subscriber to a Pub/Sub feed. You are concerned that upon deployment the subscriber may erroneously acknowledge messages, leading to message loss. Your subscriber is not set up to retain acknowledged messages. What should you do to ensure that you can recover from errors after deployment?

Options
A. Set up the Pub/Sub emulator on your local machine. Validate the behavior of your new subscriber logic before deploying it to production.
B. Create a Pub/Sub snapshot before deploying new subscriber code. Use a Seek operation to re-deliver messages that became available after the snapshot was created.
C. Use Cloud Build for your deployment. If an error occurs after deployment, use a Seek operation to locate a timestamp logged by Cloud Build at the start of the deployment.
D. Enable dead-lettering on the Pub/Sub topic to capture messages that aren't successfully acknowledged. If an error occurs after deployment, re-deliver any messages captured by the dead-letter queue.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — The emulator helps test behavior but does not provide a recovery mechanism for production mistakes after deployment.
B. Correct — Creating a Pub/Sub snapshot before deployment lets you Seek the subscription back to the snapshot to re-deliver messages that were acknowledged erroneously, enabling reliable recovery.
C. Incorrect — Using Cloud Build timestamps with Seek is unreliable; snapshots are the supported recovery mechanism.
D. Incorrect — Dead-letter topics capture messages that fail delivery (exhaust ack attempts), but they do not capture messages that were erroneously acknowledged by a buggy subscriber.
E. Incorrect — A correct option exists (B).

Question 331
You work for a large real estate firm and are preparing 6 TB of home sales data to be used for machine learning. You will use SQL to transform the data and use BigQuery ML to create a machine learning model. You plan to use the model for predictions against a raw dataset that has not been transformed. How should you set up your workflow in order to prevent skew at prediction time?

Options
A. When creating your model, use BigQuery's TRANSFORM clause to define preprocessing steps. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data.
B. When creating your model, use BigQuery's TRANSFORM clause to define preprocessing steps. Before requesting predictions, use a saved query to transform your raw input data, and then use ML.EVALUATE.
C. Use a BigQuery view to define your preprocessing logic. When creating your model, use the view as your model training data. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any transformations on the raw input data.
D. Preprocess all data using Dataflow. At prediction time, use BigQuery's ML.EVALUATE clause without specifying any further transformations on the input data.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Defining the same preprocessing inside BigQuery’s TRANSFORM clause ensures the exact same transformations are applied at training and at prediction time (the model stores the transform). This prevents train/serve skew without extra manual steps.
B. Incorrect — This can work, but it relies on you remembering and correctly applying the saved query before every prediction; it’s more error-prone than embedding transforms in the model via TRANSFORM.
C. Incorrect — Using a view for training doesn't automatically ensure the same transform is applied at prediction unless you also apply identical transformations at serve time; it’s less robust than TRANSFORM.
D. Incorrect — Preprocessing outside BigQuery (Dataflow) can work, but it requires the exact same pipeline to run for every prediction; using TRANSFORM in BigQuery is simpler and less error-prone for this scenario.
E. Incorrect — A correct option exists (A).

Question 332
You are analyzing the price of a company's stock. Every 5 seconds, you need to compute a moving average of the past 30 seconds' worth of data. You are reading data from Pub/Sub and using DataFlow to conduct the analysis. How should you set up your windowed pipeline?

Options
A. Use a fixed window with a duration of 5 seconds. Emit results by setting the following trigger: AfterProcessingTime.pastFirstElementInPane().plusDelayOf (Duration.standardSeconds(30))
B. Use a fixed window with a duration of 30 seconds. Emit results by setting the following trigger: AfterWatermark.pastEndOfWindow().plusDelayOf (Duration.standardSeconds(5))
C. Use a sliding window with a duration of 5 seconds. Emit results by setting the following trigger: AfterProcessingTime.pastFirstElementInPane().plusDelayOf (Duration.standardSeconds(30))
D. Use a sliding window with a duration of 30 seconds and a period of 5 seconds. Emit results by setting the following trigger: AfterWatermark.pastEndOfWindow ()
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Fixed 5s windows with a long processing-time trigger do not produce the intended 30s moving average every 5s in a clean, time-based sliding manner.
B. Incorrect — Fixed 30s windows with watermark + 5s delay emits only once per 30s window, not every 5s.
C. Incorrect — Sliding window of 5s duration is wrong for capturing the last 30s; the duration should be 30s with a 5s period.
D. Correct — A sliding window of duration 30s with a period (hop) of 5s produces a moving-average every 5 seconds over the last 30 seconds. Using watermark-based emission is appropriate for event-time semantics.
E. Incorrect — A correct option exists (D).

Question 333
You are designing a pipeline that publishes application events to a Pub/Sub topic. Although message ordering is not important, you need to be able to aggregate events across disjoint hourly intervals before loading the results to BigQuery for analysis. What technology should you use to process and load this data to BigQuery while ensuring that it will scale with large volumes of events?

Options
A. Create a Cloud Function to perform the necessary data processing that executes using the Pub/Sub trigger every time a new message is published to the topic.
B. Schedule a Cloud Function to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.
C. Schedule a batch Dataflow job to run hourly, pulling all available messages from the Pub/Sub topic and performing the necessary aggregations.
D. Create a streaming Dataflow job that reads continually from the Pub/Sub topic and performs the necessary aggregations using tumbling windows.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Per-message Cloud Functions are not suitable for large-scale aggregation across hourly buckets and can be cost-inefficient and unscalable for high throughput.
B. Incorrect — Pulling from Pub/Sub in scheduled Cloud Functions is inefficient and adds complexity; also Pub/Sub is optimized for streaming processing.
C. Incorrect — Batch jobs running hourly are possible but lose near-real-time capabilities and are less flexible than streaming with windowing.
D. Correct — A streaming Dataflow job with tumbling (hourly) windows scales well, provides proper aggregation semantics, and writes results to BigQuery efficiently.
E. Incorrect — A correct option exists (D).

Question 334
You work for a large financial institution that is planning to use Dialogflow to create a chatbot for the company's mobile app. You have reviewed old chat logs and tagged each conversation for intent based on each customer's stated intention for contacting customer service. About 70% of customer requests are simple requests that are solved within 10 intents. The remaining 30% of inquiries require much longer, more complicated requests. Which intents should you automate first?

Options
A. Automate the 10 intents that cover 70% of the requests so that live agents can handle more complicated requests.
B. Automate the more complicated requests first because those require more of the agents' time.
C. Automate a blend of the shortest and longest intents to be representative of all intents.
D. Automate intents in places where common words such as 'payment' appear only once so the software isn't confused.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Automating the intents that cover the majority of requests yields the greatest immediate benefit (reduce agent load) and is the usual pragmatic first step.
B. Incorrect — Focusing first on complicated, low-frequency intents gives less ROI initially and is harder to implement.
C. Incorrect — A representative blend may be useful for testing, but prioritizing high-frequency intents gives better impact.
D. Incorrect — Selecting intents by token frequency is not a principled prioritization strategy compared to coverage-based selection.
E. Incorrect — A correct option exists (A).

Question 335
Your company is implementing a data warehouse using BigQuery, and you have been tasked with designing the data model. You move your on-premises sales data warehouse with a star data schema to BigQuery but notice performance issues when querying the data of the past 30 days. Based on Google's recommended practices, what should you do to speed up the query without increasing storage costs?

Options
A. Denormalize the data.
B. Shard the data by customer ID.
C. Materialize the dimensional data in views.
D. Partition the data by transaction date.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Denormalization can help performance but may increase storage; the question calls out not increasing storage costs.
B. Incorrect — Sharding by customer ID is not the recommended approach for time-range queries like "past 30 days."
C. Incorrect — Materialized views can help some queries but may increase cost/maintenance; partitioning is the primary recommended optimization for recent time-window queries.
D. Correct — Partitioning by transaction date reduces scanned data for recent time-range queries (past 30 days) and speeds queries without increasing storage costs.
E. Incorrect — A correct option exists (D).

Question 336
You have uploaded 5 years of log data to Cloud Storage. A user reported that some data points in the log data are outside of their expected ranges, which indicates errors. You need to address this issue and be able to run the process again in the future while keeping the original data for compliance reasons. What should you do?

Options
A. Import the data from Cloud Storage into BigQuery. Create a new BigQuery table, and skip the rows with errors.
B. Create a Compute Engine instance and create a new copy of the data in Cloud Storage. Skip the rows with errors.
C. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to a new dataset in Cloud Storage.
D. Create a Dataflow workflow that reads the data from Cloud Storage, checks for values outside the expected range, sets the value to an appropriate default, and writes the updated records to the same dataset in Cloud Storage.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Importing into BigQuery and skipping erroneous rows discards data and doesn't preserve originals as required for compliance; also it makes reprocessing harder.
B. Incorrect — Manually copying and editing on Compute Engine is heavyweight and error-prone compared to a managed pipeline.
C. Correct — Using Dataflow to read from Cloud Storage, apply fixes, and write results to a new dataset (new files) preserves the original files for compliance and provides a repeatable, automated pipeline.
D. Incorrect — Overwriting the same dataset destroys the originals, violating the compliance requirement to keep originals.
E. Incorrect — A correct option exists (C).

Question 337
You want to rebuild your batch pipeline for structured data on Google Cloud. You are using PySpark to conduct data transformations at scale, but your pipelines are taking over twelve hours to run. To expedite development and pipeline run time, you want to use a serverless tool and SQL syntax. You have already moved your raw data into Cloud Storage. How should you build the pipeline on Google Cloud while meeting speed and processing requirements?

Options
A. Convert your PySpark commands into SparkSQL queries to transform the data, and then run your pipeline on Dataproc to write the data into BigQuery.
B. Ingest your data into Cloud SQL, convert your PySpark commands into SparkSQL queries to transform the data, and then use federated queries from BigQuery for machine learning.
C. Ingest your data into BigQuery from Cloud Storage, convert your PySpark commands into BigQuery SQL queries to transform the data, and then write the transformations to a new table.
D. Use Apache Beam Python SDK to build the transformation pipelines, and write the data into BigQuery.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Dataproc is not serverless; it still requires cluster management and is not the serverless SQL-first option requested.
B. Incorrect — Cloud SQL is not suitable for large-scale batch analytics and introduces unnecessary complexity; federated queries are not a direct match.
C. Correct — Loading data into BigQuery and using BigQuery SQL (serverless, highly parallel) is the recommended serverless SQL-based approach to speed up transformations and development.
D. Incorrect — Apache Beam is serverless via Dataflow but uses code (Python) rather than SQL; the question asks to use SQL syntax.
E. Incorrect — A correct option exists (C).

Question 338
You are testing a Dataflow pipeline to ingest and transform text files. The files are compressed gzip, errors are written to a dead-letter queue, and you are using SideInputs to join data. You noticed that the pipeline is taking longer to complete than expected; what should you do to expedite the Dataflow job?

Options
A. Switch to compressed Avro files.
B. Reduce the batch size.
C. Retry records that throw an error.
D. Use CoGroupByKey instead of the SideInput.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Avro can be more efficient, but switching formats may be disruptive; the primary performance issue here is join strategy.
B. Incorrect — Reducing batch size may increase overhead and slow the pipeline further.
C. Incorrect — Retrying error records increases runtime; not an optimization for normal processing.
D. Correct — Replacing SideInputs (which can cause bottlenecks when joining large SideInput data) with CoGroupByKey (a scalable join for large datasets) improves parallelism and speeds up the pipeline.
E. Incorrect — A correct option exists (D).

Question 339
You are building a real-time prediction engine that streams files, which may contain PII (personal identifiable information) data, into Cloud Storage and eventually into BigQuery. You want to ensure that the sensitive data is masked but still maintains referential integrity, because names and emails are often used as join keys. How should you use the Cloud Data Loss Prevention API (DLP API) to ensure that the PII data is not accessible by unauthorized individuals?

Options
A. Create a pseudonym by replacing the PII data with cryptogenic tokens, and store the non-tokenized data in a locked-down bucket.
B. Redact all PII data, and store a version of the unredacted data in a locked-down bucket.
C. Scan every table in BigQuery, and mask the data it finds that has PII.
D. Create a pseudonym by replacing PII data with a cryptographic format-preserving token.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Storing non-tokenized data in a locked-down bucket undermines the protection—TNO and least-privilege principles are violated.
B. Incorrect — Redaction is irreversible and will break referential integrity required for joins and lookups.
C. Incorrect — Scanning & masking existing tables is reactive; it doesn't ensure consistent referential-preserving pseudonymization at ingestion.

D. Correct — Format-preserving cryptographic tokenization (FPE) replaces PII with reversible tokens that preserve format and allow joins while protecting raw values; with keys managed securely, authorized users can reverse tokens when permitted.
E. Incorrect — A correct option exists (D).

Question 340
You are migrating an application that tracks library books and information about each book, such as author or year published, from an on-premises data warehouse to BigQuery. In your current relational database, the author information is kept in a separate table and joined to the book information on a common key. Based on Google's recommended practice for schema design, how would you structure the data to ensure optimal speed of queries about the author of each book that has been borrowed?

Options
A. Keep the schema the same, maintain the different tables for the book and each of the attributes, and query as you are doing today.
B. Create a table that is wide and includes a column for each attribute, including the author's first name, last name, date of birth, etc.
C. Create a table that includes information about the books and authors, but nest the author fields inside the author column.
D. Keep the schema the same, create a view that joins all of the tables, and always query the view.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Normalized, multi-table schemas introduce join costs in BigQuery and slow queries compared to denormalized or nested designs.
B. Incorrect — A flat wide table works, but nesting related attributes (authors) is typically preferable for hierarchical data and better storage/query semantics in BigQuery.
C. Correct — Nesting author fields inside a repeated/record column keeps related data together, reduces joins, and is aligned with BigQuery best practices for fast analytical queries.
D. Incorrect — Views that perform joins still incur the join cost at query time; nesting avoids that overhead.
E. Incorrect — A correct option exists (C).

Question 341
You need to give new website users a globally unique identifier (GUID) using a service that takes in data points and returns a GUID. This data is sourced from both internal and external systems via HTTP calls that you will make via microservices within your pipeline. There will be tens of thousands of messages per second and that can be multithreaded. You worry about the backpressure on the system. How should you design your pipeline to minimize that backpressure?

Options
A. Call out to the service via HTTP.
B. Create the pipeline statically in the class definition.
C. Create a new object in the startBundle method of DoFn.
D. Batch the job into ten-second increments.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Per-message synchronous HTTP calls will create high backpressure and latency under tens of thousands RPS.
B. Incorrect — "Create the pipeline statically in the class definition" is unrelated to mitigating external-service backpressure.
C. Incorrect — Creating objects in startBundle helps object reuse but does not address high call volume or backpressure from external GUID service.
D. Correct — Batching requests (e.g., into 10-second batches) amortizes the HTTP calls, reduces per-message overhead, and minimizes backpressure on the GUID service.
E. Incorrect — A correct option exists (D).

Question 342
You are migrating your data warehouse to Google Cloud and decommissioning your on-premises data center. Because this is a priority for your company, you know that bandwidth will be made available for the initial data load to the cloud. The files being transferred are not large in number, but each file is 90 GB. Additionally, you want your transactional systems to continually update the warehouse on Google Cloud in real time. What tools should you use to migrate the data and ensure that it continues to write to your warehouse?

Options
A. Storage Transfer Service for the migration; Pub/Sub and Cloud Data Fusion for the real-time updates
B. BigQuery Data Transfer Service for the migration; Pub/Sub and Dataproc for the real-time updates
C. gsutil for the migration; Pub/Sub and Dataflow for the real-time updates
D. gsutil for both the migration and the real-time updates
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Storage Transfer Service is for cloud-to-cloud or HTTP sources; for large file transfers from on-prem gsutil is simpler when bandwidth is available, and Cloud Data Fusion less commonly used for ongoing streaming ingestion compared to Dataflow.
B. Incorrect — BigQuery Data Transfer Service is for SaaS and scheduled transfers, not one-time large-file migrations; Dataproc is not the primary choice for continuous streaming ingestion.

C. Correct — Using gsutil to upload large files (90 GB) for the initial bulk transfer is appropriate when bandwidth is available. For continuous, low-latency updates Pub/Sub + Dataflow is the recommended streaming ingestion and transformation path into BigQuery.
D. Incorrect — gsutil is not suitable for real-time continuous streaming updates.
E. Incorrect — A correct option exists (C).

Question 343
You are using Bigtable to persist and serve stock market data for each of the major indices. To serve the trading application, you need to access only the most recent stock prices that are streaming in. How should you design your row key and tables to ensure that you can access the data with the simplest query?

Options
A. Create one unique table for all of the indices, and then use the index and timestamp as the row key design.
B. Create one unique table for all of the indices, and then use a reverse timestamp as the row key design.
C. For each index, have a separate table and use a timestamp as the row key design.
D. For each index, have a separate table and use a reverse timestamp as the row key design.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Using index+timestamp as-is will cause hot-spotting on recent timestamps (monotonic increasing), and mixing indices in one table complicates access patterns.
B. Correct — Using a reverse timestamp (e.g., inverted timestamp) in the row key causes the most recent rows to be stored at the start of the tablet, making “get latest” queries efficient even in a single table.
C. Incorrect — Separate tables per index with normal timestamp keys still suffer from monotonic hot-spotting on writes.
D. Incorrect — Separate tables per index with reverse timestamps could work, but the recommended simplest design (per question answer) is a single table with reverse timestamp to get most-recent values efficiently; the provided answer selects B.
E. Incorrect — A correct option exists (B).

Question 344
You are building a report-only data warehouse where the data is streamed into BigQuery via the streaming API. Following Google's best practices, you have both a staging and a production table for the data. How should you design your data loading to ensure that there is only one master dataset without affecting performance on either the ingestion or reporting pieces?

Options
A. Have a staging table that is an append-only model, and then update the production table every three hours with the changes written to staging.
B. Have a staging table that is an append-only model, and then update the production table every ninety minutes with the changes written to staging.
C. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every three hours.
D. Have a staging table that moves the staged data over to the production table and deletes the contents of the staging table every thirty minutes.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Periodic updates that apply changes to production (without clearing staging) may complicate deduplication and can increase storage and complexity.
B. Incorrect — Similar to (A); more frequent updates still retain staging and add complexity; not the chosen best practice here.
C. Correct — Moving staged (streamed) data into production on a cadence (e.g., every 3 hours), and then clearing staging, provides a single authoritative dataset while keeping ingestion isolated and minimizing impact on reporting.
D. Incorrect — Moving and clearing every 30 minutes may be operationally intensive; the provided answer favors the 3-hour cadence.
E. Incorrect — A correct option exists (C).

Question 345
You issue a new batch job to Dataflow. The job starts successfully, processes a few elements, and then suddenly fails and shuts down. You navigate to the Dataflow monitoring interface where you find errors related to a particular DoFn in your pipeline. What is the most likely cause of the errors?

Options
A. Job validation
B. Exceptions in worker code
C. Graph or pipeline construction
D. Insufficient permissions
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Job validation errors typically prevent the job from starting; here the job started and processed elements.
B. Correct — Runtime exceptions thrown inside DoFn worker code commonly cause processing failures after some elements have been processed — the most likely cause.
C. Incorrect — Graph/pipeline construction issues usually surface at job submission or cause different failure patterns.
D. Incorrect — Permissions issues generally fail early during resource allocation or I/O; not typical for mid-job DoFn runtime errors.
E. Incorrect — A correct option exists (B).

Question 346
Your new customer has requested daily reports that show their net consumption of Google Cloud compute resources and who used the resources. You need to quickly and efficiently generate these daily reports. What should you do?

Options
A. Do daily exports of Cloud Logging data to BigQuery. Create views filtering by project, log type, resource, and user.
B. Filter data in Cloud Logging by project, resource, and user; then export the data in CSV format.
C. Filter data in Cloud Logging by project, log type, resource, and user, then import the data into BigQuery.
D. Export Cloud Logging data to Cloud Storage in CSV format. Cleanse the data using Dataprep, filtering by project, resource, and user.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Exporting Cloud Logging (audit) data to BigQuery and building views is a standard, scalable, and efficient pattern for daily reporting and filtering by user/resource.
B. Incorrect — Exporting CSVs from Cloud Logging is less automated and less convenient than exporting to BigQuery for analysis.
C. Incorrect — Filtering then importing manually is more cumbersome; direct export to BigQuery is simpler and real-time capable.
D. Incorrect — Exporting to Cloud Storage + Dataprep adds extra steps and latency; BigQuery is the better target for reporting.
E. Incorrect — A correct option exists (A).

Question 347
The Development and External teams have the project viewer Identity and Access Management (IAM) role in a folder named Visualization. You want the Development Team to be able to read data from both Cloud Storage and BigQuery, but the External Team should only be able to read data from BigQuery. What should you do?

Options
A. Remove Cloud Storage IAM permissions to the External Team on the acme-raw-data project.
B. Create Virtual Private Cloud (VPC) firewall rules on the acme-raw-data project that deny all ingress traffic from the External Team CIDR range.
C. Create a VPC Service Controls perimeter containing both projects and BigQuery as a restricted API. Add the External Team users to the perimeter's Access Level.
D. Create a VPC Service Controls perimeter containing both projects and Cloud Storage as a restricted API. Add the Development Team users to the perimeter's Access Level.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Manually removing permissions per project is error-prone and doesn't scale; the question suggests minimizing policy complexity.
B. Incorrect — VPC firewall rules operate at network layer and won't enforce IAM-level data access restrictions by user identity.
C. Incorrect — Adding External Team to a perimeter's Access Level would grant them perimeter access; the goal is to restrict External Team from Cloud Storage, not grant them special access.
D. Correct — Use VPC Service Controls to restrict Cloud Storage access (make Cloud Storage a restricted API) and grant the Development Team an access level so they can reach Cloud Storage while External Team (without that access level) can access BigQuery only. This enforces data access separation without per-project IAM churn.
E. Incorrect — A correct option exists (D).

Question 348
Your startup has a web application that currently serves customers out of a single region in Asia. You are targeting funding that will allow your startup to serve customers globally. Your current goal is to optimize for cost, and your post-funding goal is to optimize for global presence and performance. You must use a native JDBC driver. What should you do?

Options
A. Use Cloud Spanner to configure a single region instance initially, and then configure multi-region Cloud Spanner instances after securing funding.
B. Use a Cloud SQL for PostgreSQL highly available instance first, and Bigtable with US, Europe, and Asia replication after securing funding.
C. Use a Cloud SQL for PostgreSQL zonal instance first, and Bigtable with US, Europe, and Asia after securing funding.
D. Use a Cloud SQL for PostgreSQL zonal instance first, and Cloud SQL for PostgreSQL with highly available configuration after securing funding.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Start with a single-region Cloud Spanner instance for cost-effective managed relational service with JDBC support; later move to multi-region Spanner for global low-latency reads/writes as you secure funding.
B. Incorrect — Combining Cloud SQL HA + Bigtable replication is more complex and doesn't give the same seamless SQL + global scaling that Spanner provides.
C. Incorrect — Zonal Cloud SQL won't scale globally; using Bigtable later complicates application semantics because it's NoSQL.
D. Incorrect — Upgrading Cloud SQL to HA later still leaves scaling challenges for global performance; Spanner is the more natural path for evolving global SQL needs.
E. Incorrect — A correct option exists (A).

Question 349
You need to migrate 1 PB of data from an on-premises data center to Google Cloud. Data transfer time during the migration should take only a few hours. You want to follow Google-recommended practices to facilitate the large data transfer over a secure connection. What should you do?

Options
A. Establish a Cloud Interconnect connection between the on-premises data center and Google Cloud, and then use the Storage Transfer Service.
B. Use a Transfer Appliance and have engineers manually encrypt, decrypt, and verify the data.
C. Establish a Cloud VPN connection, start gcloud compute scp jobs in parallel, and run checksums to verify the data.
D. Reduce the data into 3 TB batches, transfer the data using gsutil, and run checksums to verify the data.
E. None of the above

Answer
A

Detailed explanation
A. Correct — For extremely large transfers requiring a few hours, a high-bandwidth dedicated Cloud Interconnect connection combined with Storage Transfer or other bulk transfer tools is the recommended approach to move PB quickly and securely.
B. Incorrect — Transfer Appliance is ideal when network capacity is limited; but if you need the transfer in a few hours and can provision Interconnect, Interconnect is preferred. Manual encrypt/decrypt is discouraged—use managed encryption.
C. Incorrect — VPN and parallel scp won't provide the bandwidth and reliability needed for PB-scale transfers in a few hours.
D. Incorrect — gsutil over typical WAN links will be too slow for PB in hours and is inefficient compared to Interconnect.
E. Incorrect — A correct option exists (A).

Question 350
You are loading CSV files from Cloud Storage to BigQuery. The files have known data quality issues, including mismatched data types, such as STRINGs and INT64s in the same column, and inconsistent formatting of values such as phone numbers or addresses. You need to create the data pipeline to maintain data quality and perform the required cleansing and transformation. What should you do?

Options
A. Use Data Fusion to transform the data before loading it into BigQuery.
B. Use Data Fusion to convert the CSV files to a self-describing data format, such as AVRO, before loading the data to BigQuery.
C. Load the CSV files into a staging table with the desired schema, perform the transformations with SQL, and then write the results to the final destination table.
D. Create a table with the desired schema, load the CSV files into the table, and perform the transformations in place using SQL.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud Data Fusion provides a visual, managed ETL tool to perform data cleansing and transformations before loading into BigQuery, which is suitable for handling schema drift and formatting normalization.
B. Incorrect — Converting to Avro can help type handling, but it still requires transformation logic; Data Fusion end-to-end transformation is preferable.
C. Incorrect — Loading malformed CSV directly into a staging table with a strict schema may fail; while staging+SQL is viable, the question emphasizes maintaining data quality before load—Data Fusion is the better managed choice.
D. Incorrect — Loading directly into final table risks load failures or dirty data; in-place transformations after load are riskier for data quality.
E. Incorrect — A correct option exists (A).

Question 351
You are developing a new deep learning model that predicts a customer's likelihood to buy on your ecommerce site. After running an evaluation of the model against both the original training data and new test data, you find that your model is overfitting the data. You want to improve the accuracy of the model when predicting new data. What should you do?

Options
A. Increase the size of the training dataset, and increase the number of input features.
B. Increase the size of the training dataset, and decrease the number of input features.
C. Reduce the size of the training dataset, and increase the number of input features.
D. Reduce the size of the training dataset, and decrease the number of input features.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Increasing dataset size is good, but increasing input features often increases model complexity, making overfitting worse.
B. Correct – Increasing dataset size provides more variety for training, while reducing features helps remove noise and prevents the model from memorizing unimportant details. This directly combats overfitting.
C. Incorrect – Reducing dataset size decreases generalization power, which worsens overfitting.
D. Incorrect – Reducing both dataset size and features loses useful data and signal, worsening accuracy.
E. Incorrect – The recommended solution is already captured in option B.

Question 352
You are implementing a chatbot to help an online retailer streamline their customer service. The chatbot must be able to respond to both text and voice inquiries. You are looking for a low-code or no-code option, and you want to be able to easily train the chatbot to provide answers to keywords. What should you do?

Options
A. Use the Cloud Speech-to-Text API to build a Python application in App Engine.
B. Use the Cloud Speech-to-Text API to build a Python application in a Compute Engine instance.
C. Use Dialogflow for simple queries and the Cloud Speech-to-Text API for complex queries.
D. Use Dialogflow to implement the chatbot, defining the intents based on the most common queries collected.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Cloud Speech-to-Text only handles transcription. It does not provide chatbot intent recognition or low-code features.
B. Incorrect – Same as A, just with Compute Engine. It requires custom coding, not low-code/no-code.
C. Incorrect – Mixing APIs complicates the setup unnecessarily. Dialogflow already integrates with Speech-to-Text.
D. Correct – Dialogflow is a managed, low-code conversational AI tool that supports both text and voice, and allows easy definition of intents with keywords.
E. Incorrect – Option D is the most effective solution.

Question 353
An aerospace company uses a proprietary data format to store its flight data. You need to connect this new data source to BigQuery and stream the data into BigQuery. You want to efficiently import the data into BigQuery while consuming as few resources as possible. What should you do?

Options
A. Write a shell script that triggers a Cloud Function that performs periodic ETL batch jobs on the new data source.
B. Use a standard Dataflow pipeline to store the raw data in BigQuery, and then transform the format later when the data is used.
C. Use Apache Hive to write a Dataproc job that streams the data into BigQuery in CSV format.
D. Use an Apache Beam custom connector to write a Dataflow pipeline that streams the data into BigQuery in Avro format.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Shell script + Cloud Function works for small periodic jobs, not continuous streaming at scale.
B. Incorrect – Storing raw proprietary data directly in BigQuery is inefficient and causes extra cost/latency for later queries.
C. Incorrect – Hive and CSV add overhead and are not optimized for streaming ingestion.

D. Correct – A custom Apache Beam connector in Dataflow can parse the proprietary format and stream directly into BigQuery using Avro (optimized, compressed, schema-friendly).
E. Incorrect – Option D already provides the best scalable solution.

Question 354
An online brokerage company requires a high volume trade processing architecture. You need to create a secure queuing system that triggers jobs. The jobs will run in Google Cloud and call the company's Python API to execute trades. You need to efficiently implement a solution. What should you do?

Options
A. Use a Pub/Sub push subscription to trigger a Cloud Function to pass the data to the Python API.
B. Write an application hosted on a Compute Engine instance that makes a push subscription to the Pub/Sub topic.
C. Write an application that makes a queue in a NoSQL database.
D. Use Cloud Composer to subscribe to a Pub/Sub topic and call the Python API.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Pub/Sub with push subscription + Cloud Function is event-driven, scalable, secure, and cost-efficient for triggering jobs to the Python API.
B. Incorrect – Compute Engine adds unnecessary VM overhead compared to serverless Cloud Functions.
C. Incorrect – Building a custom queuing system in NoSQL is reinventing the wheel and less reliable than Pub/Sub.
D. Incorrect – Cloud Composer is meant for workflow orchestration, not real-time trade triggering.
E. Incorrect – The right answer is A.

Question 355
Your company wants to be able to retrieve large result sets of medical information from your current system, which has over 10 TBs in the database, and store the data in new tables for further query. The database must have a low-maintenance architecture and be accessible via SQL. You need to implement a cost-effective solution that can support data analytics for large result sets. What should you do?

Options
A. Use Cloud SQL, but first organize the data into tables. Use JOIN in queries to retrieve data.
B. Use BigQuery as a data warehouse. Set output destinations for caching large queries.
C. Use a MySQL cluster installed on a Compute Engine managed instance group for scalability.
D. Use Cloud Spanner to replicate the data across regions. Normalize the data in a series of tables.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Cloud SQL is not designed for >10 TB scale analytics; performance and cost would be poor.

B. Correct – BigQuery is a serverless, low-maintenance, cost-effective data warehouse optimized for large-scale analytics over 10 TB+ datasets.
C. Incorrect – A MySQL cluster on Compute Engine requires heavy maintenance and scaling effort.
D. Incorrect – Cloud Spanner is transactional (OLTP) and expensive for this analytical use case.
E. Incorrect – Option B is the clear fit.

Question 356
You have 15 TB of data in your on-premises data center that you want to transfer to Google Cloud. Your data changes weekly and is stored in a POSIX-compliant source. The network operations team has granted you 500 Mbps bandwidth to the public internet. You want to follow Google-recommended practices to reliably transfer your data to Google Cloud on a weekly basis. What should you do?

Options
A. Use Cloud Scheduler to trigger the gsutil command. Use the -m parameter for optimal parallelism.
B. Use Transfer Appliance to migrate your data into a Google Kubernetes Engine cluster, and then configure a weekly transfer job.
C. Install Storage Transfer Service for on-premises data in your data center, and then configure a weekly transfer job.
D. Install Storage Transfer Service for on-premises data on a Google Cloud virtual machine, and then configure a weekly transfer job.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – gsutil is manual and less reliable for weekly recurring transfers at this scale.
B. Incorrect – Transfer Appliance is meant for one-time bulk migrations, not recurring weekly updates.
C. Correct – Storage Transfer Service (on-premises agent) automates scheduled data syncs from POSIX-compliant storage to Google Cloud. Fits recurring weekly transfer use case.
D. Incorrect – Installing the agent on Google Cloud VM does not help because source is on-premises.
E. Incorrect – C is the correct Google-recommended approach.

Question 357
You are designing a system that requires an ACID-compliant database. You must ensure that the system requires minimal human intervention in case of a failure. What should you do?

Options
A. Configure a Cloud SQL for MySQL instance with point-in-time recovery enabled.
B. Configure a Cloud SQL for PostgreSQL instance with high availability enabled.
C. Configure a Bigtable instance with more than one cluster.
D. Configure a BigQuery table with a multi-region configuration.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Point-in-time recovery helps backups, but it does not provide automatic failover for high availability.
B. Correct – Cloud SQL for PostgreSQL with HA enabled is ACID-compliant and automatically handles failover with minimal manual intervention.
C. Incorrect – Bigtable is not ACID-compliant, it's eventually consistent.
D. Incorrect – BigQuery is an analytical system (OLAP), not a transactional ACID database.
E. Incorrect – Option B already satisfies requirements.

Question 358
You are implementing workflow pipeline scheduling using open source-based tools and Google Kubernetes Engine (GKE). You want to use a Google managed service to simplify and automate the task. You also want to accommodate Shared VPC networking considerations. What should you do?

Options
A. Use Dataflow for your workflow pipelines. Use Cloud Run triggers for scheduling.
B. Use Dataflow for your workflow pipelines. Use shell scripts to schedule workflows.
C. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the host project.
D. Use Cloud Composer in a Shared VPC configuration. Place the Cloud Composer resources in the service project.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Dataflow is for data processing pipelines, not workflow orchestration scheduling.
B. Incorrect – Shell scripts are not a managed service and don't scale well.
C. Incorrect – Cloud Composer resources must be in the service project, not the host project, for Shared VPC.
D. Correct – Cloud Composer is a managed Airflow service. For Shared VPC, resources go in the service project.
E. Incorrect – The correct choice is D.

Question 359
You are using BigQuery and Data Studio to design a customer-facing dashboard that displays large quantities of aggregated data. You expect a high volume of concurrent users. You need to optimize the dashboard to provide quick visualizations with minimal latency. What should you do?

Options
A. Use BigQuery BI Engine with materialized views.
B. Use BigQuery BI Engine with logical views.
C. Use BigQuery BI Engine with streaming data.
D. Use BigQuery BI Engine with authorized views.
E. None of the above

Answer
A

Detailed explanation
A. Correct – BI Engine + materialized views pre-aggregates data, reducing query latency and providing fast dashboards for many users.
B. Incorrect – Logical views do not pre-compute results, so queries still scan raw data.
C. Incorrect – Streaming data adds ingestion speed, not dashboard query optimization.
D. Incorrect – Authorized views control access, not performance.
E. Incorrect – The correct choice is A.

Question 360
Government regulations in the banking industry mandate the protection of clients' personally identifiable information (PII). Your company requires PII to be access controlled, encrypted, and compliant with major data protection standards. In addition to using Cloud Data Loss Prevention (Cloud DLP), you want to follow Google-recommended practices and use service accounts to control access to PII. What should you do?

Options
A. Assign the required Identity and Access Management (IAM) roles to every employee, and create a single service account to access project resources.
B. Use one service account to access a Cloud SQL database, and use separate service accounts for each human user.
C. Use Cloud Storage to comply with major data protection standards. Use one service account shared by all users.
D. Use Cloud Storage to comply with major data protection standards. Use multiple service accounts attached to IAM groups to grant the appropriate access to each group.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Giving IAM roles to every employee directly is risky and violates least privilege.
B. Incorrect – Service accounts are for applications/services, not human users.
C. Incorrect – A single shared service account removes accountability and is insecure.
D. Correct – Using multiple service accounts with IAM groups enforces least privilege, access control, and compliance with security standards. Cloud Storage supports encryption and compliance.
E. Incorrect – D is the best practice solution.

MJTelco Case Study

Company Overview
MJTelco is a startup that plans to build networks in rapidly growing, underserved markets around the world. The company has patents for innovative optical communications hardware. Based on these patents, they can create many reliable, high-speed backbone links with inexpensive hardware.

Company Background
Founded by experienced telecom executives, MJTelco uses technologies originally developed to overcome communications challenges in space. Fundamental to their operation, they need to create a distributed data infrastructure that drives real-time analysis and incorporates machine learning to continuously optimize their topologies. Because their hardware is inexpensive, they plan to overdeploy the network allowing them to account for the impact of dynamic regional politics on location availability and cost.

Their management and operations teams are situated all around the globe creating many-to-many relationships between data consumers and providers in their system. After careful consideration, they decided public cloud is the perfect environment to support their needs.

Solution Concept
MJTelco is running a successful proof-of-concept (PoC) project in its labs. They have two primary needs:
☞ Scale and harden their PoC to support significantly more data flows generated when they ramp to more than 50,000 installations.
☞ Refine their machine-learning cycles to verify and improve the dynamic models they use to control topology definition.

MJTelco will also use three separate operating environments — development/test, staging, and production — to meet the needs of running experiments, deploying new features, and serving production customers.

Business Requirements
☞ Scale up their production environment with minimal cost, instantiating resources when and where needed in an unpredictable, distributed telecom user community.
☞ Ensure security of their proprietary data to protect their leading-edge machine learning and analysis.
☞ Provide reliable and timely access to data for analysis from distributed research workers.
☞ Maintain isolated environments that support rapid iteration of their machine-learning models without affecting their customers.

Technical Requirements
☞ Ensure secure and efficient transport and storage of telemetry data.
☞ Rapidly scale instances to support between 10,000 and 100,000 data providers with multiple flows each.
☞ Allow analysis and presentation against data tables tracking up to 2 years of data storing approximately 100m records/day.
☞ Support rapid iteration of monitoring infrastructure focused on awareness of data pipeline problems both in telemetry flows and in production learning cycles.

CEO Statement
Our business model relies on our patents, analytics and dynamic machine learning. Our inexpensive hardware is organized to be highly reliable, which gives us cost advantages. We need to quickly stabilize our large distributed data pipelines to meet our reliability and capacity commitments.

CTO Statement
Our public cloud services must operate as advertised. We need resources that scale and keep our data secure. We also need environments in which our data scientists can carefully study and quickly adapt our models. Because we rely on automation to process our data, we also need our development and test environments to work as we iterate.

CFO Statement
The project is too large for us to maintain the hardware and software required for the data and analysis. Also, we cannot afford to staff an operations team to monitor so many data feeds, so we will rely on automation and infrastructure. Google Cloud's machine learning will allow our quantitative researchers to work on our high-value problems instead of problems with our data pipelines.

Question 360
You need to compose visualization for operations teams with the following requirements:
☞ Telemetry must include data from all 50,000 installations for the most recent 6 weeks (sampling once every minute)
☞ The report must not be more than 3 hours delayed from live data.
☞ The actionable report should only show suboptimal links.
☞ Most suboptimal links should be sorted to the top.
☞ Suboptimal links can be grouped and filtered by regional geography.
☞ User response time to load the report must be <5 seconds.

You create a data source to store the last 6 weeks of data, and create visualizations that allow viewers to see multiple date ranges, distinct geographic regions, and unique installation types. You always show the latest data without any changes to your visualizations. You want to avoid creating and updating new visualizations each month. What should you do?

Options
A. Look through the current data and compose a series of charts and tables, one for each possible combination of criteria.
B. Look through the current data and compose a small set of generalized charts and tables bound to criteria filters that allow value selection.
C. Export the data to a spreadsheet, compose a series of charts and tables, one for each possible combination of criteria, and spread them across multiple tabs.
D. Load the data into relational database tables, write a Google App Engine application that queries all rows, summarizes the data across each criteria, and then renders results using the Google Charts and visualization API.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Creating a chart/table for every combination explodes the number of visualizations and won't scale or meet the <5s load requirement.
B. Incorrect — While filters are good for flexibility, relying solely on client-side filtering over 6 weeks × 50k × 60/min rows will likely exceed the <5s UX requirement and won't guarantee timely (<3h) display without server-side aggregation.
C. Incorrect — Spreadsheets won't scale to the data volume or meet performance and freshness requirements.

D. Correct — Pre-aggregating and summarizing on the server (relational tables + a small App Engine service to fetch only necessary aggregates) lets you deliver suboptimal-link focused views, sort the worst links to the top, group by geography, keep response <5s, and avoid creating new visualizations each month. This design supports timely queries on recent data and lets you tailor APIs for the UI.
E. Incorrect — Since D is correct, “None of the above” is not valid.

Question 361
Given the record streams MJTelco is interested in ingesting per day, they are concerned about the cost of Google BigQuery increasing. MJTelco asks you to provide a design solution. They require a single large data table called tracking_table. Additionally, they want to minimize the cost of daily queries while performing fine-grained analysis of each day's events. They also want to use streaming ingestion. What should you do?

Options
A. Create a table called tracking_table and include a DATE column.
B. Create a partitioned table called tracking_table and include a TIMESTAMP column.
C. Create sharded tables for each day following the pattern tracking_table_YYYYMMDD.
D. Create a table called tracking_table with a TIMESTAMP column to represent the day.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — A single unpartitioned table with a DATE column will still require scanning large volumes and incur higher query costs.
B. Correct — Creating a partitioned table (partitioned by ingestion or by TIMESTAMP day) lets you use streaming inserts while minimizing query costs by pruning partitions for daily analysis. It supports fine-grained per-day queries without scanning the whole history.
C. Incorrect — Sharded daily tables increase management complexity and can cause metadata/maintenance overhead; BigQuery recommends partitioned tables over many sharded tables.
D. Incorrect — Similar to A: if not partitioned, a TIMESTAMP column alone won’t enable cost-efficient partition pruning.
E. Incorrect — Since B is correct, “None of the above” is not valid.

Question 362
Flowlogistic's management has determined that the current Apache Kafka servers cannot handle the data volume for their real-time inventory tracking system. You need to build a new system on Google Cloud Platform (GCP) that will feed the proprietary tracking software. The system must be able to ingest data from a variety of global sources, process and query in real-time, and store the data reliably. Which combination of GCP products should you choose?

Options
A. Cloud Pub/Sub, Cloud Dataflow, and Cloud Storage
B. Cloud Pub/Sub, Cloud Dataflow, and Local SSD
C. Cloud Pub/Sub, Cloud SQL, and Cloud Storage
D. Cloud Load Balancing, Cloud Dataflow, and Cloud Storage
E. None of the above

Answer
A

Detailed explanation
A. Correct — Cloud Pub/Sub handles global ingestion at scale, Dataflow processes streaming data in real time, and Cloud Storage provides durable storage (and can feed downstream analytics). This combination supports real-time processing and reliability.
B. Incorrect — Local SSD is ephemeral and not suitable for durable, global ingest/storage needs.
C. Incorrect — Cloud SQL is not designed for streaming at massive scale or for high-throughput time-series ingestion.
D. Incorrect — Load balancing is not a substitute for an ingestion system like Pub/Sub; it won't meet the streaming ingestion requirement.
E. Incorrect — Since A is correct, "None of the above" is not valid.

Question 363
After migrating ETL jobs to run on BigQuery, you need to verify that the output of the migrated jobs is the same as the output of the original. You've loaded a table containing the output of the original job and want to compare the contents with output from the migrated job to show that they are identical. The tables do not contain a primary key column that would enable you to join them together for comparison. What should you do?

Options
A. Select random samples from the tables using the RAND() function and compare the samples.
B. Select random samples from the tables using the HASH() function and compare the samples.
C. Use a Dataproc cluster and the BigQuery Hadoop connector to read the data from each table and calculate a hash from non-timestamp columns of the table after sorting. Compare the hashes of each table.
D. Create stratified random samples using the OVER() function and compare equivalent samples from each table.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — RAND-based sampling can miss differences and is not deterministic for full-table verification.
B. Incorrect — Simple HASH samples may not be sufficient for comprehensive verification without consistent sorting and hashing strategy across rows.
C. Correct — Using a distributed job (Dataproc) to read, sort deterministically, and compute a global hash over relevant columns is a reliable way to compare entire large tables when no primary key exists. This produces a succinct proof of equality.
D. Incorrect — Stratified samples improve coverage but still do not guarantee identical full-table equality.
E. Incorrect — Since C is correct, “None of the above” is not valid.

Question 364
You are a head of BI at a large enterprise company with multiple business units that each have different priorities and budgets. You use on-demand pricing for BigQuery with a quota of 2K concurrent on-demand slots per project. Users at your organization sometimes don't get slots to execute their query and you need to correct this. You'd like to avoid introducing new projects to your account. What should you do?

Options
A. Convert your batch BQ queries into interactive BQ queries.
B. Create an additional project to overcome the 2K on-demand per-project quota.
C. Switch to flat-rate pricing and establish a hierarchical priority model for your projects.
D. Increase the amount of concurrent slots per project at the Quotas page at the Cloud Console.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Converting from batch to interactive won’t solve the slot concurrency constraint and could worsen contention.
B. Incorrect — You wanted to avoid creating new projects; plus this is a workaround that fragments management and billing.
C. Correct — Switching to flat-rate (purchased slots) lets you allocate capacity centrally and implement priority/assignment strategies (reservation/assignment) to ensure business units get required concurrency without creating new projects.
D. Incorrect — Concurrent on-demand slot quotas are managed by Google and not simply increased arbitrarily from the console; flat-rate is the recommended approach.
E. Incorrect — Since C is correct, “None of the above” is not valid.

Question 365
You have an Apache Kafka cluster on-prem with topics containing web application logs. You need to replicate the data to Google Cloud for analysis in BigQuery and Cloud Storage. The preferred replication method is mirroring to avoid deployment of Kafka Connect plugins. What should you do?

Options
A. Deploy a Kafka cluster on GCE VM Instances. Configure your on-prem cluster to mirror your topics to the cluster running in GCE. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS.
B. Deploy a Kafka cluster on GCE VM Instances with the Pub/Sub Kafka connector configured as a Sink connector. Use a Dataproc cluster or Dataflow job to read from Kafka and write to GCS.
C. Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Source connector. Use a Dataflow job to read from Pub/Sub and write to GCS.
D. Deploy the Pub/Sub Kafka connector to your on-prem Kafka cluster and configure Pub/Sub as a Sink connector. Use a Dataflow job to read from Pub/Sub and write to GCS.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Mirroring to a Kafka cluster on GCE preserves the mirroring model and avoids installing connectors on-prem. Once replicated into GCE, Dataproc or Dataflow can read and write to GCS/BigQuery.
B. Incorrect — This requires configuring connectors and departs from the mirroring goal; also more operational overhead.
C. Incorrect — Installing the Pub/Sub Kafka connector on-prem contradicts the “avoid deployment of Kafka Connect plugins” preference.
D. Incorrect — Same as C; it installs connectors on-prem which you wanted to avoid.
E. Incorrect — Since A is correct, “None of the above” is not valid.

Question 366
You've migrated a Hadoop job from an on-prem cluster to Dataproc and GCS. Your Spark job is a complicated analytical workload that consists of many shuffling operations and initial data are parquet files (on average 200–400 MB size each). You see some degradation in performance after the migration to Dataproc, so you'd like to optimize for it. You need to keep in mind that your organization is very cost-sensitive, so you'd like to continue using Dataproc on preemptibles (with 2 non-preemptible workers only) for this workload. What should you do?

Options
A. Increase the size of your parquet files to ensure them to be 1 GB minimum.
B. Switch to TFRecords formats (appr. 200MB per file) instead of parquet files.
C. Switch from HDDs to SSDs, copy initial data from GCS to HDFS, run the Spark job and copy results back to GCS.
D. Switch from HDDs to SSDs, override the preemptible VMs configuration to increase the boot disk size.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Simply enlarging parquet files to 1 GB may help split/read patterns but can hurt parallelism and is not the most direct fix for shuffle/IO bottlenecks.
B. Incorrect — TFRecords are not inherently better than Parquet for analytical Spark workloads; Parquet is columnar and usually preferable.
C. Incorrect — Copying large data to HDFS increases costs and operational overhead; with GCS optimized connectors, this is generally unnecessary.
D. Correct — Using SSDs for worker boot disks reduces shuffle and disk I/O latency. Increasing boot disk size for preemptible VMs can avoid I/O throttling and improve stability for heavy shuffles while staying on preemptibles (cost-sensitive).
E. Incorrect — Since D is correct, “None of the above” is not valid.

Question 367
Your team is responsible for developing and maintaining ETLs in your company. One of your Dataflow jobs is failing because of some errors in the input data, and you need to improve reliability of the pipeline (incl. being able to reprocess all failing data). What should you do?

Options
A. Add a filtering step to skip these types of errors in the future, extract erroneous rows to logs, and transform the data so that it can be stored to Pub/Sub later.
B. Stop the pipeline and drop all bad records to keep pipeline steady.
C. Write a separate ad-hoc job that scans historical data and corrects erroneous rows periodically.
D. Add explicit error-handling with dead-letter side outputs that capture malformed/error records into durable storage (e.g., Cloud Storage or a dead-letter BigQuery table) and instrument the pipeline to allow reprocessing those dead records after correction.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Filtering and logging alone may lose context and doesn’t provide an easy, durable way to reprocess failed rows.
B. Incorrect — Dropping bad records loses data and is not acceptable when reprocessing is required.
C. Incorrect — An ad-hoc scanner is brittle and delays time-to-recovery; it doesn't integrate with streaming reliability.
D. Correct — Using side outputs / dead-letter queues that write malformed records to durable storage lets you inspect, fix, and re-ingest faulty rows. This improves reliability and enables deterministic reprocessing.
E. Incorrect — Since D is correct, “None of the above” is not valid.

Question 368
You're training a model to predict housing prices based on an available dataset with real estate properties. Your plan is to train a fully connected neural net, and you've discovered that the dataset contains latitude and longitude of the property. Real estate professionals have told you that the location of the property is highly influential on price, so you'd like to engineer a feature that incorporates this physical dependency. What should you do?

Options
A. Provide latitude and longitude as input vectors to your neural net.
B. Create a numeric column from a feature cross of latitude and longitude.
C. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L1 regularization during optimization.
D. Create a feature cross of latitude and longitude, bucketize it at the minute level and use L2 regularization during optimization.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Passing raw lat/long may not capture localized effects; the model might struggle to learn fine-grained locality patterns alone.
B. Incorrect — A simple numeric cross (e.g., lat*lon) is arbitrary and unlikely to capture geographic proximity semantics.
C. Correct — Feature-crossing lat/long and bucketizing to fine-grained spatial bins (minute-level) creates discrete location features that let the model learn locality-dependent behaviors; L1 regularization helps sparsify high-dimensional crossed features and control overfitting.
D. Incorrect — L2 regularization helps shrink weights but doesn't encourage sparsity as L1 does, which is often preferable with high-cardinality crossed features.
E. Incorrect — Since C is correct, "None of the above" is not valid.

Question 369
You are deploying MariaDB SQL databases on GCE VM Instances and need to configure monitoring and alerting. You want to collect metrics including network connections, disk IO and replication status from MariaDB with minimal development effort and use Stackdriver for dashboards and alerts. What should you do?

Options
A. Install the OpenCensus Agent and create a custom metric collection application with a Stackdriver exporter.
B. Place the MariaDB instances in an Instance Group with a Health Check.
C. Install the Stackdriver Logging Agent and configure fluentd in_tail plugin to read MariaDB logs.
D. Install the Stackdriver Agent and configure the MySQL plugin.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Building a custom OpenCensus exporter requires more development effort than necessary.
B. Incorrect — Health checks are useful for instance availability but do not capture DB-specific metrics like replication status.
C. Incorrect — Log collection helps with errors but does not provide structured DB metrics out of the box.
D. Correct — The Stackdriver (Ops Agent/Monitoring agent) with the MySQL plugin collects DB-specific metrics (connections, disk IO, replication info) with minimal development and integrates with Stackdriver dashboards and alerting.
E. Incorrect — Since D is correct, "None of the above" is not valid.

Question 371
You work for a bank. You have a labelled dataset that contains information on already granted loan application and whether these applications have been defaulted. You have been asked to train a model to predict default rates for credit applicants. What should you do?

Options
A. Increase the size of the dataset by collecting additional data.
B. Train a linear regression to predict a credit default risk score.
C. Remove the bias from the data and collect applications that have been declined loans.
D. Match loan applicants with their social profiles to enable feature engineering.
E. None of the above

Answer
A

Detailed explanation
A. Correct — Increasing dataset size (especially adding diverse, representative examples) generally improves model generalization and helps reduce variance. Gathering more labeled examples is a fundamental first step.
B. Incorrect — Default prediction is a classification problem (default vs. no-default); linear regression is not the appropriate primary model choice. A logistic classifier or other classification model is more suitable.
C. Incorrect — While addressing bias is important, indiscriminately collecting declined applications may introduce label/covariate issues and legal/ethical complexities; fixing bias requires careful study and possibly targeted data collection, not a blanket removal.
D. Incorrect — Matching social profiles raises serious privacy, compliance, and fairness concerns and may be illegal or unethical depending on jurisdiction; it is not an appropriate first step.
E. Incorrect — Since A is correct, "None of the above" is not valid.

Question 372
You need to migrate a 2TB relational database to Google Cloud Platform. You do not have the resources to significantly refactor the application that uses this database and cost to operate is of primary concern. Which service do you select for storing and serving your data?

Options
A. Cloud Spanner
B. Cloud Bigtable
C. Cloud Firestore
D. Cloud SQL
E. None of the above

Answer
D

Detailed explanation
A. Incorrect — Cloud Spanner provides horizontal scaling and strong consistency, but it is more expensive and requires schema/operation changes; not ideal when cost and minimal refactor are primary concerns.
B. Incorrect — Cloud Bigtable is for wide-column, high-throughput workloads (time-series, etc.) and does not provide relational semantics/SQL compatibility required by a relational app.
C. Incorrect — Cloud Firestore is a document database (NoSQL) and would require application refactoring; not suitable for a lift-and-shift relational database.
D. Correct — Cloud SQL is a managed relational database (MySQL/Postgres) that best fits a 2TB relational DB migration with minimal application changes and relatively lower operational cost compared with Spanner for this scale.
E. Incorrect — Since D is correct, “None of the above” is not valid.

Question 373
You're using Bigtable for a real-time application, and you have a heavy load that is a mix of reads and writes. You've recently identified an additional use case and need to perform an hourly analytical job to calculate certain statistics across the whole database. You need to ensure both the reliability of your production application as well as the analytical workload. What should you do?

Options
A. Export Bigtable dump to GCS and run your analytical job on top of the exported files.
B. Add a second cluster to an existing instance with a multi-cluster routing, use live-traffic app profile for your regular workload and batch-analytics profile for the analytics workload.
C. Add a second cluster to an existing instance with a single-cluster routing, use live-traffic app profile for your regular workload and batch-analytics profile for the analytics workload.
D. Increase the size of your existing cluster twice and execute your analytics workload on your new resized cluster.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Exporting dumps to GCS for each hourly job introduces latency and operational overhead; it also doesn't provide near-real-time analysis.
B. Incorrect — Multi-cluster routing distributes live traffic across clusters and can affect latency for the production app; it is not the recommended pattern for isolating analytics workloads.

C. Correct — Adding a second cluster in the same instance and using single-cluster routing lets you direct production traffic to the live-traffic cluster and run batch analytics on the second cluster without impacting production. This isolates workloads while maintaining a single logical instance.
D. Incorrect — Doubling cluster size to run analytics on the same nodes risks impacting production reliability and is inefficient compared to adding a separate cluster for analytics.
E. Incorrect — Since C is correct, “None of the above” is not valid.

Question 374
You are designing an Apache Beam pipeline to enrich data from Cloud Pub/Sub with static reference data from BigQuery. The reference data is small enough to fit in memory on a single worker. The pipeline should write enriched results to BigQuery for analysis. Which job type and transforms should this pipeline use?

Options
A. Batch job, PubSubIO, side-inputs
B. Streaming job, PubSubIO, JdbcIO, side-outputs
C. Streaming job, PubSubIO, BigQueryIO, side-inputs
D. Streaming job, PubSubIO, BigQueryIO, side-outputs
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Pub/Sub is a streaming source; choosing a batch job would not support real-time ingestion.
B. Incorrect — JdbcIO is for JDBC sources; unnecessary here since reference data is in BigQuery and side-outputs are for splitting outputs, not for providing small reference data to workers.
C. Correct — A streaming job using PubSubIO (for real-time Pub/Sub messages) and BigQueryIO (for writes) plus side-inputs (to broadcast small in-memory reference data to workers) is the appropriate design.
D. Incorrect — Side-outputs are for branching outputs (e.g., errors), not for providing reference data; side-inputs are correct for static reference data.
E. Incorrect — Since C is correct, “None of the above” is not valid.

Question 375
You have a data pipeline that writes data to Cloud Bigtable using well-designed row keys. You want to monitor your pipeline to determine when to increase the size of your Cloud Bigtable cluster. Which two actions can you take to accomplish this? (Choose two.)

Options
A. Review Key Visualizer metrics. Increase the size of the Cloud Bigtable cluster when the Read pressure index is above 100.
B. Review Key Visualizer metrics. Increase the size of the Cloud Bigtable cluster when the Write pressure index is above 100.
C. Monitor the latency of write operations. Increase the size of the Cloud Bigtable cluster when there is a sustained increase in write latency.
D. Monitor storage utilization. Increase the size of the Cloud Bigtable cluster when utilization increases above 70% of max capacity.
E. Monitor latency of read operations. Increase the size of the Cloud Bigtable cluster if read operations take longer than 100 ms.

Answer
C and **D**

Detailed explanation
A. Incorrect — Key Visualizer shows pressure but the arbitrary threshold "above 100" is not a standard guidance; pressure trends and specific metrics matter.
B. Incorrect — Same reasoning as A; simply relying on "Write pressure index >100" without contextual thresholds is not the recommended approach.
C. Correct — Sustained increases in write latency indicate the cluster is under-provisioned for throughput; scaling the cluster is appropriate.
D. Correct — Monitoring storage utilization and scaling before reaching capacity (e.g., above a conservative threshold such as ~70%) prevents issues; increasing nodes helps capacity.
E. Incorrect — While read latency matters, a single threshold like 100 ms is not universally applicable; monitoring trends and appropriate SLOs is required. (Therefore E is not one of the recommended two actions here.)

Question 376
You want to analyze hundreds of thousands of social media posts daily at the lowest cost and with the fewest steps. You have the following requirements:
☞ You will batch-load the posts once per day and run them through the Cloud Natural Language API.
☞ You will extract topics and sentiment from the posts.
☞ You must store the raw posts for archiving and reprocessing.
☞ You will create dashboards to be shared with people both inside and outside your organization.
You need to store both the data extracted from the API to perform analysis as well as the raw social media posts for historical archiving. What should you do?

Options
A. Store the social media posts and the data extracted from the API in BigQuery.
B. Store the social media posts and the data extracted from the API in Cloud SQL.
C. Store the raw social media posts in Cloud Storage, and write the data extracted from the API into BigQuery.
D. Feed the social media posts into the API directly from the source, and write the extracted data from the API into BigQuery.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Storing raw posts in BigQuery is possible but inefficient and potentially costly for large unstructured raw archives; BigQuery is best for structured analytic results.
B. Incorrect — Cloud SQL is not cost-effective or scalable for analytical queries and dashboards at this scale.
C. Correct — Raw posts in Cloud Storage (low-cost archive, reprocessable) and structured API outputs in BigQuery (fast SQL analytics and dashboards) is the recommended, cost-effective pattern.
D. Incorrect — Direct streaming from source to the API ignores the requirement to archive raw posts for reprocessing and may complicate error handling.
E. Incorrect — Since C is correct, “None of the above” is not valid.

Question 377
You store historic data in Cloud Storage. You need to perform analytics on the historic data. You want to use a solution to detect invalid data entries and perform data transformations that will not require programming or knowledge of SQL. What should you do?

Options
A. Use Cloud Dataflow with Beam to detect errors and perform transformations.
B. Use Cloud Dataprep with recipes to detect errors and perform transformations.
C. Use Cloud Dataproc with a Hadoop job to detect errors and perform transformations.
D. Use federated tables in BigQuery with queries to detect errors and perform transformations.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — Dataflow/Beam requires programming and pipeline development experience.
B. Correct — Cloud Dataprep is a visual, no-code/low-code data preparation tool that uses recipes to detect errors and perform transformations without SQL/programming.
C. Incorrect — Dataproc requires writing Hadoop/Spark jobs (programming) and is not no-code.
D. Incorrect — Federated tables require SQL queries to detect/transform data; this contradicts the “no programming/SQL knowledge” requirement.
E. Incorrect — Since B is correct, “None of the above” is not valid.

Question 378
Your company needs to upload their historic data to Cloud Storage. The security rules don't allow access from external IPs to their on-premises resources. After an initial upload, they will add new data from existing on-premises applications every day. What should they do?

Options
A. Execute gsutil rsync from the on-premises servers.
B. Use Dataflow and write the data to Cloud Storage.
C. Write a job template in Dataproc to perform the data transfer.
D. Install an FTP server on a Compute Engine VM to receive the files and move them to Cloud Storage.
E. None of the above

Answer
A

Detailed explanation
A. Correct — gsutil rsync can be run from on-prem servers (which have access to internal networks) to push files to Cloud Storage; it is straightforward and suitable for initial and incremental uploads.
B. Incorrect — Dataflow is for streaming/transform pipelines and requires connectivity; it's heavier than needed for simple file uploads.
C. Incorrect — Dataproc introduces unnecessary complexity for file transfer tasks.
D. Incorrect — Installing an FTP server exposes additional security and operational burden and violates the constraint about external access.
E. Incorrect — Since A is correct, "None of the above" is not valid.

Question 379
You have a query that filters a BigQuery table using a WHERE clause on timestamp and ID columns. By using bq query --dry_run you learn that the query triggers a full scan of the table, even though the filter on timestamp and ID select a tiny fraction of the overall data. You want to reduce the amount of data scanned by BigQuery with minimal changes to existing SQL queries. What should you do?

Options
A. Create a separate table for each ID.
B. Use the LIMIT keyword to reduce the number of rows returned.
C. Recreate the table with a partitioning column and clustering column.
D. Use the bq query --maximum_bytes_billed flag to restrict the number of bytes billed.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect — Creating a separate table per ID is unmanageable and not recommended.
B. Incorrect — LIMIT affects returned rows but does not reduce bytes scanned; BigQuery still scans the underlying data.
C. Correct — Partitioning (e.g., by DATE/TIMESTAMP) enables pruning by time, and clustering by ID allows efficient locality-based pruning; together they reduce bytes scanned with minimal SQL changes.
D. Incorrect — --maximum_bytes_billed only limits billing but doesn't reduce scanned data or improve performance.
E. Incorrect — Since C is correct, "None of the above" is not valid.

Question 380
You have a requirement to insert minute-resolution data from 50,000 sensors into a BigQuery table. You expect significant growth in data volume and need the data to be available within 1 minute of ingestion for real-time analysis of aggregated trends. What should you do?

Options
A. Use bq load to load a batch of sensor data every 60 seconds.
B. Use a Cloud Dataflow pipeline to stream data into the BigQuery table.
C. Use the INSERT statement to insert a batch of data every 60 seconds.
D. Use the MERGE statement to apply updates in batch every 60 seconds.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect — bq load is a batch load operation and introduces latency and operational overhead; it does not meet <1 minute availability reliably.
B. Correct — A Dataflow streaming pipeline can continuously ingest from sensors and write to BigQuery streaming inserts, meeting the <1 minute availability and scaling with growth.
C. Incorrect — Using INSERT in batches every 60 seconds is not scalable for high-volume, low-latency streaming and may be inefficient.
D. Incorrect — MERGE is used for updates/upsserts and is not optimal for high-throughput streaming ingestion.
E. Incorrect — Since B is correct, "None of the above" is not valid.

Question 381
You need to copy millions of sensitive patient records from a relational database to BigQuery. The total size of the database is 10 TB. You need to design a solution that is secure and time-efficient. What should you do?

Options
A. Export the records from the database as an Avro file. Upload the file to GCS using gsutil, and then load the Avro file into BigQuery using the BigQuery web UI in the GCP Console.
B. Export the records from the database as an Avro file. Copy the file onto a Transfer Appliance and send it to Google, and then load the Avro file into BigQuery using the BigQuery web UI in the GCP Console.
C. Export the records from the database into a CSV file. Create a public URL for the CSV file, and then use Storage Transfer Service to move the file to Cloud Storage. Load the CSV file into BigQuery using the BigQuery web UI in the GCP Console.
D. Export the records from the database as an Avro file. Create a public URL for the Avro file, and then use Storage Transfer Service to move the file to Cloud Storage. Load the Avro file into BigQuery using the BigQuery web UI in the GCP Console.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – gsutil upload for 10 TB will be slow and network-bound. Not ideal for such a large dataset.
B. Correct – Transfer Appliance is designed for securely and efficiently moving large datasets (10 TB+). Once in GCS, data can be loaded into BigQuery.
C. Incorrect – CSV is inefficient for large datasets (no schema, larger file size, slower load). Also, making it public violates security requirements.

D. Incorrect – Making Avro public is insecure. Also, Storage Transfer Service is not needed for local DB exports.
E. Incorrect – One valid option exists (B).

Question 382
You need to create a near real-time inventory dashboard that reads the main inventory tables in your BigQuery data warehouse. Historical inventory data is stored as inventory balances by item and location. You have several thousand updates to inventory every hour. You want to maximize performance of the dashboard and ensure that the data is accurate. What should you do?

Options
A. Leverage BigQuery UPDATE statements to update the inventory balances as they are changing.
B. Partition the inventory balance table by item to reduce the amount of data scanned with each inventory update.
C. Use the BigQuery streaming to stream changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly.
D. Use the BigQuery bulk loader to batch load inventory changes into a daily inventory movement table. Calculate balances in a view that joins it to the historical inventory balance table. Update the inventory balance table nightly.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – BigQuery UPDATE is expensive and not efficient for frequent changes.
B. Incorrect – Partitioning helps query performance, but doesn't solve the update requirement for near real-time accuracy.
C. Correct – Streaming ingestion to a movement table allows near real-time updates. Balances can be computed via a view, and nightly updates maintain efficiency.
D. Incorrect – Bulk loads introduce latency, not real-time.
E. Incorrect – A valid solution exists (C).

Question 383
You have data stored in BigQuery. The data in the BigQuery dataset must be highly available. You need to define a storage, backup, and recovery strategy of this data that minimizes cost. How should you configure the BigQuery table that have a recovery point objective (RPO) of 30 days?

Options
A. Set the BigQuery dataset to be regional. In the event of an emergency, use a point-in-time snapshot to recover the data.
B. Set the BigQuery dataset to be regional. Create a scheduled query to make copies of the data to tables suffixed with the time of the backup. In the event of an emergency, use the backup copy of the table.
C. Set the BigQuery dataset to be multi-regional. In the event of an emergency, use a point-in-time snapshot to recover the data.
D. Set the BigQuery dataset to be multi-regional. Create a scheduled query to make copies of the data to tables suffixed with the time of the backup. In the event of an emergency, use the backup copy of the table.
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Regional datasets don't provide the highest availability.
B. Incorrect – Manual scheduled copies increase cost unnecessarily and don't guarantee high availability.
C. Correct – Multi-regional datasets provide redundancy and high availability. BigQuery offers 7-day point-in-time recovery by default. For RPO within 30 days, this is cost-effective and reliable.
D. Incorrect – Scheduled queries add cost; multi-region already covers availability.
E. Incorrect – Correct solution exists (C).

Question 384
You used Dataprep to create a recipe on a sample of data in a BigQuery table. You want to reuse this recipe on a daily upload of data with the same schema, after the load job with variable execution time completes. What should you do?

Options
A. Create a cron schedule in Dataprep.
B. Create an App Engine cron job to schedule the execution of the Dataprep job.
C. Export the recipe as a Dataprep template, and create a job in Cloud Scheduler.
D. Export the Dataprep job as a Dataflow template, and incorporate it into a Composer job.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Dataprep's native cron won't handle variable execution times properly.
B. Incorrect – App Engine cron jobs don't integrate well with variable timing of load jobs.
C. Incorrect – Cloud Scheduler won't track data availability, only time.
D. Correct – Converting the recipe to a Dataflow template allows flexible orchestration. Composer (Airflow) manages dependencies and variable job timing.
E. Incorrect – A valid option exists (D).

Question 385
You want to automate execution of a multi-step data pipeline running on Google Cloud. The pipeline includes Dataproc and Dataflow jobs that have multiple dependencies on each other. You want to use managed services where possible, and the pipeline will run every day. Which tool should you use?

Options
A. cron
B. Cloud Composer
C. Cloud Scheduler
D. Workflow Templates on Dataproc
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – cron jobs are too simple for multi-step dependencies.
B. Correct – Cloud Composer (managed Apache Airflow) is designed for orchestrating dependent workflows across multiple services like Dataproc and Dataflow.
C. Incorrect – Cloud Scheduler only triggers jobs; it doesn't handle dependencies.
D. Incorrect – Workflow Templates work only for Dataproc, not for multi-service orchestration.
E. Incorrect – A valid option exists (B).

Question 386
You are managing a Cloud Dataproc cluster. You need to make a job run faster while minimizing costs, without losing work in progress on your clusters. What should you do?

Options
A. Increase the cluster size with more non-preemptible workers.
B. Increase the cluster size with preemptible worker nodes, and configure them to forcefully decommission.
C. Increase the cluster size with preemptible worker nodes, and use Cloud Stackdriver to trigger a script to preserve work.
D. Increase the cluster size with preemptible worker nodes, and configure them to use graceful decommissioning.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Non-preemptible workers are more reliable but costlier. Not cost-minimizing.
B. Incorrect – Forceful decommissioning loses in-progress work.
C. Incorrect – Stackdriver-based scripts add complexity and aren't native.
D. Correct – Preemptible nodes are cheaper, and graceful decommissioning ensures in-progress tasks are not lost when nodes are reclaimed.
E. Incorrect – Correct solution exists (D).

Question 387
You work for a shipping company that uses handheld scanners to read shipping labels. Your company has strict data privacy standards that require scanners to only transmit tracking numbers when events are sent to Kafka topics. A recent software update caused the scanners to accidentally transmit recipients' personally identifiable information (PII) to analytics systems, which violates user privacy rules. You want to quickly build a scalable solution using cloud-native managed services to prevent exposure of PII to the analytics systems. What should you do?

Options
A. Create an authorized view in BigQuery to restrict access to tables with sensitive data.
B. Install a third-party data validation tool on Compute Engine virtual machines to check the incoming data for sensitive information.
C. Use Cloud Logging to analyze the data passed through the total pipeline to identify transactions that may contain sensitive information.
D. Build a Cloud Function that reads the topics and makes a call to the Cloud Data Loss Prevention (Cloud DLP) API. Use the tagging and confidence levels to either pass or quarantine the data in a bucket for review.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Authorized views restrict query access, but PII already reached analytics systems.
B. Incorrect – Compute Engine + third-party tool adds operational overhead, not quick or fully managed.
C. Incorrect – Cloud Logging only monitors after data has passed through, not preventive.
D. Correct – Cloud Functions + Cloud DLP API inspects and filters PII in real time, preventing exposure. Fully managed and scalable.
E. Incorrect – One correct solution exists (D).

Question 388
You have developed three data processing jobs. One executes a Cloud Dataflow pipeline that transforms data uploaded to Cloud Storage and writes results to BigQuery. The second ingests data from on-premises servers and uploads it to Cloud Storage. The third is a Cloud Dataflow pipeline that gets information from third-party data providers and uploads the information to Cloud Storage. You need to be able to schedule and monitor the execution of these three workflows and manually execute them when needed. What should you do?

Options
A. Create a Directed Acyclic Graph in Cloud Composer to schedule and monitor the jobs.
B. Use Stackdriver Monitoring and set up an alert with a Webhook notification to trigger the jobs.
C. Develop an App Engine application to schedule and request the status of the jobs using GCP API calls.
D. Set up cron jobs in a Compute Engine instance to schedule and monitor the pipelines using GCP API calls.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Cloud Composer (Airflow) can orchestrate multiple jobs, monitor them, and allow manual triggering when required.
B. Incorrect – Monitoring with alerts doesn't provide scheduling or orchestration.
C. Incorrect – Building a custom App Engine app adds unnecessary complexity.
D. Incorrect – Cron jobs lack monitoring and robust orchestration.
E. Incorrect – Correct solution exists (A).

Question 389
You have Cloud Functions written in Node.js that pull messages from Cloud Pub/Sub and send the data to BigQuery. You observe that the message processing rate on the Pub/Sub topic is orders of magnitude higher than anticipated, but there is no error logged in Cloud Logging. What are the two most likely causes of this problem? (Choose two.)

Options
A. Publisher throughput quota is too small.
B. Total outstanding messages exceed the 10-MB maximum.
C. Error handling in the subscriber code is not handling run-time errors properly.
D. The subscriber code cannot keep up with the messages.
E. The subscriber code does not acknowledge the messages that it pulls.

Answer
C, E

Detailed explanation
A. Incorrect – Throughput quota would limit publishing, not cause overconsumption without logs.
B. Incorrect – 10 MB limit affects individual messages, not system-wide throughput.
C. Correct – Poor error handling may silently drop or retry messages, making processing inconsistent.
D. Incorrect – If subscriber is too slow, backlog grows, but not higher processing rate.
E. Correct – If acknowledgments are missing, Pub/Sub re-delivers messages repeatedly, inflating processing rate.

Question 390
You are creating a new pipeline in Google Cloud to stream IoT data from Cloud Pub/Sub through Cloud Dataflow to BigQuery. While previewing the data, you notice that roughly 2% of the data appears to be corrupt. You need to modify the Cloud Dataflow pipeline to filter out this corrupt data. What should you do?

Options
A. Add a SideInput that returns a Boolean if the element is corrupt.
B. Add a ParDo transform in Cloud Dataflow to discard corrupt elements.
C. Add a Partition transform in Cloud Dataflow to separate valid data from corrupt data.
D. Add a GroupByKey transform in Cloud Dataflow to group all of the valid data together and discard the rest.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – SideInput is not used for filtering, but for supplying additional static data.
B. Correct – A ParDo transform allows per-element custom logic to check and discard corrupt records. Simple and efficient.
C. Incorrect – Partitioning could separate corrupt vs valid, but more complex than necessary.
D. Incorrect – GroupByKey is for aggregation, not filtering.
E. Incorrect – Valid solution exists (B).

Question 391
You have historical data covering the last three years in BigQuery and a data pipeline that delivers new data to BigQuery daily. You have noticed that when the Data Science team runs a query filtered on a date column and limited to 30–90 days of data, the query scans the entire table. You also noticed that your bill is increasing more quickly than you expected. You want to resolve the issue as cost-effectively as possible while maintaining the ability to conduct SQL queries. What should you do?

Options
A. Re-create the tables using DDL. Partition the tables by a column containing a TIMESTAMP or DATE Type.
B. Recommend that the Data Science team export the table to a CSV file on Cloud Storage and use Cloud Datalab to explore the data by reading the files directly.
C. Modify your pipeline to maintain the last 30–90 days of data in one table and the longer history in a different table to minimize full table scans over the entire history.
D. Write an Apache Beam pipeline that creates a BigQuery table per day. Recommend that the Data Science team use wildcards on the table name suffixes to select the data they need.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Partitioning by DATE or TIMESTAMP ensures queries scan only relevant partitions, reducing cost and improving performance.
B. Incorrect – Exporting to CSV increases complexity and cost, and loses the advantages of BigQuery.
C. Incorrect – Splitting tables manually makes querying more complex, less cost-effective compared to native partitioning.
D. Incorrect – Daily table creation is messy and less efficient than partitioning.
E. Incorrect – A valid solution exists (A).

Question 392
You operate a logistics company, and you want to improve event delivery reliability for vehicle-based sensors. You operate small data centers around the world to capture these events, but leased lines that provide connectivity from your event collection infrastructure to your event processing infrastructure are unreliable, with unpredictable latency. You want to address this issue in the most cost-effective way. What should you do?

Options
A. Deploy small Kafka clusters in your data centers to buffer events.
B. Have the data acquisition devices publish data to Cloud Pub/Sub.
C. Establish a Cloud Interconnect between all remote data centers and Google.
D. Write a Cloud Dataflow pipeline that aggregates all data in session windows.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Kafka clusters add management overhead and cost.
B. Correct – Cloud Pub/Sub is managed, scalable, and cost-effective for unreliable connectivity. Devices can publish directly and buffer automatically.
C. Incorrect – Cloud Interconnect is expensive and not necessary for this use case.
D. Incorrect – Dataflow aggregation doesn't solve unreliable event delivery.
E. Incorrect – One valid option exists (B).

Question 393
You are a retailer that wants to integrate your online sales capabilities with different in-home assistants, such as Google Home. You need to interpret customer voice commands and issue an order to the backend systems. Which solutions should you choose?

Options
A. Speech-to-Text API
B. Cloud Natural Language API
C. Dialogflow Enterprise Edition
D. AutoML Natural Language
E. None of the above

Answer
C

Detailed explanation
A. Incorrect – Speech-to-Text only transcribes, doesn't handle intent or order fulfillment.
B. Incorrect – Natural Language API analyzes text but doesn't handle conversational flow.
C. Correct – Dialogflow Enterprise Edition is built for conversational AI, integrates with Google Home, and can issue orders to backends.
D. Incorrect – AutoML Natural Language is for training custom NLP models, not conversational integration.
E. Incorrect – A correct option exists (C).

Question 394
Your company has a hybrid cloud initiative. You have a complex data pipeline that moves data between cloud provider services and leverages services from each of the cloud providers. Which cloud-native service should you use to orchestrate the entire pipeline?

Options
A. Cloud Dataflow
B. Cloud Composer
C. Cloud Dataprep
D. Cloud Dataproc
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – Dataflow is for data processing, not orchestration.
B. Correct – Cloud Composer (Airflow) is built for orchestration of complex multi-cloud and multi-service pipelines.
C. Incorrect – Dataprep is for data cleaning/transformation, not orchestration.
D. Incorrect – Dataproc is for Hadoop/Spark clusters, not workflow orchestration.
E. Incorrect – One correct solution exists (B).

Question 395
You use a dataset in BigQuery for analysis. You want to provide third-party companies with access to the same dataset. You need to keep the costs of data sharing low and ensure that the data is current. Which solution should you choose?

Options
A. Use Analytics Hub to control data access, and provide third party companies with access to the dataset.
B. Use Cloud Scheduler to export the data on a regular basis to Cloud Storage, and provide third-party companies with access to the bucket.
C. Create a separate dataset in BigQuery that contains the relevant data to share, and provide third-party companies with access to the new dataset.
D. Create a Dataflow job that reads the data in frequent time intervals, and writes it to the relevant BigQuery dataset or Cloud Storage bucket for third-party companies to use.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Analytics Hub allows controlled sharing of live BigQuery datasets, cost-efficient and always up to date.
B. Incorrect – Exporting increases storage costs and requires ongoing scheduling.
C. Incorrect – Creating duplicate datasets raises storage costs.
D. Incorrect – Dataflow adds complexity and unnecessary processing cost.
E. Incorrect – Valid option exists (A).

Question 396

Your company is in the process of migrating its on-premises data warehousing solutions to BigQuery. The existing data warehouse uses trigger-based change data capture (CDC) to apply updates from multiple transactional database sources on a daily basis. With BigQuery, your company hopes to improve its handling of CDC so that changes to the source systems are available to query in BigQuery in near-real time using log-based CDC streams, while also optimizing for the performance of applying changes to the data warehouse. Which two steps should they take to ensure that changes are available in the BigQuery reporting table with minimal latency while reducing compute overhead? (Choose two.)

Options

A. Perform a DML INSERT, UPDATE, or DELETE to replicate each individual CDC record in real time directly on the reporting table.
B. Insert each new CDC record and corresponding operation type to a staging table in real time.
C. Periodically DELETE outdated records from the reporting table.
D. Periodically use a DML MERGE to perform several DML INSERT, UPDATE, and DELETE operations at the same time on the reporting table.
E. Insert each new CDC record and corresponding operation type in real time to the reporting table, and use a materialized view to expose only the newest version of each unique record.

Answer

B, D

Detailed explanation

A. Incorrect – Real-time DML on reporting tables is inefficient and costly.
B. Correct – Staging tables absorb real-time CDC inserts efficiently.
C. Incorrect – Deletes don't improve latency or reduce overhead.
D. Correct – Periodic MERGE operations apply batch changes efficiently, reducing compute overhead.
E. Incorrect – Materialized views add overhead and don't minimize latency as effectively as staging + merge.

Question 397
You are designing a data processing pipeline. The pipeline must be able to scale automatically as load increases. Messages must be processed at least once and must be ordered within windows of 1 hour. How should you design the solution?

Options
A. Use Apache Kafka for message ingestion and use Cloud Dataproc for streaming analysis.
B. Use Apache Kafka for message ingestion and use Cloud Dataflow for streaming analysis.
C. Use Cloud Pub/Sub for message ingestion and Cloud Dataproc for streaming analysis.
D. Use Cloud Pub/Sub for message ingestion and Cloud Dataflow for streaming analysis.
E. None of the above

Answer
D

Detailed explanation
A. Incorrect – Kafka + Dataproc is harder to manage and not fully cloud-native.
B. Incorrect – Kafka ingestion adds operational complexity, Cloud Pub/Sub is preferred.
C. Incorrect – Cloud Dataproc isn't ideal for streaming windowed processing compared to Dataflow.
D. Correct – Pub/Sub provides scalable ingestion with at-least-once delivery. Dataflow supports ordered processing in time windows.
E. Incorrect – A valid option exists (D).

Question 398
You need to set access to BigQuery for different departments within your company. Your solution should comply with the following requirements:

- Each department should have access only to their data.
- Each department will have one or more leads who need to be able to create and update tables and provide them to their team.
- Each department has data analysts who need to be able to query but not modify data.

How should you set access to the data in BigQuery?

Options
A. Create a dataset for each department. Assign the department leads the role of OWNER, and assign the data analysts the role of WRITER on their dataset.
B. Create a dataset for each department. Assign the department leads the role of WRITER, and assign the data analysts the role of READER on their dataset.
C. Create a table for each department. Assign the department leads the role of Owner, and assign the data analysts the role of Editor on the project the table is in.
D. Create a table for each department. Assign the department leads the role of Editor, and assign the data analysts the role of Viewer on the project the table is in.
E. None of the above

Answer
B

Detailed explanation
A. Incorrect – OWNER gives excessive privileges to leads; analysts as WRITER could modify data.
B. Correct – WRITER for leads allows creating and updating tables. READER for analysts allows querying only. Dataset-level separation ensures departmental isolation.
C. Incorrect – Project-level roles are too broad and risk access leakage.
D. Incorrect – Similar to C, roles are too broad and not aligned with requirements.
E. Incorrect – A correct solution exists (B).

Question 399
You operate a database that stores stock trades and an application that retrieves average stock price for a given company over an adjustable window of time. The data is stored in Cloud Bigtable where the datetime of the stock trade is the beginning of the row key. Your application has thousands of concurrent users, and you notice that performance is starting to degrade as more stocks are added. What should you do to improve the performance of your application?

Options
A. Change the row key syntax in your Cloud Bigtable table to begin with the stock symbol.
B. Change the row key syntax in your Cloud Bigtable table to begin with a random number per second.
C. Change the data pipeline to use BigQuery for storing stock trades, and update your application.
D. Use Cloud Dataflow to write a summary of each day's stock trades to an Avro file on Cloud Storage. Update your application to read from Cloud Storage and Cloud Bigtable to compute the responses.
E. None of the above

Answer
A

Detailed explanation
A. Correct – Starting the row key with stock symbol distributes reads/writes evenly across tablets, reducing hotspotting and improving concurrency performance.
B. Incorrect – Randomizing row keys loses natural ordering needed for range scans.
C. Incorrect – BigQuery isn’t ideal for low-latency transactional lookups.
D. Incorrect – Adding Cloud Storage and Avro introduces latency and complexity.
E. Incorrect – Correct solution exists (A).

Question 400

You are operating a Cloud Dataflow streaming pipeline. The pipeline aggregates events from a Cloud Pub/Sub subscription source, within a window, and sinks the resulting aggregation to a Cloud Storage bucket. The source has consistent throughput. You want to monitor an alert on behavior of the pipeline with Cloud Stackdriver to ensure that it is processing data. Which Stackdriver alerts should you create?

Options

A. An alert based on a decrease of subscription/num_undelivered_messages for the source and a rate of change increase of instance/storage/used_bytes for the destination.
B. An alert based on an increase of subscription/num_undelivered_messages for the source and a rate of change decrease of instance/storage/used_bytes for the destination.
C. An alert based on a decrease of instance/storage/used_bytes for the source and a rate of change increase of subscription/num_undelivered_messages for the destination.
D. An alert based on an increase of instance/storage/used_bytes for the source and a rate of change decrease of subscription/num_undelivered_messages for the destination.
E. None of the above

Answer

B

Detailed explanation

A. Incorrect – A decrease in undelivered messages means good processing, not an error.
B. Correct – If undelivered messages increase (backlog growing) while storage writes decrease, pipeline is lagging.
C. Incorrect – Storage metric decrease doesn't apply to source monitoring.
D. Incorrect – Metrics swapped incorrectly, doesn't reflect processing slowdown.
E. Incorrect – Correct option exists (B).